



GOVERNMENT OF TAMIL NADU

HIGHER SECONDARY SECOND YEAR

PHYSICS

VOLUME - I

A publication under Free Textbook Programme of Government of Tamil Nadu

Department of School Education

Untouchability is Inhuman and a Crime

Government of Tamil Nadu

First Edition - 2019

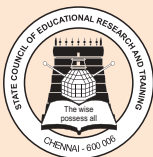
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E-book



Assessment

HOW TO USE THE BOOK ?

Scope of Physics

- Awareness on higher learning - courses, institutions and required competitive exams
- Financial assistance possible to help students to climb academic ladder



Learning Objectives:

- Overview of the unit
- Gives clarity on the goals and objective of the topics



- Additional facts related to the topics covered to facilitate curiosity driven learning

Example problems



- To ensure understanding, problems/illustrations are given at every stage before advancing to next level



ICT

- Visual representation of concepts with illustrations
- Videos, animations, and tutorials

- To harness the digital skills to class room learning and experimenting

Summary

- Recap of salient points of the lesson

Concept Map

- Schematic outline of salient learning of the unit

Evaluation

- Evaluate students' understanding and get them acquainted with the application of physical concepts to numerical and conceptual questions

Books for Reference

- List of relevant books for further reading

Practical

- List of practical and the description of each is appended for easy access.

Glossary

- Scientific terms frequently used with their Tamil equivalents

Content Focus

- Sequential understanding of the stationary charges, moving charges, electric current, magnetism and the interlink between electric and magnetic phenomena.

Back Wrapper: NIKOLA TESLA a Serbian-American Engineer (10 July 1856 – 7 January 1943)

Nikola Tesla made breakthroughs in the production, transmission and application of electric power. He invented the first alternating current (AC) motor and developed AC generation and transmission technology. In 1884, he was hired by Edison (discoverer of DC dynamos) later on Tesla became his competitor in this field.

Tesla conducted a range of experiments with mechanical oscillators/generators, electrical discharge tubes, and early X – ray imaging. Tesla in 1890 itself conducted research for wireless lighting and worldwide wireless electric power distribution in his high – voltage, high-frequency power experiments in New York. Unfortunately Tesla could not put his ideas in practical use due to lack of funds.

Scope of Physics - Higher Education



Entrance Examinations After +2

- Physics Olympiad Exam
- NEET-National Eligibility cum Entrance Test
- IIT JEE-Joint Entrance Examination(Mains & Advanced)
- NEET- National Entrance Screening Test
- KVPY-Kishore Vaigyanik Protsahan Yojana
- JEE Mains Paper II for B.Arch
- AIIMS - All Indian Institute of Medical Science's Examination
- Chemical Mathematical Institute Entrance Examination
- BITSAT- Birla Institute of Science And Technology Admission Test
- AIEEE – All India Engineering Entrance Exam
- CUET – Central Universities Common Entrance Test
- JIPMER - Jawaharlal Institute of Postgraduate Medical Education & Research
- CLAT – Common Law Admission Test
- HSEE- Humanities and Social Sciences Entrance Examination
- AIPVT -All India Pre-Veterinary Test
- NDA – National Defence Academy Examination

After Graduation

- JAM- Joint Admission Test
- JEST – Joint Entrance Screening Test
- GATE- Graduate Aptitude Test in Engineering
- CAT – Common Admission Test(for MBA)
- Exams conducted by Respective Universities

After Post Graduation

- CSIR - National Eligibility Test for JRF and Lectureship

After Completing +2

- Integrated Msc. Physics
- Central Universities through CUET
- Central Research Institutes like IISER using KVPY, JEE Advanced , IISER Aptitude Test
- Top 1% students in State board are eligible for IISER Aptitude Test
- Admission in NISER through NEST
- B.Sc Photonics
- B.Sc Hons in Mathematics and Physics in CMI
- B.Sc Hons in Mathematics and Computer Science in CMI
- Five-Year Dual degree In IIST (B.Tech + Master of Science)
- Master of Science (Astronomy and Astrophysics, Solid State Physics)

Note

- Students admitted to IISc, IIT's, NIT's
- IISER's, IIST, will get a Scholarship equivalent to INSPIRE
- Assured placement in ISRO and other divisions for the students of IIST
- Institutes and their ranking can be found in www.nirfindia.org

After B.Sc Physics

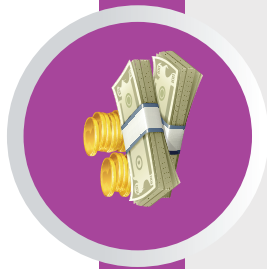
- M.Sc Physics in IIT's and NIT's through JAM
- Integrated Phd in IISER's and IISc through JAM and JEST
- M.Sc Physics in Central Universities through CUET
- M.Sc in Energy Physics, Applied Physics in IIT's through JAM
- Integrated Phd in IMSc, TIFR, JNCASR through JEST score
- Integrated Phd in TIFR through JEST and TIFR exam
- M.Sc Photonics, Reactor physics, Nuclear Engineering ,
- M.Sc Medical Physics
- M.Sc Biophysics
- Research Institutes in abroad like CERN, NASA, LIGO offer Summer internship programmes for motivated Indian students pursuing Undergraduate course in physics
- Indian Academy of Science & various other research institutes offer paid Summer Internship for science students to get an hands on experience in research.

Opportunities after B.Sc. Physics



Jobs in Government Sector

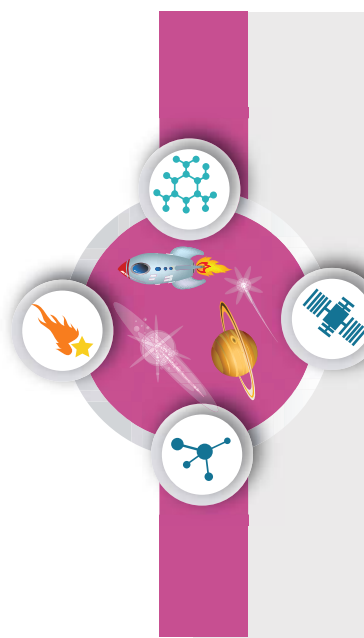
- Scientific Officer and Scientific Assistant Jobs
- CSIR Labs
- DRDO – Defence Research and Development Organisation
- DAE -Department of Atomic Energy
- DoS - Department of Science
- IMD- Indian Meteorological Department
- ONGC -Oil and Natural Gas Corporation
- ATC – Air Traffic Controller
- Teaching faculty in schools and colleges through SET, NET,TET
- Scientist post in various research institutes in India



Scholarships

- INSPIRE Scholarship - Scholarship for Higher Education(SHE) - 80000 per annum, for B.Sc/B.S/Int M.Sc/Int M.S
- Eligibility Criteria:- Top 1% students in their plus 2 board exam
- Top 10000 rank holders in JEE or NEET
- Students studying at IISER, IISER, Department of Atomic Energy Centre for Basic Science
- NTSE, KVPY, JBNSTS Scholars
- International Olympiad Medalists
- Indira Gandhi Scholarship for single girl child for full time regular Master's Degree
- Post Graduate Merit Scholarship for University rank holders in UG level
- Women Scientist Scheme (WOS -A)
- Eligibility Criteria:- Women who are pursuing M.Sc or Ph.D
- Mathematics Training and Talent Search(MTTS) Programme
- Eligibility Criteria:- Students who studied Maths at UG or PG level
- Dr. K S Krishnan Research Associateship (KSKRA)
- Eligibility Criteria:- Students who possess Master's Degree or Ph.D in science or engineering
- IGCAR -JRF
- Eligibility Criteria:- Passing JEST, GATE, NET Exams
- Promotion of Science Education (POSE) Scholarship Scheme
- Dhruvrai Ambani Scholarship Programme
- Foundation for Academic Excellence and Access Scholarship(FAEA)
- Central Sector Scheme of National Fellowship and Scholarship for Higher Education of ST students
- Pre - Matric and Post - Matric Scholarship for students belonging to minority communities to pursue their School and Collegiate education by the Ministry of Minority affairs, Government of India.
- Pre Matric and Post Matric Scholarship for students with Disabilities to pursue their School and Collegiate Education by the Department of Empowerment of Persons with Disabilities, Government of India.

Institutes in india to pursue research in physics



Research Areas

- General Relativity and Cosmology
- Astronomy and Astrophysics
- Quantum Optics and Information theory
- Plasma physics
- Meteorology and Atmospheric science
- String Theory, Quantum Gravity
- Optics and Photonics
- Condensed Matter Theory, Material Science and Spintronics
- Cryptography
- Mathematical Physics, Statistical Physics
- Crystal Growth and Crystallography
- Atomic and Molecular Physics
- Biophysics, Medical Physics
- Nuclear and High energy Particle Physics
- Energy and Environmental Studies
- Geophysics
- Quantum Biology and Quantum Thermodynamics and Cymatics

Famous Research Institutes for Physics in India		
Name of the Institution	Website	
Institute of Mathematical Sciences, Chennai (IMSc)	www.imsc.res.in	
Saha Institute of Nuclear Physics, Kolkata	www.saha.ac.in	
International Centre for Theoretical Sciences, Bangalore	www.icts.res.in	
Harish chandra Research Institute, Allahabad	www.hri.res.in	
Aryabhata Research Institute of Observational Sciences, Nainital	www.aries.res.in	
Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR)	www.jncasr.ac.in	
Institute of Physics (IOP), Bhubaneshwar	www.iopb.res.in	
Indian Association for the Cultivation of Sciences (IACS), Kolkata	www.iacs.res.in	
Vikram Sarabhai Space Centre (VSSC), Thiruvananthapuram	www.vssc.gov.in	
National Physical Laboratory (NPL), Delhi	www.nplindia.in	
National Institute of Science Education and Research (NISER), Bhubaneshwar	www.niser.ac.in	
Indian Institute of Science (IISc), Bangalore	www.iisc.ac.in	
Raman Research Institute (RRI), Bangalore	www.rri.res.in	
Tata Institute of Fundamental Research (TIFR)	www.tifr.res.in	
Bhaba Atomic Research Centre (BARC)	www.barc.gov.in	
Indira Gandhi Centre for Atomic Research (IGCAR)	www.igcar.gov.in	
Inter University Centre for Astronomy and Astrophysics (IUCAA), Pune	www.iucaa.in	
Indian Institute of Space Science and Technology (IIST), Trivandrum	www.iist.ac.in	
Institute of Plasma Research (IPR), Gujarat	www.ipr.res.in	
Physical Research Laboratory (PRL), Ahmedabad	www.prl.res.in	
Inter-University Accelerator Center (IUAC)	www.iuac.res.in	
Indian Institute of Astrophysics (IIA), Bangalore	www.iiap.res.in	
Chennai Mathematical Institute (CMI), Chennai	www.cmi.ac.in	
Liquid Propulsion Systems Centre	www.lpsc.gov.in	
S.N.Bose Centre for Basic Sciences	www.bose.res.in	
CSIR National laboratories		
Indian Institute of Technology (IIT) in various places		
IISER's in various places		
National Institute of Technology (NIT) in various places		
Indian Institute of Information Technology (IIITs) at various places		
Central and State Universities		



UNIT 1

ELECTROSTATICS

Electricity is really just organized lightning
– George Carlin

LEARNING OBJECTIVES

In this unit, student is exposed to

- Historical background of electricity and magnetism
- The role of electrostatic force in day – to-day life
- Coulomb's law and superposition principle
- The concept of electric field
- Calculation of electric field for various charge configurations
- Electrostatic potential and electrostatic potential energy
- Electric dipole and dipole moment
- Electric field and electrostatic potential for a dipole
- Electric flux
- Gauss law and its various applications
- Electrostatic properties of conductors and dielectrics
- Polarisation
- Capacitors in series and parallel combinations
- Effect of a dielectric in a capacitor
- Distribution of charges in conductors, corona discharge
- Working of a Van de Graaff generator



1.1

INTRODUCTION

Electromagnetism is one of the most important branches of physics. The technological developments of the modern 21st century are primarily due to our understanding of electromagnetism. The forces we experience in everyday life are electromagnetic in nature except gravity.

In standard XI, we studied about the gravitational force, tension, friction, normal force etc. Newton treated them to be independent of each other with each force being a separate natural force. But what is the origin of all these forces? It is now understood that except gravity, all forces which we experience in every day life (tension in the string, normal force from the surface, friction etc.) arise from electromagnetic forces within the atoms. Some examples are



(i) When an object is pushed, the atoms in our hand interact with the atoms in the object and this interaction is basically electromagnetic in nature.

(ii) When we stand on Earth's surface, the gravitational force on us acts downwards and the normal force acts upward to counter balance the gravitational force. What is the origin of this normal force?

It arises due to the electromagnetic interaction of atoms on the surface of the Earth with the atoms present in the feet of the person. Though, we are attracted by the gravitational force of the Earth, we stand on Earth only because of electromagnetic force of atoms.

(iii) When an object is moved on a surface, static friction resists the motion of the object. This static friction arises due to electromagnetic interaction between the atoms present in the object and atoms on the surface. Kinetic friction also has similar origin.

From these examples, it is clear that understanding electromagnetism is very essential to understand the universe in a holistic manner. The basic principles of electromagnetism are dealt with in volume 1 at XII standard physics. This unit deals with the behaviour and other related phenomena of charges at rest. This **branch of electricity which deals with stationary charges is called Electrostatics.**

1.1.1 Historical background of electric charges

Two millennia ago, Greeks noticed that amber (a solid, translucent material formed from the resin of a fossilized tree)

after rubbing with animal fur attracted small pieces of leaves and dust. The amber possessing this property is said to be 'charged'. It was initially thought that amber has this special property. Later people found that not only amber but even a glass rod rubbed with silk cloth, attracts pieces of papers. So glass rod also becomes 'charged' when rubbed with a suitable material.

Consider a charged rubber rod hanging from a thread as shown in Figure 1.1. Suppose another charged rubber rod is brought near the first rubber rod; the rods repel each other. Now if we bring a charged glass rod close to the charged rubber rod, they attract each other. At the same time, if a charged glass rod is brought near another charged glass rod, both the rods repel each other.

From these observations, the following inferences are made

- (i) The charging of rubber rod and that of glass rod are different from one another.
- (ii) The charged rubber rod repels another charged rubber rod, which implies that 'like charges repel each other'. We can also arrive at the same inference by observing that a charged glass rod repels another charged glass rod.
- (iii) The charged rubber rod attracts the charged glass rod, implying that the charge in the glass rod is not the same kind of charge present in the rubber. Thus unlike charges attract each other.

Therefore, two kinds of charges exist in the universe. In the 18th century, Benjamin Franklin called one type of charge as positive (+) and another type of charge as negative (-). Based on Franklin's convention, rubber and amber rods are negatively charged while the glass rod is positively charged. **If the net charge is zero in the object, it is said to be electrically neutral.**

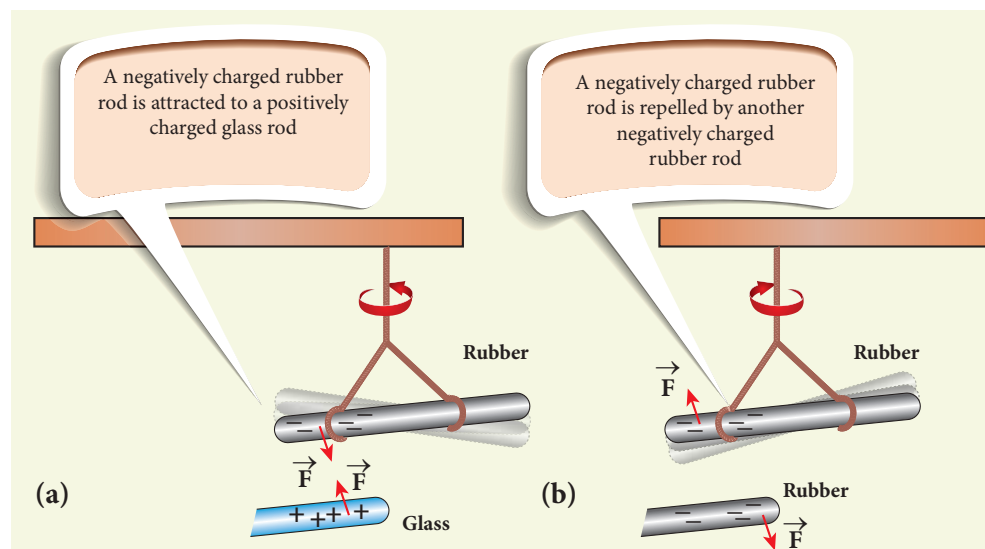


Figure 1.1 (a) Unlike charges attract each other (b) Like charges repel each other

Following the pioneering work of J. J. Thomson and E. Rutherford, in the late 19th century and in the beginning of 20th century, we now understand that the atom is electrically neutral and is made up of the negatively charged electrons, positively charged protons, and neutrons which have zero charge. The material objects made up of atoms are neutral in general. When an object is rubbed with another object (for example rubber with silk cloth), some amount of charge is transferred from one object to another due to the friction between them and the object is then said to be electrically charged. **Charging the objects through rubbing is called triboelectric charging.**

1.1.2 Basic properties of charges

(i) Electric charge

Most objects in the universe are made up of atoms, which in turn are made up of protons, neutrons and electrons. These particles have mass, an inherent property of particles. Similarly, the electric charge

is another intrinsic and fundamental property of particles. The nature of charges is understood through various experiments performed in the 19th and 20th century. The SI unit of charge is coulomb.

(ii) Conservation of charges

Benjamin Franklin argued that when one object is rubbed with another object, charges get transferred from one to the other. Before rubbing, both objects are electrically neutral and rubbing simply transfers the charges from one object to the other. (For example, when a glass rod is rubbed against silk cloth, some negative charge are transferred from glass to silk. As a result, the glass rod is positively charged and silk cloth becomes negatively charged).

From these observations, he concluded that charges are neither created nor destroyed but can only be transferred from one object to other. This is called conservation of total charges and is one of the fundamental conservation laws in physics. It is stated more generally in the following way.

The total electric charge in the universe is constant and charge can neither be created nor be destroyed. In any physical

process, the net change in charge will always be zero.

(iii) Quantisation of charges

What is the smallest amount of charge that can be found in nature? Experiments show that the charge on an electron is $-e$ and the charge on the proton is $+e$. Here, e denotes the fundamental unit of charge. The charge q on any object is equal to an integral multiple of this fundamental unit of charge e .

$$q = ne \quad (1.1)$$

Here n is any integer ($0, \pm 1, \pm 2, \pm 3, \pm 4, \dots$). This is called quantisation of electric charge.

Robert Millikan in his famous experiment found that the value of $e = 1.6 \times 10^{-19}$ C. The charge of an electron is -1.6×10^{-19} C and the charge of the proton is $+1.6 \times 10^{-19}$ C.

When a glass rod is rubbed with silk cloth, the number of charges transferred is usually very large, typically of the order of 10^{10} . So the charge quantisation is not appreciable at the macroscopic level. Hence the charges are treated to be continuous (not discrete). But at the microscopic level, quantisation of charge plays a vital role.

EXAMPLE 1.1

Calculate the number of electrons in one coulomb of negative charge.

Solution

According to the quantisation of charge,

$$q = ne$$

Here $q = 1$ C. So the number of electrons in 1 coulomb of charge is

$$n = \frac{q}{e} = \frac{1\text{C}}{1.6 \times 10^{-19}} = 6.25 \times 10^{18} \text{ electrons}$$

1.2

COULOMB'S LAW

In the year 1786, Coulomb deduced the expression for the force between two stationary point charges in vacuum or free space. Consider two point charges q_1 and q_2 at rest in vacuum, and separated by a distance of r as shown in Figure 1.2. According to Coulomb, the force on the point charge q_2 exerted by another point charge q_1 is

$$\vec{F}_{21} = k \frac{q_1 q_2}{r^2} \hat{r}_{12} \quad (1.2)$$

where $\hat{r}_{12}$ is the unit vector directed from charge q_1 to charge q_2 and k is the proportionality constant.

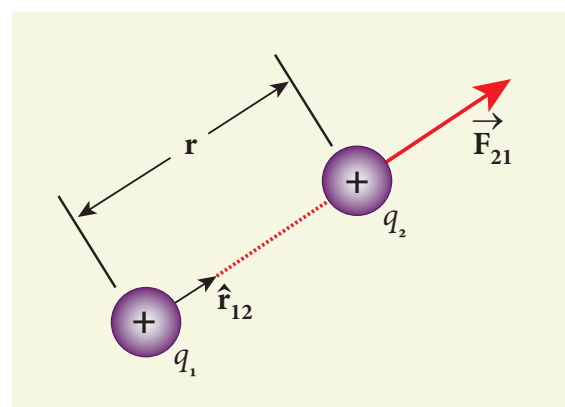


Figure 1.2 Coulomb force between two positive point charges

Important aspects of Coulomb's law

(i) Coulomb's law states that the electrostatic force is directly proportional to the product of the magnitude of the two point charges and is inversely proportional to the square of the distance between the two point charges.

(ii) The force on the charge q_2 exerted by the charge q_1 always lies along the line



joining the two charges. $\hat{r}_{12}$ is the unit vector pointing from charge q_1 to q_2 . It is shown in the Figure 1.2. Likewise, the force on the charge q_1 exerted by q_2 is along $-\hat{r}_{12}$ (i.e., in the direction opposite to $\hat{r}_{12}$).

(iii) In SI units, $k = \frac{1}{4\pi\epsilon_0}$ and its value is $9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$. Here ϵ_0 is the permittivity of free space or vacuum and its value is

$$\epsilon_0 = \frac{1}{4\pi k} = 8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}.$$

(iv) The magnitude of the electrostatic force between two charges each of one coulomb and separated by a distance of 1 m is calculated as follows:

$$|F| = \frac{9 \times 10^9 \times 1 \times 1}{1^2} = 9 \times 10^9 \text{ N}.$$

This is a huge quantity, almost equivalent to the weight of one million ton. We never come across 1 coulomb of charge in practice. Most of the electrical phenomena in day-to-day life involve electrical charges of the order of μC (micro coulomb) or nC (nano coulomb).

(v) In SI units, Coulomb's law in vacuum takes the form $\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{12}$. In a medium of permittivity ϵ , the force between two point charges is given by $\vec{F}_{21} = \frac{1}{4\pi\epsilon} \frac{q_1 q_2}{r^2} \hat{r}_{12}$. Since $\epsilon > \epsilon_0$, the force between two point charges in a medium other than vacuum is always less than that in vacuum. We define the relative permittivity for a given medium as $\epsilon_r = \frac{\epsilon}{\epsilon_0}$. For vacuum or air, $\epsilon_r = 1$ and for all other media $\epsilon_r > 1$.

(vi) Coulomb's law has same structure as Newton's law of gravitation. Both are inversely proportional to the square of the distance between the particles. The electrostatic force is directly proportional to the product of the magnitude of two point charges and gravitational force is directly proportional to the product of two masses. But there are some important differences between these two laws.

- The gravitational force between two masses is always attractive but Coulomb force between two charges can be attractive or repulsive, depending on the nature of charges.

- The value of the gravitational constant $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$. The value of the constant k in Coulomb law is $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$. Since k is much more greater than G , the electrostatic force is always greater in magnitude than gravitational force for smaller size objects.

- The gravitational force between two masses is independent of the medium. For example, if 1 kg of two masses are kept in air or inside water, the gravitational force between two masses remains the same. But the electrostatic force between the two charges depends on nature of the medium in which the two charges are kept at rest.

(vii) The force on a charge q_1 exerted by a point charge q_2 is given by

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{21}$$

Here $\hat{r}_{21}$ is the unit vector from charge q_2 to q_1 .

But $\hat{r}_{21} = -\hat{r}_{12}$,

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} (-\hat{r}_{12}) = -\frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} (\hat{r}_{12})$$

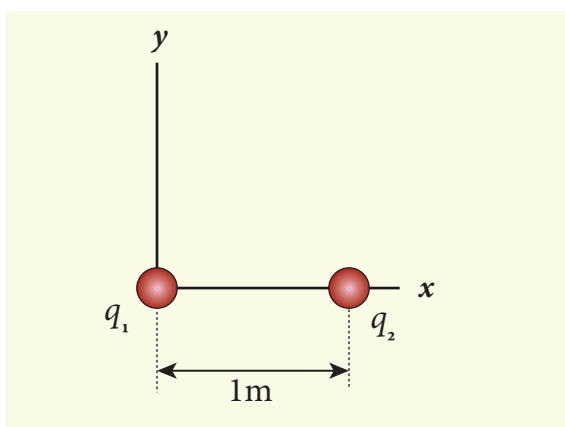
$$(\text{or}) \vec{F}_{12} = -\vec{F}_{21}$$

Therefore, the electrostatic force obeys Newton's third law.

(viii) The expression for Coulomb force is true only for point charges. But the point charge is an ideal concept. However we can apply Coulomb's law for two charged objects whose sizes are very much smaller than the distance between them. In fact, Coulomb discovered his law by considering the charged spheres in the torsion balance as point charges. The distance between the two charged spheres is much greater than the radii of the spheres.

EXAMPLE 1.2

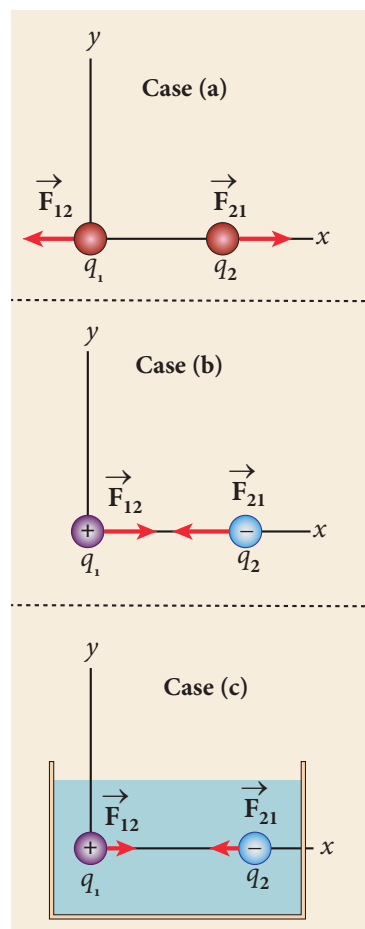
Consider two point charges q_1 and q_2 at rest as shown in the figure.



They are separated by a distance of 1m. Calculate the force experienced by the two charges for the following cases:

- (a) $q_1 = +2 \mu\text{C}$ and $q_2 = +3 \mu\text{C}$
- (b) $q_1 = +2 \mu\text{C}$ and $q_2 = -3 \mu\text{C}$
- (c) $q_1 = +2 \mu\text{C}$ and $q_2 = -3 \mu\text{C}$ kept in water ($\epsilon_r = 80$)

Solution



- (a) $q_1 = +2 \mu\text{C}$, $q_2 = +3 \mu\text{C}$, and $r = 1\text{m}$. Both are positive charges. so the force will be repulsive.

Force experienced by the charge q_2 due to q_1 is given by

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{12}$$

Here $\hat{r}_{12}$ is the unit vector from q_1 to q_2 . Since q_2 is located on the right of q_1 , we have

$$\hat{r}_{12} = \hat{i} \text{ and } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ so that}$$

$$\begin{aligned} \vec{F}_{21} &= \frac{9 \times 10^9 \times 2 \times 10^{-6} \times 3 \times 10^{-6}}{1^2} \hat{i} \\ &= 54 \times 10^{-3} \hat{i} \text{ N} \end{aligned}$$



According to Newton's third law, the force experienced by the charge q_1 due to q_2 is $\vec{F}_{12} = -\vec{F}_{21}$. Therefore,

$$\vec{F}_{12} = -54 \times 10^{-3} \hat{i} \text{ N.}$$

The directions of $\vec{F}_{21}$ and $\vec{F}_{12}$ are shown in the above figure in case (a)

(b) $q_1 = +2 \mu\text{C}$, $q_2 = -3 \mu\text{C}$, and $r = 1\text{m}$. They are unlike charges. So the force will be attractive.

Force experienced by the charge q_2 due to q_1 is given by

$$\begin{aligned}\vec{F}_{21} &= \frac{9 \times 10^9 \times (2 \times 10^{-6}) \times (-3 \times 10^{-6})}{1^2} \hat{r}_{12} \\ &= -54 \times 10^{-3} \text{ N } \hat{i} \text{ (Using } \hat{r}_{12} = \hat{i} \text{)}\end{aligned}$$

The charge q_2 will experience an attractive force towards q_1 which is in the negative x direction.

According to Newton's third law, the force experienced by the charge q_1 due to q_2 is $\vec{F}_{12} = -\vec{F}_{21}$. Therefore,

$$\vec{F}_{12} = 54 \times 10^{-3} \hat{i} \text{ N}$$

The directions of $\vec{F}_{21}$ and $\vec{F}_{12}$ are shown in the figure (case (b)).

(c) If these two charges are kept inside the water, then the force experienced by q_2 due to q_1

$$\vec{F}_{21}^w = \frac{1}{4\pi\epsilon} \frac{q_1 q_2}{r^2} \hat{r}_{12}$$

since $\epsilon = \epsilon_r \epsilon_o$

$$\text{we have } \vec{F}_{21}^w = \frac{1}{4\pi\epsilon_r \epsilon_o} \frac{q_1 q_2}{r^2} \hat{r}_{12} = \frac{\vec{F}_{21}}{\epsilon_r}$$

Therefore,

$$\vec{F}_{21}^w = -\frac{54 \times 10^{-3} \text{ N}}{80} \hat{i} = -0.675 \times 10^{-3} \text{ N } \hat{i}$$



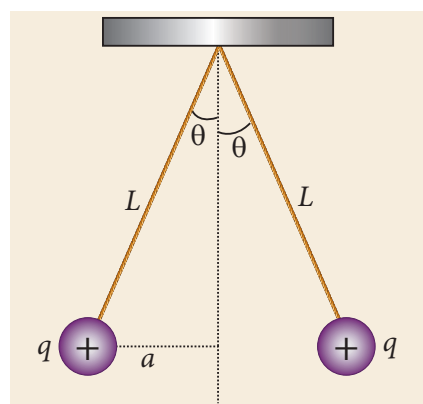
Note Note that the strength of the force between the two charges in water is reduced by 80 times compared to the force between the same two charges in vacuum.

When common salt (NaCl) is taken in water, the electrostatic force between Na and Cl ions is reduced due to the high relative permittivity of water ($\epsilon_r = 80$). This is the reason water acts as a good solvent.

EXAMPLE 1.3

Two small-sized identical equally charged spheres, each having mass 1 g are hanging in equilibrium as shown in the figure. The length of each string is 10 cm and the angle θ is 30° with the vertical. Calculate the magnitude of the charge in each sphere.

(Take $g = 10 \text{ ms}^{-2}$)

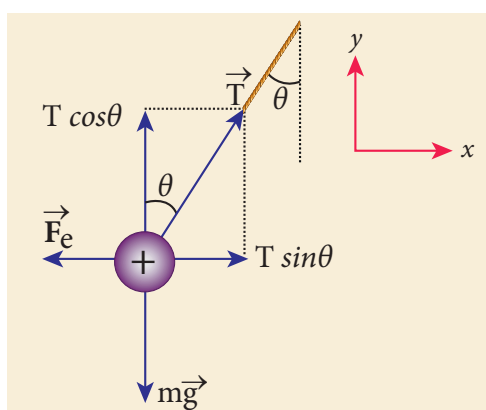


Solution

If the two spheres are neutral, the angle between them will be 0° when

hanged vertically. Since they are positively charged spheres, there will be a repulsive force between them and they will be at equilibrium with each other at an angle of 30° with the vertical. At equilibrium, each charge experiences zero net force in each direction. We can draw a free body diagram for one of the charged spheres and apply Newton's second law for both vertical and horizontal directions.

The free body diagram is shown below.



In the x -direction, the acceleration of the charged sphere is zero.

Using Newton's second law ($\vec{F}_{tot} = m\vec{a}$), we have

$$\begin{aligned} T \sin \theta \hat{i} - F_e \hat{i} &= 0 \\ T \sin \theta &= F_e \end{aligned} \quad (1)$$

Here T is the tension acting on the charge due to the string and F_e is the electrostatic force between the two charges.

In the y -direction also, the net acceleration experienced by the charge is zero.

$$\begin{aligned} T \cos \theta \hat{j} - mg \hat{j} &= 0 \\ T \cos \theta &= mg \end{aligned} \quad (2)$$

By dividing equation (1) by equation (2),

$$\tan \theta = \frac{F_e}{mg} \quad (3)$$

Since they are equally charged, the magnitude of the electrostatic force is

$$F_e = k \frac{q^2}{r^2} \quad \text{where } k = \frac{1}{4\pi\epsilon_0}$$

Here $r = 2a = 2L \sin \theta$. By substituting these values in equation (3),

$$\tan \theta = k \frac{q^2}{mg(2L \sin \theta)^2} \quad (4)$$

Rearranging the equation (4) to get q

$$\begin{aligned} q &= 2L \sin \theta \sqrt{\frac{mg \tan \theta}{k}} \\ &= 2 \times 0.1 \times \sin 30^\circ \times \sqrt{\frac{10^{-3} \times 10 \times \tan 30^\circ}{9 \times 10^9}} \\ q &= 8.01 \times 10^{-8} \text{ C} = 80.1 \text{ nC} \end{aligned}$$

EXAMPLE 1.4

Calculate the electrostatic force and gravitational force between the proton and the electron in a hydrogen atom. They are separated by a distance of $5.3 \times 10^{-11} \text{ m}$. The magnitude of charges on the electron and proton are $1.6 \times 10^{-19} \text{ C}$. Mass of the electron is $m_e = 9.1 \times 10^{-31} \text{ kg}$ and mass of proton is $m_p = 1.6 \times 10^{-27} \text{ kg}$.

Solution

The proton and the electron attract each other. The magnitude of the electrostatic force between these two particles is given by

$$\begin{aligned} F_e &= \frac{ke^2}{r^2} = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2} \\ &= \frac{9 \times 2.56}{28.09} \times 10^{-7} = 8.2 \times 10^{-8} \text{ N} \end{aligned}$$

The gravitational force between the proton and the electron is attractive. The magnitude of the gravitational force between these particles is

$$F_G = \frac{Gm_em_p}{r^2}$$

$$= \frac{6.67 \times 10^{-11} \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-27}}{(5.3 \times 10^{-11})^2}$$

$$= \frac{97.11}{28.09} \times 10^{-47} = 3.4 \times 10^{-47} \text{ N}$$

The ratio of the two forces

$$\frac{F_e}{F_G} = \frac{8.2 \times 10^{-8}}{3.4 \times 10^{-47}} = 2.41 \times 10^{39}$$

Note that $F_e \approx 10^{39} F_G$

The electrostatic force between a proton and an electron is enormously greater than the gravitational force between them. Thus the gravitational force is negligible when compared with the electrostatic force in many situations such as for small size objects and in the atomic domain. This is the reason why a charged comb attracts an uncharged piece of paper with greater force even though the piece of paper is attracted downward by the Earth. This is shown in Figure 1.3

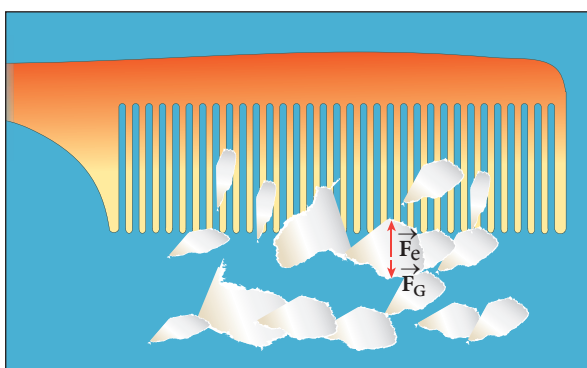


Figure 1.3 Electrostatic attraction between a comb and pieces of papers

1.2.1 Superposition principle

Coulomb's law explains the interaction between two point charges. If there are more than two charges, the force on one charge due to all the other charges needs to be calculated. Coulomb's law alone does not give the answer. The superposition principle explains the interaction between multiple charges.

According to this superposition principle, **the total force acting on a given charge is equal to the vector sum of forces exerted on it by all the other charges.**

Consider a system of n charges, namely $q_1, q_2, q_3, \dots, q_n$. The force on q_1 exerted by the charge q_2

$$\vec{F}_{12} = k \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21}$$

where $\hat{r}_{21}$ is the unit vector from q_2 to q_1 along the line joining the two charges and r_{21} is the distance between the charges q_1 and q_2 . The electrostatic force between two charges is not affected by the presence of other charges in the neighbourhood.

The force on q_1 exerted by the charge q_3 is

$$\vec{F}_{13} = k \frac{q_1 q_3}{r_{31}^2} \hat{r}_{31}$$

By continuing this, the total force acting on the charge q_1 due to all other charges is given by

$$\vec{F}_1^{tot} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14} + \dots + \vec{F}_{1n}$$

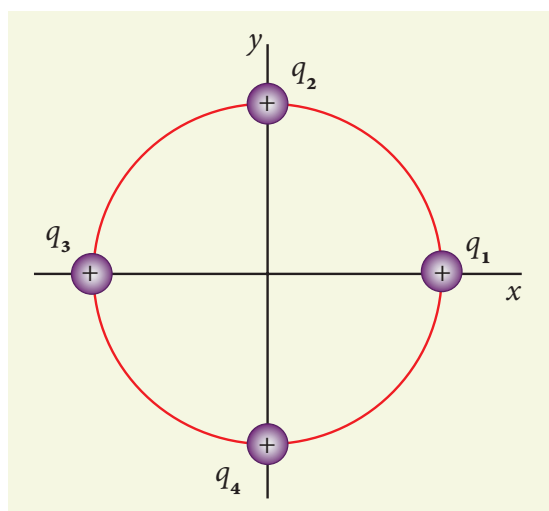
$$\vec{F}_1^{tot} = k \left\{ \frac{q_1 q_2}{r_{21}^2} \hat{r}_{21} + \frac{q_1 q_3}{r_{31}^2} \hat{r}_{31} + \frac{q_1 q_4}{r_{41}^2} \hat{r}_{41} + \dots + \frac{q_1 q_n}{r_{n1}^2} \hat{r}_{n1} \right\} \quad (1.3)$$

Note

Without the superposition principle, Coulomb's law will be incomplete when applied to more than two charges. Both the superposition principle and Coulomb's law form fundamental principles of electrostatics and explain all the phenomena in electrostatics. But they are not derivable from each other.

EXAMPLE 1.5

Consider four equal charges q_1, q_2, q_3 and $q_4 = q = +1 \mu\text{C}$ located at four different points on a circle of radius 1m, as shown in the figure. Calculate the total force acting on the charge q_1 due to all the other charges.

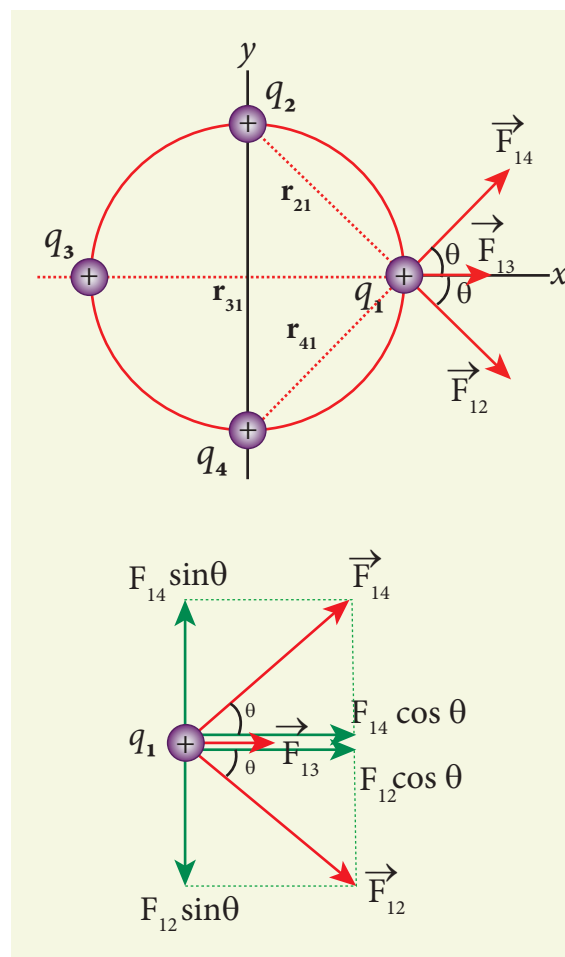


Solution

According to the superposition principle, the total electrostatic force on charge q_1 is the vector sum of the forces due to the other charges,

$$\vec{F}_1^{\text{tot}} = \vec{F}_{12} + \vec{F}_{13} + \vec{F}_{14}$$

The following diagram shows the direction of each force on the charge q_1 .



The charges q_2 and q_4 are equidistant from q_1 . As a result the strengths (magnitude) of the forces $\vec{F}_{12}$ and $\vec{F}_{14}$ are the same even though their directions are different. Therefore the vectors representing these two forces are drawn with equal lengths. But the charge q_3 is located farther compared to q_2 and q_4 . Since the strength of the electrostatic force decreases as distance increases, the strength of the force $\vec{F}_{13}$ is lesser than that of forces $\vec{F}_{12}$ and $\vec{F}_{14}$. Hence the vector representing the force $\vec{F}_{13}$ is drawn with smaller length compared to that for forces $\vec{F}_{12}$ and $\vec{F}_{14}$.

From the figure, $r_{21} = \sqrt{2} \text{ m} = r_{41}$ and $r_{31} = 2 \text{ m}$

The magnitudes of the forces are given by

$$F_{13} = \frac{kq^2}{r_{31}^2} = \frac{9 \times 10^9 \times 10^{-12}}{4}$$



$$F_{13} = 2.25 \times 10^{-3} \text{ N}$$

$$F_{12} = \frac{kq^2}{r_{21}^2} = F_{14} = \frac{9 \times 10^9 \times 10^{-12}}{2} \\ = 4.5 \times 10^{-3} \text{ N}$$

From the figure, the angle $\theta = 45^\circ$. In terms of the components, we have

$$\begin{aligned}\vec{F}_{12} &= F_{12} \cos \theta \hat{i} - F_{12} \sin \theta \hat{j} \\ &= 4.5 \times 10^{-3} \times \frac{1}{\sqrt{2}} \hat{i} - 4.5 \times 10^{-3} \times \frac{1}{\sqrt{2}} \hat{j} \\ \vec{F}_{13} &= F_{13} \hat{i} = 2.25 \times 10^{-3} \text{ N} \hat{i} \\ \vec{F}_{14} &= F_{14} \cos \theta \hat{i} + F_{14} \sin \theta \hat{j} \\ &= 4.5 \times 10^{-3} \times \frac{1}{\sqrt{2}} \hat{i} + 4.5 \times 10^{-3} \times \frac{1}{\sqrt{2}} \hat{j}\end{aligned}$$

Then the total force on q_1 is

$$\begin{aligned}\vec{F}_1^{tot} &= (F_{12} \cos \theta \hat{i} - F_{12} \sin \theta \hat{j}) + F_{13} \hat{i} \\ &\quad + (F_{14} \cos \theta \hat{i} + F_{14} \sin \theta \hat{j}) \\ \vec{F}_1^{tot} &= (F_{12} \cos \theta + F_{13} + F_{14} \cos \theta) \hat{i} \\ &\quad + (-F_{12} \sin \theta + F_{14} \sin \theta) \hat{j}\end{aligned}$$

Since $F_{12} = F_{14}$, the j^{th} component is zero. Hence we have

$$\vec{F}_1^{tot} = (F_{12} \cos \theta + F_{13} + F_{14} \cos \theta) \hat{i}$$

substituting the values in the above equation,

$$\begin{aligned}&= \left(\frac{4.5}{\sqrt{2}} + 2.25 + \frac{4.5}{\sqrt{2}} \right) \times 10^{-3} \hat{i} \\ &= (4.5\sqrt{2} + 2.25) \times 10^{-3} \hat{i} \\ \vec{F}_1^{tot} &= 8.61 \times 10^{-3} \hat{i} \text{ N}\end{aligned}$$

The resultant force is along the positive x axis.

1.3

ELECTRIC FIELD AND ELECTRIC FIELD LINES

1.3.1 Electric Field

The interaction between two charges is determined by Coulomb's law. How does the interaction itself occur? Consider a point charge kept at a point in space. If another point charge is placed at some distance from the first point charge, it experiences either an attractive force or repulsive force. This is called 'action at a distance'. But how does the second charge know about existence of the first charge which is located at some distance away from it? To answer this question, Michael Faraday introduced the **concept of field**.

According to Faraday, every charge in the universe creates an electric field in the surrounding space, and if another charge is brought into its field, it will interact with the electric field at that point and will experience a force. It may be recalled that the interaction of two masses is similarly explained using the concept of gravitational field (Refer unit 6, volume 2, XI physics). Both the electric and gravitational forces are non-contact forces, hence the field concept is required to explain action at a distance.

Consider a source point charge q located at a point in space. Another point charge q_0 (test charge) is placed at some point P which is at a distance r from the charge q . The electrostatic force experienced by the charge q_0 due to q is given by Coulomb's law.

$$\vec{F} = \frac{kqq_0}{r^2} \hat{r} = \frac{1}{4\pi\epsilon_0} \frac{qq_0}{r^2} \hat{r} \text{ where } k = \frac{1}{4\pi\epsilon_0}$$

The charge q creates an electric field in the surrounding space within which its effect can be felt by another charge. It is measured in

terms of a quantity called electric field intensity or simply called electric field $\vec{E}$. **The electric field at the point P at a distance r from the point charge q is defined as the force that would be experienced by a unit positive charge placed at that point P and is given by**

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{kq}{r^2} \hat{r} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (1.4)$$

Here $\hat{r}$ is the unit vector pointing from q to the point of interest P. The electric field is a vector quantity and its SI unit is newton per coulomb (NC^{-1}).

Important aspects of Electric field

(i) If the charge q is positive then the electric field points away from the source charge and if q is negative, the electric field points towards the source charge q . This is shown in the Figure 1.4.

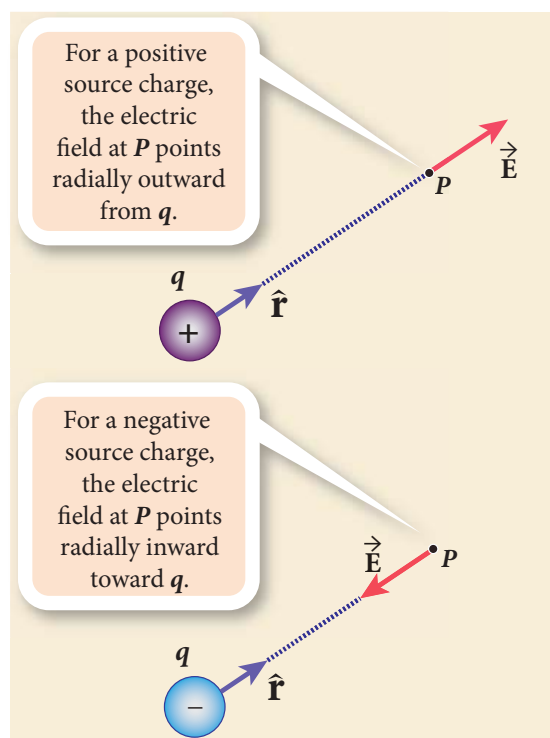


Figure 1.4 Electric field of positive and negative charges

(ii) If the electric field at a point P is $\vec{E}$, then the force experienced by the test charge q_0 placed at the point P is

$$\vec{F} = q_0 \vec{E} \quad (1.5)$$

This is Coulomb's law in terms of electric field. This is shown in Figure 1.5

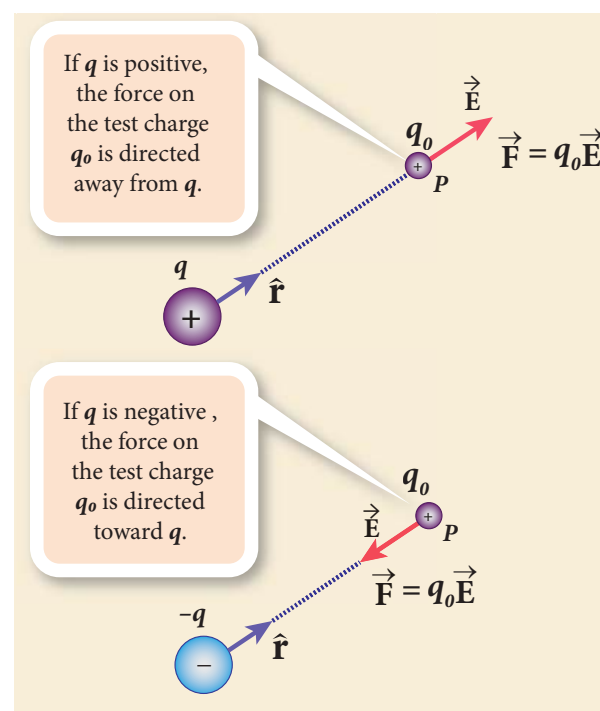


Figure 1.5 Coulomb's law in terms of electric field

(iii) The equation (1.4) implies that the electric field is independent of the test charge q_0 and it depends only on the source charge q .

(iv) Since the electric field is a vector quantity, at every point in space, this field has unique direction and magnitude as shown in Figures 1.6(a) and (b). From equation (1.4), we can infer that as distance increases, the electric field decreases in magnitude.

Note that in Figures 1.6 (a) and (b) the length of the electric field vector is shown for three different points. The strength or magnitude of the electric field at point P is stronger than at the points Q and R because the point P is closer to the source charge.

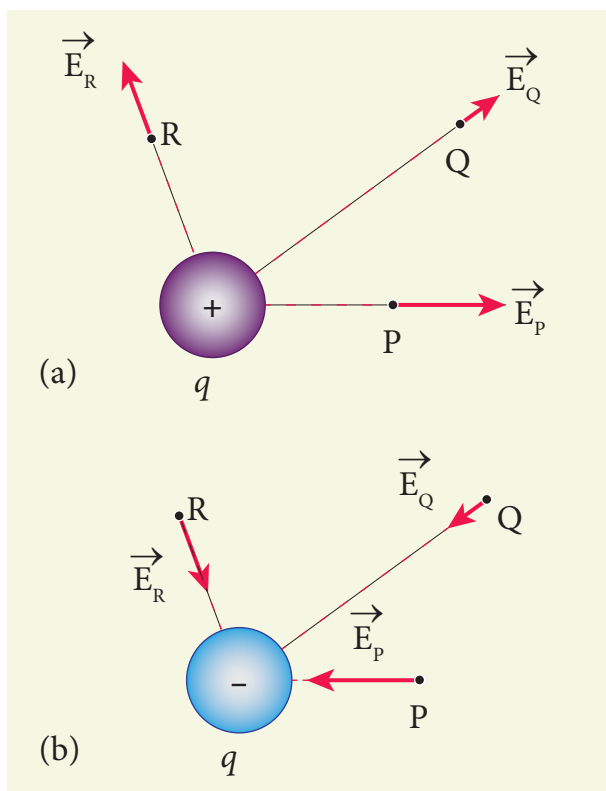


Figure 1.6 (a) Electric field due to positive charge (b) Electric field due to negative charge

(v) In the definition of electric field, it is assumed that the test charge q_0 is taken sufficiently small, so that bringing this test charge will not move the source charge. In other words, the test charge is made sufficiently small such that it will not modify the electric field of the source charge.

(vi) The expression (1.4) is valid only for point charges. For continuous and finite size charge distributions, integration techniques must be used (Refer Appendix A1.1). However, this expression can be used as an approximation for a finite-sized charge if the test point is very far away from the finite sized source charge. Note that we similarly treat the Earth as a point mass when we calculate the gravitational field of the Sun on the Earth (Refer unit 6, volume 2, XI physics).

(vii) There are two kinds of the electric field: uniform (constant) electric field and non-uniform electric field. Uniform electric field will have the same direction and constant magnitude at all points in space. Non-uniform electric field will have different directions or different magnitudes or both at different points in space. The electric field created by a point charge is basically a non uniform electric field. This non-uniformity arises, both in direction and magnitude, with the direction being radially outward (or inward) and the magnitude changes as distance increases. These are shown in Figure 1.7.

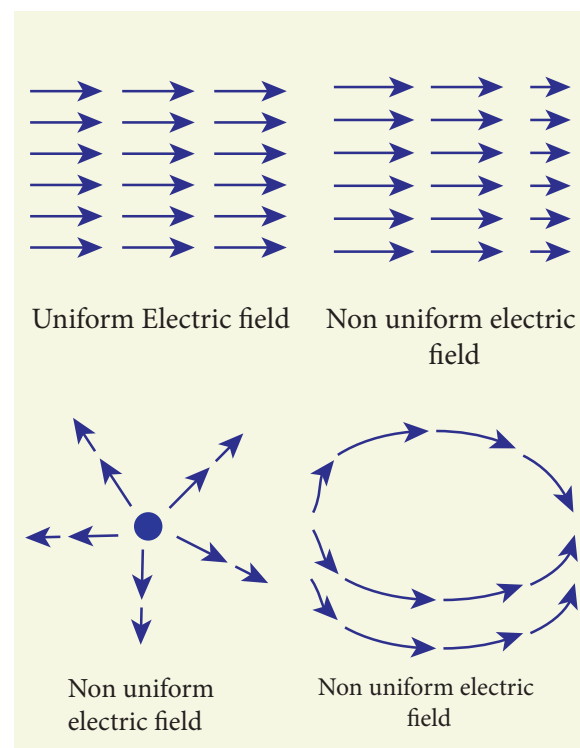


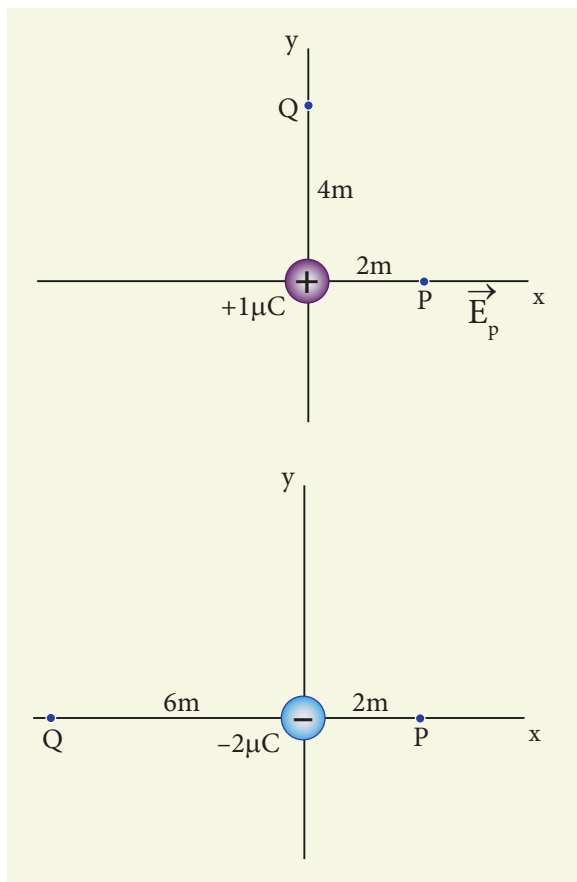
Figure 1.7 Uniform and non-uniform electric field

EXAMPLE 1.6

Calculate the electric field at points P, Q for the following two cases, as shown in the figure.



- (a) A positive point charge $+1 \mu\text{C}$ is placed at the origin
- (b) A negative point charge $-2 \mu\text{C}$ is placed at the origin



Solution

Case (a)

The magnitude of the electric field at point P is

$$E_p = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{9 \times 10^9 \times 1 \times 10^{-6}}{4} = 2.25 \times 10^3 \text{ NC}^{-1}$$

Since the source charge is positive, the electric field points away from the charge. So the electric field at the point P is given by

$$\vec{E}_p = 2.25 \times 10^3 \hat{i} \text{ NC}^{-1}$$

For the point Q

$$|\vec{E}_Q| = \frac{9 \times 10^9 \times 1 \times 10^{-6}}{16} = 0.56 \times 10^3 \text{ NC}^{-1}$$

Hence $\vec{E}_Q = 0.56 \times 10^3 \hat{j} \text{ NC}^{-1}$

Case (b)

The magnitude of the electric field at point P

$$|\vec{E}_p| = \frac{kq}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{9 \times 10^9 \times 2 \times 10^{-6}}{4} = 4.5 \times 10^3 \text{ NC}^{-1}$$

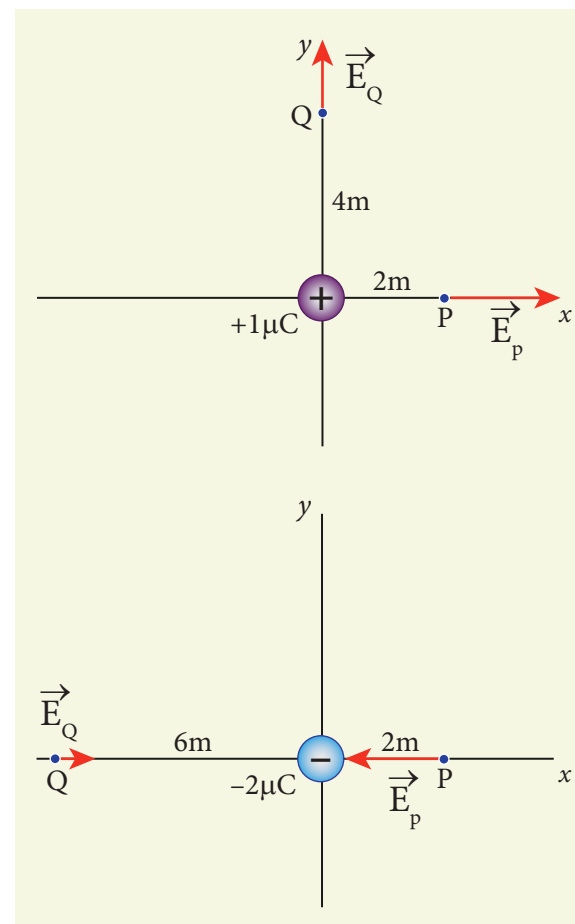
Since the source charge is negative, the electric field points towards the charge. So the electric field at the point P is given by

$$\vec{E}_p = -4.5 \times 10^3 \hat{i} \text{ NC}^{-1}$$

$$\text{For the point Q, } |\vec{E}_Q| = \frac{9 \times 10^9 \times 2 \times 10^{-6}}{36} = 0.5 \times 10^3 \text{ NC}^{-1}$$

$$\vec{E}_Q = 0.5 \times 10^3 \hat{i} \text{ NC}^{-1}$$

At the point Q the electric field is directed along the positive x-axis.



1.3.2 Electric field due to the system of point charges

Suppose a number of point charges are distributed in space. To find the electric field at some point P due to this collection of point charges, superposition principle is used. **The electric field at an arbitrary point due to a collection of point charges is simply equal to the vector sum of the electric fields created by the individual point charges. This is called superposition of electric fields.**

Consider a collection of point charges $q_1, q_2, q_3, \dots, q_n$ located at various points in space. The total electric field at some point P due to all these n charges is given by

$$\vec{E}_{tot} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots + \vec{E}_n \quad (1.6)$$

$$\vec{E}_{tot} = \frac{1}{4\pi\epsilon_0} \left\{ \frac{q_1}{r_{1P}^2} \hat{r}_{1P} + \frac{q_2}{r_{2P}^2} \hat{r}_{2P} + \frac{q_3}{r_{3P}^2} \hat{r}_{3P} + \dots + \frac{q_n}{r_{nP}^2} \hat{r}_{nP} \right\} \quad (1.7)$$

where $r_{1P}, r_{2P}, r_{3P}, \dots, r_{nP}$ are the distance of the charges $q_1, q_2, q_3, \dots, q_n$ from the point P respectively. Also $\hat{r}_{1P}, \hat{r}_{2P}, \hat{r}_{3P}, \dots, \hat{r}_{nP}$ are the corresponding unit vectors directed from $q_1, q_2, q_3, \dots, q_n$ to P. Equation (1.7) can be re-written as,

$$\vec{E}_{tot} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \left(\frac{q_i}{r_{iP}^2} \hat{r}_{iP} \right) \quad (1.8)$$

For example in Figure 1.8, the resultant electric field due to three point charges q_1, q_2, q_3 at point P is shown.

Note that the relative lengths of the electric field vectors for the charges depend on relative distances of the charges to the point P.

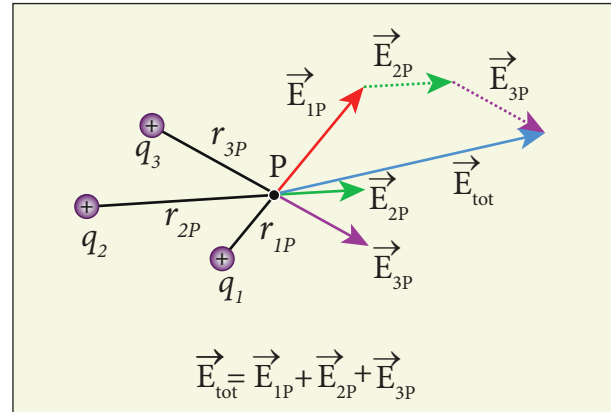
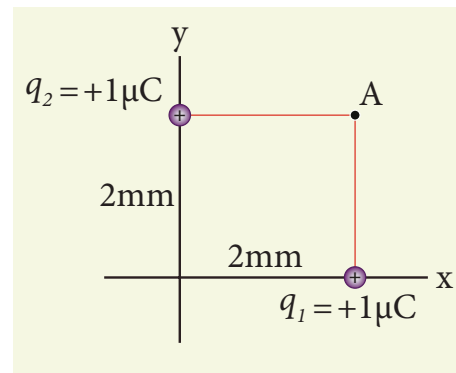


Figure 1.8 Superposition of Electric field

EXAMPLE 1.7

Consider the charge configuration as shown in the figure. Calculate the electric field at point A. If an electron is placed at points A, what is the acceleration experienced by this electron? (mass of the electron = 9.1×10^{-31} kg and charge of electron = -1.6×10^{-19} C)



Solution

By using superposition principle, the net electric field at point A is

$$\vec{E}_A = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_{1A}^2} \hat{r}_{1A} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_{2A}^2} \hat{r}_{2A}$$

where r_{1A} and r_{2A} are the distances of point A from the two charges respectively.

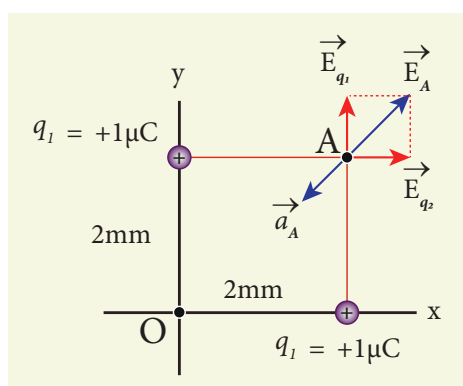
$$\vec{E}_A = \frac{9 \times 10^9 \times 1 \times 10^{-6}}{(2 \times 10^{-3})^2} (\hat{j}) + \frac{9 \times 10^9 \times 1 \times 10^{-6}}{(2 \times 10^{-3})^2} (\hat{i})$$

$$= 2.25 \times 10^9 \hat{j} + 2.25 \times 10^9 \hat{i} = 2.25 \times 10^9 (\hat{i} + \hat{j})$$

The magnitude of electric field

$$|\vec{E}_A| = \sqrt{(2.25 \times 10^9)^2 + (2.25 \times 10^9)^2} \\ = 2.25 \times \sqrt{2} \times 10^9 \text{ NC}^{-1}$$

The direction of $\vec{E}_A$ is given by $\frac{\vec{E}_A}{|\vec{E}_A|} = \frac{2.25 \times 10^9 (\hat{i} + \hat{j})}{2.25 \times \sqrt{2} \times 10^9} = \frac{(\hat{i} + \hat{j})}{\sqrt{2}}$ which is the unit vector along OA as shown in the figure.



The acceleration experienced by an electron placed at point A is

$$\vec{a}_A = \frac{\vec{F}}{m} = \frac{q\vec{E}_A}{m} \\ = \frac{(-1.6 \times 10^{-19}) \times (2.25 \times 10^9)(\hat{i} + \hat{j})}{9.1 \times 10^{-31}} \\ = -3.95 \times 10^{20} (\hat{i} + \hat{j}) \text{ N kg}^{-1}$$

The electron is accelerated in a direction exactly opposite to $\vec{E}_A$.

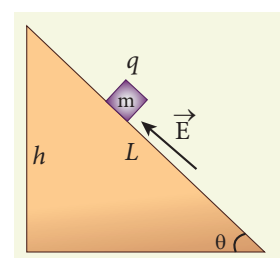
1.3.3 Electric field due to continuous charge distribution

The electric charge is quantized microscopically. The expressions (1.2), (1.3), (1.4) are applicable to only point charges. While dealing with the electric

field due to a charged sphere or a charged wire etc., it is very difficult to look at individual charges in these charged bodies. Therefore, it is assumed that charge is distributed continuously on the charged bodies and the discrete nature of charges is not considered here. The electric field due to such continuous charge distributions is found by invoking the method of calculus. (For further reading, refer Appendix A1.1).

EXAMPLE 1.8

A block of mass m carrying a positive charge q is placed on an insulated frictionless inclined plane as shown in the figure. A uniform electric field E is applied parallel to the inclined surface such that the block is at rest. Calculate the magnitude of the electric field E .

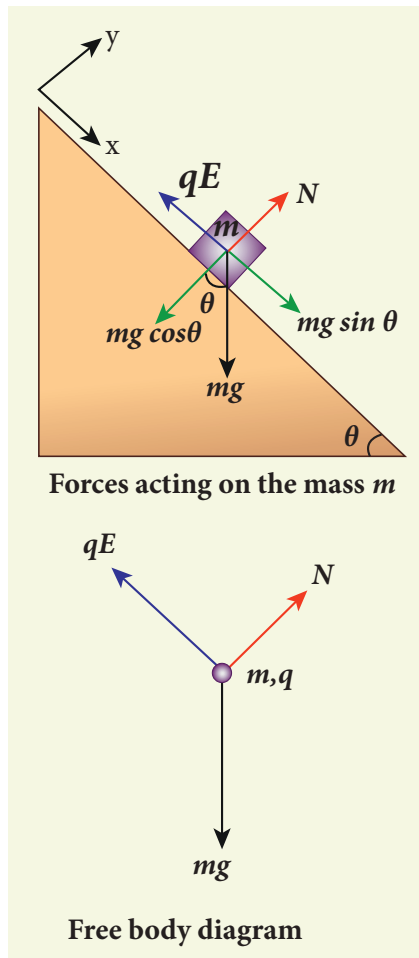


Solution

Note: A similar problem is solved in XIth Physics volume I, unit 3 section 3.3.2. There are three forces that acts on the mass m :

- The downward gravitational force exerted by the Earth (mg)
- The normal force exerted by the inclined surface (N)
- The Coulomb force given by uniform electric field (qE)

The free body diagram for the mass m is drawn below.



A convenient inertial coordinate system is located in the inclined surface as shown in the figure. The mass m has zero net acceleration both in x and y -direction.

Along x -direction, applying Newton's second law, we have

$$\begin{aligned} mg \sin \theta \hat{i} - qE \hat{i} &= 0 \\ mg \sin \theta - qE &= 0 \\ E &= \frac{mg \sin \theta}{q} \end{aligned}$$

Note that the magnitude of the electric field is directly proportional to the mass m and inversely proportional to the charge q . It implies that, if the mass is increased by keeping the charge constant, then a strong electric field is required to stop the object from sliding. If the charge is increased by

keeping the mass constant, then a weak electric field is sufficient to stop the mass from sliding down the plane.

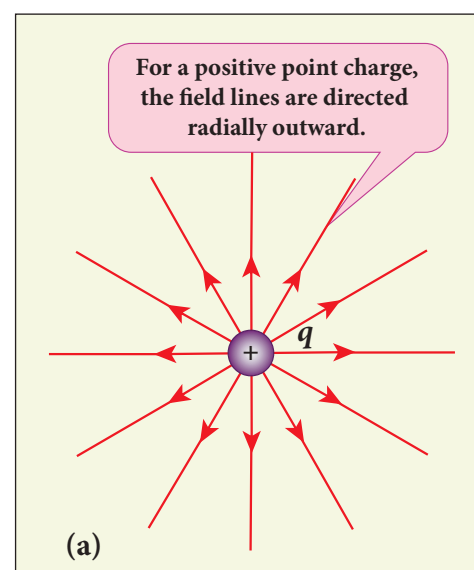
The electric field also can be expressed in terms of height and the length of the inclined surface of the plane.

$$E = \frac{mg h}{qL}$$

1.3.4 Electric field lines

Electric field vectors are visualized by the concept of electric field lines. They form a set of continuous lines which are the visual representation of the electric field in some region of space. The following rules are followed while drawing electric field lines for charges.

- The electric field lines start from a positive charge and end at negative charges or at infinity. For a positive point charge the electric field lines point radially outward and for a negative point charge, the electric field lines point radially inward. These are shown in Figure 1.9 (a) and (b).



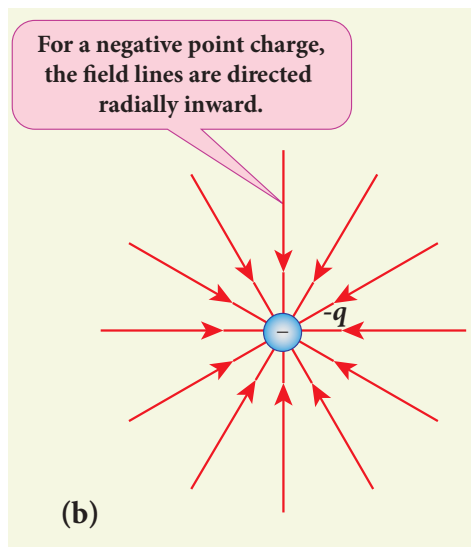


Figure 1.9 Electric field lines for isolated positive and negative charges

Note that for an isolated positive point charge the electric field line starts from the charge and ends only at infinity. For an isolated negative point charge the electric field lines start at infinity and end at the negative charge.

- The electric field vector at a point in space is tangential to the electric field line at that point. This is shown in Figure 1.10

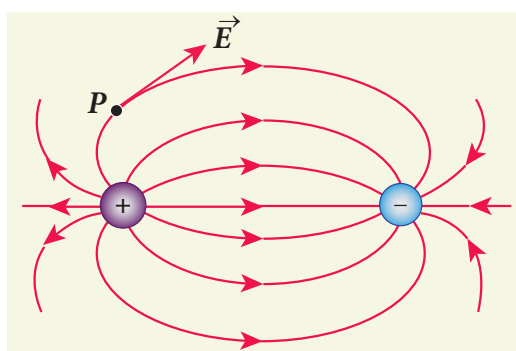


Figure 1.10 Electric field at a point P

- The electric field lines are denser (more closer) in a region where the electric field has larger magnitude and less dense in a region where the electric field is of smaller magnitude. In other words, the number of lines passing through a given surface area perpendicular to the lines is proportional to the magnitude of

the electric field in that region. This is shown in Figure 1.11

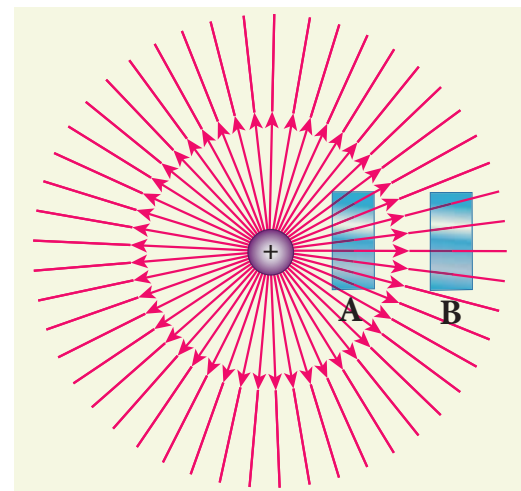


Figure 1.11 Electric field has larger magnitude at surface A than B

Figure 1.11 shows electric field lines from a positive point charge. The magnitude of the electric field for a point charge decreases as the distance increases $\left(|\vec{E}| \propto \frac{1}{r^2}\right)$. So the electric field has greater magnitude at the surface A than at B. Therefore, the number of lines crossing the surface A is greater than the number of lines crossing the surface B. Note that at surface B the electric field lines are farther apart compared to the electric field lines at the surface A.

- No two electric field lines intersect each other. If two lines cross at a point, then there will be two different electric field vectors at the same point, as shown in Figure 1.12.

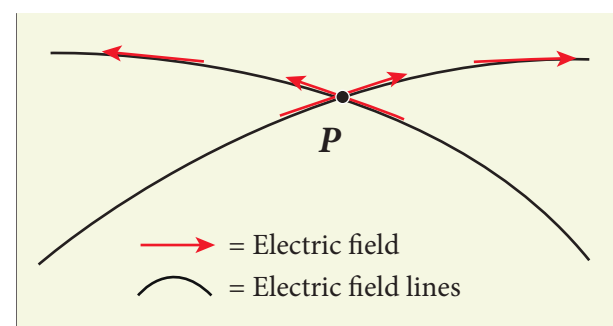


Figure 1.12 Two electric field lines never intersect each other



As a consequence, if some charge is placed in the intersection point, then it has to move in two different directions at the same time, which is physically impossible. Hence, electric field lines do not intersect.

- The number of electric field lines that emanate from the positive charge or end at a negative charge is directly proportional to the magnitude of the charges.

For example in the Figure 1.13, the electric field lines are drawn for charges $+q$ and $-2q$. Note that the number of field lines emanating from $+q$ is 8 and the number of field lines ending at $-2q$ is 16. Since the magnitude of the second charge is twice that of the first charge,

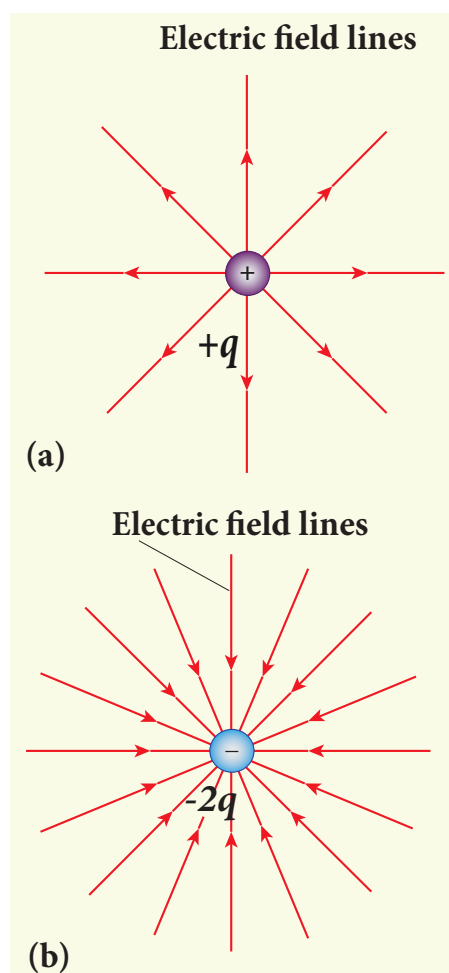
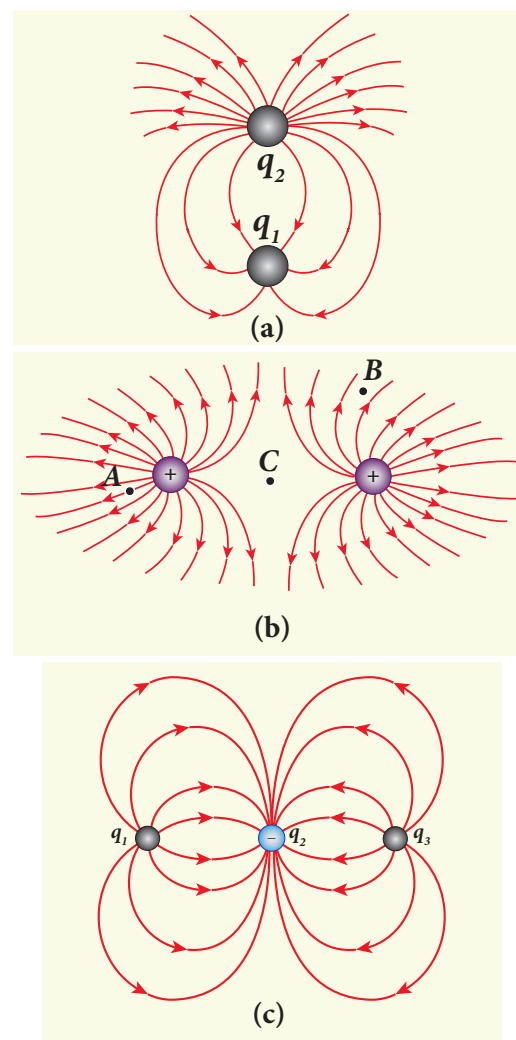


Figure 1.13 Electric field lines and magnitude of the charge

the number of field lines drawn for $-2q$ is twice in number than that for charge $+q$.

EXAMPLE 1.9

The following pictures depict electric field lines for various charge configurations.



- In figure (a) identify the signs of two charges and find the ratio $\left| \frac{q_1}{q_2} \right|$
- In figure (b), calculate the ratio of two positive charges and identify the strength of the electric field at three points A, B, and C
- Figure (c) represents the electric field lines for three charges. If $q_2 = -20 \text{ nC}$, then calculate the values of q_1 and q_3



Solution

(i) The electric field lines start at q_2 and end at q_1 . In figure (a), q_2 is positive and q_1 is negative. The number of lines starting from q_2 is 18 and number of the lines ending at q_1 is 6. So q_2 has greater magnitude. The

ratio of $\left| \frac{q_1}{q_2} \right| = \frac{N_1}{N_2} = \frac{6}{18} = \frac{1}{3}$. It implies that

$$|q_2| = 3|q_1|$$

(ii) In figure (b), the number of field lines emanating from both positive charges are equal ($N=18$). So the charges are equal. At point A, the electric field lines are denser compared to the lines at point B. So the electric field at point A is greater in magnitude compared to the field at point B. Further, no electric field line passes through C, which implies that the resultant electric field at C due to these two charges is zero.

(iii) In the figure (c), the electric field lines start at q_1 and q_3 and end at q_2 . This implies that q_1 and q_3 are positive charges. The ratio of the number of field lines is $\left| \frac{q_1}{q_2} \right| = \frac{8}{16} = \left| \frac{q_3}{q_2} \right| = \frac{1}{2}$, implying that q_1 and q_3 are half of the magnitude of q_2 . So $q_1 = q_3 = +10 \text{ nC}$.

1.4

ELECTRIC DIPOLE AND ITS PROPERTIES

1.4.1 Electric dipole

Two equal and opposite charges separated by a small distance constitute an electric dipole. In many molecules, the centres of positive and negative charge do

not coincide. Such molecules behave as permanent dipoles. Examples: CO, water, ammonia, HCl etc.

Consider two equal and opposite point charges ($+q$, $-q$) that are separated by a distance $2a$ as shown in Figure 1.14(a).

The electric dipole moment is defined as

$$\vec{p} = q\vec{r}_+ + (-q)\vec{r}_- \quad (1.9)$$

where $\vec{r}_+$ is the position vector of $+q$ from the origin and $\vec{r}_-$ is the position vector of $-q$ from the origin. Then, from Figure 1.14 (a),

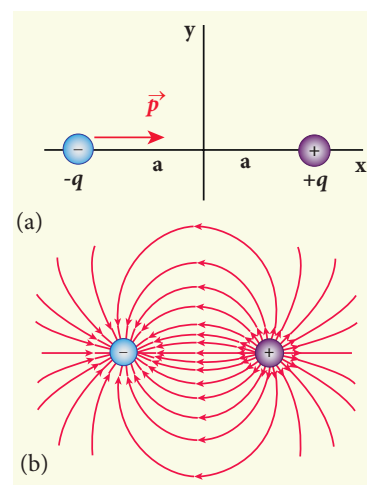


Figure 1.14 (a) Electric dipole (b) Electric field lines for the electric dipole

$$\vec{p} = qa\hat{i} - qa(-\hat{i}) = 2qa\hat{i} \quad (1.10)$$

The electric dipole moment vector lies along the line joining two charges and is directed from $-q$ to $+q$. The SI unit of dipole moment is coulomb metre (Cm). The electric field lines for an electric dipole are shown in Figure 1.14 (b).

- For simplicity, the two charges are placed on the x -axis. Even if the two charges are placed on y or z -axis, dipole moment will point from $-q$ to $+q$.
- The magnitude of the electric dipole moment is equal to the product of the magnitude of one of the charges and the distance between them,

$$|\vec{p}| = 2qa \quad (1.11)$$

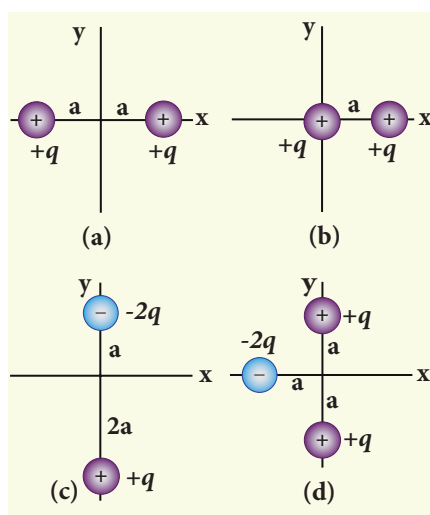
- Though the electric dipole moment for two equal and opposite charges is defined, it is possible to define and calculate the electric dipole moment for a collection of point charges. The electric dipole moment for a collection of n point charges is given by

$$\vec{p} = \sum_{i=1}^n q_i \vec{r}_i \quad (1.12)$$

where $\vec{r}_i$ is the position vector of charge q_i from the origin.

EXAMPLE 1.10

Calculate the electric dipole moment for the following charge configurations.



Solution

Case (a) The position vector for the $+q$ on the positive x -axis is $a\hat{i}$ and position vector for the $+q$ charge the negative x axis is $-a\hat{i}$. So the dipole moment is,

$$\vec{p} = (+q)(a\hat{i}) + (+q)(-a\hat{i}) = 0$$

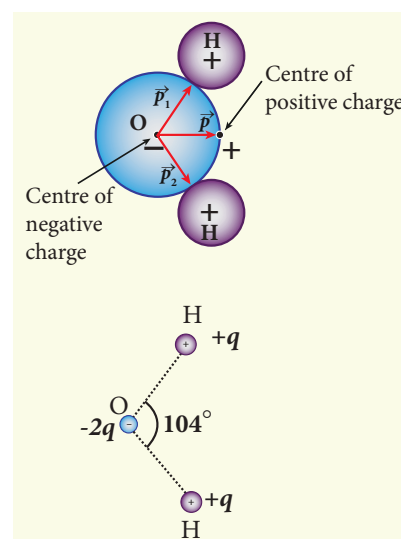
Case (b) In this case one charge is placed at the origin, so its position vector is zero. Hence only the second charge $+q$ with position vector $a\hat{i}$ contributes to the dipole moment, which is $\vec{p} = qa\hat{i}$.

From both cases (a) and (b), we can infer that in general the electric dipole moment depends on the choice of the origin and charge configuration. But for one special case, the electric dipole moment is independent of the origin. If the total charge is zero, then the electric dipole moment will be the same irrespective of the choice of the origin. It is because of this reason that the electric dipole moment of an electric dipole (total charge is zero) is always directed from $-q$ to $+q$, independent of the choice of the origin.

Case (c) $\vec{p} = (-2q)a\hat{j} + q(2a)(-\hat{j}) = -4qa\hat{j}$. Note that in this case $\vec{p}$ is directed from $-2q$ to $+q$.

Case (d) $\vec{p} = -2qa(-\hat{i}) + qa\hat{j} + qa(-\hat{j}) = 2qa\hat{i}$

The water molecule (H_2O) has this charge configuration. The water molecule has three atoms (two H atom and one O atom). The centres of positive (H) and negative (O) charges of a water molecule lie at different points, hence it possess permanent dipole moment. The electric dipole moment $\vec{p}$ is directed from centre of negative charge to the centre of positive charge, as shown in the figure.



1.4.2 Electric field due to a dipole

Case (i) Electric field due to an electric dipole at points on the axial line

Consider an electric dipole placed on the x -axis as shown in Figure 1.15. A point C is located at a distance of r from the midpoint O of the dipole on the axial line.

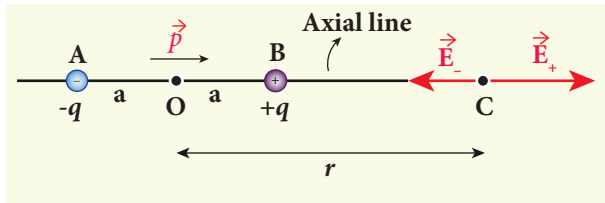


Figure 1.15 Electric field of the dipole along the axial line

The electric field at a point C due to $+q$ is $\vec{E}_+ = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2}$ along BC

Since the electric dipole moment vector $\vec{p}$ is from $-q$ to $+q$ and is directed along BC, the above equation is rewritten as

$$\vec{E}_+ = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2} \hat{p} \quad (1.13)$$

where $\hat{p}$ is the electric dipole moment unit vector from $-q$ to $+q$.

The electric field at a point C due to $-q$ is

$$\vec{E}_- = -\frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2} \hat{p} \quad (1.14)$$

Since $+q$ is located closer to the point C than $-q$, $\vec{E}_+$ is stronger than $\vec{E}_-$. Therefore, the length of the $\vec{E}_+$ vector is drawn larger than that of $\vec{E}_-$ vector.

The total electric field at point C is calculated using the superposition principle of the electric field.

$$\begin{aligned} \vec{E}_{tot} &= \vec{E}_+ + \vec{E}_- \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2} \hat{p} - \frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2} \hat{p} \end{aligned}$$

$$\vec{E}_{tot} = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} \right) \hat{p} \quad (1.15)$$

$$\vec{E}_{tot} = \frac{1}{4\pi\epsilon_0} q \left(\frac{4ra}{(r^2 - a^2)^2} \right) \hat{p} \quad (1.16)$$

Note that the total electric field is along $\vec{E}_+$, since $+q$ is closer to C than $-q$. The direction of $\vec{E}_{tot}$ is shown in Figure 1.16.

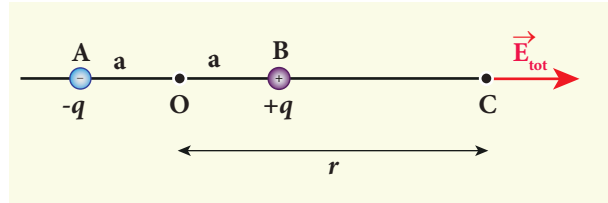


Figure 1.16 Total electric field of the dipole on the axial line

If the point C is very far away from the dipole ($r \gg a$). Then under this limit the term $(r^2 - a^2)^2 \approx r^4$. Substituting this into equation (1.16), we get

$$\begin{aligned} \vec{E}_{tot} &= \frac{1}{4\pi\epsilon_0} \left(\frac{4aq}{r^3} \right) \hat{p} \quad (r \gg a) \\ &\quad \text{since } 2aq\hat{p} = \vec{p} \\ \vec{E}_{tot} &= \frac{1}{4\pi\epsilon_0} \frac{2\vec{p}}{r^3} \quad (r \gg a) \end{aligned} \quad (1.17)$$

If the point C is chosen on the left side of the dipole, the total electric field is still in the direction of $\vec{p}$. We infer this result by examining the electric field lines of the dipole shown in Figure 1.14(b).

Case (ii) Electric field due to an electric dipole at a point on the equatorial plane

Consider a point C at a distance r from the midpoint O of the dipole on the equatorial plane as shown in Figure 1.17. Since the point C is equi-distant from $+q$ and $-q$, the magnitude of the electric fields at C due to $+q$ and $-q$ are the same. The direction

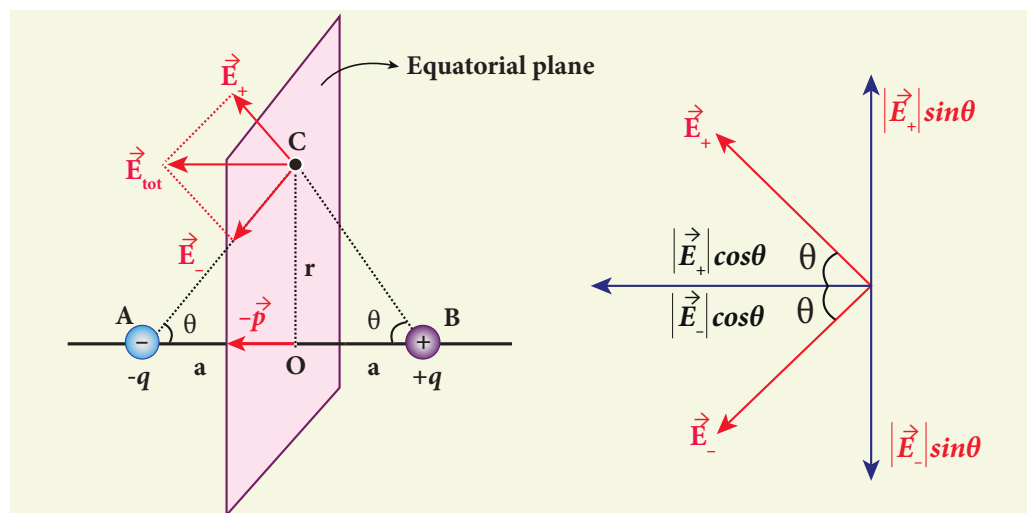


Figure 1.17 Electric field due to a dipole at a point on the equatorial plane

of $\vec{E}_+$ is along BC and the direction of $\vec{E}_-$ is along CA. $\vec{E}_+$ and $\vec{E}_-$ can be resolved into two components; one component parallel to the dipole axis and the other perpendicular to it. Since perpendicular components $|\vec{E}_+|\sin\theta$ and $|\vec{E}_-|\sin\theta$ are equal in magnitude and oppositely directed, they cancel each other. The magnitude of the total electric field at point C is the sum of the parallel components of $\vec{E}_+$ and $\vec{E}_-$ and its direction is along $-\hat{p}$ as shown in the Figure 1.17.

$$\vec{E}_{tot} = -|\vec{E}_+|\cos\theta\hat{p} - |\vec{E}_-|\cos\theta\hat{p} \quad (1.18)$$

The magnitudes $\vec{E}_+$ and $\vec{E}_-$ are the same and are given by

$$|\vec{E}_+| = |\vec{E}_-| = \frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + a^2)^{3/2}} \quad (1.19)$$

By substituting equation (1.19) into equation (1.18), we get

$$\begin{aligned} \vec{E}_{tot} &= -\frac{1}{4\pi\epsilon_0} \frac{2q\cos\theta}{(r^2 + a^2)^{3/2}} \hat{p} \\ &= -\frac{1}{4\pi\epsilon_0} \frac{2qa}{(r^2 + a^2)^{3/2}} \hat{p} \\ \text{since } \cos\theta &= \frac{a}{\sqrt{r^2 + a^2}} \end{aligned}$$

$$\vec{E}_{tot} = -\frac{1}{4\pi\epsilon_0} \frac{\vec{p}}{(r^2 + a^2)^{3/2}} \quad \text{since } \vec{p} = 2qa\hat{p} \quad (1.20)$$

At very large distances ($r \gg a$), the equation (1.20) becomes

$$\vec{E}_{tot} = -\frac{1}{4\pi\epsilon_0} \frac{\vec{p}}{r^3} \quad (r \gg a) \quad (1.21)$$

Important inferences

(i) From equations (1.17) and (1.21), it is inferred that for very large distances, the magnitude of the electric field at point on the dipole axis is twice the magnitude of the electric field at the point at the same distance on the equatorial plane. The direction of the electric field at points on the dipole axis is directed along the direction of dipole moment vector $\vec{p}$ but at points on the equatorial plane it is directed opposite to the dipole moment vector, that is along $-\vec{p}$.

(ii) At very large distances, the electric field due to a dipole varies as $\frac{1}{r^3}$. Note that for a point charge, the electric field

varies as $\frac{1}{r^2}$. This implies that the electric field due to a dipole at very large distances goes to zero faster than the electric field due to a point charge. The reason for this behavior is that at very large distance, the two charges appear to be close to each other and neutralize each other.

(iii) The equations (1.17) and (1.21) are valid only at very large distances ($r \gg a$). Suppose the distance $2a$ approaches zero and q approaches infinity such that the product of $2aq = p$ is finite, then the dipole is called a point dipole. For such point dipoles, equations (1.17) and (1.21) are exact and hold true for any r .

1.4.3 Torque experienced by an electric dipole in the uniform electric field

Consider an electric dipole of dipole moment $\vec{p}$ placed in a uniform electric field $\vec{E}$ whose field lines are equally spaced and point in the same direction. The charge $+q$ will experience a force $q\vec{E}$ in the direction of the field and charge $-q$ will experience a force $-q\vec{E}$ in a direction opposite to the field. Since the external field $\vec{E}$ is uniform, the total force acting on the dipole is zero. These two forces acting at different points will constitute a couple and the dipole experience a torque as shown in Figure 1.18. This torque tends to rotate the dipole. (Note that electric field lines of a uniform field are equally spaced and point in the same direction).

The total torque on the dipole about the point O

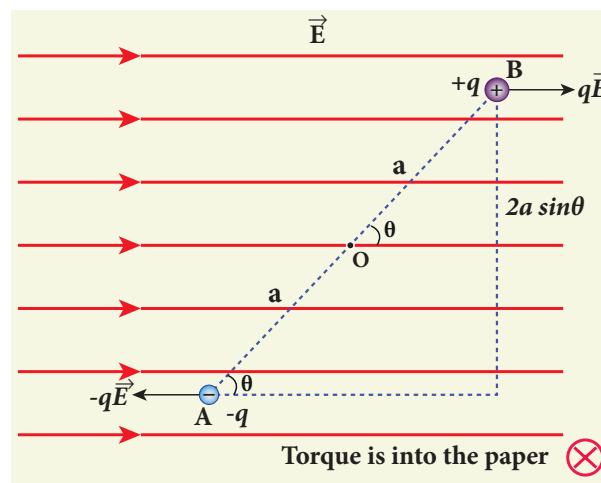


Figure 1.18 Torque on dipole

$$\vec{\tau} = \vec{OA} \times (-q\vec{E}) + \vec{OB} \times q\vec{E} \quad (1.22)$$

Using right-hand corkscrew rule (Refer XI, volume 1, unit 2), it is found that total torque is perpendicular to the plane of the paper and is directed into it.

The magnitude of the total torque

$$\tau = qE \cdot 2a \sin \theta \quad (1.23)$$

where θ is the angle made by $\vec{p}$ with $\vec{E}$. Since $p = 2aq$, the torque is written in terms of the vector product as

$$\vec{\tau} = \vec{p} \times \vec{E} \quad (1.24)$$

The magnitude of this torque is $\tau = pE \sin \theta$ and is maximum when $\theta = 90^\circ$.

This torque tends to rotate the dipole and align it with the electric field $\vec{E}$. Once $\vec{p}$ is aligned with $\vec{E}$, the total torque on the dipole becomes zero.

If the electric field is not uniform, then the force experienced by $+q$ is different from that experienced by $-q$. In addition to the torque, there will be net force acting on the dipole. This is shown in Figure 1.19.

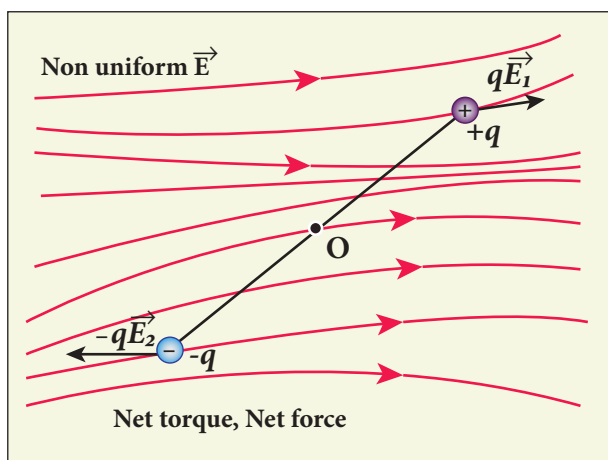


Figure 1.19 The dipole in a non-uniform electric field

EXAMPLE 1.11

A sample of HCl gas is placed in a uniform electric field of magnitude $3 \times 10^4 \text{ N C}^{-1}$. The dipole moment of each HCl molecule is $3.4 \times 10^{-30} \text{ Cm}$. Calculate the maximum torque experienced by each HCl molecule.

Solution

The maximum torque experienced by the dipole is when it is aligned perpendicular to the applied field.

$$\tau_{\max} = pE \sin 90^\circ = 3.4 \times 10^{-30} \times 3 \times 10^4$$

$$\tau_{\max} = 10.2 \times 10^{-26} \text{ Nm}$$

1.5

ELECTROSTATIC POTENTIAL AND POTENTIAL ENERGY

Introduction

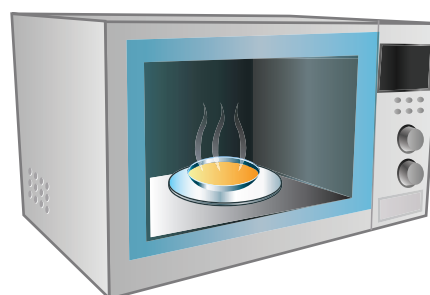
In mechanics, potential energy is defined for conservative forces. Since gravitational force is a conservative force, its gravitational potential energy is defined in XI standard physics (Unit 6). Since Coulomb force is an inverse-square-law force, its also a conservative force like gravitational force. Therefore, we can define potential energy for charge configurations.

1.5.1 Electrostatic Potential energy and Electrostatic potential

Consider a positive charge q kept fixed at the origin which produces an electric field $\vec{E}$ around it. A positive test charge q' is brought from point R to point P against the repulsive force between q and q' as shown in Figure 1.20. Work must be done to overcome the repulsion between the charges and this work done is stored as potential energy of the system.



Microwave oven works on the principle of torque acting on an electric dipole. The food we consume has water molecules which are permanent electric dipoles. Oven produces microwaves that are oscillating electromagnetic fields and produce torque on the water molecules. Due to this torque on each water molecule, the molecules rotate very fast and produce thermal energy. Thus, heat generated is used to heat the food.



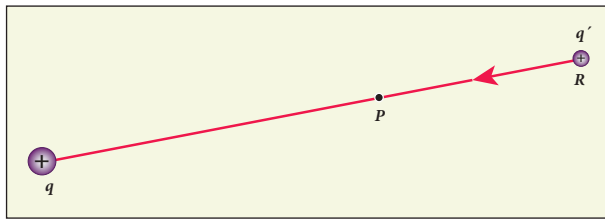


Figure 1.20 Work done is equal to potential energy

The test charge q' is brought from R to P with constant velocity which means that external force used to bring the test charge q' from R to P must be equal and opposite to the coulomb force ($\vec{F}_{ext} = -\vec{F}_{coulomb}$). The work done is

$$W = \int_R^P \vec{F}_{ext} \cdot d\vec{r} \quad (1.25)$$

Since coulomb force is conservative, work done is independent of the path and it depends only on the initial and final positions of the test charge. If potential energy associated with q' at P is U_P and that at R is U_R , then difference in potential energy is defined as the work done to bring a test charge q' from point R to P and is given as $U_P - U_R = W = \Delta U$

$$\Delta U = \int_R^P \vec{F}_{ext} \cdot d\vec{r} \quad (1.26)$$

$$\text{Since } \vec{F}_{ext} = -\vec{F}_{coulomb} = -q'\vec{E} \quad (1.27)$$

$$\Delta U = \int_R^P -(q'\vec{E}) \cdot d\vec{r} = q' \int_R^P (-\vec{E}) \cdot d\vec{r} \quad (1.28)$$

The potential energy difference per unit charge is given by

$$\frac{\Delta U}{q'} = \frac{q' \int_R^P (-\vec{E}) \cdot d\vec{r}}{q'} = - \int_R^P \vec{E} \cdot d\vec{r} \quad (1.29)$$

The above equation (1.29) is independent of q' . The quantity $\frac{\Delta U}{q'} = - \int_R^P \vec{E} \cdot d\vec{r}$ is called electric potential difference between P and R and is denoted as $V_P - V_R = \Delta V$.

In otherwords, the electric potential difference is defined as the work done by an external force to bring unit positive charge from point R to point P.

$$V_P - V_R = \Delta V = \int_R^P -\vec{E} \cdot d\vec{r} \quad (1.30)$$

The electric potential energy difference can be written as $\Delta U = q' \Delta V$. Physically potential difference between two points is a meaningful quantity. The value of the potential itself at one point is not meaningful. Therefore the point R is taken to infinity and the potential at infinity is considered as zero ($V_\infty = 0$).

Then **the electric potential at a point P is equal to the work done by an external force to bring a unit positive charge with constant velocity from infinity to the point P in the region of the external electric field $\vec{E}$.** Mathematically this is written as

$$V_P = - \int_\infty^P \vec{E} \cdot d\vec{r} \quad (1.31)$$

Important points

1. Electric potential at point P depends only on the electric field which is due to the source charge q and not on the test charge q' . Unit positive charge is brought from infinity to the point P with constant velocity because external agency should not impart any kinetic energy to the test charge.

2. From equation (1.29), the unit of electric potential is Joule per coulomb. The practical unit is volt (V) named after Alessandro Volta (1745-1827) who invented the electrical battery. The potential difference between two points is expressed in terms of volt.



Note The description of motion of objects using the concept of potential or potential energy is simpler than that using the concept of field.

1.5.2 Electric potential due to a point charge

Consider a positive charge q kept fixed at the origin. Let P be a point at distance r from the charge q . This is shown in Figure 1.21.

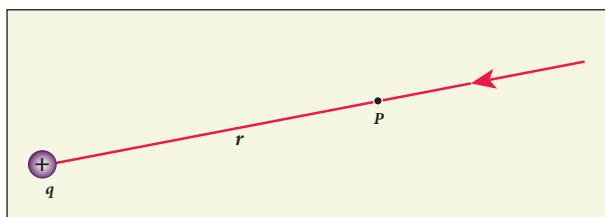


Figure 1.21 Electrostatic potential at a point P

The electric potential at the point P is

$$V = \int_{\infty}^r (-\vec{E}) \cdot d\vec{r} = - \int_{\infty}^r \vec{E} \cdot d\vec{r} \quad (1.32)$$

Electric field due to positive point charge q is

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$$

$$V = \frac{-1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r^2} \hat{r} \cdot d\vec{r}$$

The infinitesimal displacement vector, $d\vec{r} = dr\hat{r}$ and using $\hat{r} \cdot \hat{r} = 1$, we have

$$V = -\frac{1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r^2} \hat{r} \cdot d\vec{r} = -\frac{1}{4\pi\epsilon_0} \int_{\infty}^r \frac{q}{r^2} dr$$

After the integration,

$$V = -\frac{1}{4\pi\epsilon_0} q \left\{ -\frac{1}{r} \right\}_{\infty}^r = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Hence the electric potential due to a point charge q at a distance r is

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} \quad (1.33)$$

Important points

- (i) If the source charge q is positive, $V > 0$. If q is negative, then V is negative and equal

$$\text{to } V = -\frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

- (ii) From expression (1.33), it is clear that the potential due to positive charge decreases as the distance increases, but for a negative charge the potential increases as the distance is increased. At infinity ($r = \infty$) electrostatic potential is zero ($V = 0$).

In the case of gravitational force, mass moves from a point of higher gravitational potential to a point of lower

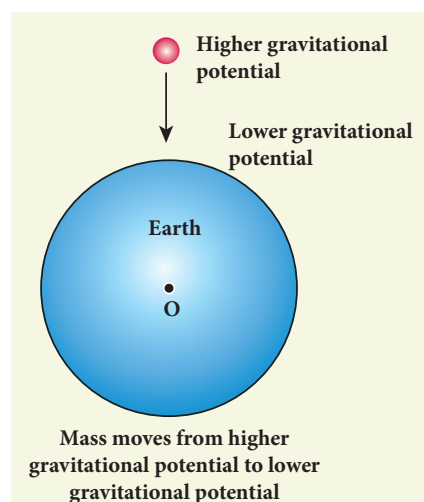


Figure 1.22 Motion of masses in terms of gravitational potential

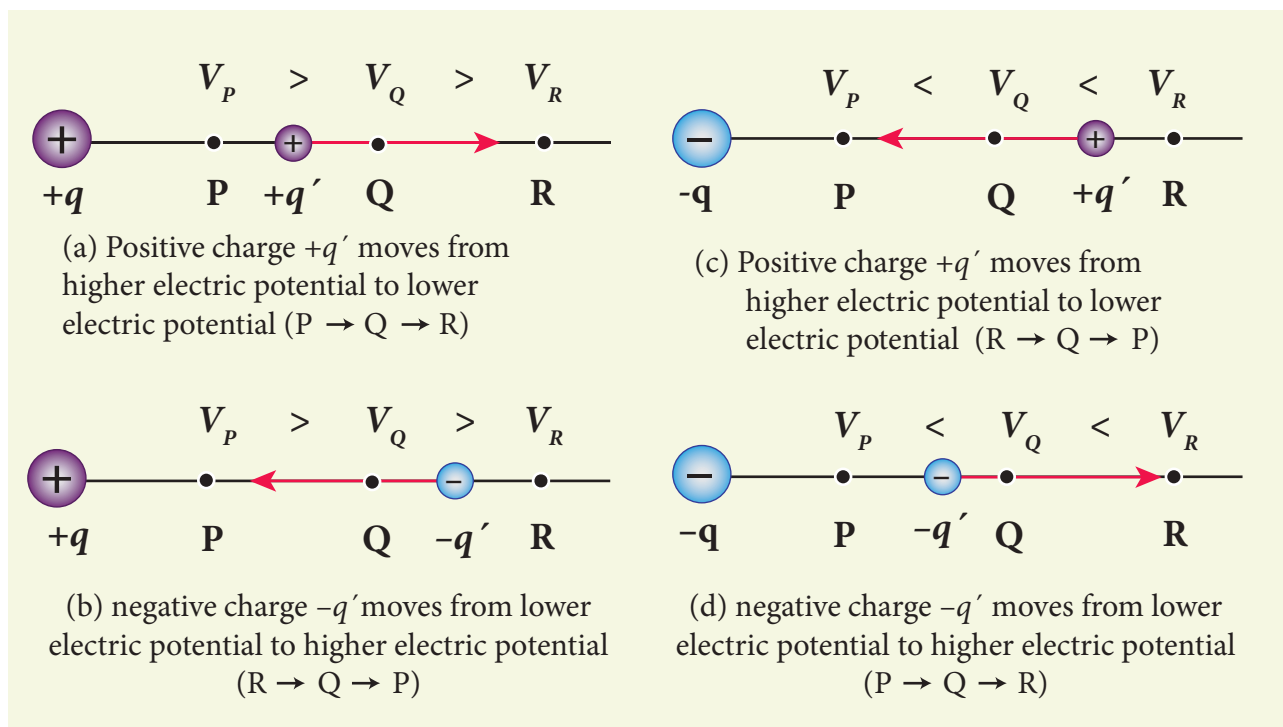


Figure 1.23 Motion of charges in terms of electric potential

gravitational potential (Figure 1.22). Similarly a positive charge moves from a point of higher electrostatic potential to a point of lower electrostatic potential. However a negative charge moves from lower electrostatic potential to higher electrostatic potential. This comparison is shown in Figure 1.23.

(iii) The electric potential at a point P due to a collection of charges $q_1, q_2, q_3, \dots, q_n$ is equal to sum of the electric potentials due to individual charges.

$$V_{tot} = \frac{kq_1}{r_1} + \frac{kq_2}{r_2} + \frac{kq_3}{r_3} + \dots + \frac{kq_n}{r_n} = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i} \quad (1.34)$$

where $r_1, r_2, r_3, \dots, r_n$ are the distances of $q_1, q_2, q_3, \dots, q_n$ respectively from P (Figure 1.24).

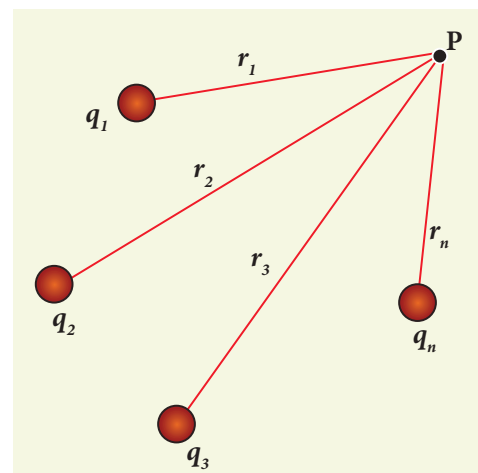
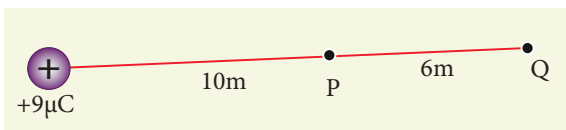


Figure 1.24 Electrostatic potential due to collection of charges

EXAMPLE 1.12

(a) Calculate the electric potential at points P and Q as shown in the figure below.

(b) Suppose the charge $+9 \mu\text{C}$ is replaced by $-9 \mu\text{C}$ find the electrostatic potentials at points P and Q



(c) Calculate the work done to bring a test charge $+2 \mu\text{C}$ from infinity to the point Q. Assume the charge $+9 \mu\text{C}$ is held fixed at origin and $+2 \mu\text{C}$ is brought from infinity to P.

Solution

(a) Electric potential at point P is given by

$$V_P = \frac{1}{4\pi\epsilon_0} \frac{q}{r_P} = \frac{9 \times 10^9 \times 9 \times 10^{-6}}{10} = 8.1 \times 10^3 \text{ V}$$

Electric potential at point Q is given by

$$V_Q = \frac{1}{4\pi\epsilon_0} \frac{q}{r_Q} = \frac{9 \times 10^9 \times 9 \times 10^{-6}}{16} = 5.06 \times 10^3 \text{ V}$$

Note that the electric potential at point Q is less than the electric potential at point P. If we put a positive charge at P, it moves from P to Q. However if we place a negative charge at P it will move towards the charge $+9 \mu\text{C}$.

The potential difference between the points P and Q is given by

$$\Delta V = V_P - V_Q = +3.04 \times 10^3 \text{ V}$$

(b) Suppose we replace the charge $+9 \mu\text{C}$ by $-9 \mu\text{C}$, then the corresponding potentials at the points P and Q are,

$$V_P = -8.1 \times 10^3 \text{ V}, V_Q = -5.06 \times 10^3 \text{ V}$$

Note that in this case electric potential at the point Q is higher than at point P.

The potential difference between the points P and Q is given by

$$\Delta V = V_P - V_Q = -3.04 \times 10^3 \text{ V}$$

(c) The electric potential V at a point Q due to some charge is defined as the work done by an external force to bring a unit positive charge from infinity to Q. So to bring the q amount of charge from infinity to the point Q, work done is given as follows.

$$W = qV$$

$$W_Q = 2 \times 10^{-6} \times 5.06 \times 10^3 = 10.12 \times 10^{-3} \text{ J.}$$

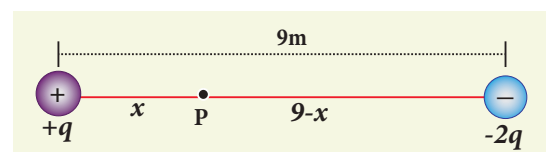
EXAMPLE 1.13

Consider a point charge $+q$ placed at the origin and another point charge $-2q$ placed at a distance of 9 m from the charge $+q$. Determine the point between the two charges at which electric potential is zero.

Solution

According to the superposition principle, the total electric potential at a point is equal to the sum of the potentials due to each charge at that point.

Consider the point at which the total potential zero is located at a distance x from the charge $+q$ as shown in the figure.



Since the total electric potential at P is zero,

$$V_{\text{tot}} = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{x} - \frac{2q}{(9-x)} \right) = 0 \text{ (or)}$$

$$\frac{q}{x} = \frac{2q}{(9-x)} \text{ (or)}$$

$$\frac{1}{x} = \frac{2}{(9-x)}$$

$$\text{Hence, } x = 3 \text{ m}$$

1.5.3 Electrostatic potential at a point due to an electric dipole

Consider two equal and opposite charges separated by a small distance $2a$ as shown in Figure 1.25. The point P is located at a distance r from the midpoint of the dipole. Let θ be the angle between the line OP and dipole axis AB.

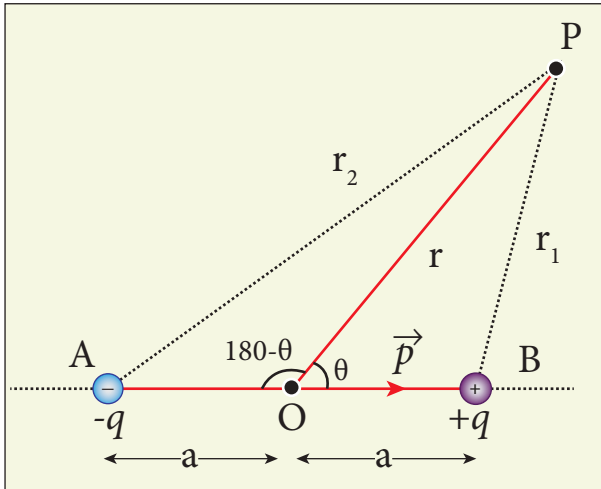


Figure 1.25 Potential due to electric dipole

Let r_1 be the distance of point P from $+q$ and r_2 be the distance of point P from $-q$.

$$\text{Potential at P due to charge } +q = \frac{1}{4\pi\epsilon_0} \frac{q}{r_1}$$

$$\text{Potential at P due to charge } -q = -\frac{1}{4\pi\epsilon_0} \frac{q}{r_2}$$

Total potential at the point P

$$V = \frac{1}{4\pi\epsilon_0} q \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \quad (1.35)$$

Suppose if the point P is far away from the dipole, such that $r \gg a$, then equation (1.35) can be expressed in terms of r .

By the cosine law for triangle BOP

$$r_1^2 = r^2 + a^2 - 2ra \cos \theta$$

$$r_1^2 = r^2 \left(1 + \frac{a^2}{r^2} - \frac{2a}{r} \cos \theta \right)$$

Since the point P is very far from the dipole ($r \gg a$). As a result the term $\frac{a^2}{r^2}$ is very small and can be neglected. Therefore

$$r_1^2 = r^2 \left(1 - \frac{2a}{r} \cos \theta \right)$$

$$\text{(or)} \quad r_1 = r \left(1 - \frac{2a}{r} \cos \theta \right)^{\frac{1}{2}}$$

$$\frac{1}{r_1} = \frac{1}{r} \left(1 - \frac{2a}{r} \cos \theta \right)^{-\frac{1}{2}}$$

Since $\frac{a}{r} \ll 1$, we can use binomial theorem and retain the terms up to first order

$$\frac{1}{r_1} = \frac{1}{r} \left(1 + \frac{a}{r} \cos \theta \right) \quad (1.36)$$

Similarly applying the cosine law for triangle AOP,

$$r_2^2 = r^2 + a^2 - 2ra \cos(180 - \theta)$$

since $\cos(180 - \theta) = -\cos \theta$ we get

$$r_2^2 = r^2 + a^2 + 2ra \cos \theta$$

Neglecting the term $\frac{a^2}{r^2}$ (because $r \gg a$)

$$r_2^2 = r^2 \left(1 + \frac{2a \cos \theta}{r} \right)$$



$$r_2 = r \left(1 + \frac{2a \cos \theta}{r} \right)^{\frac{1}{2}}$$

Using Binomial theorem, we get

$$\frac{1}{r_2} = \frac{1}{r} \left(1 - a \frac{\cos \theta}{r} \right) \quad (1.37)$$

Substituting equation (1.37) and (1.36) in equation (1.35),

$$\begin{aligned} V &= \frac{1}{4\pi\epsilon_0} q \left(\frac{1}{r} \left(1 + a \frac{\cos \theta}{r} \right) - \frac{1}{r} \left(1 - a \frac{\cos \theta}{r} \right) \right) \\ V &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r} \left(1 + a \frac{\cos \theta}{r} - 1 + a \frac{\cos \theta}{r} \right) \right) \\ V &= \frac{1}{4\pi\epsilon_0} \frac{2aq}{r^2} \cos \theta \end{aligned}$$

But the electric dipole moment $p = 2qa$ and we get,

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{p \cos \theta}{r^2} \right)$$

Now we can write $p \cos \theta = \vec{p} \cdot \hat{r}$, where $\hat{r}$ is the unit vector from the point O to point P. Hence the electric potential at a point P due to an electric dipole is given by

$$V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2} \quad (r \gg a) \quad (1.38)$$

Equation (1.38) is valid for distances very large compared to the size of the dipole. But for a point dipole, the equation (1.38) is valid for any distance.

Special cases

Case (i) If the point P lies on the axial line of the dipole on the side of $+q$, then $\theta = 0$. Then the electric potential becomes

$$V = \frac{1}{4\pi\epsilon_0} \frac{p}{r^2} \quad (1.39)$$

Case (ii) If the point P lies on the axial line of the dipole on the side of $-q$, then $\theta = 180^\circ$. Then

$$V = -\frac{1}{4\pi\epsilon_0} \frac{p}{r^2} \quad (1.40)$$

Case (iii) If the point P lies on the equatorial line of the dipole, then $\theta = 90^\circ$. Hence

$$V = 0 \quad (1.41)$$



Note

The potential due to an electric dipole falls as $\frac{1}{r^2}$ and the potential due to a single point charge falls as $\frac{1}{r}$. Thus the potential due to the dipole falls faster than that due to a monopole (point charge). As the distance increases from electric dipole, the effects of positive and negative charges nullify each other

1.5.4 Equi-potential Surface

Consider a point charge q located at some point in space and an imaginary sphere of radius r is chosen by keeping the charge q at its centre (Figure 1.26(a)). The electric potential at all points on the surface of the given sphere is the same. Such a surface is called an equipotential surface.

An equipotential surface is a surface on which all the points are at the same electric potential. For a point charge the equipotential surfaces are concentric spherical surfaces as shown in Figure 1.26(b). Each spherical surface is an equipotential surface but the

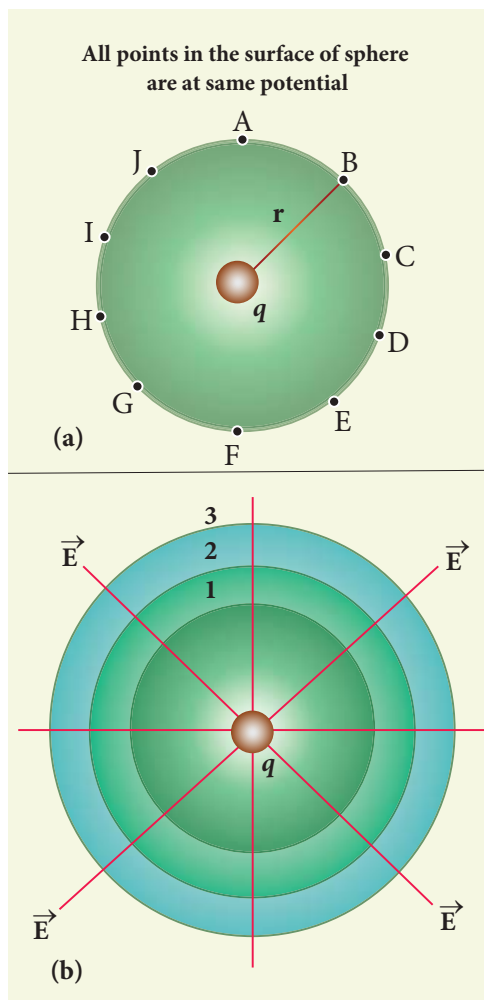


Figure 1.26 Equipotential surface of point Charge

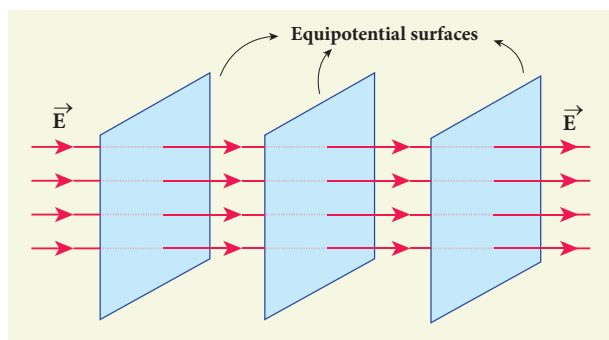


Figure 1.27 Equipotential surface for uniform electric field

value of the potential is different for different spherical surfaces.

For a uniform electric field, the equipotential surfaces form a set of planes normal to the electric field $\vec{E}$. This is shown in the Figure 1.27.

Properties of equipotential surfaces

(i) The work done to move a charge q between any two points A and B, $W = q (V_B - V_A)$. If the points A and B lie on the same equipotential surface, work done is zero because $V_A = V_B$.

(ii) The electric field is normal to an equipotential surface. If it is not normal, then there is a component of the field parallel to the surface. Then work must be done to move a charge between two points on the same surface. This is a contradiction. Therefore the electric field must always be normal to equipotential surface.

1.5.5 Relation between electric field and potential

Consider a positive charge q kept fixed at the origin. To move a unit positive charge by a small distance dx towards q in the electric field E , the work done is given by $dW = -E dx$. The minus sign implies that work is done against the electric field. This work done is equal to electric potential difference. Therefore,

$$dW = dV.$$

$$\text{(or) } dV = -E dx \quad (1.42)$$

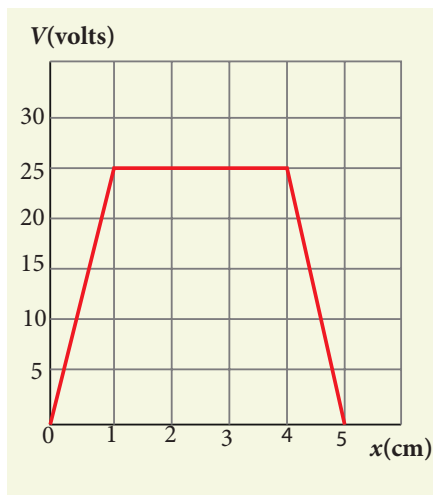
$$\text{Hence } E = -\frac{dV}{dx} \quad (1.43)$$

The electric field is the negative gradient of the electric potential. In vector form,

$$\vec{E} = -\left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right) \quad (1.44)$$

EXAMPLE 1.14

The following figure represents the electric potential as a function of x – coordinate. Plot the corresponding electric field as a function of x .



Solution

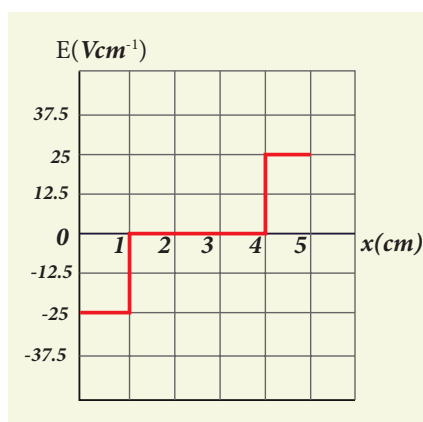
In the given problem, since the potential depends only on x , we can use $\vec{E} = -\frac{dV}{dx}\hat{i}$ (the other two terms $\frac{\partial V}{\partial y}$ and $\frac{\partial V}{\partial z}$ are zero)

From 0 to 1 cm, the slope is constant and so $\frac{dV}{dx} = 25 \text{ V cm}^{-1}$. So $\vec{E} = -25 \text{ V cm}^{-1}\hat{i}$

From 1 to 4 cm, the potential is constant, $V = 25 \text{ V}$. It implies that $\frac{dV}{dx} = 0$. So $\vec{E} = 0$

From 4 to 5 cm, the slope $\frac{dV}{dx} = -25 \text{ V cm}^{-1}$. So $\vec{E} = +25 \text{ V cm}^{-1}\hat{i}$.

The plot of electric field for the various points along the x axis is given below.



1.5.6 Electrostatic potential energy for collection of point charges

The electric potential at a point at a distance r from point charge q_1 is given by

$$V = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r}$$

This potential V is the work done to bring a unit positive charge from infinity to the point. Now if the charge q_2 is brought from infinity to that point at a distance r from q_1 , the work done is the product of q_2 and the electric potential at that point. Thus we have

$$W = q_2 V$$

This work done is stored as the electrostatic potential energy U of a system of charges q_1 and q_2 separated by a distance r . Thus we have

$$U = q_2 V = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} \quad (1.45)$$

The electrostatic potential energy depends only on the distance between the two point charges. In fact, the expression (1.45) is derived by assuming that q_1 is fixed and q_2 is brought from infinity. The equation (1.45) holds true when q_2 is fixed and q_1 is brought from infinity or both q_1 and q_2 are simultaneously brought from infinity to a distance r between them.

Three charges are arranged in the following configuration as shown in Figure 1.28.

To calculate the total electrostatic potential energy, we use the following procedure. We bring all the charges one by one and arrange them according to the configuration as shown in Figure 1.28.

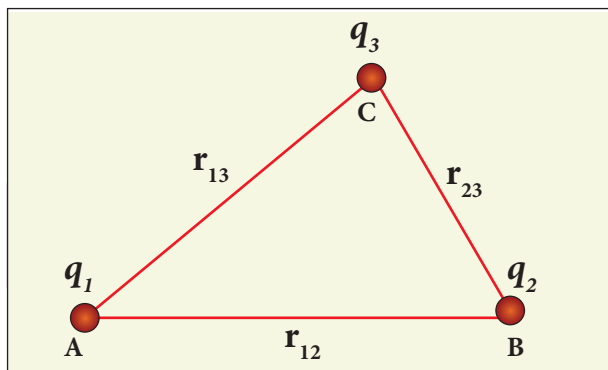


Figure 1.28 Electrostatic potential energy for collection of point charges

(i) Bringing a charge q_1 from infinity to the point A requires no work, because there are no other charges already present in the vicinity of charge q_1 .

(ii) To bring the second charge q_2 to the point B, work must be done against the electric field created by the charge q_1 . So the work done on the charge q_2 is $W = q_2 V_{1B}$. Here V_{1B} is the electrostatic potential due to the charge q_1 at point B.

$$U_I = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} \quad (1.46)$$

Note that the expression is same when q_2 is brought first and then q_1 later.

(iii) Similarly to bring the charge q_3 to the point C, work has to be done against the total electric field due to both charges q_1 and q_2 . So the work done to bring the charge q_3 is $= q_3 (V_{1C} + V_{2C})$. Here V_{1C} is the electrostatic potential due to charge q_1 at point C and V_{2C} is the electrostatic potential due to charge q_2 at point C.

The electrostatic potential energy is

$$U_{II} = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) \quad (1.47)$$

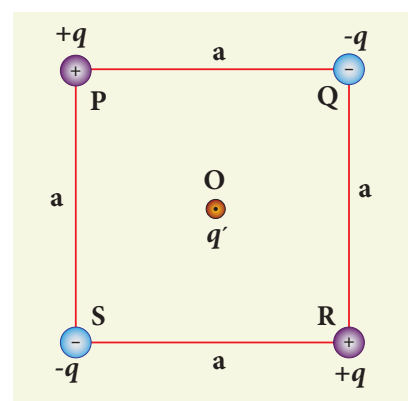
(iv) Adding equations (1.46) and (1.47), the total electrostatic potential energy for the system of three charges q_1 , q_2 and q_3 is $U = U_I + U_{II}$

$$U = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) \quad (1.48)$$

Note that this stored potential energy U is equal to the total external work done to assemble the three charges at the given locations. The expression (1.48) is same if the charges are brought to their positions in any other order. Since the Coulomb force is a conservative force, the electrostatic potential energy is independent of the manner in which the configuration of charges is arrived at.

EXAMPLE 1.15

Four charges are arranged at the corners of the square PQRS of side a as shown in the figure. (a) Find the work required to assemble these charges in the given configuration. (b) Suppose a charge q' is brought to the centre of the square, by keeping the four charges fixed at the corners, how much extra work is required for this?



Solution

(a) The work done to arrange the charges in the corners of the square is independent



of the way they are arranged. We can follow any order.

(i) First, the charge $+q$ is brought to the corner P. This requires no work since no charge is already present, $W_P = 0$

(ii) Work required to bring the charge $-q$ to the corner Q $= (-q) \times$ potential at a point Q due to $+q$ located at a point P.

$$W_Q = -q \times \frac{1}{4\pi\epsilon_0} \frac{q}{a} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{a}$$

(iii) Work required to bring the charge $+q$ to the corner R $= q \times$ potential at the point R due to charges at the point P and Q.

$$\begin{aligned} W_R &= q \times \frac{1}{4\pi\epsilon_0} \left(-\frac{q}{a} + \frac{q}{\sqrt{2}a} \right) \\ &= \frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(-1 + \frac{1}{\sqrt{2}} \right) \end{aligned}$$

(iv) Work required to bring the fourth charge $-q$ at the position S $= q \times$ potential at the point S due to all the three charges at the point P, Q and R

$$\begin{aligned} W_S &= -q \times \frac{1}{4\pi\epsilon_0} \left(\frac{q}{a} + \frac{q}{a} - \frac{q}{\sqrt{2}a} \right) \\ W_S &= -\frac{1}{4\pi\epsilon_0} \frac{q^2}{a} \left(2 - \frac{1}{\sqrt{2}} \right) \end{aligned}$$

(b) Work required to bring the charge q' to the centre of the square $= q' \times$ potential at the centre point O due to all the four charges in the four corners

The potential created by the two $+q$ charges are canceled by the potential created by the $-q$ charges which are located in the opposite corners. Therefore the net electric potential at the centre O due to all the charges in the corners is zero.

Hence no work is required to bring any charge to the point O. Physically this implies that if any charge q' when brought close to O, then it moves to the point O without any external force.

1.5.7 Electrostatic potential energy of a dipole in a uniform electric field

Consider a dipole placed in the uniform electric field $\vec{E}$ as shown in the Figure 1.29. A dipole experiences a torque when kept in a uniform electric field $\vec{E}$. This torque rotates the dipole to align it with the direction of the electric field. To rotate the dipole (at constant angular velocity) from its initial angle θ' to another angle θ against the torque exerted by the electric field, an equal and opposite external torque must be applied on the dipole.

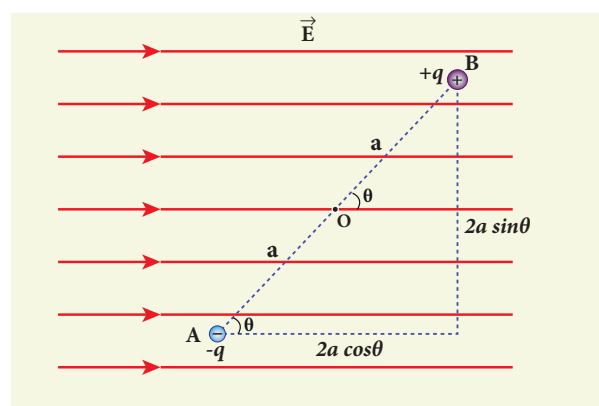


Figure 1.29 The dipole in a uniform electric field

The work done by the external torque to rotate the dipole from angle θ' to θ at constant angular velocity is

$$W = \int_{\theta'}^{\theta} \tau_{ext} d\theta \quad (1.49)$$

Since $\vec{\tau}_{ext}$ is equal and opposite to $\vec{\tau}_E = \vec{p} \times \vec{E}$, we have

$$|\vec{\tau}_{ext}| = |\vec{\tau}_E| = |\vec{p} \times \vec{E}| \quad (1.50)$$

Substituting equation (1.50) in equation (1.49), we get

$$W = \int_{\theta'}^{\theta} pE \sin \theta d\theta$$

$$W = pE (\cos \theta' - \cos \theta)$$

This work done is equal to the potential energy difference between the angular positions θ and θ' .

$$U(\theta) - U(\theta') = \Delta U = -pE \cos \theta + pE \cos \theta'$$

If the initial angle is $\theta' = 90^\circ$ and is taken as reference point, then $U(\theta') = pE \cos 90^\circ = 0$.

The potential energy stored in the system of dipole kept in the uniform electric field is given by

$$U = -pE \cos \theta = -\vec{p} \cdot \vec{E} \quad (1.51)$$

In addition to p and E , the potential energy also depends on the orientation θ of the electric dipole with respect to the external electric field.

The potential energy is maximum when the dipole is aligned anti-parallel ($\theta = \pi$) to the external electric field and minimum when the dipole is aligned parallel ($\theta = 0$) to the external electric field.

EXAMPLE 1.16

A water molecule has an electric dipole moment of 6.3×10^{-30} Cm. A sample contains 10^{22} water molecules, with all the dipole moments aligned parallel to the external electric field of magnitude 3×10^5 NC⁻¹. How much work is required to rotate all the water molecules from $\theta = 0^\circ$ to 90° ?

Solution

When the water molecules are aligned in the direction of the electric field, it has minimum potential energy. The work done to rotate the dipole from $\theta = 0^\circ$ to 90° is equal to the potential energy difference between these two configurations.

$$W = \Delta U = U(90^\circ) - U(0^\circ)$$

From the equation (1.51), we write $U = -pE \cos \theta$. Next we calculate the work done to rotate one water molecule from $\theta = 0^\circ$ to 90° .

For one water molecule

$$W = -pE \cos 90^\circ + pE \cos 0^\circ = pE$$

$$W = 6.3 \times 10^{-30} \times 3 \times 10^5 = 18.9 \times 10^{-25} \text{ J}$$

For 10^{22} water molecules, the total work done is

$$W_{tot} = 18.9 \times 10^{-25} \times 10^{22} = 18.9 \times 10^{-3} \text{ J}$$

1.6

GAUSS LAW AND ITS APPLICATIONS

1.6.1 Electric Flux

The number of electric field lines crossing a given area kept normal to the electric field lines is called electric flux. It is usually denoted by the Greek letter Φ_E and its unit is N m² C⁻¹. Electric flux is a scalar quantity and it can be positive or negative. For a simpler understanding of electric flux, the following Figure 1.30 is useful.

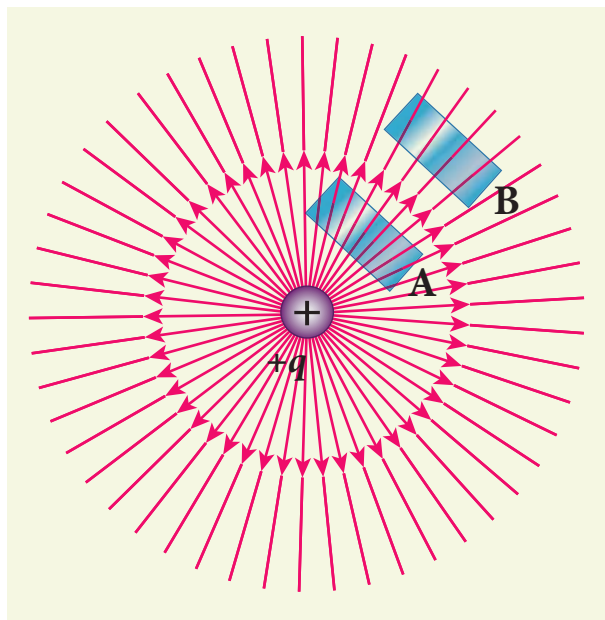


Figure 1.30 Electric flux

The electric field of a point charge is drawn in this figure. Consider two small rectangular area elements placed normal to the field at regions A and B. Even though these elements have the same area, the number of electric field lines crossing the element in region A is more than that crossing the element in region B. Therefore the electric flux in region A is more than that in region B. Since electric field strength for a point charge decreases as the distance increases, electric flux also decreases as the distance increases. The above discussion gives a qualitative idea of electric flux. However a precise definition of electric flux is needed.

Electric flux for uniform Electric field

Consider a uniform electric field in a region of space. Let us choose an area A normal to the electric field lines as shown in Figure 1.31 (a). The electric flux for this case is

$$\Phi_E = EA \quad (1.52)$$

Suppose the same area A is kept parallel to the uniform electric field, then no electric field lines pass through the area A , as shown in Figure 1.31(b). The electric flux for this case is zero.

$$\Phi_E = 0 \quad (1.53)$$

If the area is inclined at an angle θ with the field, then the component of the electric field perpendicular to the area alone contributes to the electric flux. The electric field component parallel to the surface area will not contribute to the electric flux. This is shown in Figure 1.31 (c). For this case, the electric flux

$$\Phi_E = (E \cos\theta) A \quad (1.54)$$

Further, θ is also the angle between the electric field and the direction normal to the area. Hence in general, for uniform electric field, the electric flux is defined as

$$\Phi_E = \vec{E} \cdot \vec{A} = EA \cos\theta \quad (1.55)$$

Here, note that $\vec{A}$ is the area vector $\vec{A} = A\hat{n}$. Its magnitude is simply the area A and its direction is along the unit vector $\hat{n}$ perpendicular to the area as shown in Figure 1.31. Using this definition for flux, $\Phi_E = \vec{E} \cdot \vec{A}$, equations (1.53) and (1.54) can be obtained as special cases.

In Figure 1.31 (a), $\theta = 0^\circ$. Therefore,

$$\Phi_E = \vec{E} \cdot \vec{A} = EA$$

In Figure 1.31 (b), $\theta = 90^\circ$. Therefore,

$$\Phi_E = \vec{E} \cdot \vec{A} = 0$$

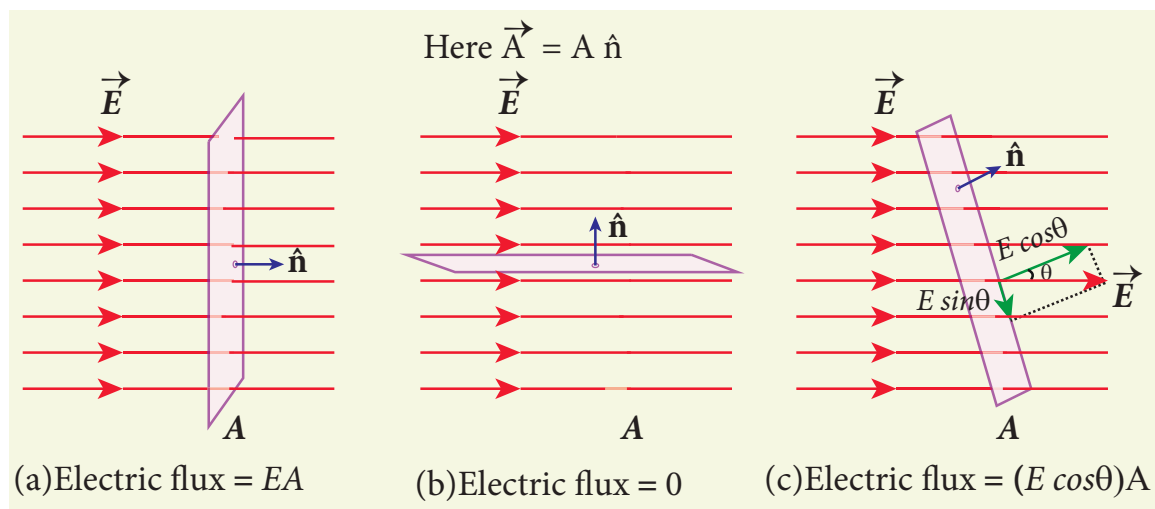
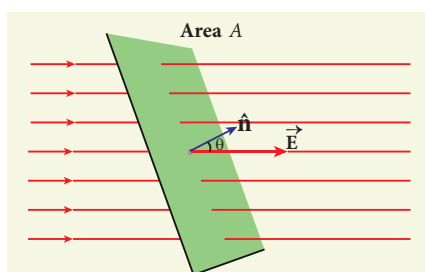


Figure 1.31 The electric flux for Uniform electric field

EXAMPLE 1.17

Calculate the electric flux through the rectangle of sides 5 cm and 10 cm kept in the region of a uniform electric field 100 NC^{-1} . The angle θ is 60° . If θ becomes zero, what is the electric flux?



Solution

The electric flux through the rectangular area

$$\begin{aligned}\Phi_E &= \vec{E} \cdot \vec{A} = EA \cos \theta \\ &= 100 \times 5 \times 10 \times 10^{-4} \times \cos 60^\circ \\ \Phi_E &= 0.25 \text{ Nm}^2 \text{ C}^{-1}\end{aligned}$$

For $\theta = 0^\circ$,

$$\begin{aligned}\Phi_E &= \vec{E} \cdot \vec{A} = EA \\ &= 100 \times 5 \times 10 \times 10^{-4} \\ &= 0.5 \text{ Nm}^2 \text{ C}^{-1}\end{aligned}$$

Electric flux through an arbitrary area kept in a non uniform electric field

Suppose the electric field is not uniform and the area A is not flat surface (Figure 1.32). Then the entire area can be divided into n small area segments $\Delta \vec{A}_1, \Delta \vec{A}_2, \Delta \vec{A}_3, \dots, \Delta \vec{A}_n$ such that each area element is almost flat and the electric field over such area element can be considered uniform.

The electric flux for the entire area A is approximately written as

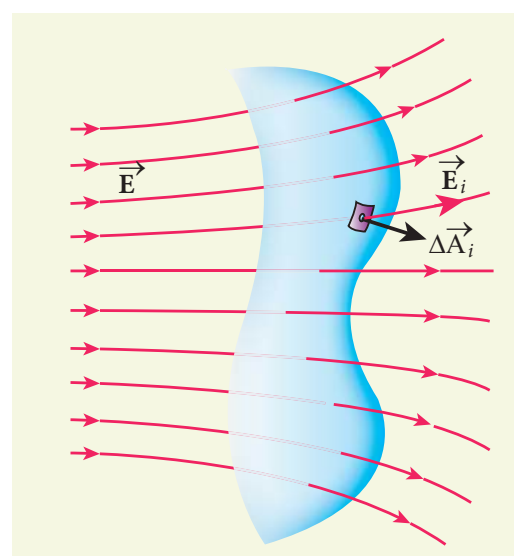


Figure 1.32 Electric flux for non-uniform electric Field



$$\Phi_E = \vec{E}_1 \cdot \Delta \vec{A}_1 + \vec{E}_2 \cdot \Delta \vec{A}_2 + \vec{E}_3 \cdot \Delta \vec{A}_3 + \dots + \vec{E}_n \cdot \Delta \vec{A}_n$$

$$= \sum_{i=1}^n \vec{E}_i \cdot \Delta \vec{A}_i \quad (1.56)$$

By taking the limit $\Delta \vec{A}_i \rightarrow 0$ (for all i) the summation in equation (1.56) becomes integration. The total electric flux for the entire area is given by

$$\Phi_E = \int \vec{E} \cdot d\vec{A} \quad (1.57)$$

From Equation (1.57), it is clear that the electric flux for a given surface depends on both the electric field pattern on the surface area and orientation of the surface with respect to the electric field.

1.6.2 Electric flux for closed surfaces

In the previous section, the electric flux for any arbitrary curved surface is discussed. Suppose a closed surface is present in the region of the non-uniform electric field as shown in Figure 1.33 (a).

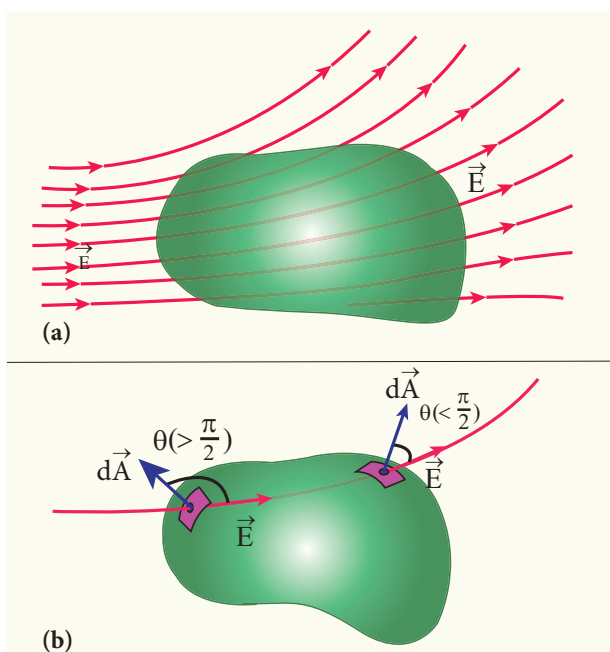


Figure 1.33 Electric flux over a closed surface

The total electric flux over this closed surface is written as

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} \quad (1.58)$$

Note the difference between equations (1.57) and (1.58). The integration in equation (1.58) is a closed surface integration and for each areal element, the outward normal is the direction of $d\vec{A}$ as shown in the Figure 1.33(b).

The total electric flux over a closed surface can be negative, positive or zero. In the Figure 1.33(b), it is shown that in one area element, the angle between $d\vec{A}$ and $\vec{E}$ is less than 90° , then the electric flux is positive and in another areal element, the angle between $d\vec{A}$ and $\vec{E}$ is greater than 90° , then the electric flux is negative.

In general, the electric flux is negative if the electric field lines enter the closed surface and positive if the electric field lines leave the closed surface.

1.6.3 Gauss law

A positive point charge Q is surrounded by an imaginary sphere of radius r as shown in Figure 1.34. We can calculate the total electric flux through the closed surface of the sphere using the equation (1.58).

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \oint E dA \cos \theta$$

The electric field of the point charge is directed radially outward at all points on the surface of the sphere. Therefore, the direction of the area element $d\vec{A}$ is along the electric field $\vec{E}$ and $\theta = 0^\circ$.

$$\Phi_E = \oint E dA \quad \text{since } \cos 0^\circ = 1 \quad (1.59)$$

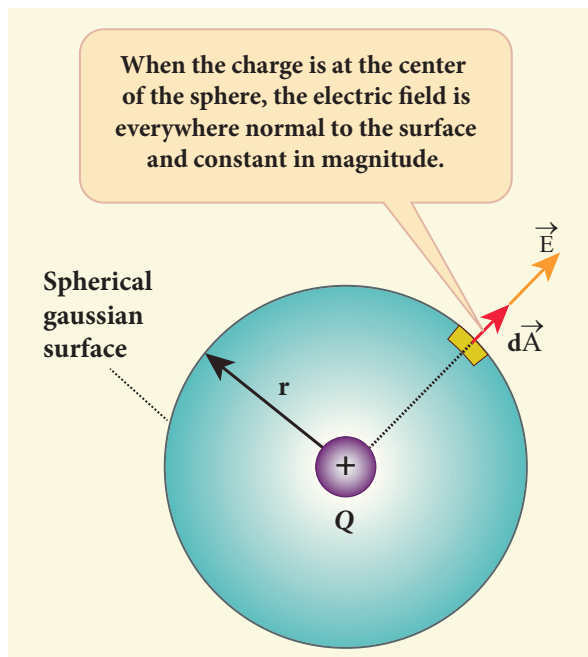


Figure 1.34 Total electric flux of point charge

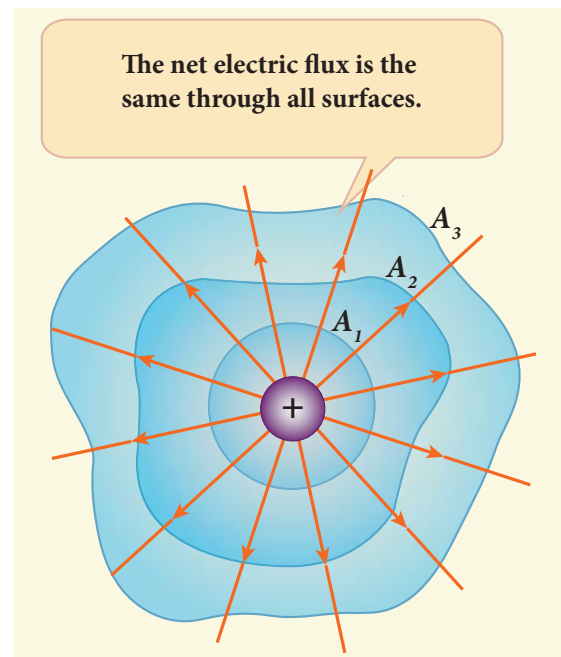


Figure 1.35 Gauss law for arbitrarily shaped surface

E is uniform on the surface of the sphere,

$$\Phi_E = E \oint dA \quad (1.60)$$

Substituting for $\oint dA = 4\pi r^2$ and $E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$ in equation (1.60), we get

$$\begin{aligned} \Phi_E &= \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \times 4\pi r^2 = 4\pi \frac{1}{4\pi\epsilon_0} Q \\ \Phi_E &= \frac{Q}{\epsilon_0} \end{aligned} \quad (1.61)$$

The equation (1.61) is called as Gauss's law.

The remarkable point about this result is that the equation (1.61) is equally true for any arbitrary shaped surface which encloses the charge Q and as shown in the Figure 1.35. It is seen that the total electric flux is the same for closed surfaces A_1 , A_2 and A_3 as shown in the Figure 1.35.

Gauss's law states that **if a charge Q is enclosed by an arbitrary closed surface, then the total electric flux Φ_E through the closed surface is**

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{encl}}{\epsilon_0} \quad (1.62)$$

where Q_{encl} denotes the charges within the closed surface.

Discussion of Gauss law

(i) The total electric flux through the closed surface depends only on the charges enclosed by the surface and the charges present outside the surface will not contribute to the flux and the shape of the closed surface which can be chosen arbitrarily.

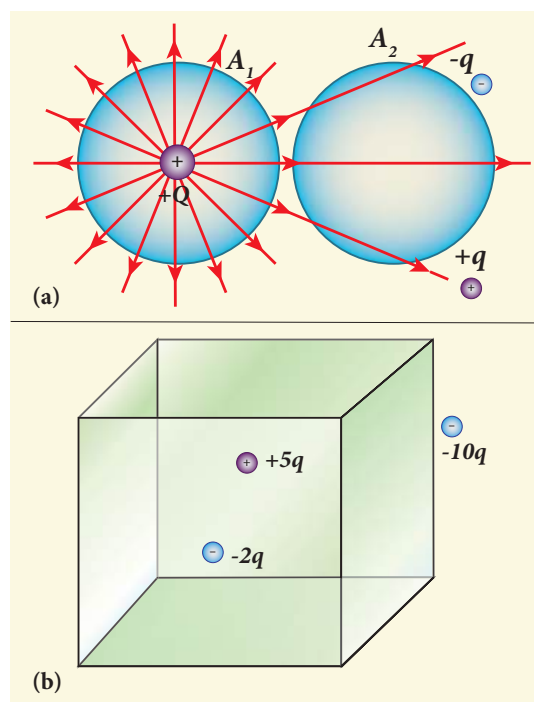
(ii) The total electric flux is independent of the location of the charges inside the closed surface.



(iii) To arrive at equation (1.62), we have chosen a spherical surface. This imaginary surface is called a Gaussian surface. The shape of the Gaussian surface to be chosen depends on the type of charge configuration and the kind of symmetry existing in that charge configuration. The electric field is spherically symmetric for a point charge, therefore spherical Gaussian surface is chosen. Cylindrical and planar Gaussian surfaces can be chosen for other kinds of charge configurations.

(iv) In the LHS of equation (1.62), the electric field $\vec{E}$ is due to charges present inside and outside the Gaussian surface but the charge Q_{encl} denotes the charges which lie only inside the Gaussian surface.

EXAMPLE 1.18



(i) In figure (a), calculate the electric flux through the closed areas A_1 and A_2 .

(ii) In figure (b), calculate the electric flux through the cube

Solution

(i) In figure (a), the area A_1 encloses the charge Q . So electric flux through this closed surface A_1 is $\frac{Q}{\epsilon_0}$. But the closed surface A_2 contains no charges inside, so electric flux through A_2 is zero.

(ii) In figure (b), the net charge inside the cube is $3q$ and the total electric flux in the cube is therefore $\Phi_E = \frac{3q}{\epsilon_0}$. Note that the charge $-10q$ lies outside the cube and it will not contribute the total flux through the surface of the cube.

1.6.4 Applications of Gauss law

Electric field due to any arbitrary charge configuration can be calculated using Coulomb's law or Gauss law. If the charge configuration possesses some kind of symmetry, then Gauss law is a very efficient way to calculate the electric field. It is illustrated in the following cases.

(i) Electric field due to an infinitely long charged wire

Consider an infinitely long straight wire having uniform linear charge density λ (charge per unit length). Let P be a point located at a perpendicular distance r from the wire (Figure 1.36(a)). The electric field at the point P can be found using Gauss law.

We choose two small charge elements A_1 and A_2 on the wire which are at equal distances from the point P. The resultant electric field due to these two charge elements points radially away from the charged wire and the magnitude of electric field is same at all points on the circle of radius r . This is shown in the Figure 1.36(b). Since the charged wire possesses

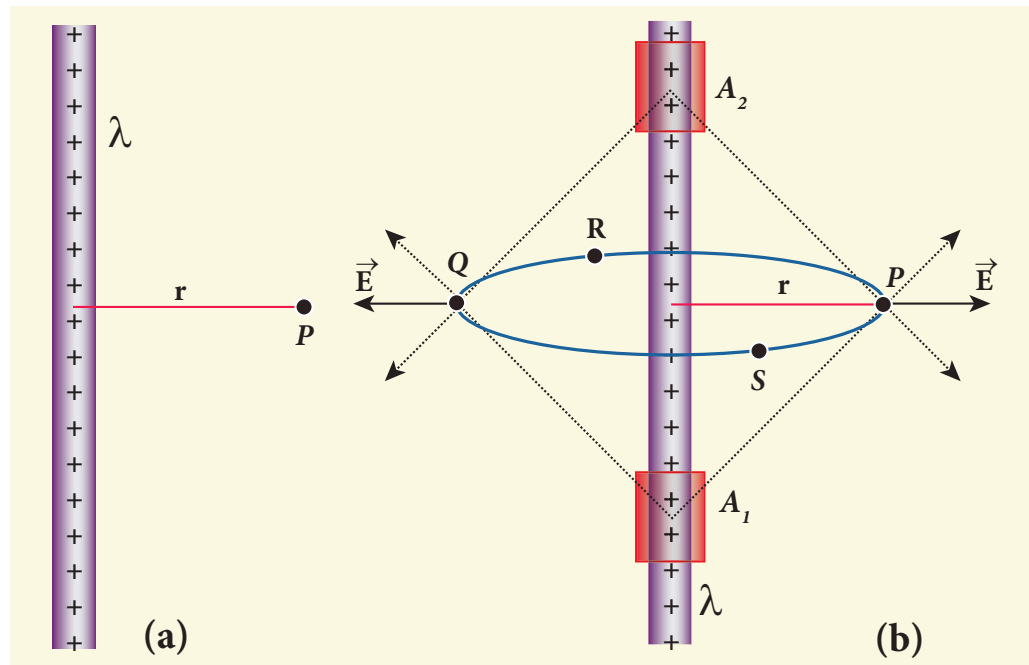


Figure 1.36 Electric field due to infinite long charged wire

a cylindrical symmetry, let us choose a cylindrical Gaussian surface of radius r and length L as shown in the Figure 1.37.

The total electric flux through this closed surface is calculated as follows.

$$\begin{aligned}\Phi_E &= \oint \vec{E} \cdot d\vec{A} \\ &= \int_{\text{Curved surface}} \vec{E} \cdot d\vec{A} + \int_{\text{top surface}} \vec{E} \cdot d\vec{A} + \int_{\text{bottom surface}} \vec{E} \cdot d\vec{A} \quad (1.63)\end{aligned}$$

It is seen from Figure (1.37) that for the curved surface, $\vec{E}$ is parallel to $\vec{A}$ and $\vec{E} \cdot d\vec{A} = E dA$. For the top and bottom surfaces, $\vec{E}$ is perpendicular to $\vec{A}$ and $\vec{E} \cdot d\vec{A} = 0$

Substituting these values in the equation (1.63) and applying Gauss law to the cylindrical surface, we have

$$\Phi_E = \int_{\text{Curved surface}} E dA = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (1.64)$$

Since the magnitude of the electric field for the entire curved surface is constant, E is

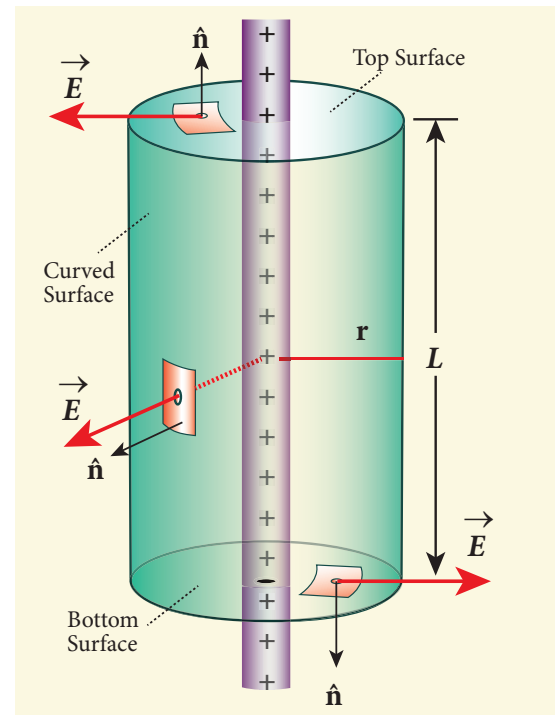


Figure 1.37 Cylindrical Gaussian surface

taken out of the integration and Q_{encl} is given by $Q_{\text{encl}} = \lambda L$, where λ is the linear charge density (charge present per unit length).

$$E \int_{\text{Curved surface}} dA = \frac{\lambda L}{\epsilon_0} \quad (1.65)$$



Here $\int_{\text{Curved surface}} dA = \text{total area of the curved surface} = 2\pi rL$. Substituting this in equation (1.65), we get

$$E \cdot 2\pi rL = \frac{\lambda L}{\epsilon_0}$$

$$E = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \quad (1.66)$$

In vector form,

$$\vec{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \hat{r} \quad (1.67)$$

The electric field due to the infinite charged wire depends on $\frac{1}{r}$ rather than $\frac{1}{r^2}$ which is for a point charge.

Equation (1.67) indicates that the electric field is always along the perpendicular direction ($\hat{r}$) to wire. In fact, if $\lambda > 0$ then $\vec{E}$ points perpendicularly outward ($\hat{r}$) from the wire and if $\lambda < 0$, then $\vec{E}$ points perpendicularly inward ($-\hat{r}$).

The equation (1.67) is true only for an infinitely long charged wire. For a charged wire of finite length, the electric field need not be radial at all points. However, equation (1.67) for such a wire is taken approximately true around the mid-point of the wire and far away from the both ends of the wire

(ii) Electric field due to charged infinite plane sheet

Consider an infinite plane sheet of charges with uniform surface charge density σ (charge present per unit area). Let P be a point at a distance of r from the sheet as shown in the Figure 1.38.

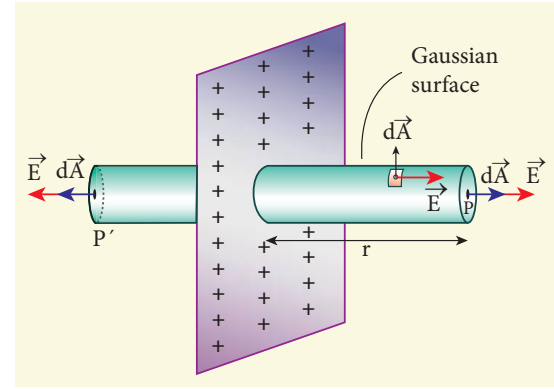


Figure 1.38 Electric field due to charged infinite planar sheet

Since the plane is infinitely large, the electric field should be same at all points equidistant from the plane and radially directed outward at all points. A cylindrical Gaussian surface of length $2r$ and two flat surfaces each of area A is chosen such that the infinite plane sheet passes perpendicularly through the middle part of the Gaussian surface.

Total electric flux linked with the cylindrical surface,

$$\Phi_E = \oint \vec{E} \cdot d\vec{A}$$

$$= \int_{\text{Curved surface}} \vec{E} \cdot d\vec{A} + \int_P \vec{E} \cdot d\vec{A} + \int_{P'} \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (1.68)$$

The electric field is perpendicular to the area element at all points on the curved surface and is parallel to the surface areas at P and P' (Figure 1.38). Then, applying Gauss' law,

$$\Phi_E = \int_P E dA + \int_{P'} E dA = \frac{Q_{\text{encl}}}{\epsilon_0} \quad (1.69)$$

Since the magnitude of the electric field at these two equal flat surfaces is uniform, E is taken out of the integration and Q_{encl} is given by $Q_{\text{encl}} = \sigma A$, we get

$$2E \int_P dA = \frac{\sigma A}{\epsilon_0}$$

The total area of surface either at P or P'

$$\int_P dA = A$$

$$\text{Hence } 2EA = \frac{\sigma A}{\epsilon_0} \text{ or } E = \frac{\sigma}{2\epsilon_0} \quad (1.70)$$

$$\text{In vector form, } \vec{E} = \frac{\sigma}{2\epsilon_0} \hat{n} \quad (1.71)$$

Here $\hat{n}$ is the outward unit vector normal to the plane. Note that the electric field due to an infinite plane sheet of charge depends on the surface charge density and is independent of the distance r .

The electric field will be the same at any point farther away from the charged plane. Equation (1.71) implies that if $\sigma > 0$ the electric field at any point P is along outward perpendicular $\hat{n}$ drawn to the plane and if $\sigma < 0$, the electric field points inward perpendicularly to the plane ($-\hat{n}$).

For a finite charged plane sheet, equation (1.71) is approximately true only in the middle region of the plane and at points far away from both ends.

(iii) Electric field due to two parallel charged infinite sheets

Consider two infinitely large charged plane sheets with equal and opposite charge

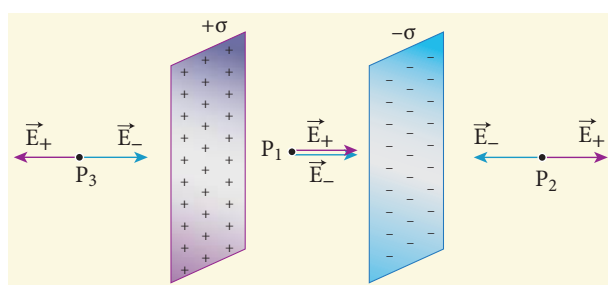


Figure 1.39 Electric field due to two parallel charged sheets

densities $+\sigma$ and $-\sigma$ which are placed parallel to each other as shown in the Figure 1.39.

The electric field between the plates and outside the plates is found using Gauss law. The magnitude of the electric field due to an infinite charged plane sheet is $\frac{\sigma}{2\epsilon_0}$ and it points perpendicularly outward if $\sigma > 0$ and points inward if $\sigma < 0$.

At the points P_2 and P_3 , the electric field due to both plates are equal in magnitude and opposite in direction (Figure 1.39). As a result, electric field at a point outside the plates is zero. But between the plates, electric fields are in the same direction i.e., towards the right and the total electric field at a point P_1 is

$$E_{\text{inside}} = \frac{\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{\sigma}{\epsilon_0} \quad (1.72)$$

The direction of the electric field between the plates is directed from positively charged plate to negatively charged plate and is uniform everywhere between the plates.

(iv) Electric field due to a uniformly charged spherical shell

Consider a uniformly charged spherical shell of radius R carrying total charge Q as shown in Figure 1.40. The electric field at points outside and inside the sphere can be found using Gauss law.

Case (a) At a point outside the shell ($r > R$)

Let us choose a point P outside the shell at a distance r from the centre as shown in Figure 1.40 (a). The charge is uniformly distributed on the surface of the sphere (spherical symmetry). Hence the electric field must point radially outward if $Q > 0$ and point radially inward if $Q < 0$. So a spherical Gaussian surface of radius r is chosen and

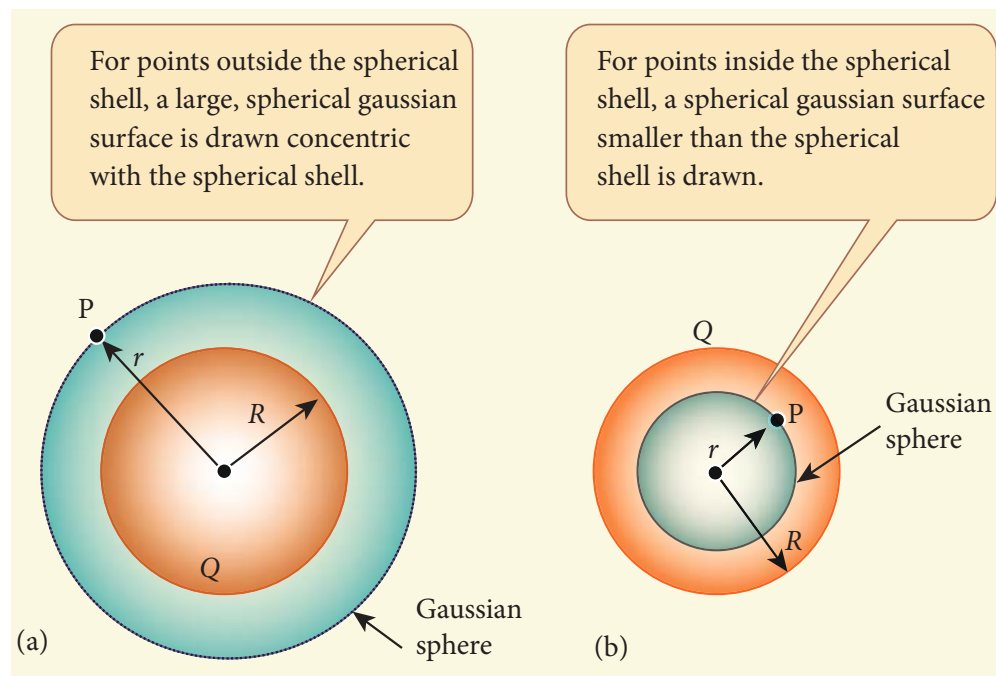


Figure 1.40 The electric field due to a charged spherical shell

the total charge enclosed by this Gaussian surface is Q . Applying Gauss law

$$\oint_{\text{Gaussian surface}} \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0} \quad (1.73)$$

The electric field $\vec{E}$ and $d\vec{A}$ point in the same direction (outward normal) at all the points on the Gaussian surface. The magnitude of $\vec{E}$ is also the same at all points due to the spherical symmetry of the charge distribution.

$$\text{Hence } E \oint_{\text{Gaussian surface}} dA = \frac{Q}{\epsilon_0} \quad (1.74)$$

But $\oint_{\text{Gaussian surface}} dA = \text{total area of Gaussian surface} = 4\pi r^2$. Substituting this value in equation (1.74)

$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$$
$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0} \quad (\text{or}) \quad E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

In vector form,

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r} \quad (1.75)$$

The electric field is radially outward if $Q > 0$ and radially inward if $Q < 0$. From equation (1.75), we infer that the electric field at a point outside the shell will be the same as if the entire charge Q is concentrated at the centre of the spherical shell. (A similar result is observed in gravitation, for gravitational force due to a spherical shell with mass M)

Case (b): At a point on the surface of the spherical shell ($r = R$)

The electrical field at points on the spherical shell ($r = R$) is given by

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 R^2} \hat{r} \quad (1.76)$$

Case (c): At a point inside the spherical shell ($r < R$)

Consider a point P inside the shell at a distance r from the centre. A Gaussian

sphere of radius r is constructed as shown in the Figure 1.40 (b). Applying Gauss law

$$\oint_{\text{Gaussian surface}} \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0} \quad (1.77)$$

Since Gaussian surface encloses no charge, $Q = 0$. The equation (1.77) becomes

$$E = 0 \quad (r < R) \quad (1.78)$$

The electric field due to the uniformly charged spherical shell is zero at all points inside the shell.

A graph is plotted between the electric field and radial distance. This is shown in Figure 1.41.

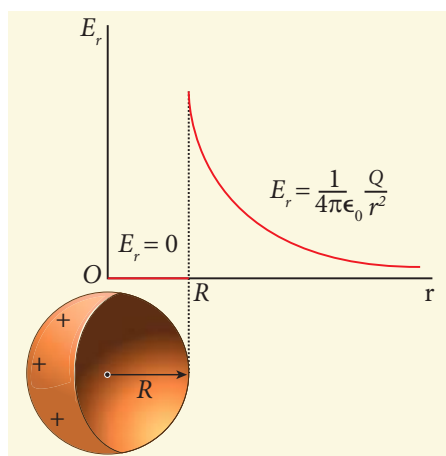


Figure 1.41 Electric field versus distance for a spherical shell of radius R

Note

Gauss law is a powerful technique whenever a given charge configuration possesses spherical, cylindrical or planar symmetry, then the electric field due to such a charge configuration can be easily found. If there is no such symmetry, the direct method (Coulomb's law and calculus) can be used. For example, it is difficult to use Gauss law to find the electric field for a dipole since it has no spherical, cylindrical or planar symmetry.

1.7

ELECTROSTATICS OF CONDUCTORS AND DIELECTRICS

1.7.1 Conductors at electrostatic equilibrium

An electrical conductor has a large number of mobile charges which are free to move in the material. In a metallic conductor, these mobile charges are free electrons which are not bound to any atom and therefore are free to move on the surface of the conductor. When there is no external electric field, the free electrons are in continuous random motion in all directions. As a result, there is no net motion of electrons along any particular direction which implies that the conductor is in electrostatic equilibrium. Thus at electrostatic equilibrium, there is no net current in the conductor. A conductor at electrostatic equilibrium has the following properties.

(i) **The electric field is zero everywhere inside the conductor. This is true regardless of whether the conductor is solid or hollow.**

This is an experimental fact. Suppose the electric field is not zero inside the metal, then there will be a force on the mobile charge carriers due to this electric field. As a result, there will be a net motion of the mobile charges, which contradicts the conductors being in electrostatic equilibrium. Thus the electric field is zero everywhere inside the conductor. We can also understand this fact by applying an external uniform electric field on the conductor. This is shown in Figure 1.42.

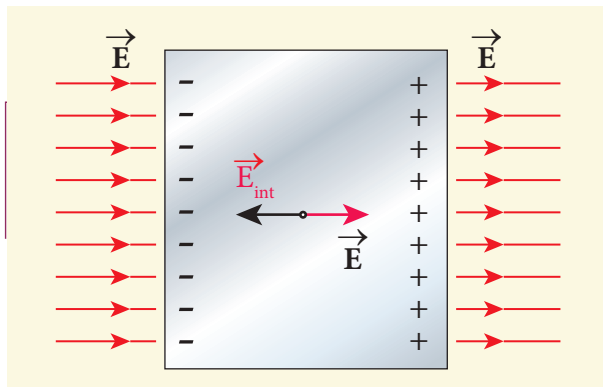


Figure 1.42 Electric field of conductors

Before applying the external electric field, the free electrons in the conductor are uniformly distributed in the conductor. When an electric field is applied, the free electrons accelerate to the left causing the left plate to be negatively charged and the right plate to be positively charged as shown in Figure 1.42.

Due to this realignment of free electrons, there will be an internal electric field created inside the conductor which increases until it nullifies the external electric field. Once the external electric field is nullified the conductor is said to be in electrostatic equilibrium. The time taken by a conductor to reach electrostatic equilibrium is in the order of 10^{-16} s, which can be taken as almost instantaneous.

(ii) There is no net charge inside the conductors. The charges must reside only on the surface of the conductors.

We can prove this property using Gauss law. Consider an arbitrarily shaped conductor as shown in Figure 1.43.

A Gaussian surface is drawn inside the conductor such that it is very close to the surface of the conductor. Since the electric field is zero everywhere inside the conductor, the net electric flux is also zero over this Gaussian surface. From Gauss's law, this implies that there is no net charge inside the conductor. Even if some charge is introduced

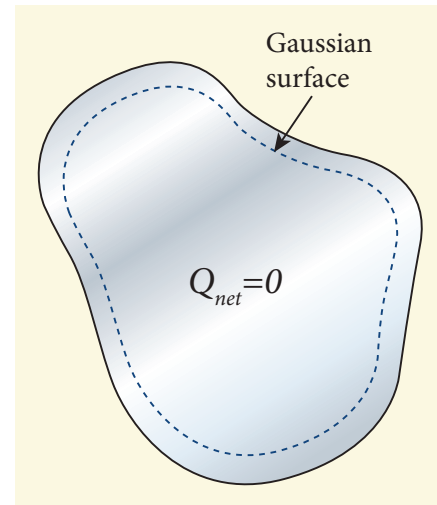


Figure 1.43 No net charge inside the conductor

inside the conductor, it immediately reaches the surface of the conductor.

(iii) The electric field outside the conductor is perpendicular to the surface of the conductor and has a magnitude of $\frac{\sigma}{\epsilon_0}$ where σ is the surface charge density at that point.

If the electric field has components parallel to the surface of the conductor, then free electrons on the surface of the conductor would experience acceleration (Figure 1.44(a)). This means that the

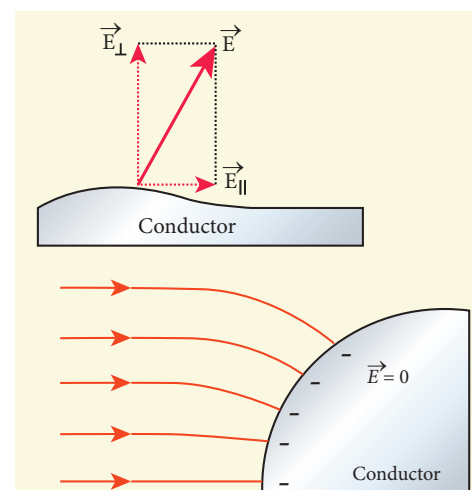


Figure 1.44 (a) Electric field is along the surface (b) Electric field is perpendicular to the surface of the conductor

conductor is not in equilibrium. Therefore at electrostatic equilibrium, the electric field must be perpendicular to the surface of the conductor. This is shown in Figure 1.44 (b).

We now prove that the electric field has magnitude $\frac{\sigma}{\epsilon_0}$ just outside the conductor's surface. Consider a small cylindrical Gaussian surface, as shown in the Figure 1.45. One half of this cylinder is embedded inside the conductor.

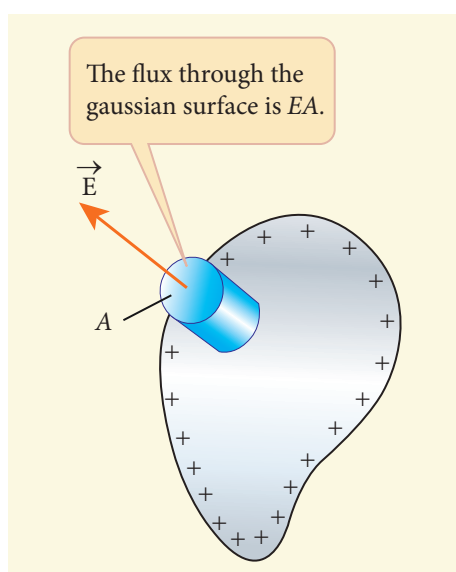


Figure 1.45 The electric field on the surface of the conductor

Since electric field is normal to the surface of the conductor, the curved part of the cylinder has zero electric flux. Also inside the conductor, the electric field is zero. Hence the bottom flat part of the Gaussian surface has no electric flux.

Therefore the top flat surface alone contributes to the electric flux. The electric field is parallel to the area vector and the total charge inside the surface is σA . By applying Gauss's law,

$$EA = \frac{\sigma A}{\epsilon_0}$$

In vector form,

$$\vec{E} = \frac{\sigma}{\epsilon_0} \hat{n} \quad (1.79)$$

where $\hat{n}$ represents the unit vector outward normal to the surface of the conductor. Suppose $\sigma < 0$, then electric field points inward perpendicular to the surface.

(iv) The electrostatic potential has the same value on the surface and inside of the conductor.

We know that the conductor has no parallel electric component on the surface which means that charges can be moved on the surface without doing any work. This is possible only if the electrostatic potential is constant at all points on the surface and there is no potential difference between any two points on the surface.

Since the electric field is zero inside the conductor, the potential is the same as the surface of the conductor. Thus at electrostatic equilibrium, the conductor is always at equipotential.

1.7.2 Electrostatic shielding

Using Gauss law, we can prove that the electric field inside the charged spherical shell is zero. Further, we can show that the electric field inside both hollow and solid conductors is zero. It is a very interesting property which has an important consequence.

Consider a cavity inside the conductor as shown in Figure 1.46 (a). Whatever be the charges at the surfaces and whatever be the electrical disturbances outside, the electric field inside the cavity is zero. A sensitive electrical instrument which is to be protected from external electrical disturbance can be kept inside this cavity. This is called electrostatic shielding.

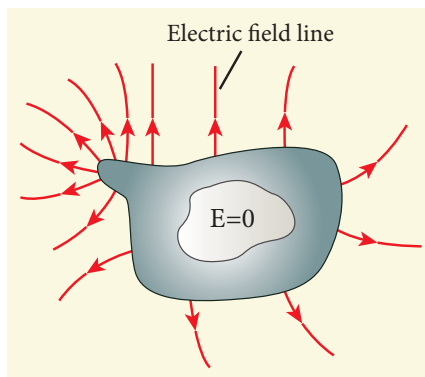


Figure 1.46 (a) Electric field inside the cavity (b) Faraday cage

Faraday cage is an instrument used to demonstrate this effect. It is made up of metal bars as shown in Figure 1.46 (b). If an artificial lightning jolt is created outside, the person inside is not affected.

During lightning accompanied by a thunderstorm, it is always safer to sit inside a bus than in open ground or under a tree. The metal body of the bus provides electrostatic shielding, since the electric field inside is zero. During lightning, the charges flow through the body of the conductor to the ground with no effect on the person inside that bus.

1.7.3 Electrostatic induction

In section 1.1, we have learnt that an object can be charged by rubbing using an appropriate material. Whenever a charged

rod is touched by another conductor, charges start to flow from charged rod to the conductor. Is it possible to charge a conductor without any contact? The answer is yes. This type of **charging without actual contact is called electrostatic induction.**

(i) Consider an uncharged (neutral) conducting sphere at rest on an insulating stand. Suppose a negatively charged rod is brought near the conductor without touching it, as shown in Figure 1.47(a).

The negative charge of the rod repels the electrons in the conductor to the opposite side. As a result, positive charges are induced near the region of the charged rod while negative charges on the farther side.

Before introducing the charged rod, the free electrons were distributed uniformly on the surface of the conductor and the net charge is zero. Once the charged rod is brought near the conductor, the distribution is no longer uniform with more electrons located on the farther side of the rod and positive charges are located closer to the rod. But the total charge is zero.

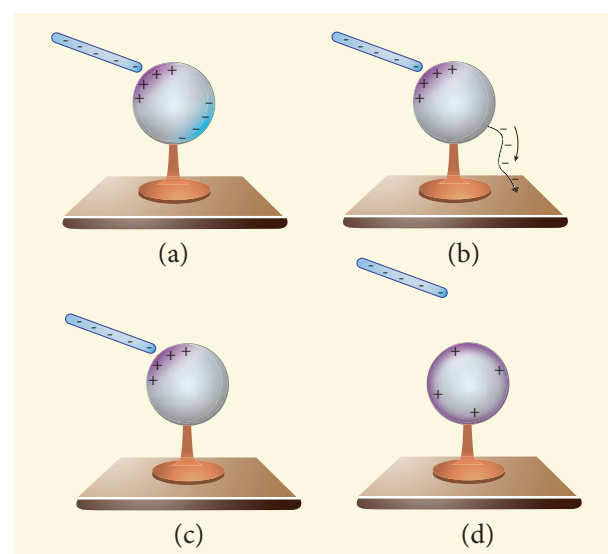


Figure 1.47 Various steps in electrostatic induction



(ii) Now the conducting sphere is connected to the ground through a conducting wire. This is called grounding. Since the ground can always receive any amount of electrons, grounding removes the electron from the conducting sphere. Note that positive charges will not flow to the ground because they are attracted by the negative charges of the rod (Figure 1.47(b)).

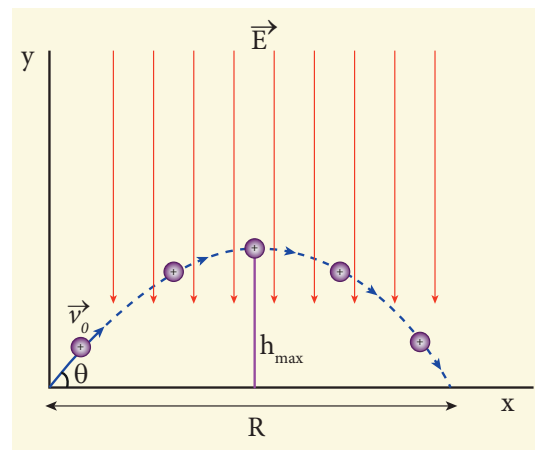
(iii) When the grounding wire is removed from the conductor, the positive charges remain near the charged rod (Figure 1.47(c)).

(iv) Now the charged rod is taken away from the conductor. As soon as the charged rod is removed, the positive charge gets distributed uniformly on the surface of the conductor (Figure 1.47(d)). By this process, the neutral conducting sphere becomes positively charged.

For an arbitrary shaped conductor, the intermediate steps and conclusion are the same except the final step. The distribution of positive charges is not uniform for arbitrarily-shaped conductors. Why is it not uniform? The reason for it is discussed in the section 1.9

EXAMPLE 1.19

A small ball of conducting material having a charge $+q$ and mass m is thrown upward at an angle θ to horizontal surface with an initial speed v_0 as shown in the figure. There exists a uniform electric field E downward along with the gravitational field g . Calculate the range, maximum height and time of flight in the motion of this charged ball. Neglect the effect of air and treat the ball as a point mass.



Solution

If the conductor has no net charge, then its motion is the same as usual projectile motion of a mass m which we studied in Kinematics (unit 2, vol-1 XI physics). Here, in this problem, in addition to downward gravitational force, the charge also will experience a downward uniform electrostatic force.

The acceleration of the charged ball due to gravity $= -g\hat{j}$

The acceleration of the charged ball due to uniform electric field $= -\frac{qE}{m}\hat{j}$

The total acceleration of charged ball in downward direction $\vec{a} = -\left(g + \frac{qE}{m}\right)\hat{j}$

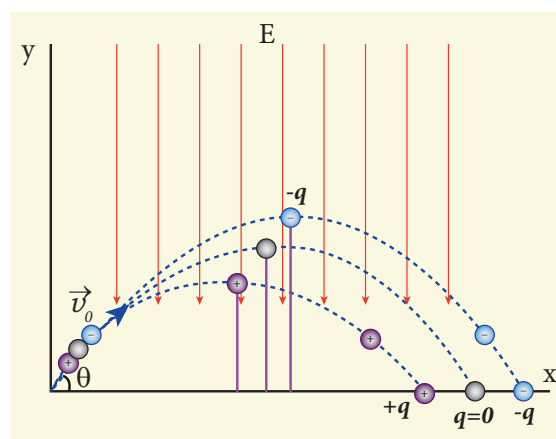
It is important here to note that the acceleration depends on the mass of the object. Galileo's conclusion that all objects fall at the same rate towards the Earth is true only in a uniform gravitational field. When a uniform electric field is included, the acceleration of a charged object depends on both mass and charge.

But still the acceleration $a = \left(g + \frac{qE}{m}\right)$ is constant throughout the motion. Hence we use kinematic equations to calculate the

range, maximum height and time of flight. In fact we can simply replace g by $g + \frac{qE}{m}$ in the usual expressions of range, maximum height and time of flight of a projectile.

	Without charge	With the charge $+q$
Time of flight T	$\frac{2v_0 \sin \theta}{g}$	$\frac{2v_0 \sin \theta}{\left(g + \frac{qE}{m}\right)}$
Maximum height $h_{\max}$	$\frac{v_0^2 \sin^2 \theta}{2g}$	$\frac{v_0^2 \sin^2 \theta}{2\left(g + \frac{qE}{m}\right)}$
Range R	$\frac{v_0^2 \sin 2\theta}{g}$	$\frac{v_0^2 \sin 2\theta}{\left(g + \frac{qE}{m}\right)}$

Note that the time of flight, maximum height, range are all inversely proportional to the acceleration of the object. Since $\left(g + \frac{qE}{m}\right) > g$ for charge $+q$, the quantities T , $h_{\max}$ and R will decrease when compared to the motion of an object of mass m and zero net charge. Suppose the charge is $-q$, then $\left(g - \frac{qE}{m}\right) < g$, and the quantities T , $h_{\max}$ and R will increase. Interestingly the trajectory is still parabolic as shown in the figure.



1.7.4 Dielectrics or insulators

A dielectric is a non-conducting material and has no free electrons. The electrons in a dielectric are bound within the atoms. Ebonite, glass and mica are some examples of dielectrics. When an external electric field is applied, the electrons are not free to move anywhere but they are realigned in a specific way. A dielectric is made up of either polar molecules or non-polar molecules.

Non-polar molecules

A non-polar molecule is one in which centres of positive and negative charges coincide. As a result, it has no permanent dipole moment. Examples of non-polar molecules are hydrogen (H_2), oxygen (O_2), and carbon dioxide (CO_2) etc.

When an external electric field is applied, the centres of positive and negative charges are separated by a small distance which induces dipole moment in the direction of the external electric field. Then the dielectric is said to be polarized by an external electric field. This is shown in Figure 1.48.

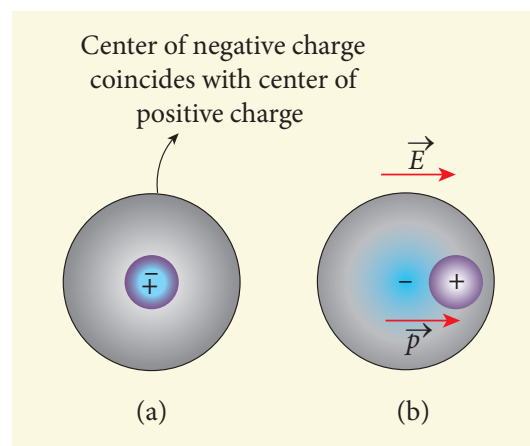


Figure 1.48 Non polar molecules (a) without external field (b) with the external field

Polar molecules

In polar molecules, the centres of the positive and negative charges are separated even in the absence of an external electric field. They have a permanent dipole moment. Due to thermal motion, the direction of each dipole moment is oriented randomly (Figure 1.49(a)). Hence the net dipole moment is zero in the absence of an external electric field. Examples of polar molecules are H_2O , N_2O , HCl , NH_3 .

When an external electric field is applied, the dipoles inside the material tend to align in the direction of the electric field. Hence a net dipole moment is induced in it. Then the dielectric is said to be polarized by an external electric field (Figure 1.49(b)).

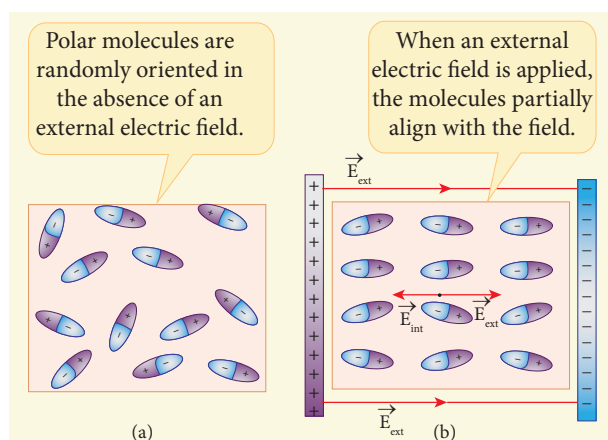


Figure 1.49 (a) Randomly oriented polar molecules (b) Align with the external electric field

Polarisation

In the presence of an external electric field, the dipole moment is induced in the dielectric material. **Polarisation $\vec{P}$ is defined as the total dipole moment per unit volume of the dielectric.** For most dielectrics (linear isotropic), the Polarisation is directly proportional to the strength of the external electric field. This is written as

$$\vec{P} = \chi_e \vec{E}_{\text{ext}} \quad (1.80)$$

where χ_e is a constant called the electric susceptibility which is a characteristic of each dielectric.

1.7.5 Induced Electric field inside the dielectric

When an external electric field is applied on a conductor, the charges are aligned in such a way that an internal electric field is created which tends to cancel the external electric field. But in the case of a dielectric, which has no free electrons, the external electric field only realigns the charges so that an internal electric field is produced. The magnitude of the internal electric field is smaller than that of external electric field. Therefore the net electric field inside the dielectric is not zero but is parallel to an external electric field with magnitude less than that of the external electric field. For example, let us consider a rectangular dielectric slab placed between two oppositely charged plates (capacitor) as shown in the Figure 1.50.

The uniform electric field between the plates acts as an external electric field $\vec{E}_{\text{ext}}$ which polarizes the dielectric placed between plates. The positive charges are induced on one side surface and negative charges are induced on the other side of surface.

But inside the dielectric, the net charge is zero even in a small volume. So the dielectric in the external field is equivalent to two oppositely charged sheets with the surface charge densities $+\sigma_b$ and $-\sigma_b$. These charges are called bound charges. They are not free to move like free electrons in

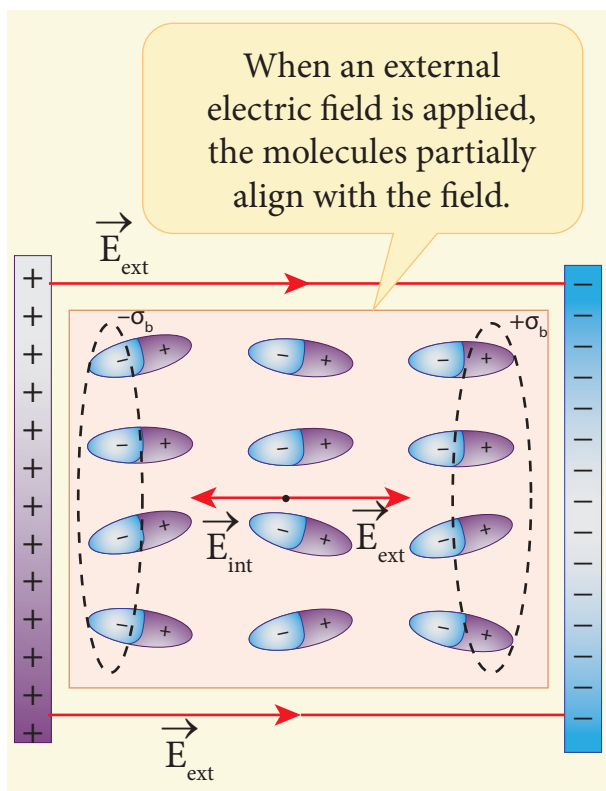


Figure 1.50 Induced electric field lines inside the dielectric

conductors. This is shown in the Figure 1.50.

For example, the charged balloon after rubbing sticks onto a wall. The reason is that the negatively charged balloon is brought near the wall, it polarizes opposite charges on the surface of the wall, which attracts the balloon. This is shown in Figure 1.51.

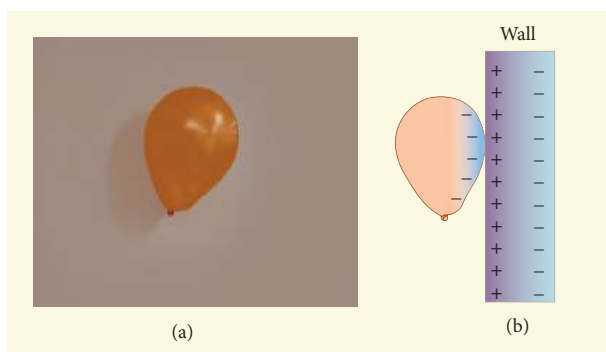


Figure 1.51 (a) Balloon sticks to the wall (b) Polarisation of wall due to the electric field created by the balloon

1.7.6 Dielectric strength

When the external electric field applied to a dielectric is very large, it tears the atoms apart so that the bound charges become free charges. Then the dielectric starts to conduct electricity. This is called **dielectric breakdown**. The maximum electric field the dielectric can withstand before it breakdown is called dielectric strength. For example, the dielectric strength of air is $3 \times 10^6 \text{ V m}^{-1}$. If the applied electric field increases beyond this, a spark is produced in the air. The dielectric strengths of some dielectrics are given in the Table 1.1.

Table 1.1 Dielectric strength

Substance	Dielectric strength (Vm^{-1})
Mica	100×10^6
Teflon	60×10^6
Paper	16×10^6
Air	3×10^6
Pyrex glass	14×10^6

1.8

CAPACITORS AND CAPACITANCE

1.8.1 Capacitors

Capacitor is a device used to store electric charge and electrical energy. It consists of two conducting objects (usually plates or sheets) separated by some distance. Capacitors are widely used in many electronic circuits and have applications in many areas of science and technology.

A simple capacitor consists of two parallel metal plates separated by a small distance as shown in Figure 1.52 (a).

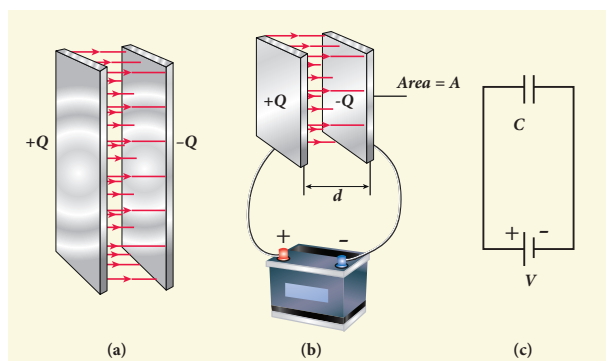


Figure 1.52 (a) Parallel plate capacitor
(b) Capacitor connected with a battery
(c) Symbolic representation of capacitor.

When a capacitor is connected to a battery of potential difference V , the electrons are transferred from one plate to the other plate by battery so that one plate becomes negatively charged with a charge of $-Q$ and the other plate positively charged with $+Q$. The potential difference between the plates is equivalent to the battery's terminal voltage. This is shown in Figure 1.52 (b). If the battery voltage is increased, the amount of charges stored in the plates also increase. In general, the charge stored in the capacitor is proportional to the potential difference between the plates.

$$Q \propto V$$

so that $Q = CV$

where the C is the proportionality constant called capacitance. **The capacitance C of a capacitor is defined as the ratio of the magnitude of charge on either of the conductor plates to the potential difference existing between them.**

$$C = \frac{Q}{V} \quad (1.81)$$

The SI unit of capacitance is *coulomb per volt* or *farad* (F) in honor of Michael Faraday. Farad is a larger unit of capacitance. In practice, capacitors are available in the range of microfarad ($1\mu\text{F} = 10^{-6}\text{F}$) to picofarad ($1\text{pF} = 10^{-12}\text{F}$). A capacitor is represented by the symbol $\text{||} \text{---} \text{||}$ or $\text{||} \text{---} \text{||}$. Note that the total charge stored in the capacitor is zero ($Q - Q = 0$). When we say the capacitor stores charges, it means the amount of charge that can be stored in any one of the plates.

Nowadays there are capacitors available in various shapes (cylindrical, disk) and types (tantalum, ceramic and electrolytic), as shown in Figure 1.53. These capacitors are extensively used in various kinds of electronic circuits.



Figure 1.53 Various types of capacitors

Capacitance of a parallel plate capacitor

Consider a capacitor with two parallel plates each of cross-sectional area A and separated by a distance d as shown in Figure 1.54.

The electric field between two infinite parallel plates is uniform and is given by $E = \frac{\sigma}{\epsilon_0}$ where σ is the surface charge density on either plates $\left(\sigma = \frac{Q}{A}\right)$. If the separation distance d is very much smaller than the size of the plate ($d^2 \ll A$), then the above result can be used even for finite-sized parallel plate capacitor.

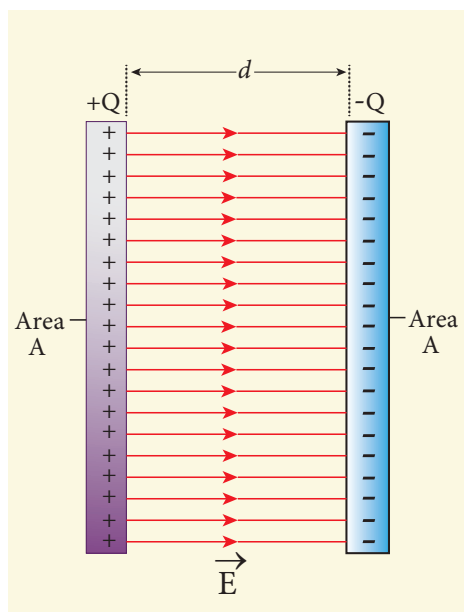


Figure 1.54 Capacitance of a parallel plate capacitor

The electric field between the plates is

$$E = \frac{Q}{A\epsilon_0} \quad (1.82)$$

Since the electric field is uniform, the electric potential difference between the plates having separation d is given by

$$V = Ed = \frac{Qd}{A\epsilon_0} \quad (1.83)$$

Therefore the capacitance of the capacitor is given by

$$C = \frac{Q}{V} = \frac{Q}{\left(\frac{Qd}{A\epsilon_0}\right)} = \frac{\epsilon_0 A}{d} \quad (1.84)$$

From equation (1.84), it is evident that capacitance is directly proportional to the area of cross section and is inversely proportional to the distance between the plates. This can be understood from the following.

(i) If the area of cross-section of the capacitor plates is increased, more charges can be distributed for the same potential difference. As a result, the capacitance is increased.

(ii) If the distance d between the two plates is reduced, the potential difference between the plates ($V = Ed$) decreases with E constant. As a result, voltage difference between the terminals of the battery increases which in turn leads to an additional flow of charge to the plates from the battery, till the voltage on the capacitor equals to the battery's terminal voltage. Suppose the distance is increased, the capacitor voltage increases and becomes greater than the battery voltage. Then, the charges flow from capacitor plates to battery till both voltages become equal.

EXAMPLE 1.20

A parallel plate capacitor has square plates of side 5 cm and separated by a distance of 1 mm. (a) Calculate the capacitance of this capacitor. (b) If a 10 V battery is connected to the capacitor, what is the charge stored in any one of the plates? (The value of $\epsilon_0 = 8.85 \times 10^{-12} \text{ N}^{-1}\text{m}^{-2} \text{ C}^2$)

Solution

(a) The capacitance of the capacitor is

$$\begin{aligned} C &= \frac{\epsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 25 \times 10^{-4}}{1 \times 10^{-3}} \\ &= 221.2 \times 10^{-13} \text{ F} \\ C &= 22.12 \times 10^{-12} \text{ F} = 22.12 \text{ pF} \end{aligned}$$

(b) The charge stored in any one of the plates is $Q = CV$, Then

$$Q = 22.12 \times 10^{-12} \times 10 = 221.2 \times 10^{-12} \text{ C} = 221.2 \text{ pC}$$



Sometimes we notice that the ceiling fan does not start rotating as soon as it is switched on. But when we rotate the blades, it starts to rotate as usual. Why it is so? We know that to rotate any object, there must be a torque applied on the object. For the ceiling fan, the initial torque is given by the capacitor widely known as a condenser. If the condenser is faulty, it will not give sufficient initial torque to rotate the blades when the fan is switched on.

1.8.2 Energy stored in the capacitor

Capacitor not only stores the charge but also it stores energy. When a battery is connected to the capacitor, electrons of total charge $-Q$ are transferred from one plate to the other plate. To transfer the charge, work is done by the battery. This work done is stored as electrostatic potential energy in the capacitor.

To transfer an infinitesimal charge dQ for a potential difference V , the work done is given by

$$dW = V dQ \quad (1.85)$$

where $V = \frac{Q}{C}$

The total work done to charge a capacitor is

$$W = \int_0^Q \frac{Q}{C} dQ = \frac{Q^2}{2C} \quad (1.86)$$

This work done is stored as electrostatic potential energy (U_E) in the capacitor.

$$U_E = \frac{Q^2}{2C} = \frac{1}{2} CV^2 \quad (\because Q = CV) \quad (1.87)$$

where $Q = CV$ is used. This stored energy is thus directly proportional to the capacitance of the capacitor and the square of the voltage between the plates of the capacitor.

But where is this energy stored in the capacitor? To understand this question, the equation (1.87) is rewritten as follows using the results $C = \frac{\epsilon_0 A}{d}$ and $V = Ed$

$$U_E = \frac{1}{2} \left(\frac{\epsilon_0 A}{d} \right) (Ed)^2 = \frac{1}{2} \epsilon_0 (Ad) E^2 \quad (1.88)$$

where Ad = volume of the space between the capacitor plates. **The energy stored per unit volume of space is defined as energy density**

$u_E = \frac{U}{\text{Volume}}$. From equation (1.88), we get

$$u_E = \frac{1}{2} \epsilon_0 E^2 \quad (1.89)$$

From equation (1.89), we infer that the energy is stored in the electric field existing between the plates of the capacitor. Once the capacitor is allowed to discharge, the energy is retrieved.

It is important to note that the energy density depends only on the electric field and not on the size of the plates of the capacitor. In fact, expression (1.89) is true for the electric field due to any type of charge configuration.

1.8.3 Applications of capacitors

Capacitors are used in various electronics circuits. A few of the applications.

(a) Flash capacitors are used in digital cameras for taking photographs. The flash which comes from the camera when we take photographs is due to the energy released from the capacitor, called a flash capacitor (Figure 1.55 (a))

(b) During cardiac arrest, a device called heart defibrillator is used to give a sudden surge of a large amount of electrical energy to the patient's chest to retrieve the normal heart function. This is shown in Figure 1.55 (b).

(c) Capacitors are used in the ignition system of automobile engines to eliminate sparking

(d) Capacitors are used to reduce power fluctuations in power supplies and to increase the efficiency of power transmission.

However, capacitors have disadvantage as well. Even after the battery or power supply is removed, the capacitor stores charges and energy for some time. For example if the TV is switched off, it is always advisable to not touch the back side of the TV panel.

1.8.4 Effect of dielectrics in capacitors

In earlier discussions, we assumed that the space between the parallel plates of a capacitor is either empty or filled with air. Suppose dielectrics like mica, glass or paper are introduced between the plates, then the capacitance of the capacitor is altered. The dielectric can be inserted into the plates in two different ways. (i) when the capacitor is disconnected from the battery. (ii) when the capacitor is connected to the battery.

(i) when the capacitor is disconnected from the battery

Consider a capacitor with two parallel plates each of cross-sectional area A and are separated by a distance d . The capacitor is charged by a battery of voltage V_0 and the charge stored is Q_0 . The capacitance of the capacitor without the dielectric is

$$C_0 = \frac{Q_0}{V_0} \quad (1.90)$$

The battery is then disconnected from the capacitor and the dielectric is inserted between the plates. This is shown in Figure 1.56.



(a)



(b)

Figure 1.55 (a) Flash capacitor in camera (b) Heart defibrillator

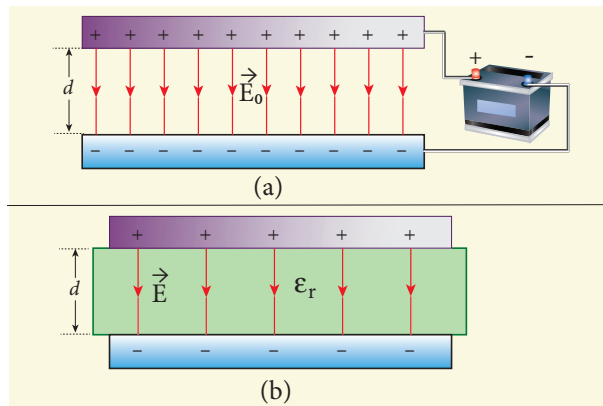


Figure 1.56 (a) Capacitor is charged with a battery (b) Dielectric is inserted after the battery is disconnected

The introduction of dielectric between the plates will decrease the electric field. Experimentally it is found that the modified electric field is given by

$$E = \frac{E_0}{\epsilon_r} \quad (1.91)$$

where E_0 is the electric field inside the capacitors when there is no dielectric and ϵ_r is the relative permittivity of the dielectric or simply known as the dielectric constant. Since $\epsilon_r > 1$, the electric field $E < E_0$.

As a result, the electrostatic potential difference between the plates ($V = Ed$) is also reduced. But at the same time, the charge Q_0 will remain constant once the battery is disconnected.

Hence the new potential difference is

$$V = Ed = \frac{E_0}{\epsilon_r} d = \frac{V_0}{\epsilon_r} \quad (1.92)$$

We know that capacitance is inversely proportional to the potential difference. Therefore as V decreases, C increases.

Thus new capacitance in the presence of a dielectric is

$$C = \frac{Q_0}{V} = \epsilon_r \frac{Q_0}{V_0} = \epsilon_r C_0 \quad (1.93)$$

Since $\epsilon_r > 1$, we have $C > C_0$. Thus insertion of the dielectric increases the capacitance.

Using equation (1.84),

$$C = \frac{\epsilon_r \epsilon_0 A}{d} = \frac{\epsilon A}{d} \quad (1.94)$$

where $\epsilon = \epsilon_r \epsilon_0$ is the permittivity of the dielectric medium.

The energy stored in the capacitor before the insertion of a dielectric is given by

$$U_0 = \frac{1}{2} \frac{Q_0^2}{C_0} \quad (1.95)$$

After the dielectric is inserted, the charge Q_0 remains constant but the capacitance is increased. As a result, the stored energy is decreased.

$$U = \frac{1}{2} \frac{Q_0^2}{C} = \frac{1}{2} \frac{Q_0^2}{\epsilon_r C_0} = \frac{U_0}{\epsilon_r} \quad (1.96)$$

Since $\epsilon_r > 1$ we get $U < U_0$. There is a decrease in energy because, when the dielectric is inserted, the capacitor spends some energy in pulling the dielectric inside.

(ii) When the battery remains connected to the capacitor

Let us now consider what happens when the battery of voltage V_0 remains connected to the capacitor when the dielectric is inserted into the capacitor. This is shown in Figure 1.57.

The potential difference V_0 across the plates remains constant. But it is found experimentally (first shown by Faraday) that when dielectric is inserted, the charge stored in the capacitor is increased by a factor ϵ_r .

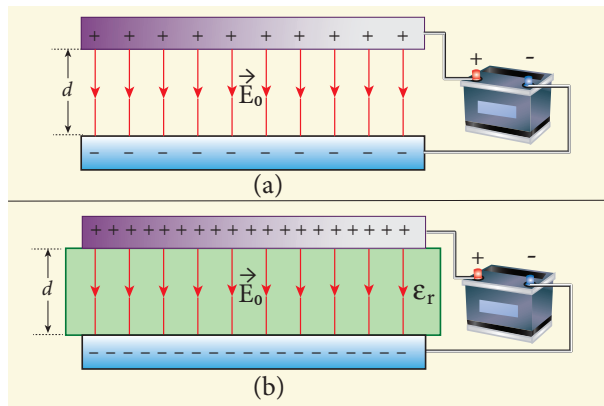


Figure 1.57 (a) Capacitor is charged through a battery (b) Dielectric is inserted when the battery is connected.

$$Q = \epsilon_r Q_0 \quad (1.97)$$

Due to this increased charge, the capacitance is also increased. The new capacitance is

$$C = \frac{Q}{V_0} = \epsilon_r \frac{Q_0}{V_0} = \epsilon_r C_0 \quad (1.98)$$

However the reason for the increase in capacitance in this case when the battery remains connected is different from the case when the battery is disconnected before introducing the dielectric.

$$\begin{aligned} \text{Now, } C_0 &= \frac{\epsilon_0 A}{d} \\ \text{and } C &= \frac{\epsilon A}{d} \end{aligned} \quad (1.99)$$

The energy stored in the capacitor before the insertion of a dielectric is given by

$$U_0 = \frac{1}{2} C_0 V_0^2 \quad (1.100)$$

Note that here we have not used the expression $U_0 = \frac{1}{2} \frac{Q_0^2}{C_0}$ because here, both charge and capacitance are changed, whereas in equation (1.100), V_0 remains constant.

After the dielectric is inserted, the capacitance is increased; hence the stored energy is also increased.

$$U = \frac{1}{2} C V_0^2 = \frac{1}{2} \epsilon_r C_0 V_0^2 = \epsilon_r U_0 \quad (1.101)$$

Since $\epsilon_r > 1$ we have $U > U_0$.

It may be noted here that since voltage between the capacitor V_0 is constant, the electric field between the plates also remains constant.

The energy density is given by

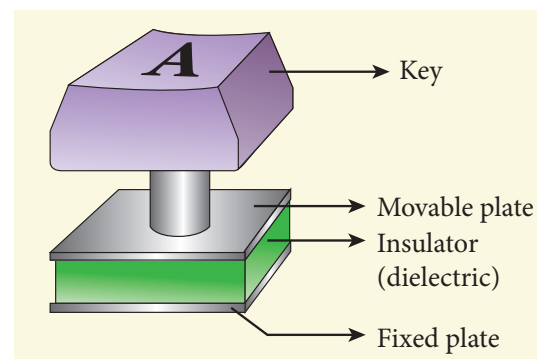
$$u = \frac{1}{2} \epsilon E_0^2 \quad (1.102)$$

where ϵ is the permittivity of the given dielectric material.

The results of the above discussions are summarised in the following Table 1.2



Computer keyboard keys are constructed using capacitors with a dielectric as shown in the figure.



When the key is pressed, the separation between the plates decreases leading to an increase in the capacitance. This in turn triggers the electronic circuits in the computer to identify which key is pressed.

Table 1.2 Effect of dielectrics in capacitors

S. No	Dielectric is inserted	Charge Q	Voltage V	Electric field E	Capacitance C	Energy U
1	When the battery is disconnected	Constant	decreases	Decreases	Increases	Decreases
2	When the battery is connected	Increases	Constant	Constant	Increases	Increases

EXAMPLE 1.21

A parallel plate capacitor filled with mica having $\epsilon_r = 5$ is connected to a 10 V battery. The area of each parallel plate is 6 cm² and separation distance is 6 mm. (a) Find the capacitance and stored charge.

(b) After the capacitor is fully charged, the battery is disconnected and the dielectric is removed carefully.

Calculate the new values of capacitance, stored energy and charge.

Solution

(a) The capacitance of the capacitor in the presence of dielectric is

$$C = \frac{\epsilon_r \epsilon_0 A}{d} = \frac{5 \times 8.85 \times 10^{-12} \times 6 \times 10^{-4}}{6 \times 10^{-3}} \\ = 44.25 \times 10^{-13} \text{ F} = 4.425 \text{ pF}$$

The stored charge is

$$Q = CV = 44.25 \times 10^{-13} \times 10 \\ = 442.5 \times 10^{-13} \text{ C} = 44.25 \text{ pC}$$

The stored energy is

$$U = \frac{1}{2} CV^2 = \frac{1}{2} \times 44.25 \times 10^{-13} \times 100 \\ = 2.21 \times 10^{-10} \text{ J}$$

(b) After the removal of the dielectric, since the battery is already disconnected the total charge will not change. But the potential

difference between the plates increases. As a result, the capacitance is decreased.

New capacitance is

$$C_0 = \frac{C}{\epsilon_r} = \frac{4.425 \times 10^{-12}}{5} \\ = 0.885 \times 10^{-12} \text{ F} = 0.885 \text{ pF}$$

The stored charge remains same and 44.25 pC. Hence newly stored energy is

$$U_0 = \frac{Q^2}{2C_0} = \frac{Q^2 \epsilon_r}{2C} = \epsilon_r U \\ = 5 \times 2.21 \times 10^{-10} \text{ J} = 11.05 \times 10^{-10} \text{ J}$$

The increased energy is

$$\Delta U = (11.05 - 2.21) \times 10^{-10} \text{ J} = 8.84 \times 10^{-10} \text{ J}$$

When the dielectric is removed, it experiences an inward pulling force due to the plates. To remove the dielectric, an external agency has to do work on the dielectric which is stored as additional energy. This is the source for the extra energy $8.84 \times 10^{-10} \text{ J}$.

1.8.5 Capacitor in series and parallel**(i) Capacitor in series**

Consider three capacitors of capacitance C_1 , C_2 and C_3 connected in series with a battery of voltage V as shown in the Figure 1.58 (a).

As soon as the battery is connected to the capacitors in series, the electrons of charge

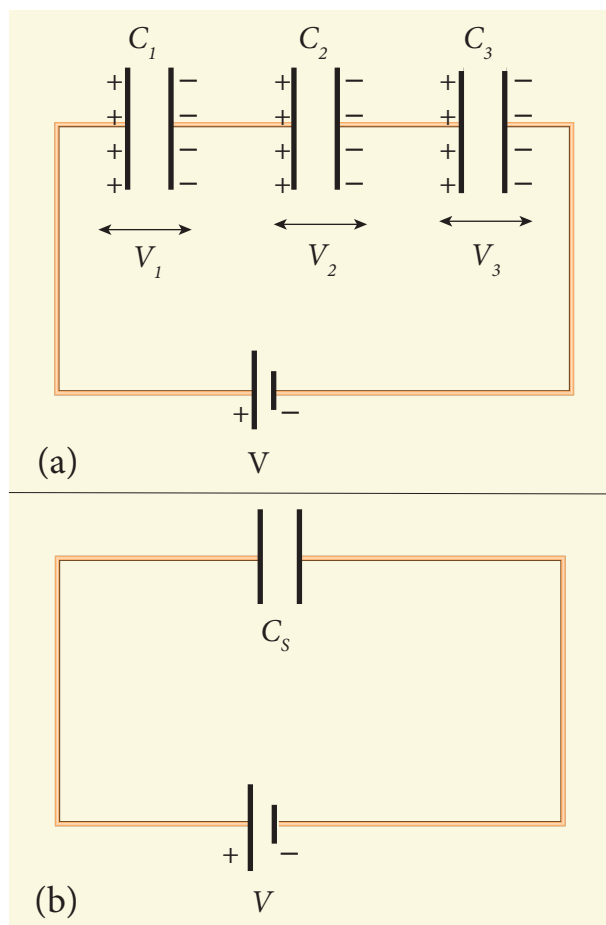


Figure 1.58 (a) Capacitors connected in series (b) Equivalent capacitors C_s

$-Q$ are transferred from negative terminal to the right plate of C_3 which pushes the electrons of same amount $-Q$ from left plate of C_3 to the right plate of C_2 due to electrostatic induction. Similarly, the left plate of C_2 pushes the charges of $-Q$ to the right plate of C_1 which induces the positive charge $+Q$ on the left plate of C_1 . At the same time, electrons of charge $-Q$ are transferred from left plate of C_1 to positive terminal of the battery.

By these processes, each capacitor stores the same amount of charge Q . The capacitances of the capacitors are in general different, so that the voltage across each capacitor is also different and are denoted as V_1 , V_2 and V_3 respectively.

The sum of the voltages across the capacitor must be equal to the voltage of the battery.

$$V = V_1 + V_2 + V_3 \quad (1.103)$$

Since, $Q = CV$, we have $V = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$

$$= Q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right) \quad (1.104)$$

If three capacitors in series are considered to form an equivalent single capacitor C_s shown in Figure 1.58(b), then we have $V = \frac{Q}{C_s}$. Substituting this expression into equation (1.104), we get

$$\frac{Q}{C_s} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \right)$$
$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \quad (1.105)$$

Thus, the inverse of the equivalent capacitance C_s of three capacitors connected in series is equal to the sum of the inverses of each capacitance. This equivalent capacitance C_s is always less than the smallest individual capacitance in the series.

(ii) Capacitance in parallel

Consider three capacitors of capacitance C_1 , C_2 and C_3 connected in parallel with a battery of voltage V as shown in Figure 1.59 (a).

Since corresponding sides of the capacitors are connected to the same positive and negative terminals of the battery, the voltage across each capacitor is equal to the battery's voltage. Since capacitances of the capacitors are different,

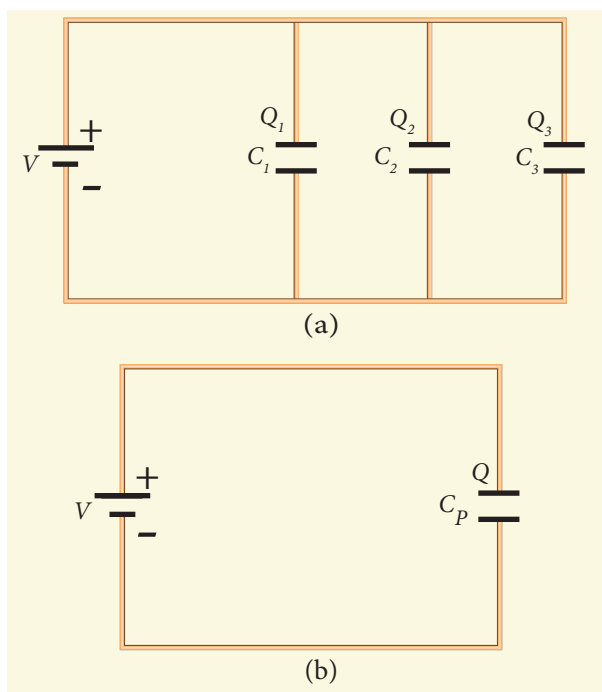


Figure 1.59 (a) capacitors in parallel
(b) equivalent capacitance with the same total charge

the charge stored in each capacitor is not the same. Let the charge stored in the three capacitors be Q_1 , Q_2 , and Q_3 respectively. According to the law of conservation of total charge, the sum of these three charges is equal to the charge Q transferred by the battery,

$$Q = Q_1 + Q_2 + Q_3 \quad (1.106)$$

Since $Q = CV$, we have

$$Q = C_1 V + C_2 V + C_3 V \quad (1.107)$$

If these three capacitors are considered to form a single equivalent capacitance C_p which stores the total charge Q as shown in the Figure 1.59(b), then we can write $Q = C_p V$. Substituting this in equation (1.107), we get

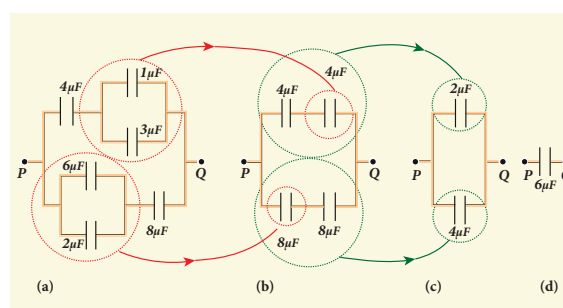
$$C_p V = C_1 V + C_2 V + C_3 V$$

$$C_p = C_1 + C_2 + C_3 \quad (1.108)$$

Thus, the equivalent capacitance of capacitors connected in parallel is equal to the sum of the individual capacitances. The equivalent capacitance C_p in a parallel connection is always greater than the largest individual capacitance. In a parallel connection, it is equivalent as area of each capacitance adds to give more effective area such that total capacitance increases.

EXAMPLE 1.22

Find the equivalent capacitance between P and Q for the configuration shown below in the figure (a).



Solution

The capacitors $1 \mu\text{F}$ and $3 \mu\text{F}$ are connected in parallel and $6 \mu\text{F}$ and $2 \mu\text{F}$ are also separately connected in parallel. So these parallel combinations reduced to equivalent single capacitances in their respective positions, as shown in the figure (b).

$$C_{eq} = 1 + 3 = 4 \mu\text{F}$$

$$C_{eq} = 6 + 2 = 8 \mu\text{F}$$

From the figure (b), we infer that the two $4 \mu\text{F}$ capacitors are connected in series and the two $8 \mu\text{F}$ capacitors are connected in series. By using formula for the series, we can reduce to their equivalent capacitances as shown in figure (c).

$$\frac{1}{C_{eq}} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} \quad \Rightarrow C_{eq} = 2 \mu\text{F}$$

and

$$\frac{1}{C_{eq}} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4} \quad \Rightarrow C_{eq} = 4 \mu\text{F}$$

From the figure (c), we infer that $2 \mu\text{F}$ and $4 \mu\text{F}$ are connected in parallel. So the equivalent capacitance is given in the figure (d).

$$C_{eq} = 2 + 4 = 6 \mu\text{F}$$

Thus the combination of capacitances in figure (a) can be replaced by a single capacitance $6 \mu\text{F}$.

1.9

DISTRIBUTION OF CHARGES IN A CONDUCTOR AND ACTION AT POINTS

1.9.1 Distribution of charges in a conductor

Consider two conducting spheres A and B of radii r_1 and r_2 respectively connected to each other by a thin conducting wire as shown in the Figure 1.60. The distance between the spheres is much greater than the radii of either spheres.

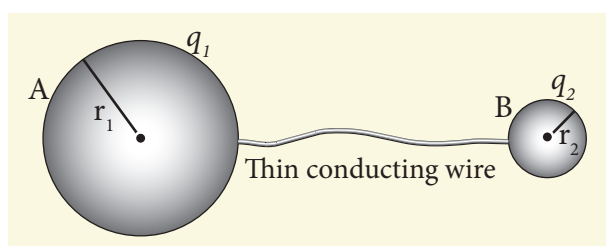


Figure 1.60 Two conductors are connected through conducting wire

If a charge Q is introduced into any one of the spheres, this charge Q is redistributed into both the spheres such that the electrostatic potential is same in both the spheres. They are now uniformly charged and attain electrostatic equilibrium. Let q_1 be the charge residing on the surface of sphere A and q_2 is the charge residing on the surface of sphere B such that $Q = q_1 + q_2$. The charges are distributed only on the surface and there is no net charge inside the conductor.

The electrostatic potential at the surface of the sphere A is given by

$$V_A = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} \quad (1.109)$$

The electrostatic potential at the surface of the sphere B is given by

$$V_B = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2} \quad (1.110)$$

The surface of the conductor is an equipotential. Since the spheres are connected by the conducting wire, the surfaces of both the spheres together form an equipotential surface. This implies that

$$V_A = V_B$$

$$\text{or } \frac{q_1}{r_1} = \frac{q_2}{r_2} \quad (1.111)$$

Let the charge density on the surface of sphere A be σ_1 and that on the surface of sphere B be σ_2 . This implies that $q_1 = 4\pi r_1^2 \sigma_1$ and

$q_2 = 4\pi r_2^2 \sigma_2$. Substituting these values into equation (1.112), we get

$$\sigma_1 r_1 = \sigma_2 r_2 \quad (1.112)$$

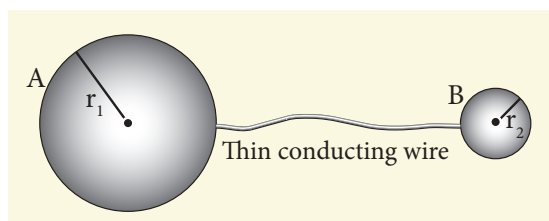
from which we conclude that

$$\sigma r = \text{constant} \quad (1.113)$$

Thus the surface charge density σ is inversely proportional to the radius of the sphere. For a smaller radius, the charge density will be larger and vice versa.

EXAMPLE 1.23

Two conducting spheres of radius $r_1 = 8 \text{ cm}$ and $r_2 = 2 \text{ cm}$ are separated by a distance much larger than 8 cm and are connected by a thin conducting wire as shown in the figure. A total charge of $Q = +100 \text{ nC}$ is placed on one of the spheres. After a fraction of a second, the charge Q is redistributed and both the spheres attain electrostatic equilibrium.



- Calculate the charge and surface charge density on each sphere.
- Calculate the potential at the surface of each sphere.

Solution

- The electrostatic potential on the surface of the sphere A is $V_A = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1}$

The electrostatic potential on the surface of the sphere B is $V_B = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2}$

Since $V_A = V_B$. We have

$$\frac{q_1}{r_1} = \frac{q_2}{r_2} \Rightarrow q_1 = \left(\frac{r_1}{r_2}\right) q_2$$

But from the conservation of total charge, $Q = q_1 + q_2$, we get $q_1 = Q - q_2$. By substituting this in the above equation,

$$Q - q_2 = \left(\frac{r_1}{r_2}\right) q_2$$

$$\text{so that } q_2 = Q \left(\frac{r_2}{r_1 + r_2} \right)$$

Therefore,

$$q_2 = 100 \times 10^{-9} \times \left(\frac{2}{10} \right) = 20 \text{ nC}$$

$$\text{and } q_1 = Q - q_2 = 80 \text{ nC}$$

The electric charge density on sphere A is $\sigma_1 = \frac{q_1}{4\pi r_1^2}$

The electric charge density on sphere B is $\sigma_2 = \frac{q_2}{4\pi r_2^2}$

Therefore,

$$\sigma_1 = \frac{80 \times 10^{-9}}{4\pi \times 64 \times 10^{-4}} = 0.99 \times 10^{-6} \text{ C m}^{-2}$$

and

$$\sigma_2 = \frac{20 \times 10^{-9}}{4\pi \times 4 \times 10^{-4}} = 3.9 \times 10^{-6} \text{ C m}^{-2}$$

Note that the surface charge density is greater on the smaller sphere compared to the larger sphere ($\sigma_2 \approx 4\sigma_1$) which confirms the result $\frac{\sigma_1}{\sigma_2} = \frac{r_2}{r_1}$.

The potential on both spheres is the same. So we can calculate the potential on any one of the spheres.

$$V_A = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} = \frac{9 \times 10^9 \times 80 \times 10^{-9}}{8 \times 10^{-2}} = 9 \text{ kV}$$

1.9.2 Action of points or Corona discharge

Consider a charged conductor of irregular shape as shown in Figure 1.61 (a).

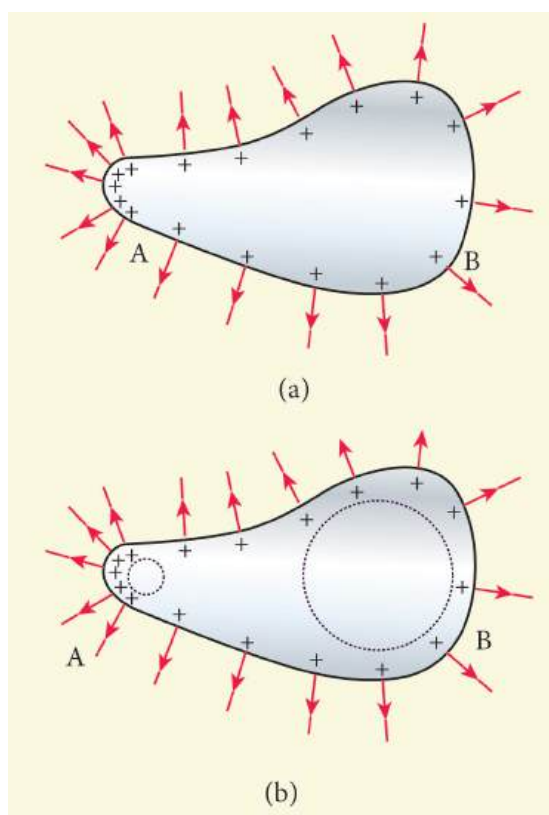


Figure 1.61 Action of points or corona discharge

We know that smaller the radius of curvature, the larger is the charge density. The end of the conductor which has larger curvature (smaller radius) has a large charge accumulation as shown in Figure 1.61 (b).

As a result, the electric field near this edge is very high and it ionizes the surrounding air. The positive ions are repelled at the sharp edge and negative ions are attracted towards the sharper edge. This reduces the total charge of the conductor near the sharp edge. This is called action of points or corona discharge.

1.9.3 Lightning arrester or lightning conductor

This is a device used to protect tall buildings from lightning strikes. It works on the principle of action at points or corona discharge.

This device consists of a long thick copper rod passing from top of the building to the ground. The upper end of the rod has a sharp spike or a sharp needle as shown in Figure 1.62 (a) and (b).

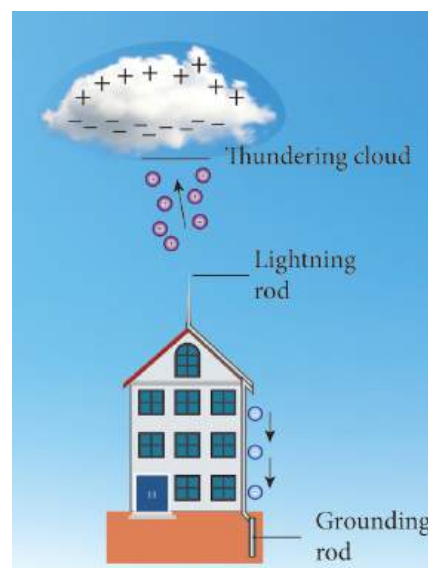


Figure 1.62 (a) Schematic diagram of a lightning arrester. (b) A house with a lightning arrester

The lower end of the rod is connected to copper plate which is buried deep into the ground. When a negatively charged cloud is passing above the building, it induces a positive charge on the spike. Since the induced charge density on thin sharp spike is large, it results in a corona discharge. This positive charge ionizes the surrounding air which in turn neutralizes the negative charge in the cloud. The negative charge pushed to the spikes passes through the copper rod and

is safely diverted to the Earth. The lightning arrester does not stop the lightning; rather it diverts the lightning to the ground safely.

1.9.4 Van de Graaff Generator

In the year 1929, Robert Van de Graaff designed a machine which produces a large amount of electrostatic potential difference, up to several million volts (10^7 V). This Van de Graff generator works on the principle of electrostatic induction and action at points.

A large hollow spherical conductor is fixed on the insulating stand as shown in Figure 1.63. A pulley B is mounted at the centre of the hollow sphere and another pulley C is fixed at the bottom. A belt made up of insulating materials like silk or rubber runs over both pulleys. The pulley C is driven continuously by the electric motor. Two comb shaped metallic conductors E and D are fixed near the pulleys. The pulley C is driven continuously by the electric motor. Two comb shaped metallic conductors E and D are fixed near the pulleys.

The comb D is maintained at a positive potential of 10^4 V by a power supply. The upper comb E is connected to the inner side of the hollow metal sphere.

Due to the high electric field near comb D, air between the belt and comb D gets ionized by the action of points. The positive charges are pushed towards the belt and negative charges are attracted towards the comb D. The positive charges stick to the belt and move up. When the positive charges on the belt reach the point near the comb E, the comb E acquires negative charge and the sphere acquires positive charge due to electrostatic induction. As a result, the positive charges are pushed away from the comb E and they reach the outer surface of the sphere. Since the sphere is a conductor,

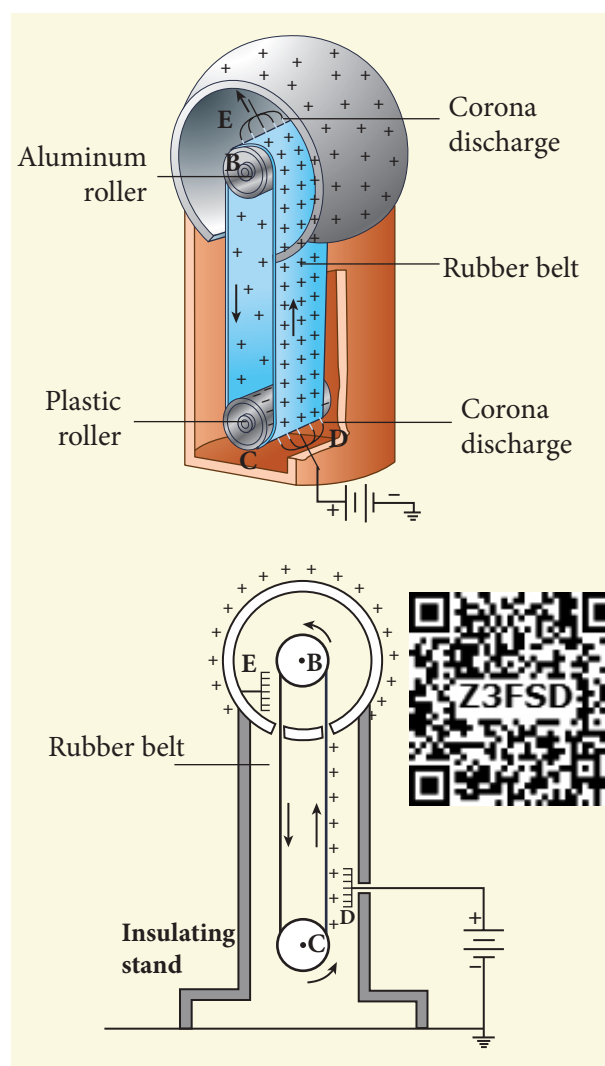


Figure 1.63 Van de Graaff generator

the positive charges are distributed uniformly on the outer surface of the hollow sphere. At the same time, the negative charges nullify the positive charges in the belt due to corona discharge before it passes over the pulley.

When the belt descends, it has almost no net charge. At the bottom, it again gains a large positive charge. The belt goes up and delivers the positive charges to the outer surface of the sphere. This process continues until the outer surface produces the potential difference of the order of 10^7 which is the limiting value. We cannot store charges beyond this limit since the extra charge starts leaking to the surroundings due to ionization of air. The leakage of charges can

be reduced by enclosing the machine in a gas filled steel chamber at very high pressure.

The high voltage produced in this Van de Graaff generator is used to accelerate positive ions (protons and deuterons) for nuclear disintegrations and other applications.

EXAMPLE 1.24

Dielectric strength of air is $3 \times 10^6 \text{ V m}^{-1}$. Suppose the radius of a hollow sphere in the Van de Graff generator is $R = 0.5 \text{ m}$, calculate the maximum potential difference created by this Van de Graaff generator.

Solution

The electric field on the surface of the sphere is given by (by Gauss law)

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$$

The potential on the surface of the hollow metallic sphere is given by

$$V = \frac{1}{4\pi\epsilon_0} \frac{Q}{R} = ER$$

Since $V_{\max} = E_{\max} R$

Here $E_{\max} = 3 \times 10^6 \text{ V m}^{-1}$. So the maximum potential difference created is given by

$$\begin{aligned} V_{\max} &= 3 \times 10^6 \times 0.5 \\ &= 1.5 \times 10^6 \text{ V (or) 1.5 million volt} \end{aligned}$$

SUMMARY

- Like charges repel and unlike charges attract.
- The total charge in the universe is conserved.
- Charge is quantized. Total charge in an object $q = ne$ where $n = 0, 1, 2, 3, \dots$ and e is charge of the electron.
- Coulomb's law in vector form: $\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}$ ($\hat{r}$ is unit vector along joining q_1, q_2).
- Electrostatic force obeys the superposition principle.
- Electric field at a distance r from a point charge: $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$.
- Electric field lines starts at a positive charge and end at a negative charge or at infinity.
- Electric field due to electric dipole at points on the axial line: $\vec{E}_{tot} = \frac{1}{4\pi\epsilon_0} \left(\frac{2\vec{p}}{r^3} \right)$.
- Electric field due to electric dipole at points on the equatorial line: $\vec{E}_{tot} = -\frac{1}{4\pi\epsilon_0} \left(\frac{\vec{p}}{r^3} \right)$.
- Torque experienced by a dipole in a uniform electric field: $\vec{\tau} = \vec{p} \times \vec{E}$.
- Electrostatic potential at a distance r from the point charge: $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$.
- Electrostatic potential due to an electric dipole: $V = \frac{1}{4\pi\epsilon_0} \frac{\vec{p} \cdot \hat{r}}{r^2}$.
- The electrostatic potential is the same at all points on an equipotential surface.
- The relation between electric field and electrostatic potential:

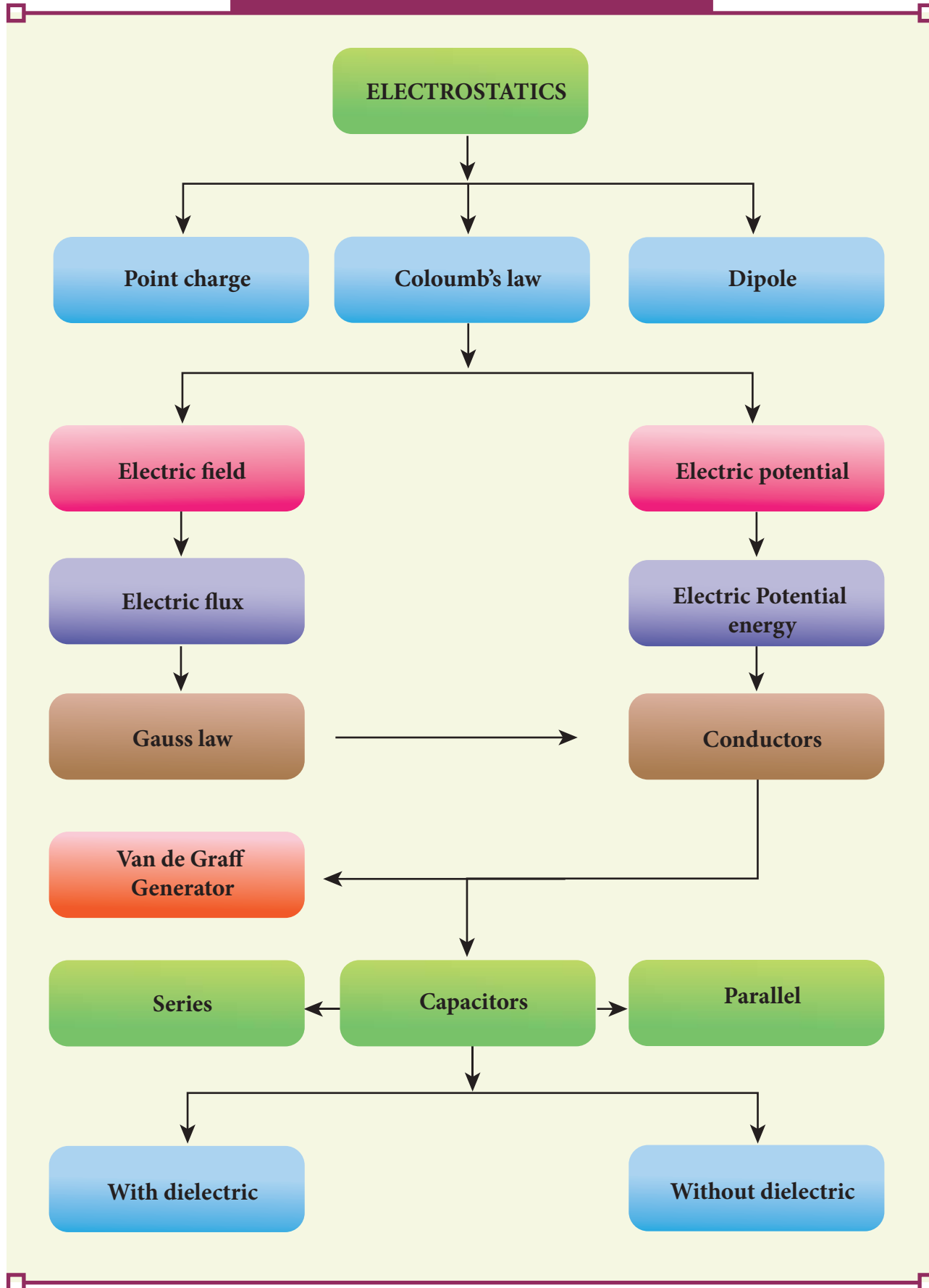
$$\vec{E} = -\left(\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right).$$

- Electrostatic potential energy for system of charges is equal to the work done to arrange the charges in the given configuration.
- Electrostatic potential energy of a dipole system in a uniform electric field: $U = -\vec{p} \cdot \vec{E}$.
- The total electric flux through a closed surface: $\Phi_E = \frac{Q}{\epsilon_0}$ where Q is the net charge enclosed by the surface.
- Electric field due to a charged infinite wire: $\vec{E} = \frac{1}{2\pi\epsilon_0} \frac{\lambda}{r} \hat{r}$.
- Electric field due to a charged infinite plane: $\vec{E} = \frac{\sigma}{2\epsilon_0} \hat{n}$ ($\hat{n}$ is normal to the plane).
- Electric field inside a charged spherical shell is zero. For points outside: $\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$.



- Electric field inside a conductor is zero. The electric field at the surface of the conductor is normal to the surface and has magnitude $E = \frac{\sigma}{\epsilon_0}$.
- The surface of the conductor has the same potential, at all points on the surface.
- Conductor can be charged using the process of induction.
- A dielectric or insulator has no free electrons. When an electric field is applied, the dielectric is polarised.
- Capacitance of a conductor is given by $C = \frac{Q}{V}$.
- Capacitance of a parallel plate capacitor: $C = \frac{\epsilon_0 A}{d}$.
- Electrostatic energy stored in a capacitor: $U = \frac{1}{2} CV^2$.
- The equivalent capacitance for parallel combination is equal to the sum of individual capacitance of the capacitors.
- For a series combination: The inverse of equivalent capacitance is equal to sum of inverse of individual capacitance of capacitors.
- The distribution of charges in the conductors depends on the shape of conductor. For sharper edge, the surface charge density is greater. This principle is used in the lightning arrestor.
- Van de Graaff generator is used to produce large potential difference ($\sim 10^7 \text{ V}$).

CONCEPT MAP



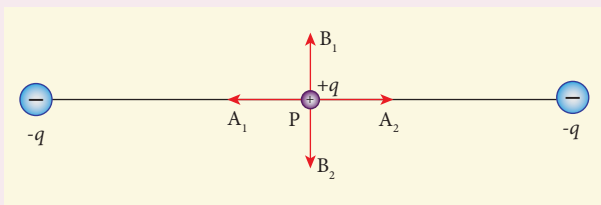


EVALUATION

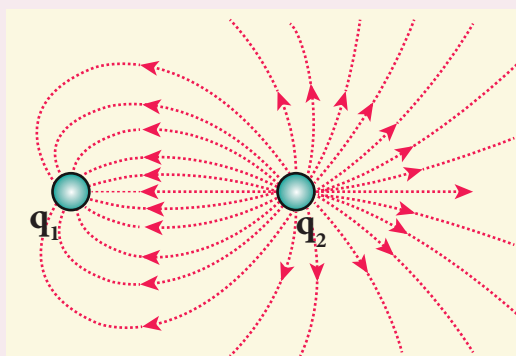


I Multiple choice questions

1. Two identical point charges of magnitude $-q$ are fixed as shown in the figure below. A third charge $+q$ is placed midway between the two charges at the point P. Suppose this charge $+q$ is displaced a small distance from the point P in the directions indicated by the arrows, in which direction(s) will $+q$ be stable with respect to the displacement?

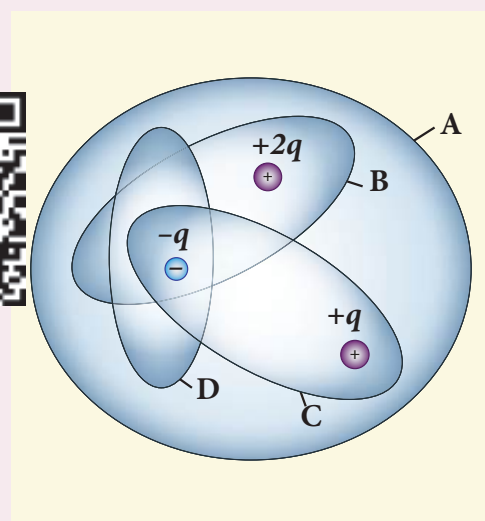


- (a) A_1 and A_2 (b) B_1 and B_2
(c) both directions (d) No stable
2. Which charge configuration produces a uniform electric field?
- (a) point charge
(b) uniformly charged infinite line
(c) uniformly charged infinite plane
(d) uniformly charged spherical shell
3. What is the ratio of the charges $\left| \frac{q_1}{q_2} \right|$ for the following electric field line pattern?

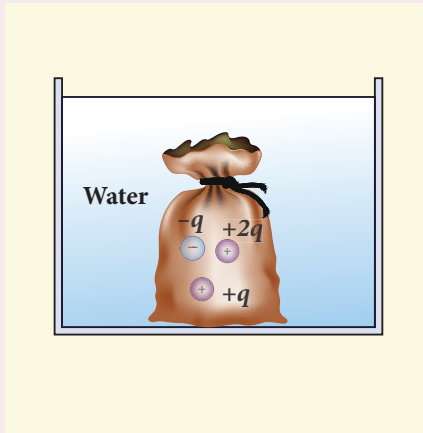


- (a) $\frac{1}{5}$ (b) $\frac{25}{11}$
(c) 5 (d) $\frac{11}{25}$

4. An electric dipole is placed at an alignment angle of 30° with an electric field of $2 \times 10^5 \text{ N C}^{-1}$. It experiences a torque equal to 8 N m . The charge on the dipole if the dipole length is 1 cm is
- (a) 4 mC (b) 8 mC
(c) 5 mC (d) 7 mC
5. Four Gaussian surfaces are given below with charges inside each Gaussian surface. Rank the electric flux through each Gaussian surface in increasing order.

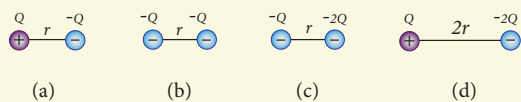


- (a) $D < C < B < A$
(b) $A < B = C < D$
(c) $C < A = B < D$
(d) $D > C > B > A$
6. The total electric flux for the following closed surface which is kept inside water



- (a) $\frac{80q}{\epsilon_0}$ (b) $\frac{q}{40\epsilon_0}$
 (c) $\frac{q}{80\epsilon_0}$ (d) $\frac{q}{160\epsilon_0}$

7. Two identical conducting balls having positive charges q_1 and q_2 are separated by a centre to centre distance r . If they are made to touch each other and then separated to the same distance, the force between them will be (NSEP 04-05)
- (a) less than before
 (b) same as before
 (c) more than before
 (d) zero
8. Rank the electrostatic potential energies for the given system of charges in increasing order.

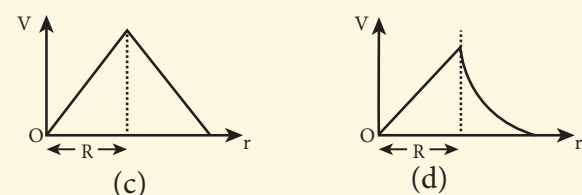
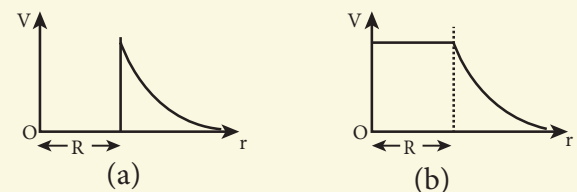


- (a) $1 = 4 < 2 < 3$ (b) $2 = 4 < 3 < 1$
 (c) $2 = 3 < 1 < 4$ (d) $3 < 1 < 2 < 4$
9. An electric field $\vec{E} = 10x\hat{i}$ exists in a certain region of space. Then the

potential difference $V = V_o - V_A$, where V_o is the potential at the origin and V_A is the potential at $x = 2$ m is:

- (a) 10 V (b) -20 V
 (c) +20 V (d) -10 V

10. A thin conducting spherical shell of radius R has a charge Q which is uniformly distributed on its surface. The correct plot for electrostatic potential due to this spherical shell is



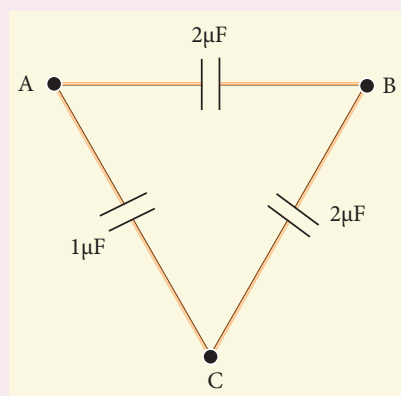
11. Two points A and B are maintained at a potential of 7 V and -4 V respectively. The work done in moving 50 electrons from A to B is
- (a) 8.80×10^{-17} J
 (b) -8.80×10^{-17} J
 (c) 4.40×10^{-17} J
 (d) 5.80×10^{-17} J
12. If voltage applied on a capacitor is increased from V to $2V$, choose the correct conclusion.
- (a) Q remains the same, C is doubled
 (b) Q is doubled, C doubled
 (c) C remains same, Q doubled
 (d) Both Q and C remain same



13. A parallel plate capacitor stores a charge Q at a voltage V . Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled then which is the quantity that will change?

(a) Capacitance
(b) Charge
(c) Voltage
(d) Energy density

14. Three capacitors are connected in triangle as shown in the figure. The equivalent capacitance between the points A and C is



(a) $1\mu\text{F}$ (b) $2\mu\text{F}$
(c) $3\mu\text{F}$ (d) $\frac{1}{4}\mu\text{F}$

15. Two metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2} C and 5×10^{-2} C respectively. If these are connected by a conducting wire, the final charge on the bigger sphere is (AIIPMT -2012)

(a) 3×10^{-2} C
(b) 4×10^{-2} C
(c) 1×10^{-2} C
(d) 2×10^{-2} C

Answers

- 1) b 2) c 3) d 4) b 5) a
6) b 7) c 8) a 9) c 10) b
11) a 12) c 13) d 14) b 15) a

II Short Answer Questions

- What is meant by quantisation of charges?
- Write down Coulomb's law in vector form and mention what each term represents.
- What are the differences between Coulomb force and gravitational force?
- Write a short note on superposition principle.
- Define 'electric field'.
- What is meant by 'electric field lines'?
- The electric field lines never intersect. Justify.
- Define 'electric dipole'. Give the expression for the magnitude of its electric dipole moment and the direction.
- Write the general definition of electric dipole moment for a collection of point charge.
- Define 'electrostatic potential'.
- What is an equipotential surface?
- What are the properties of an equipotential surface?
- Give the relation between electric field and electric potential.
- Define 'electrostatic potential energy'.
- Define 'electric flux'.
- What is meant by electrostatic energy density?

17. Write a short note on 'electrostatic shielding'.
18. What is polarisation?
19. What is dielectric strength?
20. Define 'capacitance'. Give its unit.
21. What is corona discharge?

III Long Answer questions

1. Discuss the basic properties of electric charges.
2. Explain in detail Coulomb's law and its various aspects.
3. Define 'electric field' and discuss its various aspects.
4. Calculate the electric field due to a dipole on its axial line and equatorial plane.
5. Derive an expression for the torque experienced by a dipole due to a uniform electric field.
6. Derive an expression for electrostatic potential due to a point charge.
7. Derive an expression for electrostatic potential due to an electric dipole.
8. Obtain an expression for potential energy due to a collection of three point charges which are separated by finite distances.
9. Derive an expression for electrostatic potential energy of the dipole in a uniform electric field.
10. Obtain Gauss law from Coulomb's law.
11. Obtain the expression for electric field due to an infinitely long charged wire.
12. Obtain the expression for electric field due to an charged infinite plane sheet.
13. Obtain the expression for electric field due to an uniformly charged spherical shell.
14. Discuss the various properties of conductors in electrostatic equilibrium.
15. Explain the process of electrostatic induction.
16. Explain dielectrics in detail and how an electric field is induced inside a dielectric.
17. Obtain the expression for capacitance for a parallel plate capacitor.
18. Obtain the expression for energy stored in the parallel plate capacitor.
19. Explain in detail the effect of a dielectric placed in a parallel plate capacitor.
20. Derive the expression for resultant capacitance, when capacitors are connected in series and in parallel.
21. Explain in detail how charges are distributed in a conductor, and the principle behind the lightning conductor.
22. Explain in detail the construction and working of a Van de Graaff generator.

Exercises

1. When two objects are rubbed with each other, approximately a charge of 50 nC can be produced in each object. Calculate the number of electrons that must be transferred to produce this charge.

Ans: 31.25×10^{10} electrons

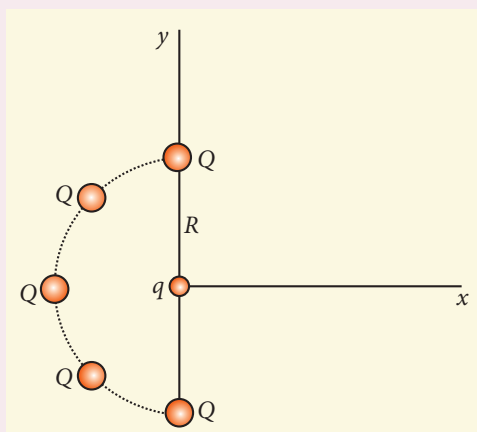
2. The total number of electrons in the human body is typically in the order of 10^{28} . Suppose, due to some reason, you

and your friend lost 1% of this number of electrons. Calculate the electrostatic force between you and your friend separated at a distance of 1m. Compare this with your weight. Assume mass of each person is 60 kg and use point charge approximation.

Ans: $F_e = 23 \times 10^{23} \text{ N}$, $W = 588 \text{ N}$,

$$\frac{F_e}{W} = 3.9 \times 10^{21}$$

3. Five identical charges Q are placed equidistant on a semicircle as shown in the figure. Another point charge q is kept at the centre of the circle of radius R . Calculate the electrostatic force experienced by the charge q .



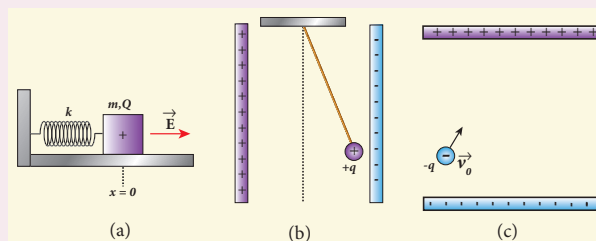
Ans: $\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{R^2} (1 + \sqrt{2}) \hat{i} \text{ N}$

4. Suppose a charge $+q$ on Earth's surface and another $+q$ charge is placed on the surface of the Moon. (a) Calculate the value of q required to balance the gravitational attraction between Earth and Moon (b) Suppose the distance between the Moon and Earth is halved, would the charge q change?

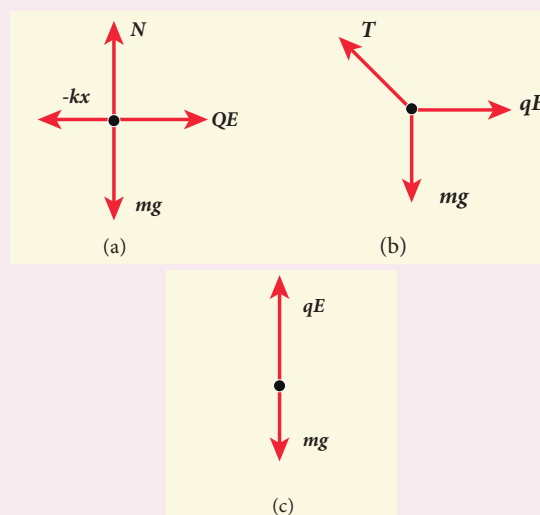
(Take $m_E = 5.9 \times 10^{24} \text{ kg}$, $m_M = 7.9 \times 10^{22} \text{ kg}$)

Ans: (a) $q \approx +5.87 \times 10^{13} \text{ C}$,
(b) no change

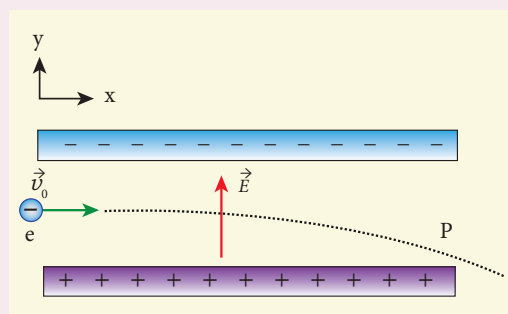
5. Draw the free body diagram for the following charges as shown in the figure (a), (b) and (c).



Ans:



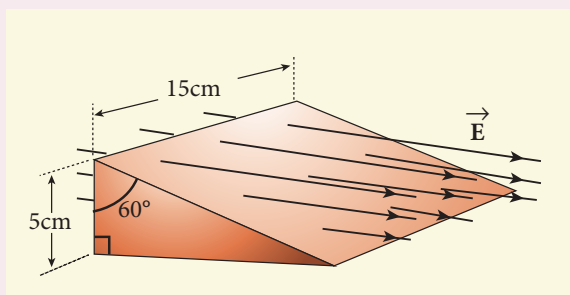
6. Consider an electron travelling with a speed v_0 and entering into a uniform electric field $\vec{E}$ which is perpendicular to $\vec{v}_0$ as shown in the Figure. Ignoring gravity, obtain the electron's acceleration, velocity and position as functions of time.



Ans :

$$\vec{a} = -\frac{eE}{m} \hat{j}, \vec{v} = v_0 \hat{i} - \frac{eE}{m} t \hat{j}, \vec{r} = v_0 t \hat{i} - \frac{1}{2} \frac{eE}{m} t^2 \hat{j}$$

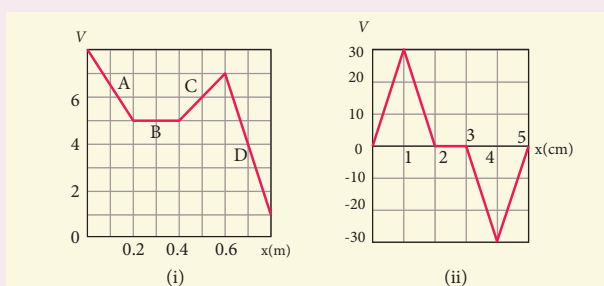
7. A closed triangular box is kept in an electric field of magnitude $E = 2 \times 10^3 \text{ N C}^{-1}$ as shown in the figure.



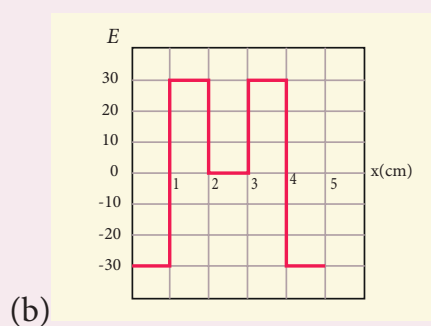
Calculate the electric flux through the (a) vertical rectangular surface (b) slanted surface and (c) entire surface.

Ans: (a) $-15 \text{ Nm}^2 \text{ C}^{-1}$ (b) $15 \text{ Nm}^2 \text{ C}^{-1}$ (c) zero

8. The electrostatic potential is given as a function of x in figure (i) and (ii). (a) Calculate the corresponding electric fields in regions A, B, C and D for the Figure (i). (b) Plot the electric field as a function of x for the figure (ii).



Ans: (a) $E_x = 15 \text{ Vm}^{-1}$ (region A), $E_x = -10 \text{ Vm}^{-1}$ (region C)
 $E_x = 0$ (region B), $E_x = 30 \text{ Vm}^{-1}$ (region D)



(b)

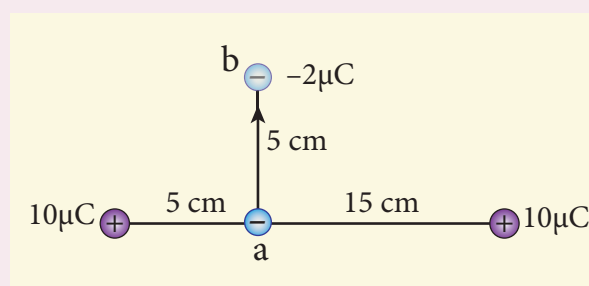
9. A spark plug in a bike or a car is used to ignite the air-fuel mixture in the engine. It consists of two electrodes separated by a gap of around 0.6 mm gap as shown in the figure.



To create the spark, an electric field of magnitude $3 \times 10^6 \text{ Vm}^{-1}$ is required. (a) What potential difference must be applied to produce the spark? (b) If the gap is increased, does the potential difference increase, decrease or remains the same? (c) find the potential difference if the gap is 1 mm.

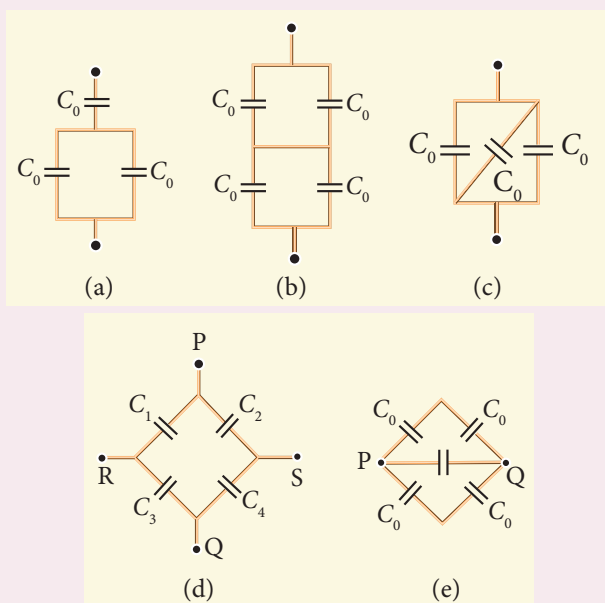
Ans: (a) 1800 V, (b) increases (c) 3000 V

10. A point charge of $+10 \mu\text{C}$ is placed at a distance of 20 cm from another identical point charge of $+10 \mu\text{C}$. A point charge of $-2 \mu\text{C}$ is moved from point a to b as shown in the figure. Calculate the change in potential energy of the system? Interpret your result.



Ans: $\Delta U = +1.12 \text{ J}$, positive sign implies that to move the charge $-2\mu\text{C}$ external work is required.

11. Calculate the resultant capacitances for each of the following combinations of capacitors.



Ans: (a) $\frac{2}{3}C_0$ (b) C_0 (c) $3C_0$

(d) across PQ:

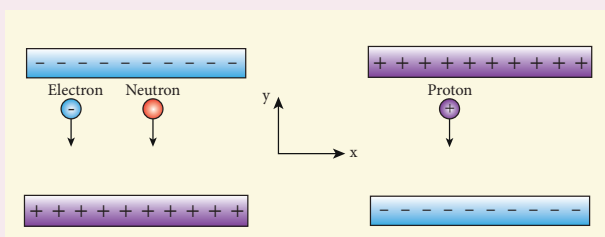
$$\frac{C_1C_2C_3 + C_2C_3C_4 + C_1C_2C_4 + C_1C_3C_4}{(C_1 + C_3)(C_2 + C_4)}$$

across RS:

$$\frac{C_1C_2C_3 + C_2C_3C_4 + C_1C_2C_4 + C_1C_3C_4}{(C_1 + C_2)(C_3 + C_4)}$$

(e) across PQ: $2C_0$

12. An electron and a proton are allowed to fall through the separation between the plates of a parallel plate capacitor of voltage 5 V and separation distance $h = 1 \text{ mm}$ as shown in the figure.



- (a) Calculate the time of flight for both electron and proton (b) Suppose if a neutron is allowed to fall, what is the time of flight? (c) Among the three, which one will reach the bottom first? (Take $m_p = 1.6 \times 10^{-27} \text{ kg}$, $m_e = 9.1 \times 10^{-31} \text{ kg}$ and $g = 10 \text{ m s}^{-2}$)

Ans:

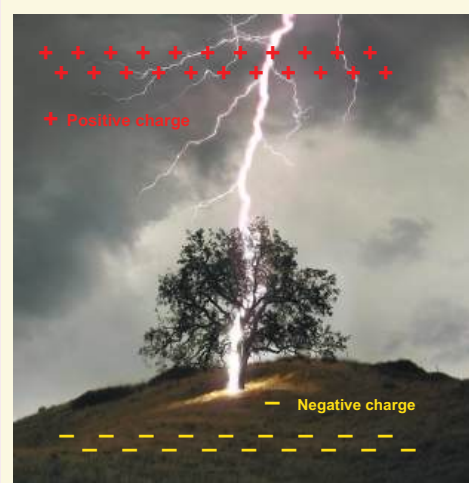
(a) $t_e = \sqrt{\frac{2hm_e}{eE}} \approx 1.5 \text{ ns}$ (ignoring the gravity),

$t_p = \sqrt{\frac{2hm_p}{eE}} \approx 63 \text{ ns}$ (ignoring the gravity)

(b) $t_n = \sqrt{\frac{2h}{g}} \approx 14.1 \text{ ms}$

(c) electron will reach first

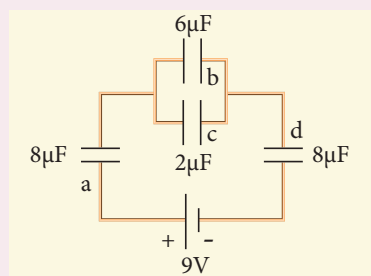
13. During a thunder storm, the movement of water molecules within the clouds creates friction, partially causing the bottom part of the clouds to become negatively charged. This implies that the bottom of the cloud and the ground act as a parallel plate capacitor. If the electric field between the cloud and ground exceeds the dielectric breakdown of the air ($3 \times 10^6 \text{ Vm}^{-1}$), lightning will occur.



- (a) If the bottom part of the cloud is 1000 m above the ground, determine the electric potential difference that exists between the cloud and ground.
- (b) In a typical lightning phenomenon, around 25 C of electrons are transferred from cloud to ground. How much electrostatic potential energy is transferred to the ground?

Ans: (a) $V = 3 \times 10^9$ V, (b) $U = 75 \times 10^9$ J

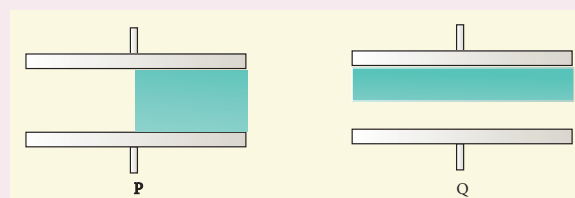
14. For the given capacitor configuration
- Find the charges on each capacitor
 - potential difference across them
 - energy stored in each capacitor



Ans:

$$\begin{aligned} Q_a &= 24 \mu\text{C}, & Q_b &= 18 \mu\text{C}, \\ Q_c &= 6 \mu\text{C}, & Q_d &= 24 \mu\text{C} \\ V_a &= 3\text{V}, & V_b &= 3\text{V}, \\ V_c &= 3\text{V}, & V_d &= 3\text{V}, \\ U_a &= 36 \mu\text{J}, & U_b &= 27 \mu\text{J}, \\ U_c &= 9 \mu\text{J}, & U_d &= 36 \mu\text{J} \end{aligned}$$

15. Capacitors P and Q have identical cross sectional areas A and separation d . The space between the capacitors is filled with a dielectric of dielectric constant ϵ_r as shown in the figure. Calculate the capacitance of capacitors P and Q.



Ans : $C_P = \frac{\epsilon_0 A}{2d} (1 + \epsilon_r)$, $C_Q = \frac{2\epsilon_0 A}{d} \left(\frac{\epsilon_r}{1 + \epsilon_r} \right)$

BOOKS FOR REFERENCE

1. Douglas C. Giancoli, , “*Physics for Scientist & Engineers with Modern Physics*”, Pearson Prentice Hall, Fourth edition.
2. James Walker, “*Physics*”, Pearson- Addison Wesley Publishers, Fourth Edition.
3. Purcell, Morin, “*Electricity and Magnetism*”, “Cambridge University Press, Third Edition.
4. Serway and Jewett, “*Physics for Scientist and Engineers with Modern Physics*”, Brook/Cole Publishers, Eighth Edition.
5. Tipler, Mosca, “*Physics for scientist and Engineers with Modern Physics*”, Freeman and Company, Sixth Edition.
6. Tarasov and Tarasova, “*Questions and problems in School Physics*”, Mir Publishers.
7. H.C.Verma, “*Concepts of Physics: Vol 2*, Bharthi Bhawan Publishers.
8. Eric Roger, *Physics for the Inquiring Mind*, Princeton University Press.



ICT CORNER

Electrostatics

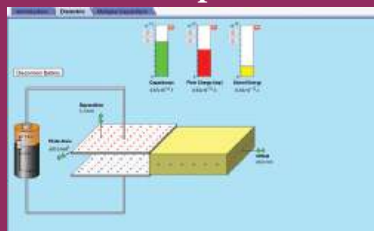
In this activity you will be able to learn about capacitor and the factors affecting capacitance.

Topic: Capacitor lab

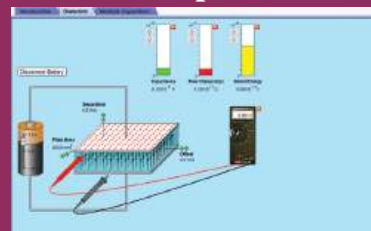
STEPS:

- Open the browser and type “phet.colorado.edu/en/simulation/legacy/capacitor-lab” in the address bar. Go to the tab ‘Dielectric’.
- Change the plate area, distance between the plate and dielectric. Identify what you would maximize or minimize to make a capacitor with the greatest capacitance.
- Explore the relationships between charge, voltage, and stored energy for a capacitor. Design a capacitor system to store the greatest energy.
- Charge the capacitor with 1.0 v using the battery. Disconnect the battery. Now insert a dielectric between the plates. Discuss how electric field changes in between the plates when dielectric is introduced.
- What is the effect of introducing a dielectric between plates? (Change dielectric materials)

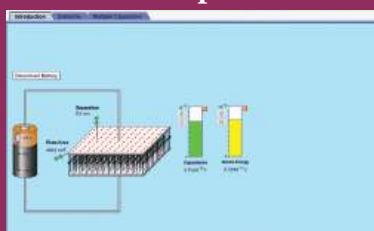
Step1



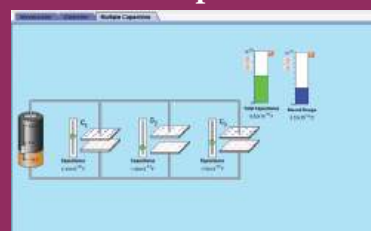
Step2



Step3



Step4



Connect capacitors parallel and series combination and find the effective capacitance.

Note:

Install Java application if it is not in your system. You can download all the phet simulation and works in off line from <https://phet.colorado.edu/en/offline-access>.

URL:

<https://phet.colorado.edu/en/simulation/legacy/capacitor-lab>

* Pictures are indicative only.

* If browser requires, allow **Flash Player** or **Java Script** to load the page.



UNIT 2

CURRENT ELECTRICITY

We will make electricity so cheap that only the rich will burn candles
– Thomas A. Edison

LEARNING OBJECTIVES

In this unit, the student is exposed to

- Flow of charges in a metallic conductor
- Ohm's law, electrical resistance, V-I characteristics
- Carbon resistors and combination of resistors
- Kirchhoff's laws - Wheatstone's bridge and its applications
- Electric power and Electric energy
- Heating effect - Joule's law – applications
- Thermoelectric effects – Seebeck effect – Peltier effect – Thomson effect



INTRODUCTION



In unit 1, we studied the properties of charges when they are at rest. In reality, the charges are always moving within the materials. For example, the electrons in a copper wire are never at rest and are continuously in random motion. Therefore it is important to analyse the behaviour of charges when they are in motion. The motion of charges constitutes 'electric current'. Current electricity is the study of flow of electric charges. It owes its origin to Alessandro Volta (1745-1827), who invented the electric battery which produced the first steady flow of electric current. Modern world depends heavily on the use of electricity. It is used to operate machines, communication systems, electronic devices, home appliances etc., In this unit, we will study about the electric current, resistance and related phenomenon in materials.

2.1

ELECTRIC CURRENT

Matter is made up of atoms. Each atom consists of a positively charged nucleus with negatively charged electrons moving around the nucleus. Atoms in metals have one or more electrons which are loosely bound to the nucleus. These electrons are called free electrons and can be easily detached from the atoms. The substances which have an abundance of these free electrons are called conductors. These free electrons move randomly throughout the conductor at a given temperature. In general due to this random motion, there is no net transfer of charges from one end of the

conductor to other end and hence no current in the conductor. When a potential difference is applied by the battery across the ends of the conductor, the free electrons drift towards the positive terminal of the battery, producing a net electric current. This is easily understandable from the analogy given in the Figure 2.1.

In the XI Volume 2, unit 6, we studied, that the mass move from higher gravitational potential to lower gravitational potential. Likewise, positive charge flows from region of higher electric potential to region of lower electric potential and negative charge flows from region of lower electric potential to region of higher electric potential. So battery or electric cell simply creates potential difference across the conductor.

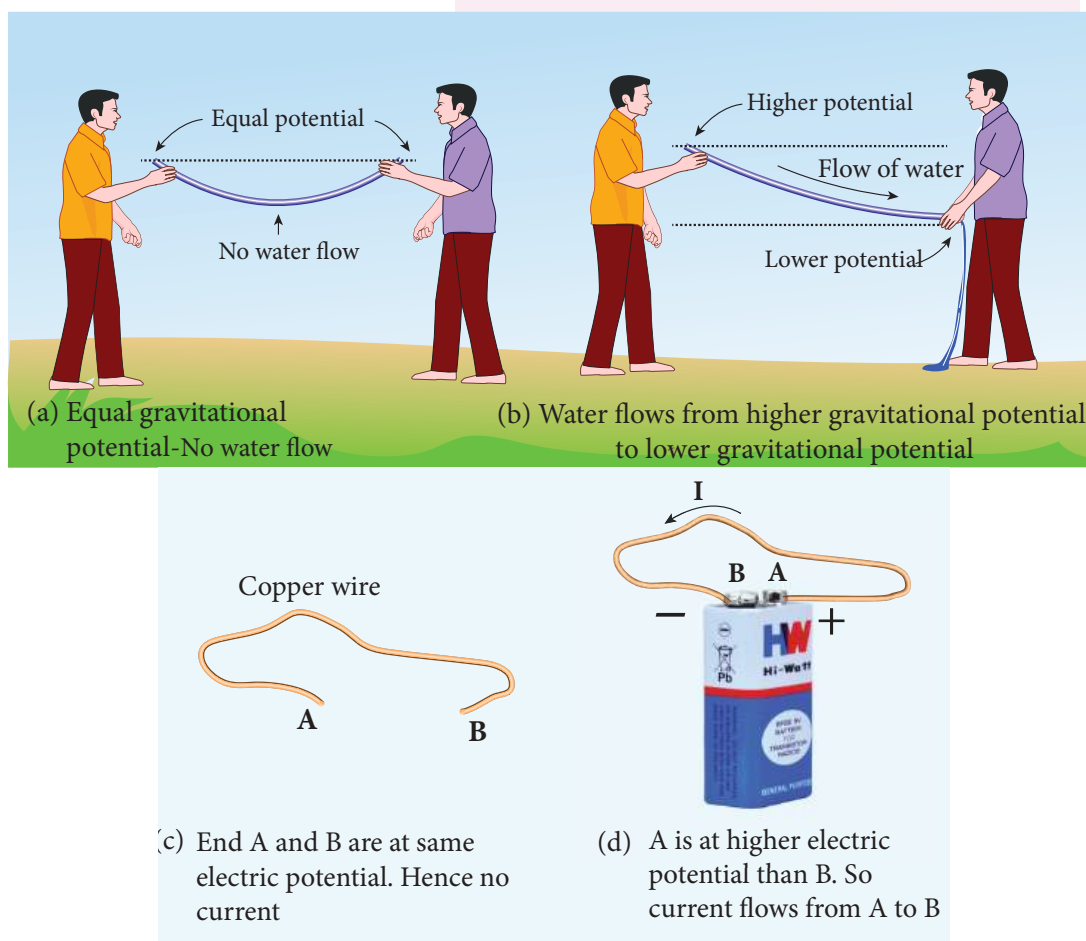


Figure 2.1 Water current and Electric current

The electric current in a conductor is defined as the rate of flow of charges through a given cross-sectional area A . It is shown in the Figure 2.2.

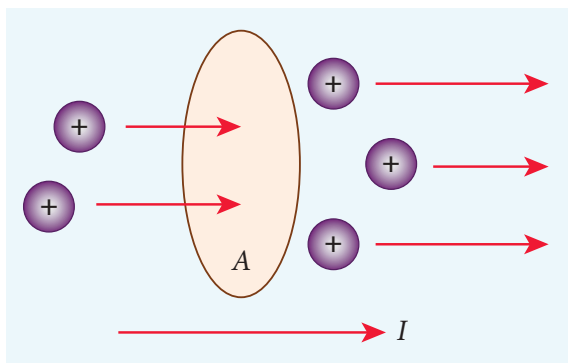


Figure 2.2 Charges flow across the area A

If a net charge Q passes through any cross section of a conductor in time t , then the current is defined as $I = \frac{Q}{t}$. But charge flow is not always constant. Hence current can more generally be defined as

$$I_{avg} = \frac{\Delta Q}{\Delta t} \quad (2.1)$$

Where ΔQ is the amount of charge that passes through the conductor at any cross section during the time interval Δt . If the rate at which charge flows changes with time, the current also changes. The instantaneous current I is defined as the limit of the average current, as $\Delta t \rightarrow 0$

$$I = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt} \quad (2.2)$$

The SI unit of current is the **ampere** (A)

$$1\text{A} = \frac{1\text{C}}{1\text{s}}$$

That is, 1A of current is equivalent to 1 coulomb of charge passing through a perpendicular cross section in a conductor in one second. The electric current is a scalar quantity.

EXAMPLE 2.1

Compute the current in the wire if a charge of 120 C is flowing through a copper wire in 1 minute.

Solution

The current (rate of flow of charge) in the wire is

$$I = \frac{Q}{t} = \frac{120}{60} = 2\text{A}$$

2.1.1 Conventional Current

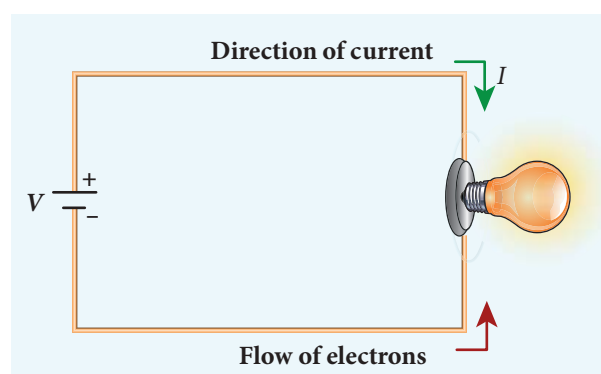


Figure 2.3 Direction of conventional current and electron flow

In an electric circuit, arrow heads are used to indicate the direction of flow of current. By convention, this flow in the circuit should be from the positive terminal of the battery to the negative terminal. This current is called the conventional current or simply current and is in the direction in which a positive test charge would move. In typical circuits the charges that flow are actually electrons, from the negative terminal of the battery to the positive terminal. As a result, the flow of electrons and the direction of conventional current point in opposite direction as shown in Figure 2.3. Mathematically, a transfer of positive charge is the



same as a transfer of negative charge in the opposite direction.



Electric current is not only produced by batteries. In nature, lightning bolt produces enormous electric current in a short time. During lightning, very high potential difference is created between the clouds and ground and hence charges flow between the clouds and ground.

2.1.2 Drift velocity

In a conductor the charge carriers are free electrons. These electrons move freely through the conductor and collide repeatedly with the positive ions. If there is no electric field, the electrons move in random directions, and hence their velocities are also randomly oriented. On an average, the number of electrons travelling in any direction will be equal to the number of electrons travelling in the opposite direction. As a result, there is no net flow of electrons in any direction and hence there will not be any current.

Suppose a potential difference is set across the conductor by connecting a battery, an electric field $\vec{E}$ is created in the conductor. This electric field exerts a force on the electrons, producing a current. The

electric field accelerates the electrons, while ions scatter the electrons and change their direction of motion. Thus, we see zigzag motion of electrons. In addition to the zigzag motion due to the collisions, the electrons move slowly along the conductor in a direction opposite to that of $\vec{E}$ as shown in the Figure 2.4.

Ions

Any material is made up of neutral atoms with equal number of electrons and protons. If the outermost electrons leave the atoms, they become free electrons and are responsible for electric current. The atoms after losing their outer most electrons will have more positive charges and hence are called positive ions. These ions will not move freely within the material like the free electrons. Hence the positive ions will not give rise to current.

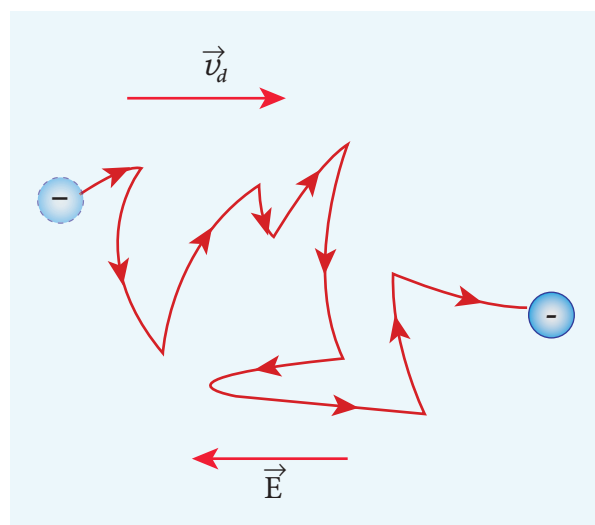


Figure 2.4 Zig-zag motion and drift velocity

This velocity is called drift velocity $\vec{v}_d$. The drift velocity is the average velocity acquired by the electrons inside the conductor when

it is subjected to an electric field. The average time between two successive collisions is called the mean free time denoted by τ . The acceleration $\vec{a}$ experienced by the electron in an electric field $\vec{E}$ is given by

$$\vec{a} = \frac{-e\vec{E}}{m} \quad (\text{since } \vec{F} = -e\vec{E}) \quad (2.3)$$

The drift velocity $\vec{v}_d$ is given by

$$\vec{v}_d = \vec{a}\tau$$

$$\vec{v}_d = -\frac{e\tau}{m}\vec{E} \quad (2.4)$$

$$\vec{v}_d = -\mu\vec{E} \quad (2.5)$$

Here $\mu = \frac{e\tau}{m}$ is the mobility of the electron and it is defined as the magnitude of the drift velocity per unit electric field.

$$\mu = \frac{|\vec{v}_d|}{|\vec{E}|} \quad (2.6)$$

The SI unit of mobility is $\text{m}^2\text{V}^{-1}\text{s}^{-1}$.

Note

The typical drift velocity of electrons in the wire is 10^{-4} m s^{-1} . If an electron drifts with this speed, then the electrons leaving the battery will take hours to reach the light bulb. Then how electric bulbs glow as soon as we switch on the battery? When battery is switched on, the electrons begin to move away from the negative terminal of the battery and this electron exerts force on the nearby electrons. This process creates a propagating influence (electric field) that travels through the wire at the speed of light. In other words, the energy is transported from the battery to bulb at the speed of light through propagating influence (electric field). Due to this reason, the bulb glows as soon as the battery is switched on.

EXAMPLE 2.2

If an electric field of magnitude 570 N C^{-1} , is applied in the copper wire, find the acceleration experienced by the electron.

Solution:

$E = 570 \text{ N C}^{-1}$, $e = 1.6 \times 10^{-19} \text{ C}$,
 $m = 9.11 \times 10^{-31} \text{ kg}$ and $a = ?$

$$F = ma = eE$$

$$a = \frac{eE}{m} = \frac{570 \times 1.6 \times 10^{-19}}{9.11 \times 10^{-31}}$$

$$= \frac{912 \times 10^{-19} \times 10^{31}}{9.11}$$

$$= 1.001 \times 10^{14} \text{ m s}^{-2}$$

Misconception

- There is a common misconception that the battery is the source of electrons. It is not true. When a battery is connected across the given wire, the electrons in the closed circuit resulting the current. Battery sets the potential difference (electrical energy) due to which these electrons in the conducting wire flow in a particular direction. The resulting electrical energy is used by electric bulb, electric fan etc. Similarly the electricity board is supplying the electrical energy to our home.
- We often use the phrases like 'charging the battery in my mobile' and 'my mobile phone battery has no charge' etc. These sentences are not correct.



When we say ‘battery has no charge,’ it means, that the battery has lost ability to provide energy or provide potential difference to the electrons in the circuit. When we say ‘mobile is charging,’ it implies that the battery is receiving energy from AC power supply and not electrons.

2.1.3 Microscopic model of current

Consider a conductor with area of cross section A and let an electric field $\vec{E}$ be applied to it from right to left. Suppose there are n electrons per unit volume in the conductor and assume that all the electrons move with the same drift velocity $\vec{v}_d$ as shown in Figure 2.5.

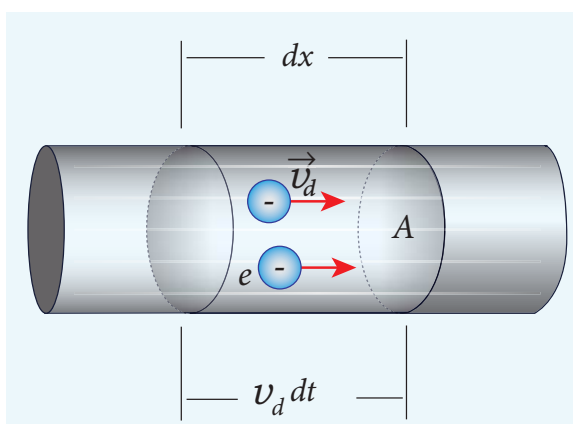


Figure 2.5 Microscopic model of current

The drift velocity of the electrons $= v_d$
If the electrons move through a distance dx within a small interval of dt , then

$$v_d = \frac{dx}{dt}; \quad dx = v_d dt \quad (2.7)$$

Since A is the area of cross section of the conductor, the electrons available in the volume of length dx is

= volume $\times$ number of electrons per unit volume

$$= A dx \times n \quad (2.8)$$

Substituting for dx from equation (2.7) in (2.8)

$$= (A v_d dt) n$$

Total charge in the volume element $dQ =$ (charge) $\times$ (number of electrons in the volume element)

$$dQ = (e)(A v_d dt) n$$

$$\text{Hence the current } I = \frac{dQ}{dt}$$

$$I = ne A v_d \quad (2.9)$$

Current density (J)

The current density (J) is defined as the current per unit area of cross section of the conductor.

$$J = \frac{I}{A}$$

The S.I unit of current density is $\frac{A}{m^2}$ (or) $A m^{-2}$

$$J = \frac{ne A v_d}{A} \quad (\text{from equation 2.9})$$

$$J = ne v_d \quad (2.10)$$

The above expression is valid only when the direction of the current is perpendicular to the area A . In general, the current density is a vector quantity and it is given by

$$\vec{J} = ne \vec{v}_d$$

Substituting $\vec{v}_d$ from equation (2.4)

$$\vec{J} = -\frac{n \cdot e^2 \tau}{m} \vec{E} \quad (2.11)$$

$$\vec{J} = -\sigma \vec{E}$$

But conventionally, we take the direction of (conventional) current density as the direction of electric field. So the above equation becomes

$$\vec{J} = \sigma \vec{E} \quad (2.12)$$

where $\sigma = \frac{ne^2 \tau}{m}$ is called conductivity.

The equation (2.12) is called microscopic form of ohm's law.

The inverse of conductivity is called resistivity (ρ) [Refer section 2.2.1].

$$\rho = \frac{1}{\sigma} = \frac{m}{ne^2\tau} \quad (2.13)$$

EXAMPLE 2.3

A copper wire of cross-sectional area 0.5 mm^2 carries a current of 0.2 A . If the free electron density of copper is $8.4 \times 10^{28} \text{ m}^{-3}$ then compute the drift velocity of free electrons.

Solution

The relation between drift velocity of electrons and current in a wire of cross-sectional area A is

$$v_d = \frac{I}{neA} = \frac{0.2}{8.4 \times 10^{28} \times 1.6 \times 10^{-19} \times 0.5 \times 10^{-6}}$$

$$v_d = 0.03 \times 10^{-3} \text{ m s}^{-1}$$



Note

Why current density is a vector but current is a scalar?

In general, the current I is defined as the scalar product of the current density and area vector in which the charges cross.

$$I = \vec{J} \cdot \vec{A}$$

The current I can be positive or negative depending on the choice of the unit vector normal to the surface area A .

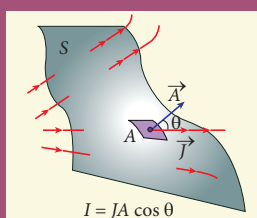


Figure 2.6 Current is a scalar

EXAMPLE 2.4

Determine the number of electrons flowing per second through a conductor, when a current of 32 A flows through it.

Solution

$$I = 32 \text{ A}, t = 1 \text{ s}$$

$$\text{Charge of an electron, } e = 1.6 \times 10^{-19} \text{ C}$$

The number of electrons flowing per second, $n = ?$

$$I = \frac{q}{t} = \frac{ne}{t}$$

$$n = \frac{It}{e}$$

$$n = \frac{32 \times 1}{1.6 \times 10^{-19} \text{ C}}$$

$$n = 20 \times 10^{19} = 2 \times 10^{20} \text{ electrons}$$

2.2

OHM'S LAW

The ohm's law can be derived from the equation $J = \sigma E$. Consider a segment of wire of length l and cross sectional area A as shown in Figure 2.7.

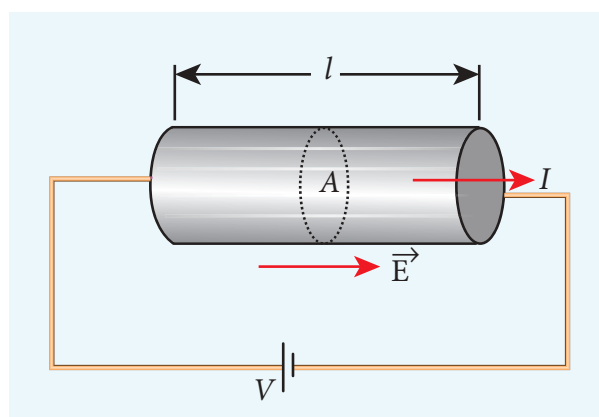


Figure 2.7 Current through the conductor

When a potential difference V is applied across the wire, a net electric field is created in the wire which constitutes the current in the wire. For simplicity, we assume that the electric field is uniform in the entire length of the wire, then the potential difference (voltage V) can be written as

$$V = El$$

As we know, the magnitude of current density

$$J = \sigma E = \sigma \frac{V}{l} \quad (2.14)$$

But $J = \frac{I}{A}$, so we write the equation (2.14) as

$$\frac{I}{A} = \sigma \frac{V}{l}.$$

By rearranging the above equation, we get

$$V = I \left(\frac{l}{\sigma A} \right) \quad (2.15)$$

The quantity $\frac{l}{\sigma A}$ is called resistance of the conductor and it is denoted as R . Note that the resistance is directly proportional to the length of the conductor and inversely proportional to area of cross section.

Therefore, the macroscopic form of Ohm's law can be stated as **"the potential difference across a given conductor is directly proportional to the current passing through it when the temperature remains constant"**.

$$V = IR \quad (2.16)$$

From the above equation, **the resistance is the ratio of potential difference across the given conductor to the current passing through the conductor.**

$$R = \frac{V}{I} \quad (2.17)$$

The SI unit of resistance is ohm (Ω). From the equation (2.16), we infer that the

graph between current versus voltage is straight line with a slope equal to the inverse of resistance R of the conductor. It is shown in the Figure 2.8 (a).

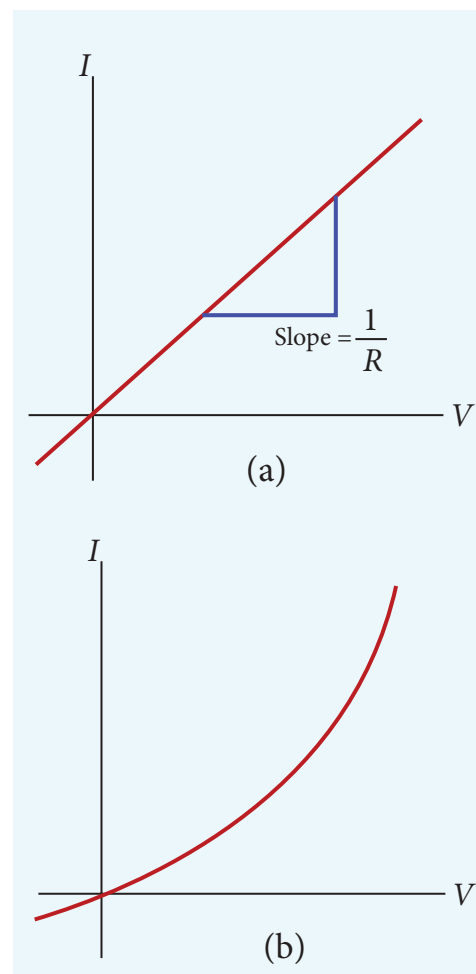


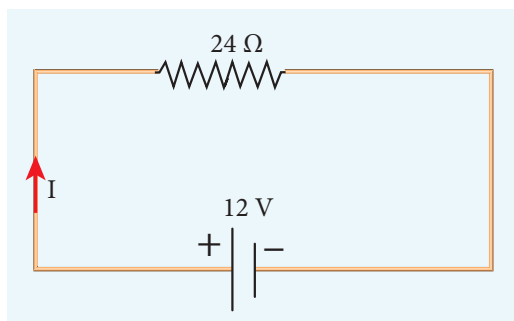
Figure 2.8 Current against voltage for (a) a conductor which obeys Ohm's law and (b) for a non-ohmic device (Diode given in XII physics, unit 10 is an example of a non-ohmic device)

Materials for which the current versus voltage graph is a straight line through the origin, are said to obey Ohm's law and their behaviour is said to be ohmic as shown in Figure 2.8(a). Materials or devices that do not follow Ohm's law are said to be non-ohmic. These materials have more complex relationships between voltage and current. A plot of I versus V for a non-ohmic material is non-linear and they do not have a constant resistance (Figure 2.8(b)).

EXAMPLE 2.5

A potential difference across $24\ \Omega$ resistor is 12 V . What is the current through the resistor?

Solution



$$V = 12\text{ V and } R = 24\ \Omega$$

Current, $I = ?$

$$\text{From Ohm's law, } I = \frac{V}{R} = \frac{12}{24} = 0.5\text{ A}$$

2.2.1 Resistivity

In the previous section, we have seen that the resistance R of any conductor is given by

$$R = \frac{l}{\sigma A} \quad (2.18)$$

where σ is called the conductivity of the material and it depends only on the type of the material used and not on its dimension.

The resistivity of a material is equal to the reciprocal of its conductivity.

$$\rho = \frac{1}{\sigma} \quad (2.19)$$

Now we can rewrite equation (2.18) using equation (2.19)

$$R = \rho \frac{l}{A} \quad (2.20)$$

The resistance of a material is directly proportional to the length of the conductor and inversely proportional to the area of cross section of the conductor. The

proportionality constant ρ is called the resistivity of the material.

If $l = 1\text{ m}$ and $A = 1\text{ m}^2$, then the resistance $R = \rho$. In other words, the **electrical resistivity of a material is defined as the resistance offered to current flow by a conductor of unit length having unit area of cross section**. The SI unit of ρ is ohm-metre ($\Omega\text{ m}$). Based on the resistivity, materials are classified as conductors, insulators and semiconductors. The conductors have lowest resistivity, insulators have highest resistivity and semiconductors have resistivity greater than conductors but less than insulators. The typical resistivity values of some conductors, insulators and semiconductors are given in the Table 2.1

Table 2.1 Resistivity for various materials

Material	Resistivity, ρ ($\Omega\text{ m}$) at 20°C
Insulators	
Pure Water	2.5×10^5
Glass	$10^{10} - 10^{14}$
Hard Rubber	$10^{13} - 10^{16}$
NaCl	10^{14}
Fused Quartz	10^{16}
Semiconductors	
Germanium	0.46
Silicon	640
Conductors	
Silver	1.6×10^{-8}
Copper	1.7×10^{-8}
Aluminium	2.7×10^{-8}
Tungsten	5.6×10^{-8}
Iron	10×10^{-8}

EXAMPLE 2.6

The resistance of a wire is $20\ \Omega$. What will be new resistance, if it is stretched uniformly 8 times its original length?

Solution

$$R_1 = 20\ \Omega, R_2 = ?$$

Let the original length of the wire (l_1) be l .

$$\text{New length, } l_2 = 8l_1 \text{ (i.e.) } l_2 = 8l$$

$$\text{Original resistance, } R_1 = \rho \frac{l_1}{A_1}$$

$$\text{New resistance } R_2 = \rho \frac{l_2}{A_2} = \frac{\rho(8l)}{A_2}$$

Though the wire is stretched, its volume remains unchanged.

Initial volume = Final volume

$$A_1 l_1 = A_2 l_2, \quad A_1 l = A_2 (8l)$$

$$\frac{A_1}{A_2} = \frac{8l}{l} = 8$$

By dividing equation for R_2 by equation for R_1 , we get

$$\frac{R_2}{R_1} = \frac{\rho(8l)}{A_2} \times \frac{A_1}{\rho l}$$

$$\frac{R_2}{R_1} = \frac{A_1}{A_2} \times 8$$

Substituting the value of $\frac{A_1}{A_2}$, we get

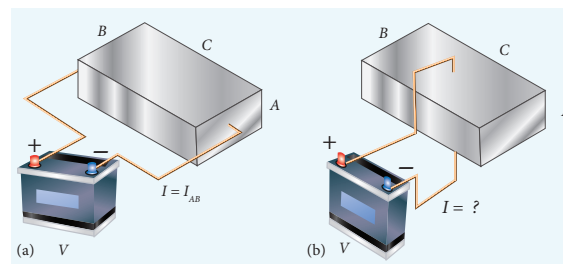
$$\frac{R_2}{R_1} = 8 \times 8 = 64$$

$$R_2 = 64 \times 20 = 1280\ \Omega$$

Hence, stretching the length of the wire has increased its resistance.

EXAMPLE 2.7

Consider a rectangular block of metal of height A , width B and length C as shown in the figure.



If a potential difference of V is applied between the two faces A and B of the block (figure (a)), the current I_{AB} is observed. Find the current that flows if the same potential difference V is applied between the two faces B and C of the block (figure (b)). Give your answers in terms of I_{AB} .

Solution

In the first case, the resistance of the block

$$R_{AB} = \rho \frac{\text{length}}{\text{Area}} = \rho \frac{C}{AB}$$

$$\text{The current } I_{AB} = \frac{V}{R_{AB}} = \frac{V}{\rho} \cdot \frac{AB}{C} \quad (1)$$

In the second case, the resistance of the block $R_{BC} = \rho \frac{A}{BC}$

$$\text{The current } I_{BC} = \frac{V}{R_{BC}} = \frac{V}{\rho} \cdot \frac{BC}{A} \quad (2)$$

To express I_{BC} in terms of I_{AB} , we multiply and divide equation (2) by AC , we get

$$I_{BC} = \frac{V}{\rho} \cdot \frac{BC}{A} \cdot \frac{AC}{AC} = \left(\frac{V}{\rho} \cdot \frac{AB}{C} \right) \cdot \frac{C^2}{A^2} = \frac{C^2}{A^2} \cdot I_{AB}$$

Since $C > A$, the current $I_{BC} > I_{AB}$



The human body contains a large amount of water which has low resistance of around $200\ \Omega$ and the dry skin has high resistance of around $500\ \text{k}\Omega$. But when the skin is wet, the resistance is reduced to around $1000\ \Omega$. This is the reason why repairing the electrical connection with the wet skin is always dangerous.

2.2.2 Resistors in series and parallel

An electric circuit may contain a number of resistors which can be connected in different ways. For each type of circuit, we can calculate the equivalent resistance produced by a group of individual resistors.

Resistors in series

When two or more resistors are connected end to end, they are said to be in series. The resistors could be simple resistors or bulbs or heating elements or other devices. Figure 2.9 (a) shows three resistors R_1 , R_2 and R_3 connected in series.

The amount of charge passing through resistor R_1 must also pass through resistors R_2 and R_3 since the charges cannot accumulate anywhere in the circuit. Due to

this reason, the current I passing through all the three resistors is the same. According to Ohm's law, if same current pass through different resistors of different values, then the potential difference across each resistor must be different. If V_1 , V_2 and V_3 be the potential differences (voltage) across each of the resistors R_1 , R_2 and R_3 respectively, then we can write $V_1 = IR_1$, $V_2 = IR_2$ and $V_3 = IR_3$. But the supply voltage V must be equal to the sum of voltages (potential differences) across each resistor.

$$V = V_1 + V_2 + V_3 = IR_1 + IR_2 + IR_3 \quad (2.21)$$

$$V = I(R_1 + R_2 + R_3)$$

$$V = IR_s \quad (2.22)$$

where R_s is the equivalent resistance.

$$R_s = R_1 + R_2 + R_3 \quad (2.23)$$

When several resistors are connected in series, the total or equivalent resistance is the sum of the individual resistances as shown in the Figure 2.9 (b).

Note: The value of equivalent resistance in series connection will be greater than each individual resistance.

EXAMPLE 2.8

Calculate the equivalent resistance for the circuit which is connected to 24 V battery and also find the potential difference across each resistors in the circuit.

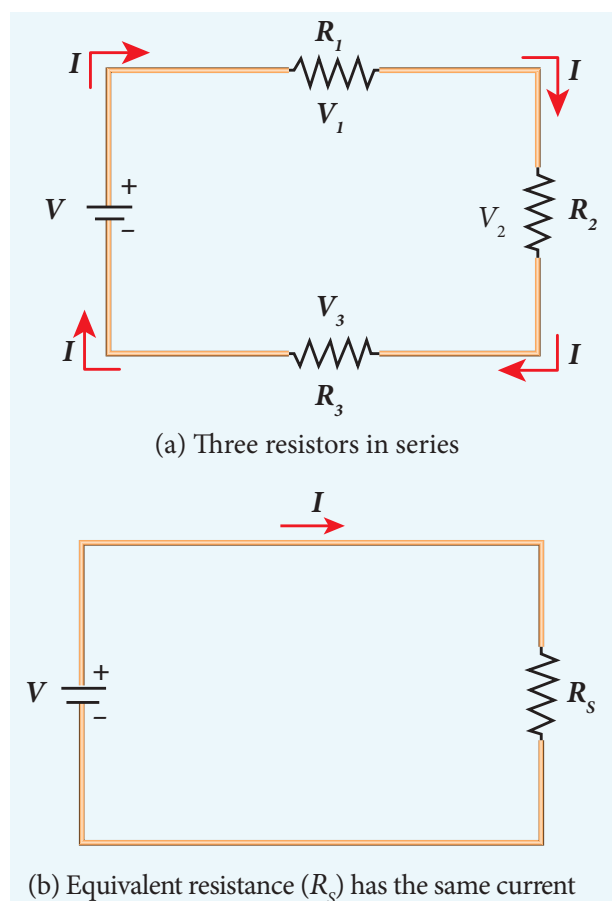
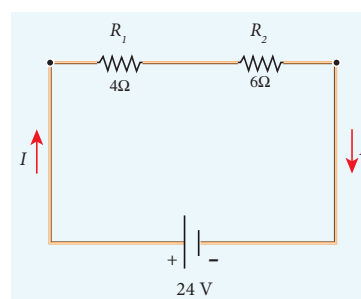


Figure 2.9 Resistors in series

Solution

Since the resistors are connected in series, the effective resistance in the circuit

$$= 4\ \Omega + 6\ \Omega = 10\ \Omega$$

$$\text{current } I \text{ in the circuit} = \frac{V}{R_{eq}} = \frac{24}{10} = 2.4\ \text{A}$$

Voltage across $4\ \Omega$ resistor

$$V_1 = IR_1 = 2.4\ \text{A} \times 4\ \Omega = 9.6\ \text{V}$$

Voltage across $6\ \Omega$ resistor

$$V_2 = IR_2 = 2.4\ \text{A} \times 6\ \Omega = 14.4\ \text{V}$$

Resistors in parallel

Resistors are in parallel when they are connected across the same potential difference as shown in Figure 2.10 (a).

In this case, the total current I that leaves the battery is split into three separate components. Let I_1 , I_2 and I_3 be the current through the resistors R_1 , R_2 and R_3 respectively. Due to the conservation of charge, total current in the circuit I is equal to sum of the currents through each of the three resistors.

$$I = I_1 + I_2 + I_3 \quad (2.24)$$

Since the voltage across each resistor is the same, applying Ohm's law to each resistor, we have

$$I_1 = \frac{V}{R_1}, I_2 = \frac{V}{R_2}, I_3 = \frac{V}{R_3} \quad (2.25)$$

Substituting these values in equation (2.24), we get

$$I = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3} = V \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right]$$

$$I = \frac{V}{R_p}$$

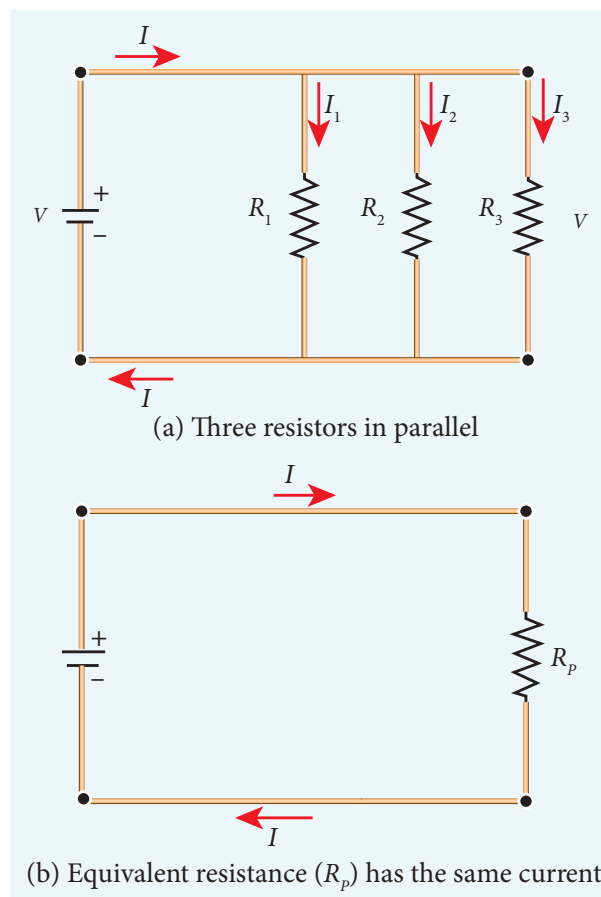


Figure 2.10 Resistors in parallel

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad (2.26)$$

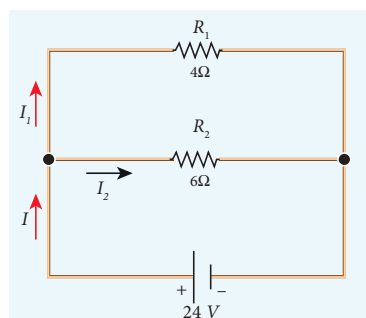
Here R_p is the equivalent resistance of the parallel combination of the resistors. Thus, when a number of resistors are connected in parallel, the sum of the reciprocals of resistance of the individual resistors is equal to the reciprocal of the effective resistance of the combination as shown in the Figure 2.10 (b).

Note: The value of equivalent resistance in parallel connection will be lesser than each individual resistance.

House hold appliances are always connected in parallel so that even if one is switched off, the other devices could function properly.

EXAMPLE 2.9

Calculate the equivalent resistance in the following circuit and also find the values of current I , I_1 and I_2 in the given circuit.



Solution

Since the resistances are connected in parallel, the equivalent resistance in the circuit is

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{6}$$
$$\frac{1}{R_p} = \frac{5}{12} \Omega \quad \text{or} \quad R_p = \frac{12}{5} \Omega$$

The resistors are connected in parallel, the potential difference (voltage) across them is the same.

$$I_1 = \frac{V}{R_1} = \frac{24V}{4\Omega} = 6A$$
$$I_2 = \frac{V}{R_2} = \frac{24}{6} = 4A$$

The current I is the sum of the currents in the two branches. Then,

$$I = I_1 + I_2 = 6A + 4A = 10A$$

EXAMPLE 2.10

Two resistors when connected in series and parallel, their equivalent resistances are 15Ω and $\frac{56}{15} \Omega$ respectively. Find the values of the resistances.

Solution

$$R_s = R_1 + R_2 = 15 \Omega \quad (1)$$

$$R_p = \frac{R_1 R_2}{R_1 + R_2} = \frac{56}{15} \Omega \quad (2)$$

From equation (1) substituting for $R_1 + R_2$ in equation (2)

$$\frac{R_1 R_2}{15} = \frac{56}{15} \Omega$$

$$\therefore R_1 R_2 = 56$$

$$R_2 = \frac{56}{R_1} \Omega \quad (3)$$

Substituting for R_2 in equation (1) from equation (3)

$$R_1 + \frac{56}{R_1} = 15$$

$$\text{Then, } \frac{R_1^2 + 56}{R_1} = 15$$

$$R_1^2 + 56 = 15 R_1$$

$$R_1^2 - 15 R_1 + 56 = 0$$

The above equation can be solved using factorisation.

$$R_1 = 8 \Omega \text{ (or) } R_1 = 7 \Omega$$

$$\text{If } R_1 = 8 \Omega$$

Substituting in equation (1)

$$8 + R_2 = 15$$

$$R_2 = 15 - 8 = 7 \Omega,$$

$$R_2 = 7 \Omega \text{ i.e., (when } R_1 = 8 \Omega; R_2 = 7 \Omega)$$

$$\text{If } R_1 = 7 \Omega$$

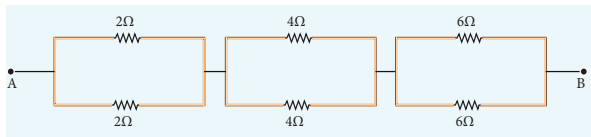
Substituting in equation (1)

$$7 + R_2 = 15$$

$$R_2 = 8 \Omega, \text{ i.e., (when } R_1 = 7 \Omega; R_2 = 8 \Omega)$$

EXAMPLE 2.11

Calculate the equivalent resistance between A and B in the given circuit.



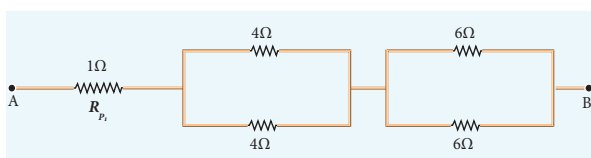
Solution

In all the sections, the resistors are connected in parallel.

Section 1

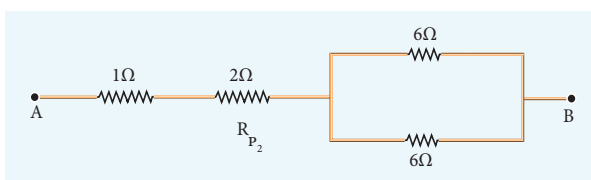
$$\frac{1}{R_{p_1}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{p_1}} = \frac{1}{2} + \frac{1}{2} = \frac{2}{2} \quad R_{p_1} = 1\Omega$$



Section II

$$\frac{1}{R_{p_2}} = \frac{1}{4} + \frac{1}{4} = \frac{2}{4}, \quad \frac{1}{R_{p_2}} = \frac{1}{2}, \quad R_{p_2} = 2\Omega$$



Section III

$$\frac{1}{R_{p_3}} = \frac{1}{6} + \frac{1}{6} = \frac{2}{6}$$

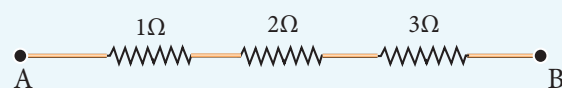
$$\frac{1}{R_{p_3}} = \frac{1}{3}, \quad R_{p_3} = 3\Omega$$

Equivalent resistance is given by

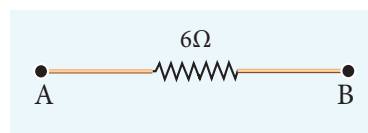
$$R = R_{p_1} + R_{p_2} + R_{p_3}$$

$$R = 1\Omega + 2\Omega + 3\Omega = 6\Omega$$

The circuit becomes,

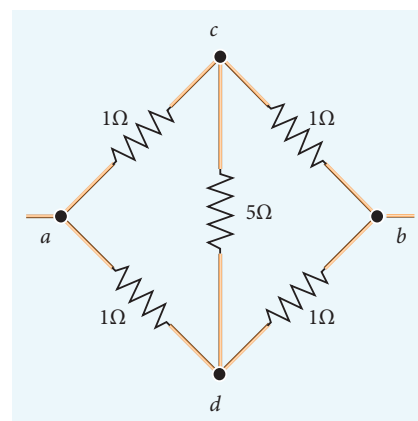


Equivalent resistance between A and B is



EXAMPLE 2.12

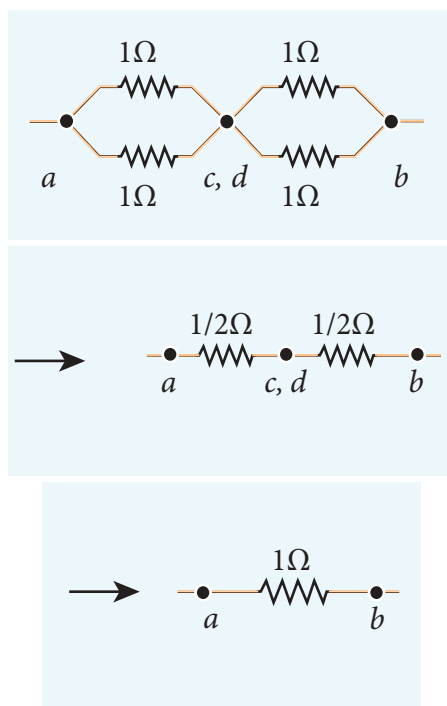
Five resistors are connected in the configuration as shown in the figure. Calculate the equivalent resistance between the points a and b.



Solution

Case (a)

To find the equivalent resistance between the points a and b, we assume that a current is entering the junction at a. Since all the resistances in the outside loop are the same (1Ω), the current in the branches ac and ad must be equal. Hence the points C and D are at the same potential and no current through 5Ω . It implies that the 5Ω has no role in determining the equivalent resistance and it can be removed. So the circuit is simplified as shown in the figure.



The equivalent resistance of the circuit between a and b is $R_{eq} = 1\ \Omega$

2.2.3 Colour code for Carbon resistors



Figure 2.11 Resistances used in laboratory

Carbon resistors consist of a ceramic core, on which a thin layer of crystalline carbon is deposited as shown in Figure 2.11. These resistors are inexpensive, stable and compact in size. Colour rings are used to indicate the value of the resistance according to the rules given in the Table 2.2.

Three coloured rings are used to indicate the values of a resistor: the first two rings are significant figures of resistances, the third ring indicates the decimal multiplier after them. The fourth colour, silver or gold,

Table 2.2 Colour Coding for Resistors

Colour	Number	Multiplier	Tolerance
Black	0	1	
Brown	1	10^1	
Red	2	10^2	
Orange	3	10^3	
Yellow	4	10^4	
Green	5	10^5	
Blue	6	10^6	
Violet	7	10^7	
Gray	8	10^8	
White	9	10^9	
Gold		10^{-1}	5%
Silver		10^{-2}	10%
Colourless			20%

shows the tolerance of the resistor at 10% or 5% as shown in the Figure 2.12. If there is no fourth ring, the tolerance is 20%.

For the resistor shown in Figure 2.12, the first digit = 5 (green), the second digit = 6 (blue), decimal multiplier = 10^3 (orange) and tolerance = 5% (gold). The value of resistance = $56 \times 10^3\ \Omega$ or 56 k Ω with the tolerance value 5%.

Note While reading the colour code, hold the resistor with colour bands to your left. Resistors never start with a metallic band on the left.

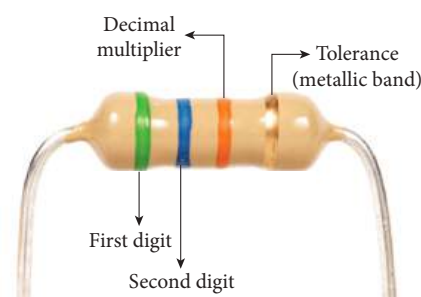


Figure 2.12 Resistor colour coding

2.2.4 Temperature dependence of resistivity

The resistivity of a material is dependent on temperature. It is experimentally found that for a wide range of temperatures, the resistivity of a conductor increases with increase in temperature according to the expression,

$$\rho_T = \rho_0 [1 + \alpha(T - T_0)] \quad (2.27)$$

where ρ_T is the resistivity of a conductor at $T^\circ\text{C}$, ρ_0 is the resistivity of the conductor at some reference temperature T_0 (usually at 20°C) and α is the temperature coefficient of resistivity. **It is defined as the ratio of increase in resistivity per degree rise in temperature to its resistivity at T_0 .**

From the equation (2.27), we can write

$$\begin{aligned} \rho_T - \rho_0 &= \alpha \rho_0 (T - T_0) \\ \therefore \alpha &= \frac{\rho_T - \rho_0}{\rho_0 (T - T_0)} = \frac{\Delta \rho}{\rho_0 \Delta T} \end{aligned}$$

where $\Delta \rho = \rho_T - \rho_0$ is change in resistivity for a change in temperature $\Delta T = T - T_0$. Its unit is per $^\circ\text{C}$.

α of conductors

For conductors α is positive. If the temperature of a conductor increases, the average kinetic energy of electrons in the conductor increases. This results in more frequent collisions and hence the resistivity increases. The graph of the equation (2.27) is shown in Figure 2.13.

Even though, the resistivity of conductors like metals varies linearly for wide range of temperatures, there also exists a non-linear region at very low temperatures. The resistivity approaches some finite value as the temperature approaches absolute zero as shown in Figure 2.13(b).

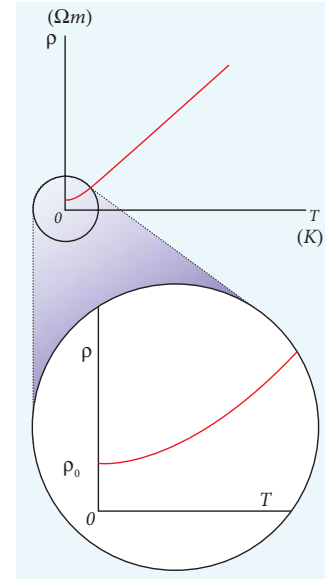


Figure 2.13 (a) Temperature dependence of resistivity for a conductor (b) Non linear region at low temperature

Using the equation $\rho = R \frac{A}{l}$ in equation (2.27), we get the expression for the resistance of a conductor at temperature $T^\circ\text{C}$ as

$$R_T = R_0 [1 + \alpha(T - T_0)] \quad (2.28)$$

The temperature coefficient of resistivity can also be obtained from the equation (2.28),

$$\begin{aligned} R_T - R_0 &= \alpha R_0 (T - T_0) \\ \therefore \alpha &= \frac{R_T - R_0}{R_0 (T - T_0)} = \frac{1}{R_0} \frac{\Delta R}{\Delta T} \\ \alpha &= \frac{1}{R_0} \frac{\Delta R}{\Delta T} \end{aligned} \quad (2.29)$$

where $\Delta R = R_T - R_0$ is change in resistance during the change in temperature $\Delta T = T - T_0$

α of semiconductors

For semiconductors, the resistivity decreases with increase in temperature. As the temperature increases, more electrons will be liberated from their atoms (Refer unit 9 for conduction in semi conductors).

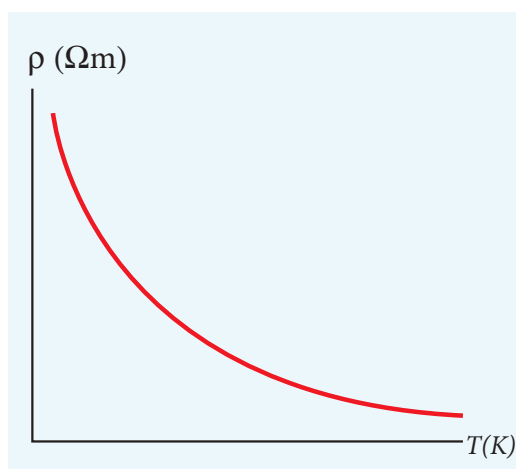


Figure 2.14 Temperature dependence of resistivity for a semiconductor

Hence the current increases and therefore the resistivity decreases as shown in Figure 2.14. A semiconductor with a negative temperature coefficient of resistivity is called a thermistor.

The typical values of temperature coefficients of various materials are given in table 2.3.

Table 2.3

Material	Temperature Coefficient of resistivity α [$^{\circ}\text{C}^{-1}$]
Silver	3.8×10^{-3}
Copper	3.9×10^{-3}
Gold	3.4×10^{-3}
Aluminum	3.9×10^{-3}
Tungsten	4.5×10^{-3}
Iron	5.0×10^{-3}
Platinum	3.92×10^{-3}
Lead	3.9×10^{-3}
Nichrome	0.4×10^{-3}
Carbon	-0.5×10^{-3}
Germanium	-48×10^{-3}
Silicon	-75×10^{-3}

We can understand the temperature dependence of resistivity in the following way. In section 2.1.3, we have shown that the electrical conductivity, $\sigma = \frac{ne^2\tau}{m}$. As the resistivity is inverse of σ , it can be written as

$$\rho = \frac{m}{ne^2\tau} \quad (2.30)$$

The resistivity of materials is

- inversely proportional to the number density (n) of the electrons
- inversely proportional to the average time between the collisions (τ).

In metals, if the temperature increases, the average time between the collision (τ) decreases and n is independent of temperature. In semiconductors when temperature increases, n increases and τ decreases, but increase in n is dominant than decreasing τ , so that overall resistivity decreases.



The resistance of certain materials become zero below certain temperature T_c . This temperature is known as critical temperature or transition temperature. The materials which exhibit this property are known as superconductors. This phenomenon was first observed by Kammerlingh Onnes in 1911. He found that mercury exhibits superconductor behaviour at 4.2 K. Since $R = 0$, current once induced in a superconductor persists without any potential difference.

EXAMPLE 2.13

If the resistance of coil is 3Ω at 20°C and $\alpha = 0.004/^{\circ}\text{C}$ then determine its resistance at 100°C .



Solution

$$R_0 = 3 \, \Omega, \quad T = 100^\circ\text{C}, \quad T_0 = 20^\circ\text{C}$$

$$\alpha = 0.004/^\circ\text{C}, \quad R_T = ?$$

$$R_T = R_0(1 + \alpha(T - T_0))$$

$$R_{100} = 3(1 + 0.004 \times 80)$$

$$R_{100} = 3.96 \, \Omega$$

EXAMPLE 2.14

Resistance of a material at 20°C and 40°C are $45 \, \Omega$ and $85 \, \Omega$ respectively. Find its temperature coefficient of resistivity.

Solution

$$T_0 = 20^\circ\text{C}, \quad T = 40^\circ\text{C}, \quad R_0 = 45 \, \Omega, \quad R = 85 \, \Omega$$

$$\alpha = \frac{1}{R_0} \frac{\Delta R}{\Delta T}$$

$$\alpha = \frac{1}{45} \left(\frac{85 - 45}{40 - 20} \right) = \frac{1}{45} (2)$$

$$\alpha = 0.044 \, \text{per}^\circ\text{C}$$

2.3

ENERGY AND POWER IN ELECTRICAL CIRCUITS

When a battery is connected between the ends of a conductor, a current is established. The battery is supplying energy to the device which is connected in the circuit. Consider a circuit in which a battery of voltage V is connected to the resistor as shown in Figure 2.15.

Assume that a positive charge of dQ moves from point a to b through the battery and moves from point c to d through the resistor and back to point a . When the charge

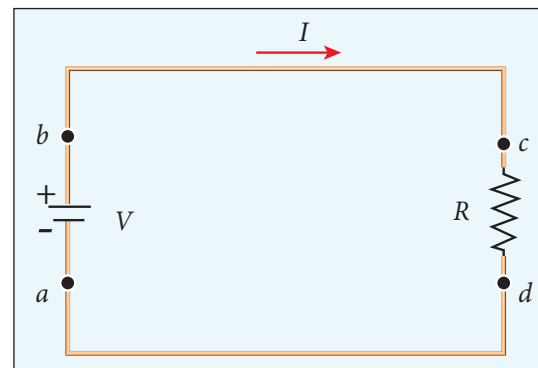


Figure 2.15 Energy given by the battery

moves from point a to b , it gains potential energy $dU = V \cdot dQ$ and the chemical potential energy of the battery decreases by the same amount. When this charge dQ passes through resistor it loses the potential energy $dU = V \cdot dQ$ due to collision with atoms in the resistor and again reaches the point a . This process occurs continuously till the battery is connected in the circuit. The rate at which the charge loses its electrical potential energy in the resistor can be calculated.

The electrical power P is the rate at which the electrical potential energy is delivered,

$$P = \frac{dU}{dt} = \frac{(V \cdot dQ)}{dt} = V \frac{dQ}{dt} \quad (2.31)$$

Since the electric current $I = \frac{dQ}{dt}$, the equation (2.31) can be rewritten as

$$P = VI \quad (2.32)$$

This expression gives the power delivered by the battery to any electrical system, where I is the current passing through it and V is the potential difference across it. The SI unit of electrical power is watt ($1\text{W} = 1 \text{ J s}^{-1}$). Commercially, the electrical bulbs used in houses come with the power and voltage rating of 5W-220V, 30W-220V, 60W-220V etc. (Figure 2.16).



Figure 2.16 Electrical bulbs with power rating

Usually these voltage rating refers AC RMS voltages. For a given bulb, if the voltage drop across the bulb is greater than voltage rating, the bulb will fuse.

Using Ohm's law, power delivered to the resistance R is expressed in other forms

$$P = IV = I(IR) = I^2R \quad (2.33)$$

$$P = IV = \frac{V}{R}V = \frac{V^2}{R} \quad (2.34)$$



The electrical power produced (dissipated) by a resistor is I^2R . It depends on the square of the current. Hence, if current is doubled, the power will increase by four times. Similar explanation holds true for voltage also.

The total electrical energy used by any device is obtained by multiplying the power and duration of the time when it is ON. If the power is in watts and the time is in seconds, the energy will be in joules. In practice, electrical energy is measured in kilowatt hour (kWh). 1 kWh is known as 1 unit of electrical energy.

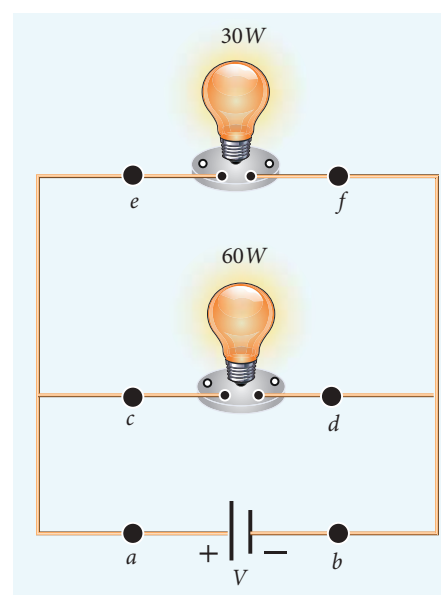
$$(1 \text{ kWh} = 1000 \text{ Wh} = (1000 \text{ W}) (3600 \text{ s}) = 3.6 \times 10^6 \text{ J})$$



The Tamilnadu Electricity Board is charging for the amount of energy you use and not for the power. A current of 1A flowing through a potential difference of 1V produces a power of 1W.

EXAMPLE 2.15

A battery of voltage V is connected to 30 W bulb and 60 W bulb as shown in the figure. (a) Identify brightest bulb (b) which bulb has greater resistance? (c) Suppose the two bulbs are connected in series, which bulb will glow brighter?



Solution

- (a) The power delivered by the battery $P = VI$. Since the bulbs are connected in parallel, the voltage drop across each bulb is the same. If the voltage is kept fixed, then the power is directly proportional to current ($P \propto I$). So 60 W bulb draws twice as much as current as 30 W and it will glow brighter than 30 W bulb.



- (b) To calculate the resistance of the bulbs, we use the relation $P = \frac{V^2}{R}$. In both the bulbs, the voltage drop is the same. So the power is inversely proportional to the resistance or resistance is inversely proportional to the power $\left(R \propto \frac{1}{P}\right)$. It implies that, the 30W has twice as much as resistance as 60 W bulb.
- (c) When the bulbs are connected in series, the current passing through each bulb is the same. It is equivalent to two resistors connected in series. The bulb which has higher resistance has higher voltage drop. So 30W bulb will glow brighter than 60W bulb. So the higher power rating does not always imply more brightness and it depends whether bulbs are connected in series or parallel.

EXAMPLE 2.16

Two electric bulbs marked 20 W – 220 V and 100 W – 220 V are connected in series to 440 V supply. Which bulb will get fused?

Solution

To check which bulb will get fused, the voltage drop across each bulb has to be calculated.

The resistance of the bulb,

$$R = \frac{V^2}{P} = \frac{(\text{Rated voltage})^2}{\text{Rated power}}$$

For 20W-220V bulb,

$$R_1 = \frac{(220)^2}{20} \Omega = 2420 \Omega$$



For 100W-220V bulb,

$$R_2 = \frac{(220)^2}{100} \Omega = 484 \Omega$$

Both the bulbs are connected in series. So same current will pass through both the bulbs. The current that passes through the circuit, $I = \frac{V}{R_{\text{tot}}}$.

$$R_{\text{tot}} = (R_1 + R_2)$$

$$R_{\text{tot}} = (484 + 2420) \Omega = 2904 \Omega$$

$$I = \frac{440 \text{ V}}{2904 \Omega} \approx 0.151 \text{ A}$$

The voltage drop across the 20W bulb is

$$V_1 = IR_1 = \frac{440}{2904} \times 2420 \approx 366.6 \text{ V}$$

The voltage drop across the 100W bulb is

$$V_2 = IR_2 = \frac{440}{2904} \times 484 \approx 73.3 \text{ V}$$

The 20 W bulb will get fused because the voltage across it is more than the voltage rating.

2.4

ELECTRIC CELLS AND BATTERIES

An electric cell converts chemical energy into electrical energy to produce electricity. It contains two electrodes (carbon and zinc) immersed in an electrolyte (sulphuric acid) as shown in Figure 2.17.

Several electric cells connected together form a battery. When a cell or battery is connected to a circuit, electrons flow from the negative terminal to the positive terminal through the circuit. By using chemical reactions, a battery

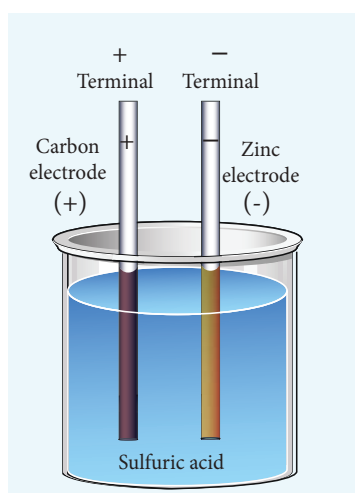


Figure 2.17 Simple electric cell

produces potential difference across its terminals. This potential difference provides the energy to move the electrons through the circuit. Commercially available electric cells and batteries are shown in Figure 2.18.



Figure 2.18 Electric cells and Batteries

2.4.1 Electromotive force and internal resistance

A battery or cell is called a source of electromotive force (emf). The term 'electromotive force' is a misnomer since it does not really refer to a force but describes a potential difference in volts. The emf of a battery or cell is the voltage provided by the battery when no current flows in the external circuit. It is shown in Figure 2.19.

Electromotive force determines the amount of work a battery or cell has to do



Figure 2.19 Measuring the emf of a cell

move a certain amount of charge around the circuit. It is denoted by the symbol ε . An ideal battery has zero internal resistance and the potential difference (terminal voltage) across the battery equals to its emf. In reality, the battery is made of electrodes and electrolyte, there is resistance to the flow of charges within the battery. This resistance is called internal resistance r . For a real battery, the terminal voltage is not equal to the emf of the battery. A freshly prepared cell has low internal resistance and it increases with ageing.

2.4.2 Determination of internal resistance

The circuit connections are made as shown in Figure 2.20.

The emf of cell ε is measured by connecting a high resistance voltmeter across it without connecting the external resistance R as shown in Figure 2.20(a). Since the voltmeter draws very little current for deflection, the circuit may be considered as open. Hence the voltmeter reading gives the emf of the cell. Then, external resistance R is included in the circuit and current I is established in the circuit. The potential difference across



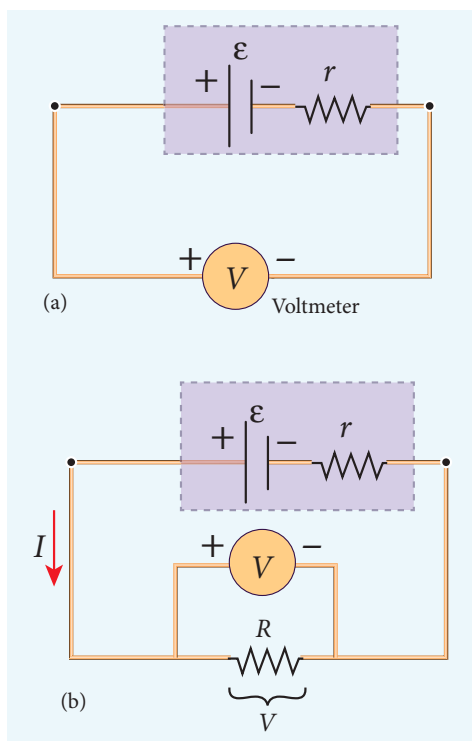


Figure 2.20 Internal resistance of the cell

R is equal to the potential difference across the cell (V) as shown in Figure 2.20(b).

The potential drop across the resistor R is

$$V = IR \quad (2.35)$$

Due to internal resistance r of the cell, the voltmeter reads a value V , which is less than the emf of cell ϵ . It is because, certain amount of voltage (Ir) has dropped across the internal resistance r .

Then $V = \epsilon - Ir$

$$Ir = \epsilon - V \quad (2.36)$$

Dividing equation (2.36) by equation (2.35), we get

$$\frac{Ir}{IR} = \frac{\epsilon - V}{V}$$

$$r = \left[\frac{\epsilon - V}{V} \right] R \quad (2.37)$$

Since ϵ , V and R are known, internal resistance r can be determined. We can also find the total current that flows in the circuit.

Due to this internal resistance, the power delivered to the circuit is not equal to power rating mentioned in the battery. For a battery of emf ϵ , with an internal resistance r , the power delivered to the circuit of resistance R is given by

$$P = I\epsilon = I(V + Ir) \quad (\text{from equation 2.36})$$

Here V is the voltage drop across the resistance R and it is equal to IR .

Therefore, $P = I(IR + Ir)$

$$P = I^2 R + I^2 r \quad (2.38)$$

Here $I^2 r$ is the power delivered to the internal resistance and $I^2 R$ is the power delivered to the electrical device (here it is the resistance R). For a good battery, the internal resistance r is very small, then $I^2 r \ll I^2 R$ and almost entire power is delivered to the external resistance.

EXAMPLE 2.17

A battery has an emf of 12 V and connected to a resistor of 3Ω . The current in the circuit is 3.93 A. Calculate (a) terminal voltage and the internal resistance of the battery (b) power delivered by the battery and power delivered to the resistor

Solution

The given values $I = 3.93 \text{ A}$, $\epsilon = 12 \text{ V}$, $R = 3 \Omega$

- (a) The terminal voltage of the battery is equal to voltage drop across the resistor

$$V = IR = 3.93 \times 3 = 11.79 \text{ V}$$

The internal resistance of the battery,

$$r = \left[\frac{\epsilon - V}{V} \right] R = \left[\frac{12 - 11.79}{11.79} \right] \times 3 = 0.05 \, \Omega$$

- (b) The power delivered by the battery $P = I\epsilon = 3.93 \times 12 = 47.1 \, \text{W}$

The power delivered to the resistor $= I^2 R = 46.3 \, \text{W}$

The remaining power $P = (47.1 - 46.3) = 0.8 \, \text{W}$ is delivered to the internal resistance and cannot be used to do useful work. (It is equal to $I^2 r$).

2.4.3 Cells in series

Several cells can be connected to form a battery. In series connection, the negative terminal of one cell is connected to the positive terminal of the second cell, the negative terminal of second cell is connected to the positive terminal of the third cell and so on. The free positive terminal of the first cell and the free negative terminal of the last cell become the terminals of the battery.

Suppose n cells, each of emf ϵ volts and internal resistance r ohms are connected in series with an external resistance R as shown in Figure 2.21

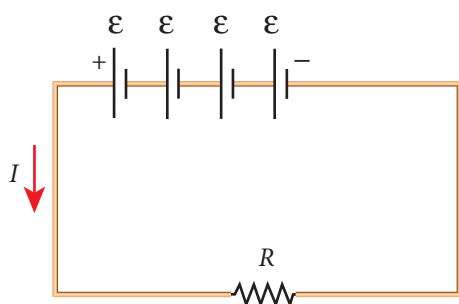


Figure 2.21 cells in series

The total emf of the battery $= n\epsilon$

The total resistance in the circuit $= nr + R$

By Ohm's law, the current in the circuit is

$$I = \frac{\text{total emf}}{\text{total resistance}} = \frac{n\epsilon}{nr + R} \quad (2.39)$$

Case (a) If $r \ll R$, then,

$$I = \frac{n\epsilon}{R} \approx nI_1 \quad (2.40)$$

where, I_1 is the current due to a single cell

$$\left(I_1 = \frac{\epsilon}{R} \right)$$

Thus, if r is negligible when compared to R the current supplied by the battery is n times that supplied by a single cell.

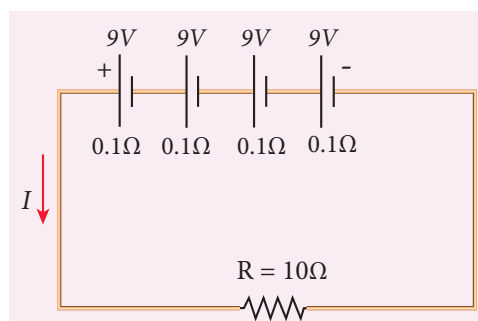
$$\text{Case (b) If } r \gg R, I = \frac{n\epsilon}{nr} \approx \frac{\epsilon}{r} \quad (2.41)$$

It is the current due to a single cell. That is, current due to the whole battery is the same as that due to a single cell and hence there is no advantage in connecting several cells.

Thus series connection of cells is advantageous only when the effective internal resistance of the cells is negligibly small compared with R .

EXAMPLE 2.18

From the given circuit,



Find

- Equivalent emf of the combination
- Equivalent internal resistance
- Total current
- Potential difference across external resistance
- Potential difference across each cell

Solution

- Equivalent emf of the combination
 $\varepsilon_{eq} = n\varepsilon = 4 \times 9 = 36 \text{ V}$
- Equivalent internal resistance $r_{eq} = nr$
 $= 4 \times 0.1 = 0.4 \Omega$
- Total current $I = \frac{n\varepsilon}{R + nr}$

$$= \frac{4 \times 9}{10 + (4 \times 0.1)}$$

$$= \frac{4 \times 9}{10 + 0.4} = \frac{36}{10.4}$$

$$I = 3.46 \text{ A}$$
- Potential difference across external resistance $V = IR = 3.46 \times 10 = 34.6 \text{ V}$. The remaining 1.4 V is dropped across the internal resistances of cells.
- Potential difference across each cell

$$\frac{V}{n} = \frac{34.6}{4} = 8.65 \text{ V}$$

2.4.4 Cells in parallel

In parallel connection all the positive terminals of the cells are connected to one point and all the negative terminals to a second point. These two points form the positive and negative terminals of the battery.

Let n cells be connected in parallel between the points A and B and a resistance R is connected between the points A and B as shown in Figure 2.22. Let ε be the emf and r the internal resistance of each cell.

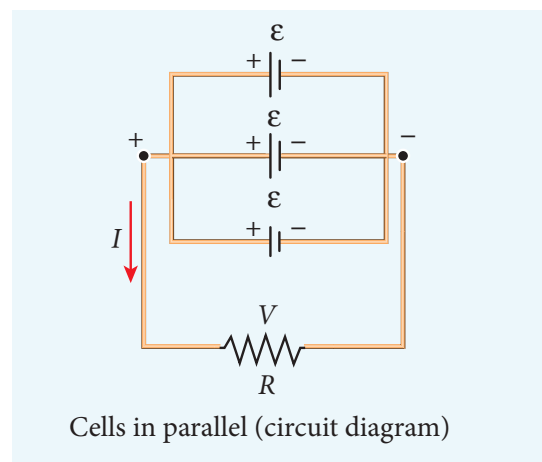


Figure 2.22 Cells in parallel

The equivalent internal resistance of the battery is $\frac{1}{r_{eq}} = \frac{1}{r} + \frac{1}{r} + \dots + \frac{1}{r} (n \text{ terms}) = \frac{n}{r}$. So $r_{eq} = \frac{r}{n}$ and the total resistance in the circuit $= R + \frac{r}{n}$. The total emf is the potential difference between the points A and B, which is equal to ε . The current in the circuit is given by

$$I = \frac{\varepsilon}{\frac{r}{n} + R}$$

$$I = \frac{n\varepsilon}{r + nR} \quad (2.42)$$

Case (a) If $r \gg R$, $I = \frac{n\varepsilon}{r} = nI_1 \quad (2.43)$

where I_1 is the current due to a single cell $\left(\frac{\varepsilon}{r}\right)$ when R is negligible. Thus, the current through the external resistance due to the whole battery is n times the current due to a single cell.

Case (b) If $r \ll R$, $I = \frac{\varepsilon}{R} \quad (2.44)$

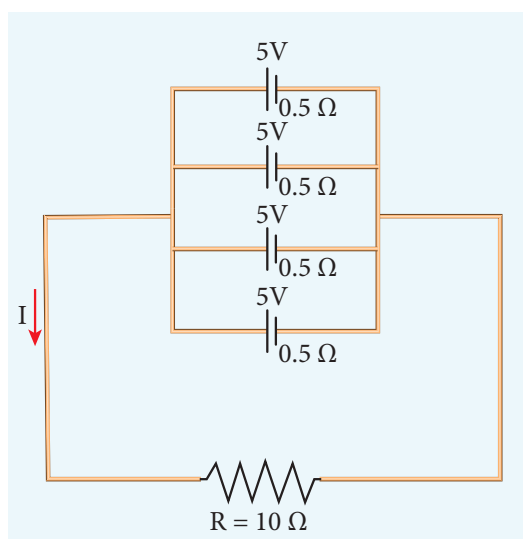
The above equation implies that current due to the whole battery is the same as that due to a single cell. Hence it is advantageous to connect cells in parallel when the external resistance is very small compared to the internal resistance of the cells.



When the car engine is started with headlights turned on, they sometimes become dim. This is due to the internal resistance of the car battery.

EXAMPLE 2.19

For the given circuit



Find

- Equivalent emf
- Equivalent internal resistance
- Total current (I)
- Potential difference across each cell
- Current from each cell

Solution

- Equivalent emf $\varepsilon_{eq} = 5 \text{ V}$
- Equivalent internal resistance,

$$R_{eq} = \frac{r}{n} = \frac{0.5}{4} = 0.125 \Omega$$

$$\text{iii) total current, } I = \frac{\varepsilon}{R + \frac{r}{n}}$$

$$I = \frac{5}{10 + 0.125} = \frac{5}{10.125}$$

$$I \approx 0.5 \text{ A}$$

$$\text{iv) Potential difference across each cell}$$

$$V = IR = 0.5 \times 10 = 5 \text{ V}$$

$$\text{v) Current from each cell, } I' = \frac{I}{n}$$

$$I' = \frac{0.5}{4} = 0.125 \text{ A}$$

2.5

KIRCHHOFF'S RULES

Ohm's law is useful only for simple circuits. For more complex circuits, Kirchhoff's rules can be used to find current and voltage. There are two generalized rules: i) Kirchhoff's current rule ii) Kirchhoff's voltage rule.

2.5.1 Kirchhoff's first rule (Current rule or Junction rule)

It states that the algebraic sum of the currents at any junction of a circuit is zero. It is a statement of law of conservation of electric charge. The charges that enter a given junction in a circuit must leave that junction since charge cannot build up or disappear at a junction. By convention, current entering the junction is taken as positive and current leaving the junction is taken as negative.

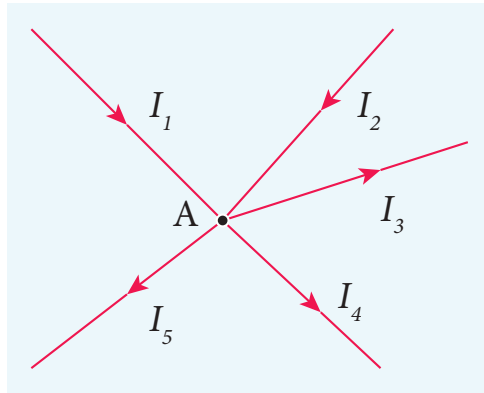


Figure 2.23 Kirchhoff's current rule

Applying this law to the junction A in Figure 2.23

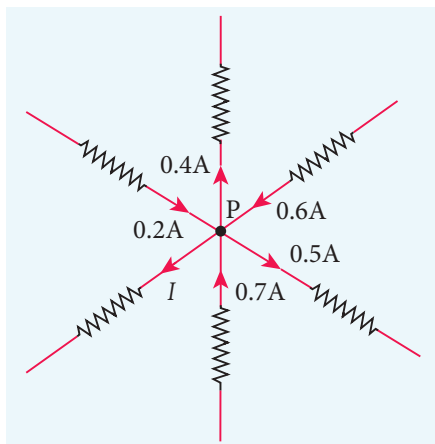
$$I_1 + I_2 - I_3 - I_4 - I_5 = 0$$

(or)

$$I_1 + I_2 = I_3 + I_4 + I_5$$

EXAMPLE 2.20

For the given circuit find the value of I .



Solution

Applying Kirchhoff's rule to the point P in the circuit,

The arrows pointing towards P are positive and away from P are negative.

$$\text{Therefore, } 0.2\text{A} - 0.4\text{A} + 0.6\text{A} - 0.5\text{A} + 0.7\text{A} - I = 0$$

$$1.5\text{A} - 0.9\text{A} - I = 0$$

$$0.6\text{A} - I = 0$$

$$I = 0.6\text{ A}$$

2.5.2 Kirchhoff's Second rule (Voltage rule or Loop rule)

It states that in a closed circuit the algebraic sum of the products of the current and resistance of each part of the circuit is equal to the total emf included in the circuit. This rule follows from the law of conservation of energy for an isolated system (The energy supplied by the emf sources is equal to the sum of the energy delivered to all resistors). The product of current and resistance is taken as positive when the direction of the current is followed. Suppose if the direction of current is opposite to the direction of the loop, then product of current and voltage across the resistor is negative. It is shown in Figure 2.24 (a) and (b). The emf is considered positive when proceeding from the negative to the positive terminal of the cell. It is shown in Figure 2.24 (c) and (d).

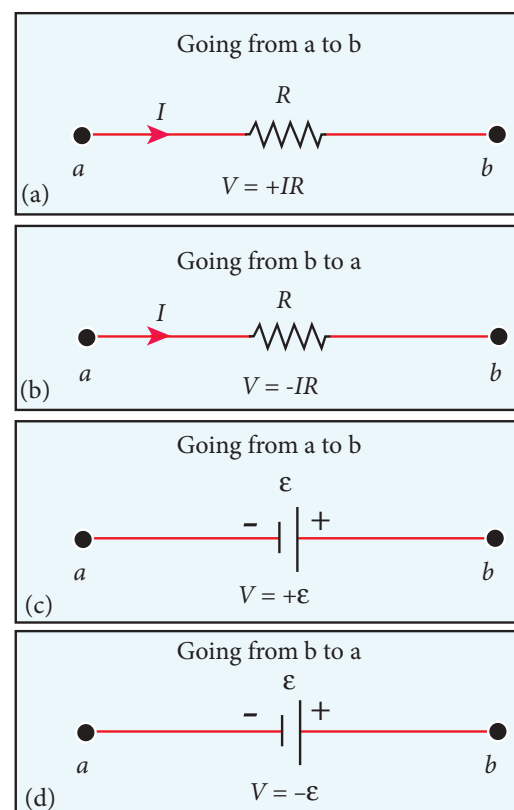
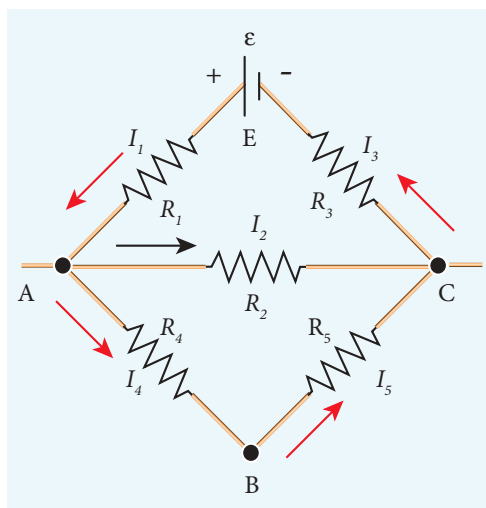


Figure 2.24 Kirchhoff voltage rule

Kirchhoff voltage rule has to be applied only when all currents in the circuit reach a steady state condition (the current in various branches are constant).

EXAMPLE 2.21

The following figure shows a complex network of conductors which can be divided into two closed loops like EACE and ABCA. Apply Kirchhoff's voltage rule (KVR),



Solution

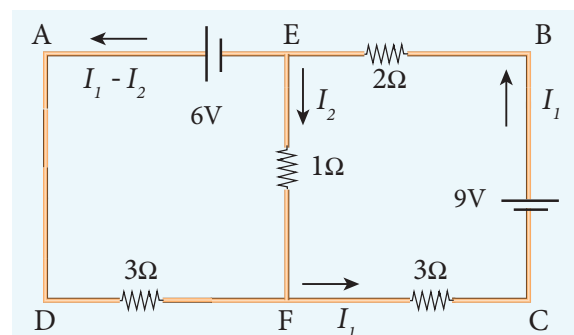
Thus applying Kirchhoff's second law to the closed loop EACE

$$I_1 R_1 + I_2 R_2 + I_3 R_3 = \varepsilon$$

and for the closed loop ABCA

$$I_4 R_4 + I_5 R_5 - I_2 R_2 = 0$$

Solution



We can denote the current that flows from 9V battery as I_1 and it splits up into I_2 and $(I_1 - I_2)$ at the junction E according to Kirchhoff's current rule (KCR).

Now consider the loop EFCBE and apply KVR, we get

$$\begin{aligned} 1I_2 + 3I_1 + 2I_1 &= 9 \\ 5I_1 + I_2 &= 9 \end{aligned} \quad (1)$$

Applying KVR to the loop EADFE, we get

$$\begin{aligned} 3(I_1 - I_2) - 1I_2 &= 6 \\ 3I_1 - 4I_2 &= 6 \end{aligned} \quad (2)$$

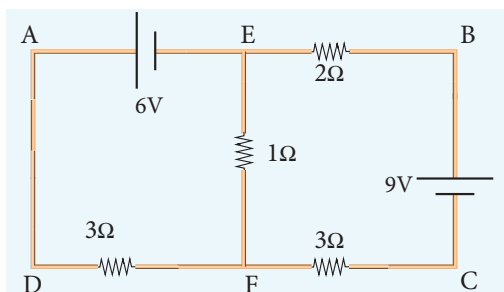
Solving equation (1) and (2), we get

$$I_1 = 1.83 \text{ A and } I_2 = -0.13 \text{ A}$$

It implies that the current in the 1 ohm resistor flows from F to E.

EXAMPLE 2.22

Calculate the current that flows in the 1Ω resistor in the following circuit.



2.5.3 Wheatstone's bridge

An important application of Kirchhoff's rules is the Wheatstone's bridge. It is used to compare resistances and in determining the unknown resistance in electrical network. The bridge consists of four resistances P , Q , R and S connected as shown in Figure 2.25. A galvanometer G is connected between the points B and D . The battery is connected between the points A and C . The current

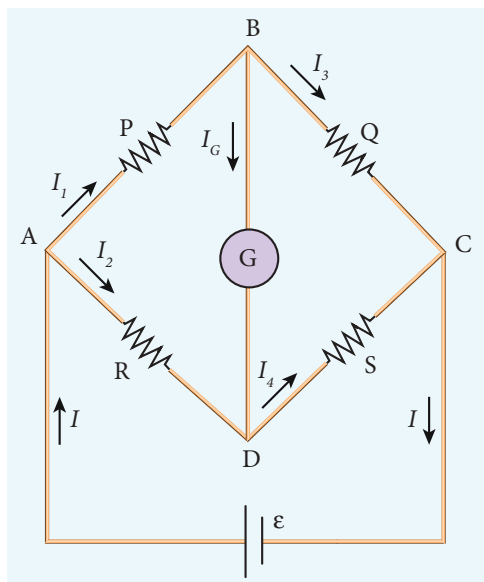


Figure 2.25 Wheatstone's bridge

through the galvanometer is I_G and its resistance is G .

Applying Kirchhoff's current rule to junction B and D respectively.

$$I_1 - I_G - I_3 = 0 \quad (2.45)$$

$$I_2 + I_G - I_4 = 0 \quad (2.46)$$

Applying Kirchhoff's voltage rule to loop $ABDA$,

$$I_1 P + I_G G - I_2 R = 0 \quad (2.47)$$

Applying Kirchhoff's voltage rule to loop $ABCD$,

$$I_1 P + I_3 Q - I_4 S - I_2 R = 0 \quad (2.48)$$

When the points B and D are at the same potential, the bridge is said to be balanced. As there is no potential difference between B and D , no current flows through galvanometer ($I_G = 0$). Substituting $I_G = 0$ in equation (2.45), (2.46) and (2.47), we get

$$I_1 = I_3 \quad (2.49)$$

$$I_2 = I_4 \quad (2.50)$$

$$I_1 P = I_2 R \quad (2.51)$$

Using equation (2.51) in equation (2.48)

$$I_3 Q = I_4 S \quad (2.52)$$

Dividing equation (2.52) by equation (2.51), we get

$$\frac{P}{Q} = \frac{R}{S} \quad (2.53)$$

This is the condition for bridge balance. Only under this condition, galvanometer shows null deflection. Suppose we know the values of two adjacent resistances, the other two resistances can be compared. If three of the resistances are known, the value of unknown resistance (fourth one) can be determined.



A galvanometer is an instrument used for detecting and measuring even very small electric currents. It is extensively useful to compare the potential difference between various parts of the circuit.

EXAMPLE 2.23

In a Wheatstone's bridge $P = 100 \, \Omega$, $Q = 1000 \, \Omega$ and $R = 40 \, \Omega$. If the galvanometer shows zero deflection, determine the value of S .

Solution

$$\frac{P}{Q} = \frac{R}{S}$$

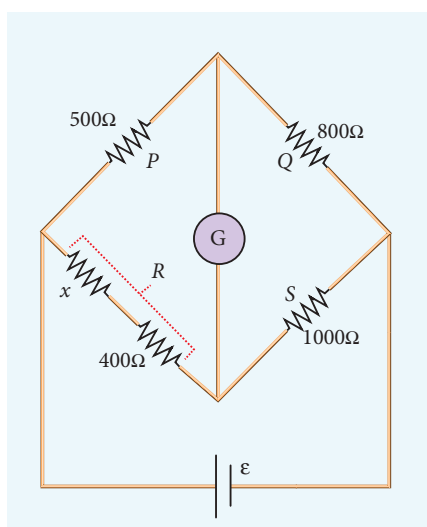
$$S = \frac{Q}{P} \times R$$

$$S = \frac{1000}{100} \times 40 \quad S = 400 \, \Omega$$

EXAMPLE 2.24

What is the value of x when the Wheatstone's network is balanced?

$$P = 500 \, \Omega, Q = 800 \, \Omega, R = x + 400, \\ S = 1000 \, \Omega$$



Solution

$$\frac{P}{Q} = \frac{R}{S}, \text{ when the network is balanced}$$

$$\frac{500}{800} = \frac{x + 400}{1000}$$

$$x + 400 = \frac{5}{8} \times 1000$$

$$x + 400 = 625$$

$$x = 625 - 400$$

$$x = 225 \, \Omega$$

2.5.4 Meter bridge

The meter bridge is another form of Wheatstone's bridge. It consists of a uniform wire of manganin AB of one meter length. This wire is stretched along a metre scale on a wooden board between two copper strips C and D. Between these two copper strips another copper strip E is mounted to enclose two gaps G_1 and G_2 as shown in Figure 2.26.

An unknown resistance P is connected in G_1 and a standard resistance Q is connected in G_2 . A jockey (conducting wire-contact maker) is connected to the terminal E on the central copper strip through a galvanometer (G) and a high resistance (HR). The exact position of jockey on the wire can be read on the scale. A Leclanche cell and a key (K) are connected between the ends of the bridge wire.

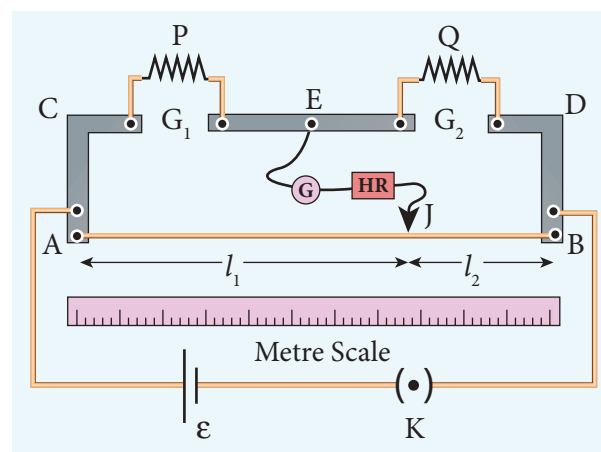


Figure 2.26 Meter bridge

The position of the jockey on the wire is adjusted so that the galvanometer shows zero deflection. Let the position of jockey at the wire be at J. The resistances corresponding to AJ and JB of the bridge wire form the resistances R and S of the Wheatstone's bridge. Then for the bridge balance

$$\frac{P}{Q} = \frac{R}{S} = \frac{r \cdot AJ}{r \cdot JB} \quad (2.54)$$

where r is the resistance per unit length of wire.

$$\frac{P}{Q} = \frac{AJ}{JB} = \frac{l_1}{l_2} \quad (2.55)$$

$$P = Q \frac{l_1}{l_2} \quad (2.56)$$

The bridge wire is soldered at the ends of the copper strips. Due to imperfect contact, some resistance might be introduced at the contact. These are called end resistances. This error can be eliminated, if another set of readings is taken with P and Q interchanged and the average value of P is found.

To find the specific resistance of the material of the wire in the coil P , the radius a and length l of the wire are measured. The specific resistance or resistivity ρ can be calculated using the relation.

$$\text{Resistance} = \rho \frac{l}{A}$$

By rearranging the above equation, we get

$$\rho = \text{Resistance} \times \frac{A}{l} \quad (2.57)$$

If P is the unknown resistance equation (2.57) becomes,

$$\rho = P \frac{\pi a^2}{l}$$

EXAMPLE 2.25

In a meter bridge experiment with a standard resistance of 15Ω in the right gap, the ratio of balancing length is 3:2. Find the value of the other resistance.

Solution

$$Q = 15 \Omega, \quad l_1:l_2 = 3:2$$

$$\frac{l_1}{l_2} = \frac{3}{2}$$

$$\frac{P}{Q} = \frac{l_1}{l_2}$$

$$P = Q \frac{l_1}{l_2}$$

$$P = 15 \times \frac{3}{2} = 22.5 \Omega$$

EXAMPLE 2.26

In a meter bridge experiment, the value of resistance in the resistance box connected in the right gap is 10Ω . The balancing length is $l_1 = 55 \text{ cm}$. Find the value of unknown resistance.

Solution

$$Q = 10 \Omega$$

$$\frac{P}{Q} = \frac{l_1}{100-l_1} = \frac{l_1}{l_2}$$

$$P = Q \times \frac{l_1}{100-l_1}$$

$$P = \frac{10 \times 55}{100-55}$$

$$P = \frac{550}{45} = 12.2 \Omega$$

2.5.5 Potentiometer

Potentiometer is used for the accurate measurement of potential differences, current and resistances. It consists of ten meter long uniform wire of manganin or constantan stretched in parallel rows each of 1 meter length, on a wooden board. The two free ends A and B are brought to the same side and fixed to copper strips with binding screws. A meter scale is fixed parallel to the wire. A jockey is provided for making contact.

The principle of the potentiometer is illustrated in Figure 2.27. A steady current is maintained across the wire CD by a battery Bt .

The battery, key and the potentiometer wire connected in series form the primary circuit. The positive terminal of a primary cell of emf ε is connected to the point C and negative terminal is connected to the jockey through a galvanometer G and a high resistance HR. This forms the secondary circuit.

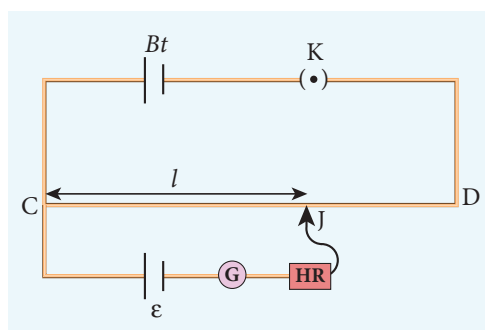


Figure 2.27 Potentiometer

Let the contact be made at any point J on the wire by jockey. If the potential difference across CJ is equal to the emf of the cell ε , then no current will flow through the galvanometer and it will show zero deflection. CJ is the balancing length l . The potential difference across CJ is equal to Irl where I is the current flowing through the wire and r is the resistance per unit length of the wire.

$$\text{Hence } \varepsilon = Irl \quad (2.58)$$

Since I and r are constants, $\varepsilon \propto l$. The emf of the cell is directly proportional to the balancing length.

2.5.6 Comparison of emf of two cells with a potentiometer

To compare the emf of two cells, the circuit connections are made as shown in Figure 2.28. Potentiometer wire CD is connected to a battery Bt and a key K in

series. This is the primary circuit. The end C of the wire is connected to the terminal M of a DPDT (Double Pole Double Throw) switch and the other terminal N is connected to a jockey through a galvanometer G and a high resistance HR. The cells whose emf ε_1 and ε_2 to be compared are connected to the terminals M_1, N_1 and M_2, N_2 of the DPDT switch. The positive terminals of Bt , ε_1 and ε_2 should be connected to the same end C.

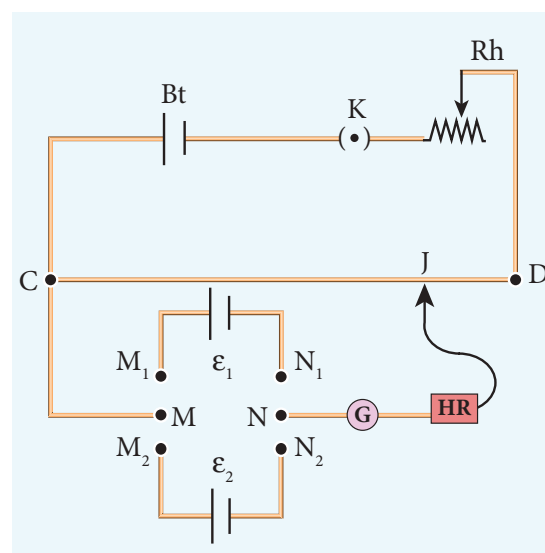


Figure 2.28 Comparison of emf of two cells

The DPDT switch is pressed towards M_1, N_1 so that cell ε_1 is included in the secondary circuit and the balancing length l_1 is found by adjusting the jockey for zero deflection. Then the second cell ε_2 is included in the circuit and the balancing length l_2 is determined. Let r be the resistance per unit length of the potentiometer wire and I be the current flowing through the wire.

$$\text{we have } \varepsilon_1 = Irl_1 \quad (2.59)$$

$$\varepsilon_2 = Irl_2 \quad (2.60)$$

By dividing equation (2.59) by (2.60)

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2} \quad (2.61)$$

By including a rheostat (Rh) in the primary circuit, the experiment can be repeated several times by changing the current flowing through it.

2.5.7 Measurement of internal resistance of a cell by potentiometer

To measure the internal resistance of a cell, the circuit connections are made as shown in Figure 2.29. The end C of the potentiometer wire is connected to the positive terminal of the battery Bt and the negative terminal of the battery is connected to the end D through a key K_1 . This forms the primary circuit.

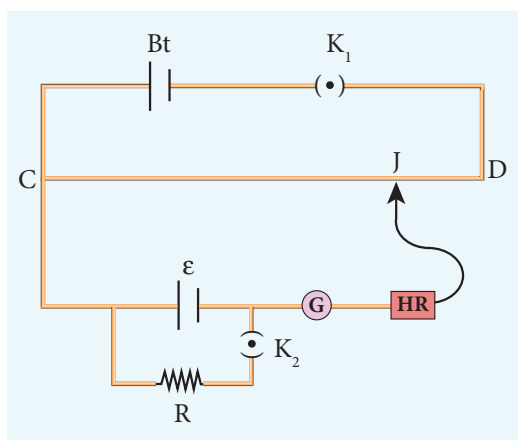


Figure 2.29 measurement of internal resistance

The positive terminal of the cell of emf ε whose internal resistance is to be determined is also connected to the end C of the wire. The negative terminal of the cell ε is connected to a jockey through a galvanometer and a high resistance. A resistance box R and key K_2 are connected across the cell ε . With K_2 open, the balancing point J is obtained and the balancing length CJ = l_1 is measured. Since the cell is in open circuit, its emf is

$$\varepsilon \propto l_1 \quad (2.62)$$

A suitable resistance (say, 10Ω) is included in the resistance box and key K_2 is closed. Let r be the internal resistance of the cell. The current passing through the cell and the resistance R is given by

$$I = \frac{\varepsilon}{R + r}$$

The potential difference across R is

$$V = \frac{\varepsilon R}{R + r}$$

When this potential difference is balanced on the potentiometer wire, let l_2 be the balancing length.

$$\text{Then } \frac{\varepsilon R}{R + r} \propto l_2 \quad (2.63)$$

From equations (2.62) and (2.63)

$$\frac{R + r}{R} = \frac{l_1}{l_2} \quad (2.64)$$

$$1 + \frac{r}{R} = \frac{l_1}{l_2};$$

$$r = R \left[\frac{l_1}{l_2} - 1 \right]$$

$$\therefore r = R \left(\frac{l_1 - l_2}{l_2} \right) \quad (2.65)$$

Substituting the values of the R , l_1 and l_2 , the internal resistance of the cell is determined. The experiment can be repeated for different values of R . It is found that the internal resistance of the cell is not constant but increases with increase of external resistance connected across its terminals.

2.6

HEATING EFFECT OF ELECTRIC CURRENT

When current flows through a resistor, some of the electrical energy delivered to the resistor is converted into heat energy and it is dissipated. This heating effect of

current is known as Joule's heating effect. Just as current produces thermal energy, thermal energy may also be suitably used to produce an electromotive force. This is known as thermoelectric effect.

2.6.1 Joule's law

If a current I flows through a conductor kept across a potential difference V for a time t , the work done or the electric potential energy spent is

$$W = VIt \quad (2.66)$$

In the absence of any other external effect, this energy is spent in heating the conductor. The amount of heat (H) produced is

$$H = VIt \quad (2.67)$$

For a resistance R ,

$$H = I^2Rt \quad (2.68)$$

This relation was experimentally verified by Joule and is known as Joule's law of heating. It states that **the heat developed in an electrical circuit due to the flow of current varies directly as**

- (i) **the square of the current**
- (ii) **the resistance of the circuit and**
- (iii) **the time of flow.**

EXAMPLE 2.27

Find the heat energy produced in a resistance of $10\ \Omega$ when 5 A current flows through it for 5 minutes.

Solution

$$R = 10\ \Omega, I = 5\text{ A}, t = 5\text{ minutes} = 5 \times 60\text{ s}$$

$$H = I^2Rt$$

$$= 5^2 \times 10 \times 5 \times 60$$

$$= 25 \times 10 \times 300$$

$$= 25 \times 3000$$

$$= 75000\text{ J (or) } 75\text{ kJ}$$

2.6.2 Application of Joule's heating effect

1. Electric heaters

Electric iron, electric heater, electric toaster shown in Figure 2.30 are some of the home appliances that utilize the heating effect of current. In these appliances, the heating elements are made of nichrome, an alloy of nickel and chromium. Nichrome has a high specific resistance and can be heated to very high temperatures without oxidation.

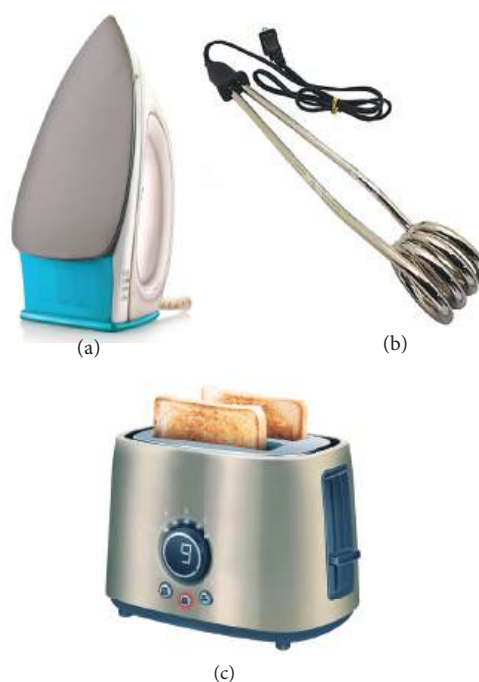


Figure 2.30 (a) Electric Iron box, (b) electric heater (c) electric Toaster

EXAMPLE 2.28

An electric heater of resistance $10\ \Omega$ connected to 220 V power supply is immersed in the water of 1 kg . How long the electrical heater has to be switched on to increase its temperature from 30°C to 60°C . (Specific heat capacity of water is $s = 4200\text{ J kg}^{-1}\text{ K}^{-1}$)



Solution

According to Joule's heating law $H = I^2Rt$

The current passed through the electrical

$$\text{heater} = \frac{220\text{V}}{10\Omega} = 22\text{ A}$$

Heat produced in one second by the electrical heater $H = I^2R$

Heat produced in one second

$$H = (22)^2 \times 10 = 4840\text{ J} = 4.84\text{ k J. In fact the power rating of this electrical heater is } 4.84\text{ k W.}$$

The amount of heat energy to increase the temperature of 1kg water from 30°C to 60°C is

$$Q = ms \Delta T \quad (\text{Refer XI physics vol 2, unit 8})$$

Here $m = 1\text{ kg}$,

$$s = 4200\text{ J kg}^{-1}\text{ K}^{-1},$$

$$\Delta T = 30\text{ K},$$

$$\text{so } Q = 1 \times 4200 \times 30 = 126\text{ kJ}$$

The time required to produce this heat

$$\text{energy } t = \frac{Q}{I^2R} = \frac{126 \times 10^3}{4840} \approx 26.03\text{ s}$$

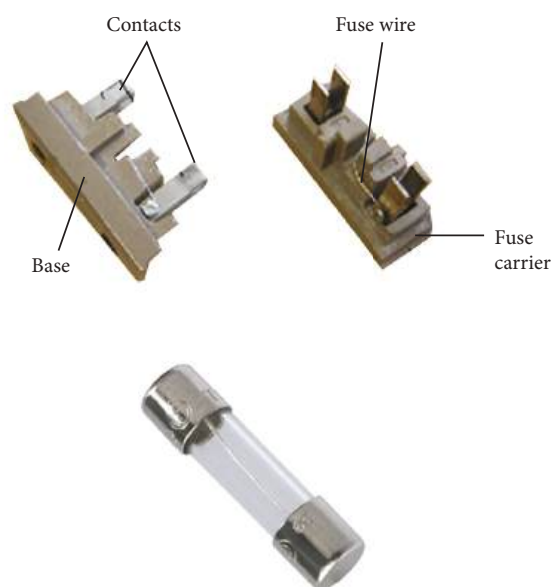


Figure 2.31 Electric Fuse

(trippers) are also used instead of fuses. Whenever there is an excessive current produced due to faulty wire connection, the circuit breaker switch opens. After repairing the faulty connection, we can close the circuit breaker switch. It is shown in the Figure 2.32.



Figure 2.32 circuit breakers

2. Electric fuses

Fuses as shown in Figure 2.31, are connected in series in a circuit to protect the electric devices from the heat developed by the passage of excessive current. It is a short length of a wire made of a low melting point material. It melts and breaks the circuit if current exceeds a certain value. An alloy of lead - tin is used for fuses when current rating is below 15 A and when current rating is above 15 A, copper fuse wires are used.

The only disadvantage with the above fuses is that once fuse wire is burnt due to excessive current, they need to be replaced. Nowadays in houses, circuit breakers

3. Electric furnace

Furnaces as shown in Figure 2.33 are used to manufacture a large number of technologically important materials such as steel, silicon carbide, quartz, gallium arsenide, etc. To produce temperatures up to 1500°C, molybdenum-nichrome wire wound on a silica tube is used. Carbon arc furnaces produce temperatures up to 3000°C.



Figure 2.33 Electric furnace

4. Electrical lamp

It consists of a tungsten filament (melting point 3380°C) kept inside a glass bulb and heated to incandescence by current. In incandescent electric lamps only about 5% of electrical energy is converted into light and the rest is wasted as heat. Electric discharge lamps, electric welding and electric arc also utilize the heating effect of current as shown in Figure 2.34.



Figure 2.34 Electric bulb, electric arc and electric welding

2.7

THERMOELECTRIC EFFECT

Conversion of temperature differences into electrical voltage and vice versa is known as thermoelectric effect. A thermoelectric device generates voltage when there is a temperature difference on each side. If a voltage is applied, it generates a temperature difference.

2.7.1 Seebeck effect

Seebeck discovered that in a closed circuit consisting of two dissimilar metals, when the junctions are maintained at different temperatures an emf (potential difference) is developed. The current that flows due to the emf developed is called thermoelectric current. The two dissimilar metals connected to form two junctions is known as thermocouple (Figure 2.35).

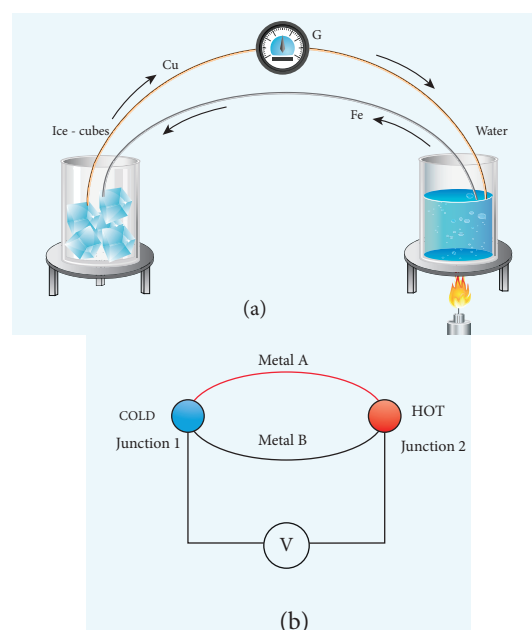


Figure 2.35 Seebeck effect (Thermocouple)

If the hot and cold junctions are interchanged, the direction of current also reverses. Hence the effect is reversible.

The magnitude of the emf developed in a thermocouple depends on (i) the nature of the metals forming the couple and (ii) the temperature difference between the junctions.

Applications of Seebeck effect

1. Seebeck effect is used in thermoelectric generators (Seebeck generators). These thermoelectric generators are used in power plants to convert waste heat into electricity.

2. This effect is utilized in automobiles as automotive thermoelectric generators for increasing fuel efficiency.
3. Seebeck effect is used in thermocouples and thermopiles to measure the temperature difference between the two objects.

2.7.2 Peltier effect

In 1834, Peltier discovered that when an electric current is passed through a circuit of a thermocouple, heat is evolved at one junction and absorbed at the other junction. This is known as Peltier effect.

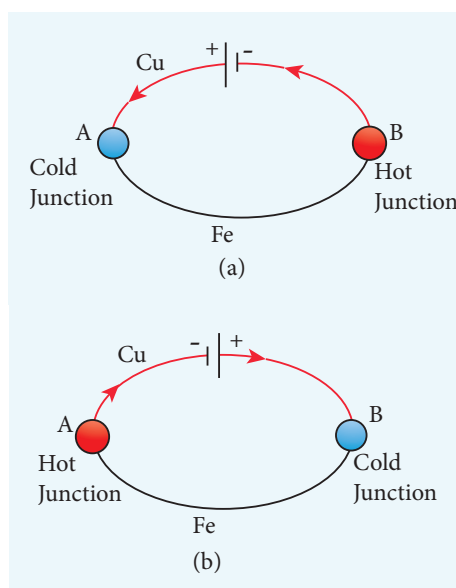


Figure 2.36 Peltier effect: Cu – Fe thermocouple

In the Cu-Fe thermocouple the junctions A and B are maintained at the same temperature. Let a current from a battery flow through the thermocouple (Figure 2.36 (a)). At the junction A, where the current flows from Cu to Fe, heat is absorbed and the junction A becomes cold. At the junction B, where the current flows from Fe to Cu heat is liberated and it becomes hot. When the direction of current is reversed, junction A

gets heated and junction B gets cooled as shown in the Figure 2.36(b). Hence Peltier effect is reversible.

2.7.3 Thomson effect

Thomson showed that if two points in a conductor are at different temperatures, the density of electrons at these points will differ and as a result the potential difference is created between these points. Thomson effect is also reversible.

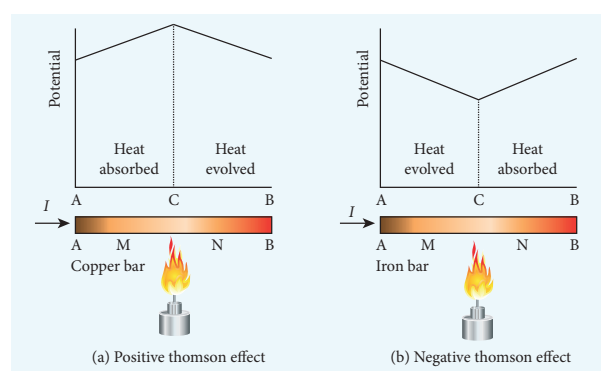


Figure 2.37 (a) Positive Thomson effect
(b) Negative Thomson effect

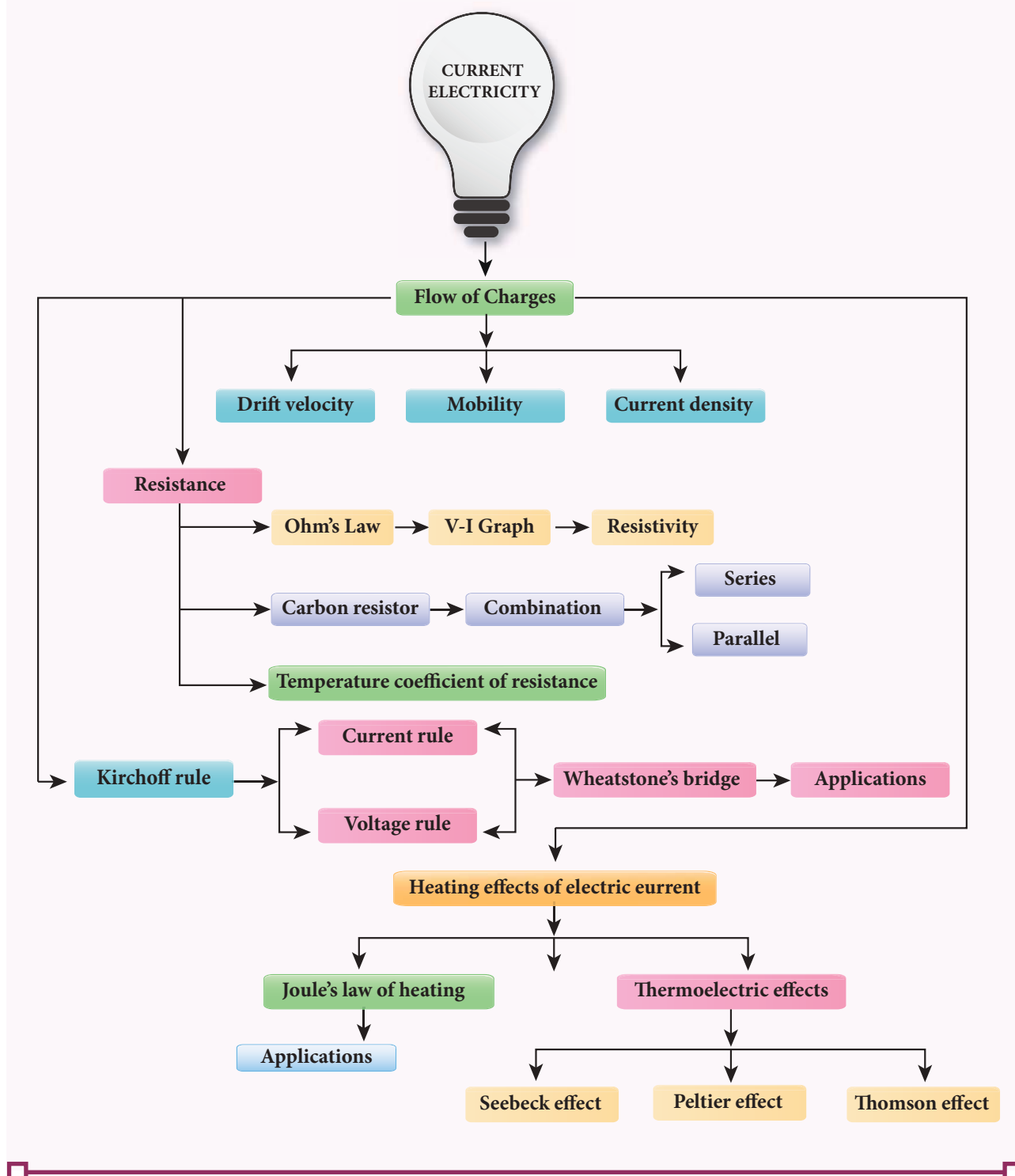
If current is passed through a copper bar AB which is heated at the middle point C, the point C will be at higher potential. This indicates that the heat is absorbed along AC and evolved along CB of the conductor as shown in Figure 2.37(a). Thus heat is transferred due to the current flow in the direction of the current. It is called positive Thomson effect. Similar effect is observed in metals like silver, zinc, and cadmium.

When the copper bar is replaced by an iron bar, heat is evolved along CA and absorbed along BC. Thus heat is transferred due to the current flow in the direction opposite to the direction of current. It is called negative Thomson effect as shown in the Figure 2.37(b). Similar effect is observed in metals like platinum, nickel, cobalt, and mercury.

SUMMARY

- The current, I flowing in a conductor $I = \frac{dQ}{dt}$, where dQ is the charge that flows through a cross-section in a time interval dt . SI unit of current is ampere (A).
 $1\text{ A} = 1\text{ C s}^{-1}$.
- The current density J in a conductor is the current flowing per unit area. $\left(J = \frac{I}{A} \right)$
- Current is a scalar but current density is a vector.
- The general form of Ohm's law $\vec{J} = \sigma \vec{E}$
- Practical form of Ohm's law states that $V \propto I$, or $V = IR$ where I is the current and R is the resistance, V potential difference between ends of the conductor.
- The resistance R of a conductor is $R = \frac{V}{I}$. SI unit of resistance is ohm (Ω) and
 $1\ \Omega = \frac{1\text{ V}}{1\text{ A}}$
- The resistance of a conductor $R = \rho \frac{l}{A}$ where l is length of the conductor and A is its area of cross section.
- The resistivity of a material determines how much resistance it offers to the flow of current.
- The equivalent resistance (R_s) of several resistances ($R_1, R_2, R_3, \dots$) connected in series is $R_s = (R_1 + R_2 + R_3 + \dots)$
- The equivalent resistance (R_p) of several resistances ($R_1, R_2, R_3, \dots$) connected in parallel is $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots$
- Kirchhoff's first rule (Current rule or junction rule): The algebraic sum of the currents at any junction is zero.
- Kirchhoff's second rule (Voltage rule or loop rule): In a closed circuit the algebraic sum of the products of the current and resistance of each part of the circuit is equal to the total emf included in the circuit.
- Electric power is the rate at which electric energy is delivered.
- If a current I flows across a potential difference V , the power delivered to the circuit is $P = IV$.
- In a resistor R , the electrical power converted to heat is $P = I^2 R = \frac{V^2}{R}$
- The energy equivalent of one kilowatt-hour (kWh) is $1\text{ kWh} = 3.6 \times 10^6\text{ J}$.
- Metre bridge is one form of Wheatstone's network.
- Potentiometer is used to compare emf of cells.
- Joule's law of heating is $H = VIt$ (or) $H = I^2 Rt$.

CONCEPT MAP



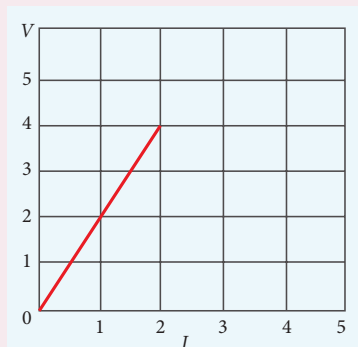


EVALUATION

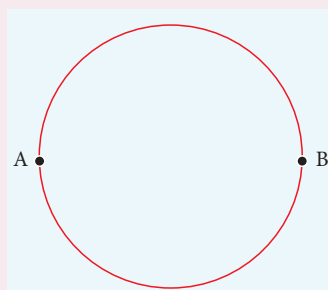


I Multiple Choice Questions

1. The following graph shows current versus voltage values of some unknown conductor. What is the resistance of this conductor?



- (a) 2 ohm (b) 4 ohm
(c) 8 ohm (d) 1 ohm
2. A wire of resistance 2 ohms per meter is bent to form a circle of radius 1m. The equivalent resistance between its two diametrically opposite points, A and B as shown in the figure is



- (a) $\pi \Omega$ (c) $\frac{\pi}{2} \Omega$
(b) $2\pi \Omega$ (d) $\frac{\pi}{4} \Omega$
3. A toaster operating at 240 V has a resistance of 120 Ω . Its power is
- a) 400 W b) 2 W
c) 480 W d) 240 W

4. A carbon resistor of $(47 \pm 4.7) \text{ k}\Omega$ to be marked with rings of different colours for its identification. The colour code sequence will be
- a) Yellow – Green – Violet – Gold
b) Yellow – Violet – Orange – Silver
c) Violet – Yellow – Orange – Silver
d) Green – Orange – Violet – Gold
5. What is the value of resistance of the following resistor?



- (a) 100 k Ω (b) 10 k Ω
(c) 1 k Ω (d) 1000 k Ω
6. Two wires of A and B with circular cross section are made up of the same material with equal lengths. Suppose $R_A = 3 R_B$, then what is the ratio of radius of wire A to that of B?
- (a) 3 (b) $\sqrt{3}$
(c) $\frac{1}{\sqrt{3}}$ (d) $\frac{1}{3}$
7. A wire connected to a power supply of 230 V has power dissipation P_1 . Suppose the wire is cut into two equal pieces and connected parallel to the same power supply. In this case power dissipation is P_2 . The ratio $\frac{P_2}{P_1}$ is
- (a) 1 (b) 2
(c) 3 (d) 4





8. In India electricity is supplied for domestic use at 220 V. It is supplied at 110 V in USA. If the resistance of a 60W bulb for use in India is R , the resistance of a 60W bulb for use in USA will be

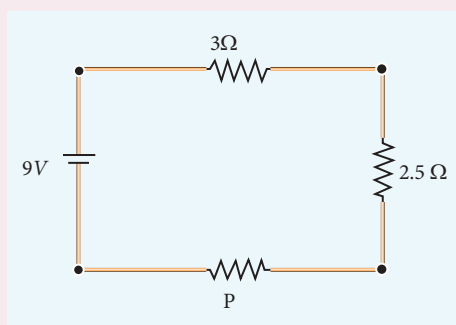
(a) R (b) $2R$
(c) $\frac{R}{4}$ (d) $\frac{R}{2}$

9. In a large building, there are 15 bulbs of 40 W, 5 bulbs of 100 W, 5 fans of 80 W and 1 heater of 1kW are connected. The voltage of electric mains is 220 V. The maximum capacity of the main fuse of the building will be

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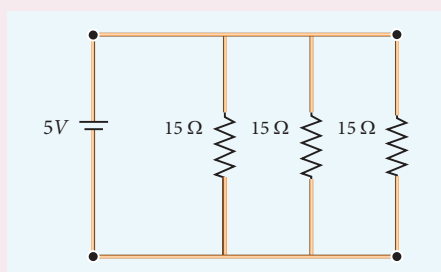
(a) 14 A (b) 8 A
(c) 10 A (d) 12 A

10. There is a current of 1.0 A in the circuit shown below. What is the resistance of P ?



a) 1.5 Ω (b) 2.5 Ω
c) 3.5 Ω (d) 4.5 Ω

11. What is the current drawn out from the battery?



a) 1A (b) 2A
c) 3A (d) 4A

12. The temperature coefficient of resistance of a wire is 0.00125 per $^{\circ}\text{C}$. At 20°C , its resistance is 1Ω . The resistance of the wire will be 2Ω at

a) 800°C (b) 700°C
c) 850°C (d) 820°C

13. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is

a) 0.2 Ω (b) 0.5 Ω
c) 0.8 Ω (d) 1.0 Ω

14. A piece of copper and another of germanium are cooled from room temperature to 80 K. The resistance of

a) each of them increases
b) each of them decreases
c) copper increases and germanium decreases
d) copper decreases and germanium increases

15. In Joule's heating law, when R and t are constant, if the H is taken along the y axis and I^2 along the x axis, the graph is

a) straight line (b) parabola
c) circle (d) ellipse

Answers

- 1) a 2) a 3) c 4) b 5) a
6) c 7) d 8) c 9) d 10) c
11) a 12) d 13) b 14) d 15) a

II Short Answer Questions

1. Why current is a scalar?
2. Define current density.
3. Distinguish between drift velocity and mobility.

4. State microscopic form of Ohm's law.
5. State macroscopic form of Ohm's law.
6. What are ohmic and non ohmic devices?
7. Define electrical resistivity.
8. Define temperature coefficient of resistance.
9. Write a short note on superconductors?
10. What is electric power and electric energy?
11. Derive the expression for power $P=VI$ in electrical circuit.
12. Write down the various forms of expression for power in electrical circuit.
13. State Kirchhoff's current rule.
14. State Kirchhoff's voltage rule.
15. State the principle of potentiometer.
16. What do you mean by internal resistance of a cell?
17. State Joule's law of heating.
18. What is Seebeck effect?
19. What is Thomson effect?
20. What is Peltier effect?
21. State the applications of Seebeck effect.

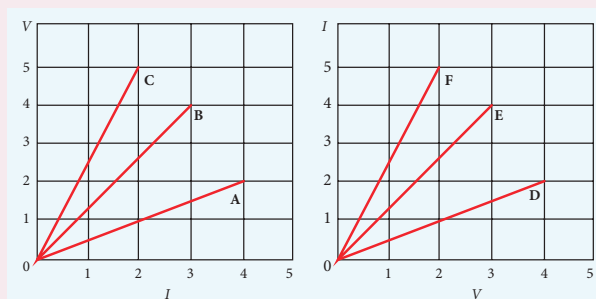
III Long Answer Questions

1. Describe the microscopic model of current and obtain microscopic form of Ohm's law.
2. Obtain the macroscopic form of Ohm's law from its microscopic form and discuss its limitation.
3. Explain the equivalent resistance of a series and parallel resistor network.
4. Explain the determination of the internal resistance of a cell using voltmeter.

5. State and explain Kirchhoff's rules.
6. Obtain the condition for bridge balance in Wheatstone's bridge.
7. Explain the determination of unknown resistance using meter bridge.
8. How the emf of two cells are compared using potentiometer?

IV Numerical problems

1. The following graphs represent the current versus voltage and voltage versus current for the six conductors A,B,C,D,E and F. Which conductor has least resistance and which has maximum resistance?



Ans: Least: $R_F = 0.4 \Omega$, maximum $R_C = 2.5 \Omega$

2. Lightning is very good example of natural current. In typical lightning, there is 10^9 J energy transfer across the potential difference of $5 \times 10^7 \text{ V}$ during a time interval of 0.2 s .



Using this information, estimate the following quantities (a) total amount of charge transferred between cloud and ground (b) the current in the lightning bolt (c) the power delivered in 0.2 s.

Ans: charge = 20 C, $I = 100$ A, $P = 5$ GW

3. A copper wire of 10^{-6} m^2 area of cross section, carries a current of 2 A. If the number of free electrons per cubic meter in the wire is 8×10^{28} , calculate the current density and average drift velocity of electrons.

Ans: $J = 2 \times 10^6 \text{ Am}^{-2}$
 $v_d = 15.6 \times 10^{-5} \text{ ms}^{-1}$

4. The resistance of a nichrome wire at 20°C is 10Ω . If its temperature coefficient of resistivity of nichrom is $0.004/^\circ\text{C}$, find the resistance of the wire at boiling point of water. Comment on the result.

Ans: $R_T = 13.2 \Omega$.

As the temperature increases the resistance of the wire also increases.

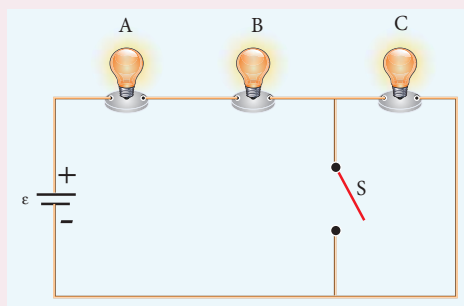
5. The rod given in the figure is made up of two different materials.



Both have square cross sections of 3 mm side. The resistivity of the first material is $4 \times 10^{-3} \Omega\text{m}$ and that of second material has resistivity of $5 \times 10^{-3} \Omega\text{m}$. What is the resistance of rod between its ends?

Ans: 500Ω

6. Three identical lamps each having a resistance R are connected to the battery of emf ϵ as shown in the figure.



Suddenly the switch S is closed. (a) Calculate the current in the circuit when S is open and closed (b) What happens to the intensities of the bulbs A, B and C. (c) Calculate the voltage across the three bulbs when S is open and closed (d) Calculate the power delivered to the circuit when S is opened and closed (e) Does the power delivered to the circuit decrease, increase or remain same?

Ans:

Electrical quantities	Switch S is open	Switch S is closed
a) Current	$\frac{\epsilon}{3R}$	$\frac{\epsilon}{2R}$
b) Voltage	$V_A = \frac{\epsilon}{3}$ $V_B = \frac{\epsilon}{3}$ $V_C = \frac{\epsilon}{3}$	$V_A = \frac{\epsilon}{2}$ $V_B = \frac{\epsilon}{2}$ $V_C = 0$
c) Power	$P_A = \frac{\epsilon^2}{9R}$ $P_B = \frac{\epsilon^2}{9R}$ $P_C = \frac{\epsilon^2}{9R}$	$P_A = \frac{\epsilon^2}{4R}$ $P_B = \frac{\epsilon^2}{4R}$ $P_C = 0$ Total power increases
d) Intensity	All the bulbs glow with equal intensity	The intensities of the bulbs A and B equally increase. Bulb C will not glow since no current pass through it.
e) Increases		

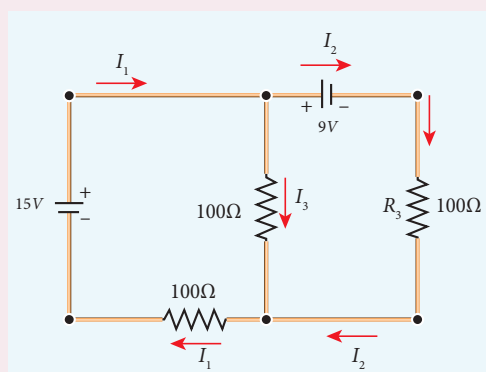
7. An electronics hobbyist is building a radio which requires $150\ \Omega$ in her circuit. But she has only $220\ \Omega$, $79\ \Omega$ and $92\ \Omega$ resistors available. How can she connect the available resistors to get the desired value of resistance?

Ans: Parallel combination of $220\ \Omega$ and $79\ \Omega$ in series with $92\ \Omega$

8. A cell supplies a current of $0.9\ \text{A}$ through a $2\ \Omega$ resistor and a current of $0.3\ \text{A}$ through a $7\ \Omega$ resistor. Calculate the internal resistance of the cell.

Ans: $0.5\ \Omega$

9. Calculate the currents in the following circuit.

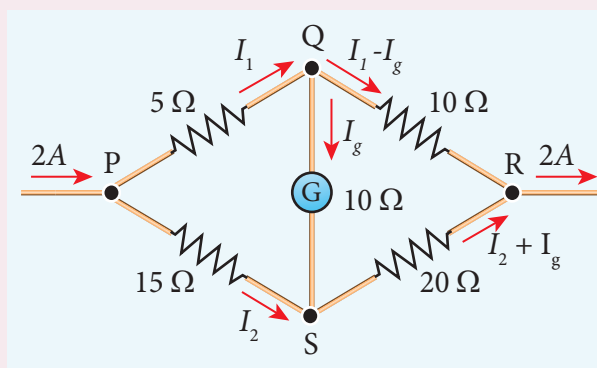


Ans : $I_1 = 0.070\ \text{A}$, $I_2 = -0.010\ \text{A}$ and $I_3 = 0.080\ \text{A}$

10. A potentiometer wire has a length of $4\ \text{m}$ and resistance of $20\ \Omega$. It is connected in series with resistance of $2980\ \Omega$ and a cell of emf $4\ \text{V}$. Calculate the potential gradient along the wire.

Ans: Potential = $0.65 \times 10^{-2}\ \text{V m}^{-1}$.

11. Determine the current flowing through the galvanometer (G) as shown in the figure.



Ans: $I_g = \frac{1}{11}\ \text{A}$

12. Two cells each of $5\ \text{V}$ are connected in series with a $8\ \Omega$ resistor and three parallel resistors of $4\ \Omega$, $6\ \Omega$ and $12\ \Omega$. Draw a circuit diagram for the above arrangement. Calculate i) the current drawn from the cells (ii) current through each resistor

Ans:

i) Current through $8\ \Omega = 1\ \text{A}$

ii) The current through $4\ \Omega$, $I = \frac{2}{4} = 0.5\ \text{A}$,

the current through $6\ \Omega$, $I = \frac{2}{6} = 0.33\ \text{A}$,

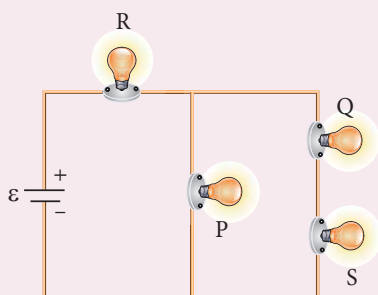
the current through $12\ \Omega$, $I = \frac{2}{12} = 0.17\ \text{A}$

13. Four bulbs P, Q, R, S are connected in a circuit of unknown arrangement. When each bulb is removed one at a time and replaced, the following behavior is observed.

	P	Q	R	S
P removed	*	on	on	on
Q removed	on	*	on	off
R removed	off	off	*	off
S removed	on	off	on	*

Draw the circuit diagram for these bulbs.

Ans:



14. In a potentiometer arrangement, a cell of emf 1.25 V gives a balance point at 35 cm length of the wire. If the cell is replaced by another cell and the balance point shifts to 63 cm, what is the emf of the second cell?

Ans: emf of the second cell is 2.25 V

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ICT CORNER

Electric current

In this activity you will be able to

- measure the potential difference of cells
- measure the internal resistance of a given primary cell

Topic: Potentiometer

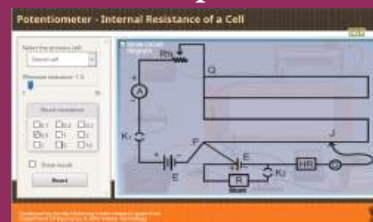
STEPS:

- Open the browser and type “olabs.edu.in” in the address bar. Click physics tab and then click “Potentiometer-Internal Resistance of a Cell” in class 12 section. Go to “simulator” tab to do the experiment.
- Construct the electric circuit as per the connection diagram by clicking “show circuit diagram” tab. You can connect wires between electric component by dragging the mouse between the component.
- To check whether the connections are correct or not, drag the jockey and place it at the two end points of the wire. If the galvanometer shows opposite deflections, the connections are correct. (keep both keys on)

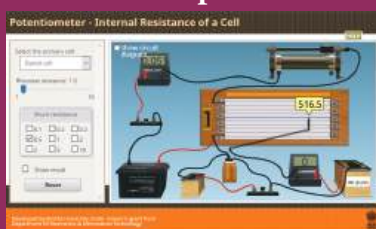
Step1



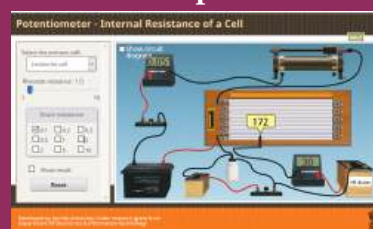
Step2



Step3



Step4



Find the balancing length. Calculate the internal resistance for the observed balancing lengths. Repeat the experiment for five times and take the average.

Note:

- One time sign up is needed to do simulation. Then login using that username and password.
- Read theory, procedure and animation to get the theory by clicking the corresponding tab.

URL:

<http://amrita.olabs.edu.in/?sub=1&brch=6&sim=147&cnt=4>

* Pictures are indicative only.

* If browser requires, allow **Flash Player** or **Java Script** to load the page.





UNIT 3

MAGNETISM AND MAGNETIC EFFECTS OF ELECTRIC CURRENT

“The magnetic force is animate, or imitates a soul; in many respects it surpasses the human soul while it is united to an organic body” – William Gilbert

LEARNING OBJECTIVES

In this unit, the student is exposed to

- Earth's magnetic field and magnetic elements
- Basic properties of magnets
- Statement of Coulomb inverse square law of magnetism
- Magnetic dipole
- Magnetic field due to a dipole along axial line and equatorial line
- Torque acting on a bar magnet in a uniform magnetic field
- Potential energy of a bar magnet placed in a uniform magnetic field
- Magnetic properties – permeability, susceptibility etc
- Classification of magnetic materials – dia, para and ferro magnetic materials
- Concept of Hysteresis
- Magnetic effects of electric current – long straight conductor and circular coil
- Right hand thumb rule and Maxwell's right hand cork screw rule
- Biot-Savart's law – applications
- Tangent law and Tangent Galvanometer
- Current loop as a magnetic dipole
- Magnetic dipole moment of revolving electron
- Ampère's circuital law – applications
- Solenoid and toroid
- Lorentz force – charged particle moving in an electromagnetic field
- Cyclotron
- Force on a current carrying conductor in a magnetic field
- Force between two long parallel current carrying conductor
- Torque on a current loop in a magnetic field
- Moving Coil Galvanometer



3.1

INTRODUCTION TO MAGNETISM

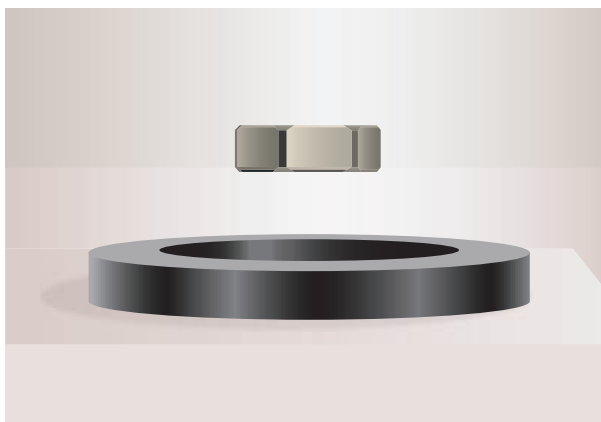


Figure 3.1: Magnetic levitation

Magnets! No doubt, their behaviour will attract everyone. The world enjoys their benefits, to lead a modern luxurious life. The study of magnets fascinated scientists around our globe for many centuries and even now, door for research on magnets is still open (Figure 3.1).



Many birds and animals have magnetic sense in their eyes using Earth's magnetic field for navigation.



Magnetic sensing in eyes - for Zebra finch bird, due to protein cryptochromes Cry4 present in retina; the bird uses Earth's magnetic field for navigation

Magnetism exists everywhere from tiny particles like electrons to the entire universe. Historically the word 'magnetism' was derived from iron ore magnetite (Fe_3O_4). In olden days, magnets were used as magnetic compass for navigation, magnetic therapy for treatment and also used in magic shows.

In modern days, many things we use in our daily life contain magnets (Figure 3.2). Motors, cycle dynamo, loudspeakers, magnetic tapes used in audio and video recording, mobile phones, head phones, CD, pen-drive, hard disc of laptop, refrigerator door, generator are a few examples.

Earlier, both electricity and magnetism were thought to be two independent branches in physics. In 1820, H.C. Oersted observed the deflection of magnetic compass needle kept near a current carrying wire. This unified the two different branches, electricity and magnetism as a single subject 'electromagnetism' in physics.

In this unit, basics of magnets and their properties are given. Later, how a current carrying conductor (here only steady current, not time-varying current is considered) behaves like a magnet is presented.

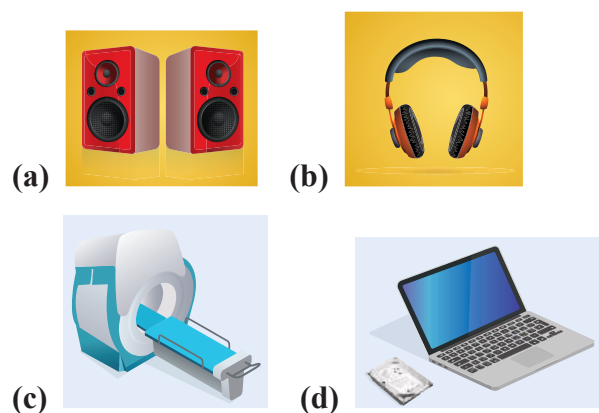


Figure 3.2 Uses of magnets in modern world – (a) speakers (b) head phones (c) MRI scan (d) Hard disc of laptop

3.1.1 Earth's magnetic field and magnetic elements

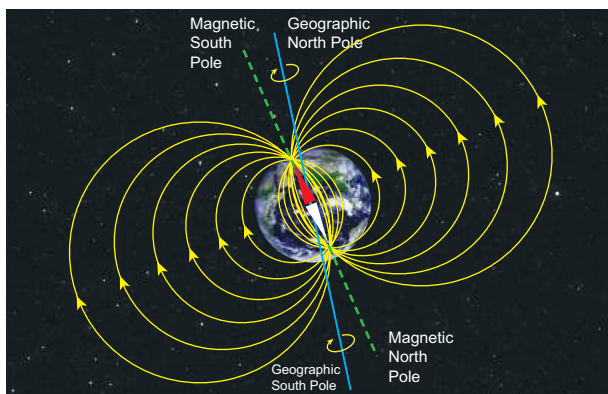


Figure 3.3 Earth's magnetic field

From the activities performed in lower classes, you might have noticed that the needle in a magnetic compass or freely suspended magnet comes to rest in a position which is approximately along the geographical north-south direction of the Earth.



William Gilbert in 1600 proposed that Earth itself behaves like a gigantic powerful bar magnet. But this theory is not successful because the temperature inside the Earth is very high and so it will not be possible for a magnet to retain its magnetism.

Gover suggested that the Earth's magnetic field is due to hot rays coming out from the Sun. These rays will heat up the air near equatorial region. Once air becomes hotter, it rises above and will move towards northern and southern hemispheres and get electrified. This may be responsible to magnetize the ferromagnetic materials near the Earth's surface. Till date, so many theories have been proposed. But none of the theorium completely explains the cause for the Earth's magnetism.

The north pole of magnetic compass needle is attracted towards the magnetic south pole of the Earth which is near the geographic north pole (Figure 3.3). Similarly, the south pole of magnetic compass needle is attracted towards the magnetic north-pole of the Earth which is near the geographic south pole. **The branch of physics which deals with the Earth's magnetic field is called Geomagnetism or Terrestrial magnetism.**

There are three quantities required to specify the magnetic field of the Earth on its surface, which are often called as the elements of the Earth's magnetic field. They are

- magnetic declination (D)
- magnetic dip or inclination (I)
- the horizontal component of the Earth's magnetic field (B_H)

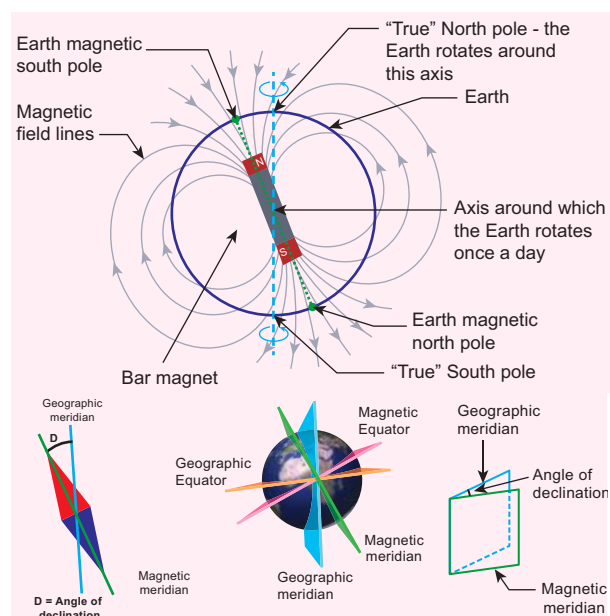


Figure 3.4 Declination angle

Day and night occur because Earth spins about an axis called geographic axis. A vertical plane passing through the geographic axis is called geographic meridian and a great circle perpendicular to Earth's geographic axis is called geographic equator.

The straight line which connects magnetic poles of Earth is known as magnetic axis.

A vertical plane passing through magnetic axis is called magnetic meridian and a great circle perpendicular to Earth's magnetic axis is called magnetic equator.

When a magnetic needle is freely suspended, the alignment of the magnet does not exactly lie along the geographic meridian as shown in Figure 3.4. **The angle between magnetic meridian at a point and geographical meridian is called the declination or magnetic declination (D).** At higher latitudes, the declination is greater whereas near the equator, the declination is smaller. In India, declination angle is very small and for Chennai, magnetic declination angle is $-1^{\circ}16'$ (which is negative (west)).

The angle subtended by the Earth's total magnetic field $\vec{B}$ with the horizontal direction in the magnetic meridian is called dip or magnetic inclination (I) at that point (Figure 3.5). For Chennai, inclination angle is $14^{\circ}28'$. **The component of Earth's magnetic field along the horizontal direction in the magnetic meridian is called horizontal component of Earth's magnetic field, denoted by B_H .**

Let B_E be the net Earth's magnetic field at any point on the surface of the Earth. B_E can be resolved into two perpendicular components.

$$\text{Horizontal component } B_H = B_E \cos I \quad (3.1)$$

$$\text{Vertical component } B_V = B_E \sin I \quad (3.2)$$

Dividing equation (3.2) and (3.1), we get

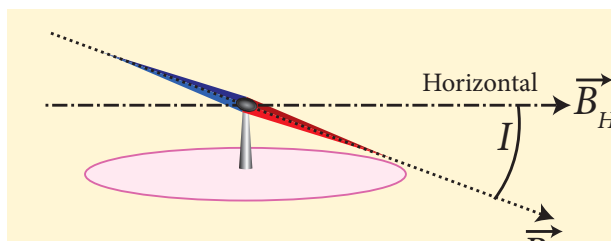


Figure 3.5 Inclination angle

$$\tan I = \frac{B_V}{B_H} \quad (3.3)$$

(i) At magnetic equator

The Earth's magnetic field is parallel to the surface of the Earth (i.e., horizontal) which implies that the needle of magnetic compass rests horizontally at an angle of dip, $I = 0^{\circ}$.

$$\begin{aligned} B_H &= B_E \\ B_V &= 0 \end{aligned}$$

This implies that the horizontal component is maximum and vertical component is zero at the equator.

(ii) At magnetic poles

The Earth's magnetic field is perpendicular to the surface of the Earth (i.e., vertical) which implies that the needle of magnetic compass rests vertically at an angle of dip, $I = 90^{\circ}$. Hence,

$$\begin{aligned} B_H &= 0 \\ B_V &= B_E \end{aligned}$$

This implies that the vertical component is maximum at poles and horizontal component is zero at poles.

EXAMPLE 3.1

The horizontal component and vertical component of Earth's magnetic field at a place are 0.15 G and 0.26 G respectively. Calculate the angle of dip and resultant magnetic field. (G-gauss, cgs unit for magnetic field $1\text{G} = 10^{-4}\text{ T}$)

Solution:

$$B_H = 0.15\text{ G and } B_V = 0.26\text{ G}$$

$$\tan I = \frac{0.26}{0.15} \Rightarrow I = \tan^{-1}(1.732) = 60^{\circ}$$

The resultant magnetic field of the Earth is

$$B = \sqrt{B_H^2 + B_V^2} = 0.3\text{ G}$$



Aurora Borealis and Aurora Australis

People living at high latitude regions (near Arctic or Antarctic) might experience dazzling coloured natural lights across the night sky. This ethereal display on the sky is known as aurora borealis (northern lights) or aurora australis (southern lights). These lights are often called as polar lights. The lights are seen above the magnetic poles of the northern and southern hemispheres. They are called as “Aurora borealis” in the north and “Aurora australis” in the south. This occurs as a result of interaction between the gaseous particles in the Earth’s atmosphere with highly charged particles released from the Sun’s atmosphere through solar wind. These particles emit light due to collision and variations in colour are due to the type of the gas particles that take part in the collisions. A pale yellowish – green colour is produced when the ionized oxygen takes part in the collision and a blue or purplish – red aurora is produced due to ionized nitrogen molecules.



3.1.2 Basic properties of magnets

Some basic terminologies and properties used in describing bar magnet.

(a) Magnetic dipole moment

Consider a bar magnet as shown in Figure 3.6. Let q_m be the pole strength of the magnetic pole and let l be the distance between the geometrical centre of bar magnet O and one end of the pole. **The magnetic dipole moment is defined as the product of its pole strength and magnetic length.** It is a vector quantity, denoted by $\vec{p}_m$.

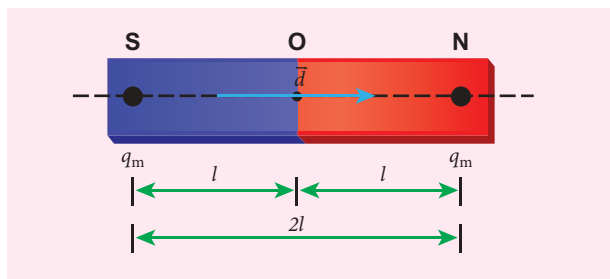


Figure 3.6 A bar magnet

$$\vec{p}_m = q_m \vec{d} \quad (3.4)$$

where $\vec{d}$ is the vector drawn from south pole to north pole and its magnitude $|\vec{d}| = 2l$.

The magnitude of magnetic dipole moment is $p_m = 2q_m l$

The SI unit of magnetic moment is $A \, m^2$. The direction of magnetic moment is from south pole to north pole.

(b) Magnetic field

Magnetic field is the region or space around every magnet within which its influence will be felt by keeping another magnet in that region. **The magnetic field $\vec{B}$ at a point is defined as a force experienced by the bar magnet of unit pole strength.**

$$\vec{B} = \frac{1}{q_m} \vec{F} \quad (3.5)$$

Its unit is $N \, A^{-1} \, m^{-1}$.

(c) Types of magnets

Magnets are classified into natural magnets and artificial magnets. For example, iron, cobalt, nickel, etc. are natural magnets. Strengths of natural magnets are very weak and the shapes of the magnet are irregular. Artificial magnets are made in order to have desired shape and strength. If the magnet is in the form of rectangular shape or cylindrical shape, then it is known as bar magnet.

Properties of magnet

The following are the properties of bar magnet (Figure 3.7),

1. A freely suspended bar magnet will always point along the north-south direction.
2. A magnet attracts or repels another magnet or magnetic substances towards itself. The attractive or repulsive force is maximum near the end of the bar magnet. When a bar magnet is dipped into iron filling, they cling to the ends of the magnet.
3. When a magnet is broken into pieces, each piece behaves like a magnet with poles at its ends.
4. Two poles of a magnet have pole strength equal to one another.
5. The length of the bar magnet is called geometrical length and the length between two magnetic poles in a bar magnet is called magnetic length. Magnetic length is always slightly smaller than geometrical length. The ratio of magnetic length and geometrical length is $\frac{5}{6}$.

$$\frac{\text{Magnetic length}}{\text{Geometrical length}} = \frac{5}{6} = 0.833$$

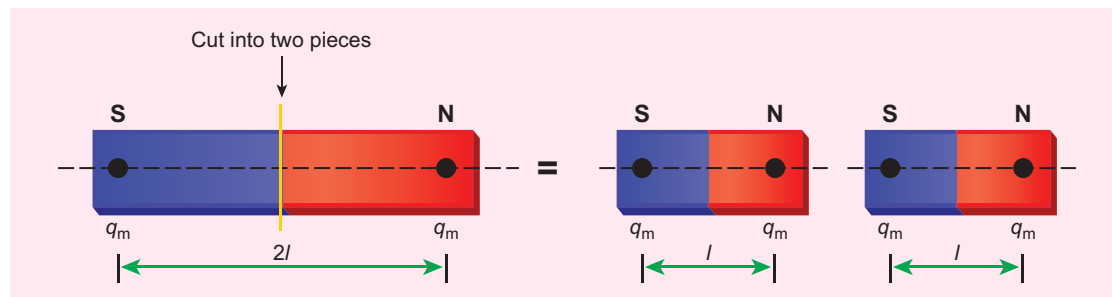
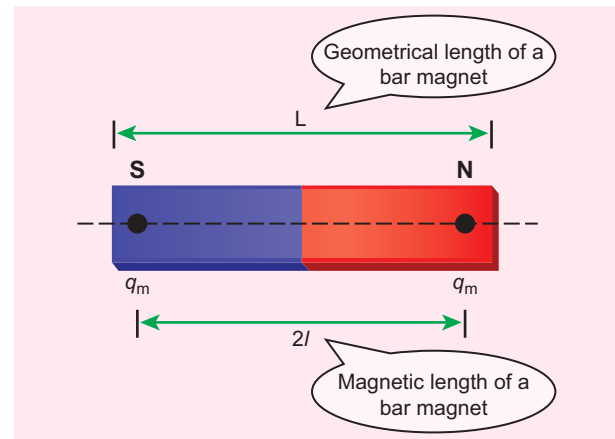
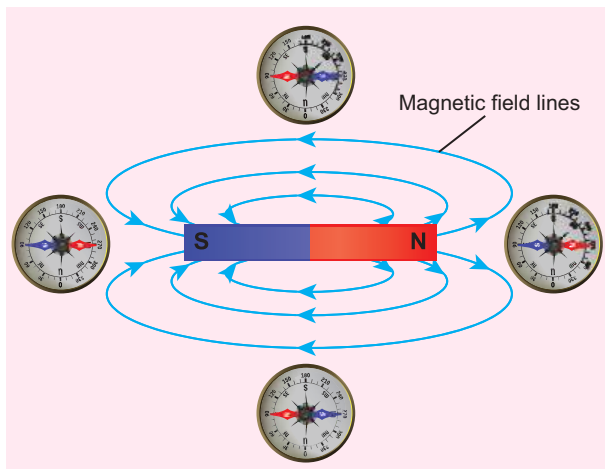
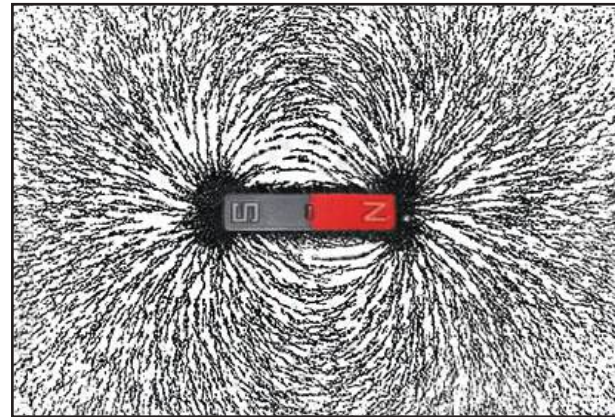
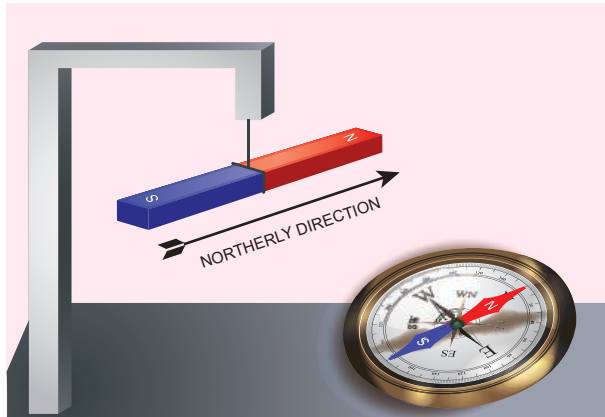


Figure 3.7 Properties of bar magnet

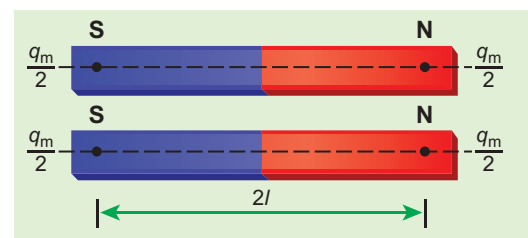
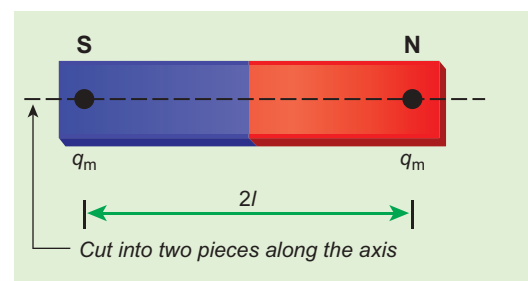
EXAMPLE 3.2

Let the magnetic moment of a bar magnet be $\vec{p}_m$ whose magnetic length is $d = 2l$ and pole strength is q_m . Compute the magnetic moment of the bar magnet when it is cut into two pieces

- along its length
- perpendicular to its length.

Solution

- a bar magnet cut into two pieces along its length:





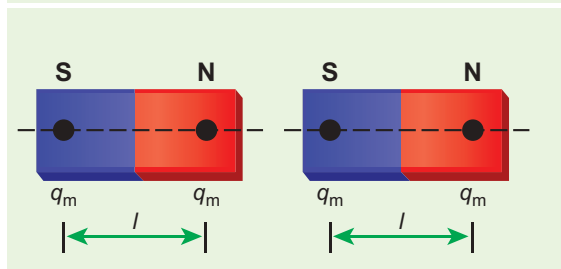
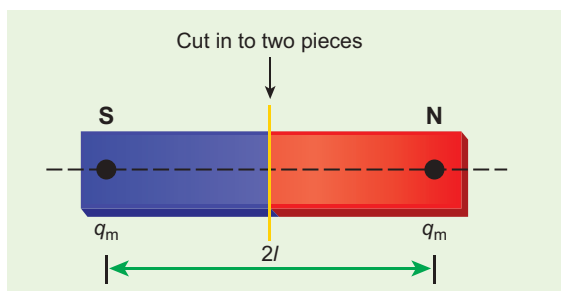
When the bar magnet is cut along the axis into two pieces, new magnetic pole strength is $q'_m = \frac{q_m}{2}$ but magnetic length does not change. So, the magnetic moment is

$$p'_m = q'_m 2l$$

$$p'_m = \frac{q_m}{2} 2l = \frac{1}{2} (q_m 2l) = \frac{1}{2} p_m$$

$$\text{In vector notation, } \vec{p}'_m = \frac{1}{2} \vec{p}_m$$

(b) a bar magnet cut into two pieces perpendicular to the axis:



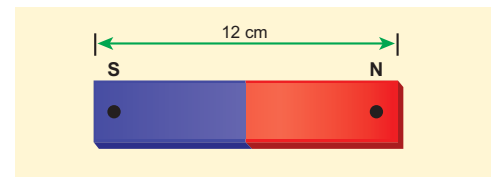
When the bar magnet is cut perpendicular to the axis into two pieces, magnetic pole strength will not change but magnetic length will be halved. So the magnetic moment is

$$p'_m = q_m \times \frac{1}{2} (2l) = \frac{1}{2} (q_m \cdot 2l) = \frac{1}{2} p_m$$

$$\text{In vector notation, } \vec{p}'_m = \frac{1}{2} \vec{p}_m$$

EXAMPLE 3.3

Compute the magnetic length of a uniform bar magnet if the geometrical length of the magnet is 12 cm. Mark the positions of magnetic pole points.

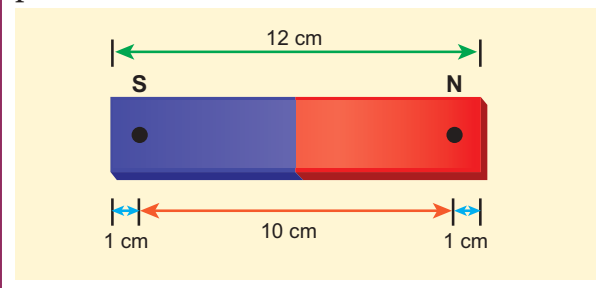


Solution

Geometrical length of the bar magnet is 12 cm

$$\begin{aligned} \text{Magnetic length} &= \frac{5}{6} \times (\text{geometrical length}) \\ &= \frac{5}{6} \times 12 = 10 \text{ cm} \end{aligned}$$

In this figure, the dot implies the pole points.



Note

- (i) Pole strength is a scalar quantity with dimension $[M^0 L^1 T^0 A]$. Its SI unit is NT^{-1} (newton per tesla) or A m (ampere-metre).
- (ii) Like positive and negative charges in electrostatics, north pole of a magnet experiences a force in the direction of magnetic field while south pole of a magnet experiences force opposite to the magnetic field.
- (iii) Pole strength depends on the nature of materials of the magnet, area of cross-section and the state of magnetization.
- (iv) If a magnet is cut into two equal halves along the length then pole strength is reduced to half.
- (v) If a magnet is cut into two equal halves perpendicular to the length, then pole strength remains same.
- (vi) If a magnet is cut into two pieces, we will not get separate north and south poles. Instead, we get two magnets. In other words, isolated monopole does not exist in nature.

Magnetic field lines

1. Magnetic field lines are continuous closed curves. The direction of magnetic field lines is from North pole to South pole outside the magnet and from South pole to North pole inside the magnet.
2. The direction of magnetic field at any point on the curve is known by drawing tangent to the magnetic field lines at that point.
3. Magnetic field lines never intersect each other. Otherwise, the magnetic compass needle would point towards two different directions, which is not possible.
4. The degree of closeness of the field lines determines the relative strength of the magnetic field. The magnetic field is strong where magnetic field lines crowd and weak where magnetic field lines are well separated.

(d) Magnetic flux

The number of magnetic field lines crossing any area normally is defined as **magnetic flux** Φ_B through the area. Mathematically, the magnetic flux through a surface of area $\vec{A}$ in a uniform magnetic field $\vec{B}$ is defined as

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta = B_{\perp} A \quad (3.6)$$

where θ is the angle between $\vec{B}$ and $\vec{A}$ as shown in Figure 3.8.

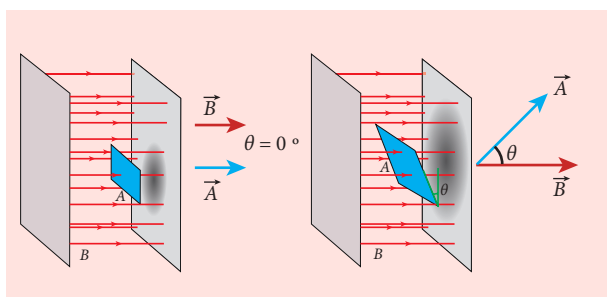


Figure 3.8 Magnetic flux

Special cases

- (a) When $\vec{B}$ is normal to the surface i.e., $\theta = 0^\circ$, the magnetic flux is $\Phi_B = BA$ (maximum).
- (b) When $\vec{B}$ is parallel to the surface i.e., $\theta = 90^\circ$, the magnetic flux is $\Phi_B = 0$.

Suppose the magnetic field is not uniform over the surface, the equation (3.6) can be written as

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

Magnetic flux is a scalar quantity. The SI unit for magnetic flux is weber, which is denoted by symbol Wb. Dimensional formula for magnetic flux is $[ML^2T^{-2}A^{-1}]$. The CGS unit of magnetic flux is maxwell.

$$1 \text{ weber} = 10^8 \text{ maxwell}$$

The magnetic flux density is defined as the number of magnetic field lines crossing per unit area kept normal to the direction of lines of force. Its unit is Wb m^{-2} or tesla (T).

(e) Uniform magnetic field and Non-uniform magnetic field

Uniform magnetic field

Magnetic field is said to be uniform if it has same magnitude and direction at all the

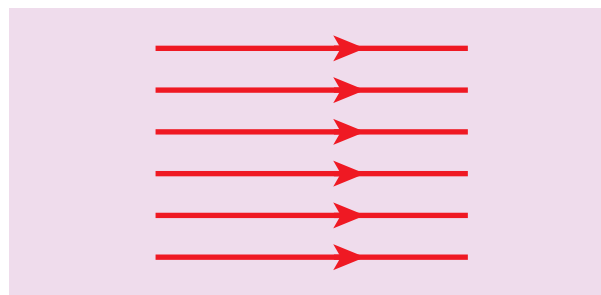


Figure 3.9 Uniform magnetic field

points in a given region. Example, locally Earth's magnetic field is uniform.

The magnetic field of Earth has same value over the entire area of your school!

Non-uniform magnetic field

Magnetic field is said to be non-uniform if the magnitude or direction or both vary at different points in a region. Example: magnetic field of a bar magnet

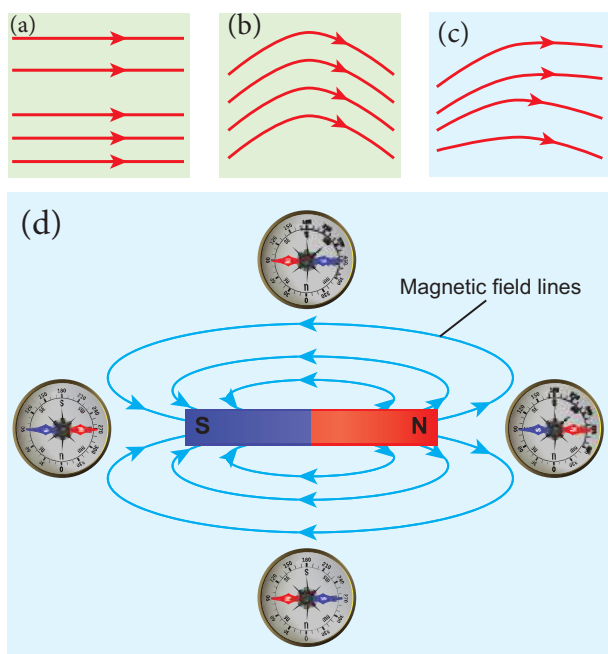
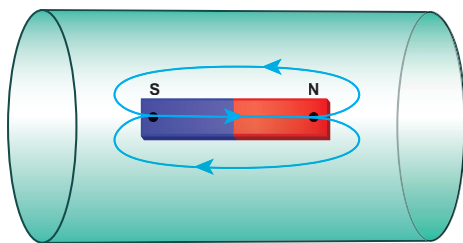


Figure 3.10 Non-uniform magnetic field (a) direction is constant (b) direction is not a constant (c) both magnitude and direction are not constant (d) magnetic field of a bar magnet

EXAMPLE 3.4

Calculate the magnetic flux coming out from closed surface containing magnetic dipole (say, a bar magnet) as shown in figure.



Solution

The total flux emanating from the closed surface S enclosing the dipole is zero. So,

$$\Phi_B = \oint \vec{B} \cdot d\vec{A} = 0$$

Here the integral is taken over closed surface. Since no isolated magnetic pole (called magnetic monopole) exists, this integral is always zero,

$$\oint \vec{B} \cdot d\vec{A} = 0$$

This is similar to Gauss's law in electrostatics.

3.2

COULOMB'S INVERSE SQUARE LAW OF MAGNETISM

Consider two bar magnets A and B as shown in Figure 3.11. When the north pole of magnet A and the north pole of magnet B or the south pole of magnet A and the south pole of magnet B are brought closer, they repel each other.

On the other hand, when the north pole of magnet A and the south pole of magnet B or the south pole of magnet A and the north pole of magnet B are brought closer, their poles attract each other.

This looks similar to Coulomb's law for static charges studied in Unit I (opposite charges attract and like charges repel each other). So analogous to Coulomb's law in electrostatics, we can state Coulomb's law for magnetism (Figure 3.12) as follows:

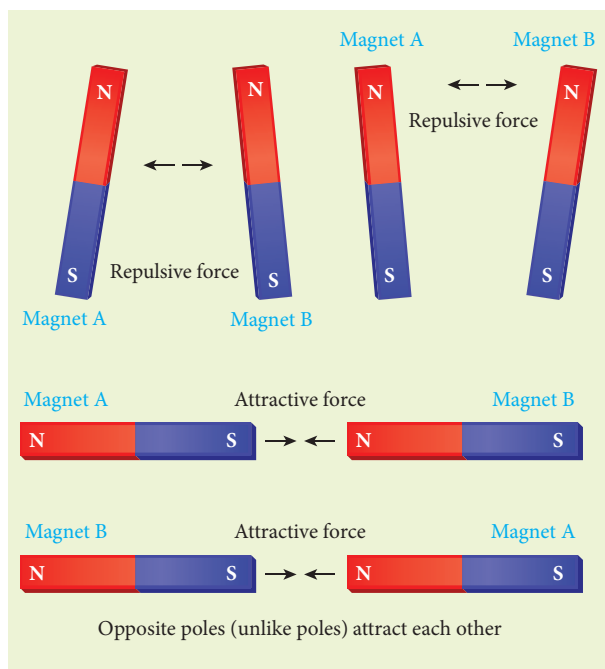


Figure 3.11: Magnetic poles behave like electric charges – like poles repel and unlike poles attract

The force of attraction or repulsion between two magnetic poles is directly proportional to the product of their pole strengths and inversely proportional to the square of the distance between them.

Mathematically, we can write

$$\vec{F} \propto \frac{q_{m_A} q_{m_B}}{r^2} \hat{r}$$

where q_{m_A} and q_{m_B} are pole strengths of two poles and r is the distance between two magnetic poles.

$$\vec{F} = k \frac{q_{m_A} q_{m_B}}{r^2} \hat{r} \quad (3.7)$$

$$\text{In magnitude, } F = k \frac{q_{m_A} q_{m_B}}{r^2} \quad (3.8)$$

where k is a proportionality constant whose value depends on the surrounding medium. In SI unit, the value of k for free space is $k = \frac{\mu_0}{4\pi} \approx 10^{-7} \text{ H m}^{-1}$, where μ_0 is the

absolute permeability of free space (air or vacuum) and H stands for henry.

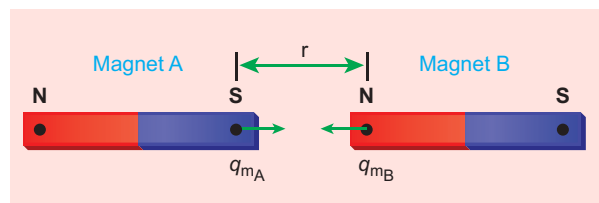


Figure 3.12 Coulomb's law – force between two magnetic poles

EXAMPLE 3.5

The repulsive force between two magnetic poles in air is $9 \times 10^{-3} \text{ N}$. If the two poles are equal in strength and are separated by a distance of 10 cm, calculate the pole strength of each pole.

Solution:

The magnitude of the force between two poles is given by

$$F = k \frac{q_{m_A} q_{m_B}}{r^2}$$

Given: $F = 9 \times 10^{-3} \text{ N}$, $r = 10 \text{ cm} = 10 \times 10^{-2} \text{ m}$

Since $q_{m_A} = q_{m_B} = q_m$, we have

$$9 \times 10^{-3} = 10^{-7} \times \frac{q_m^2}{(10 \times 10^{-2})^2} \Rightarrow q_m = 30 \text{ NT}^{-1}$$

3.2.1 Magnetic field at a point along the axial line of the magnetic dipole (bar magnet)

Consider a bar magnet NS as shown in Figure 3.13. Let N be the north pole and S be the south pole of the bar magnet, each of pole strength q_m and are separated by a distance of $2l$. The magnetic field at a point C (lies along the axis of the magnet)

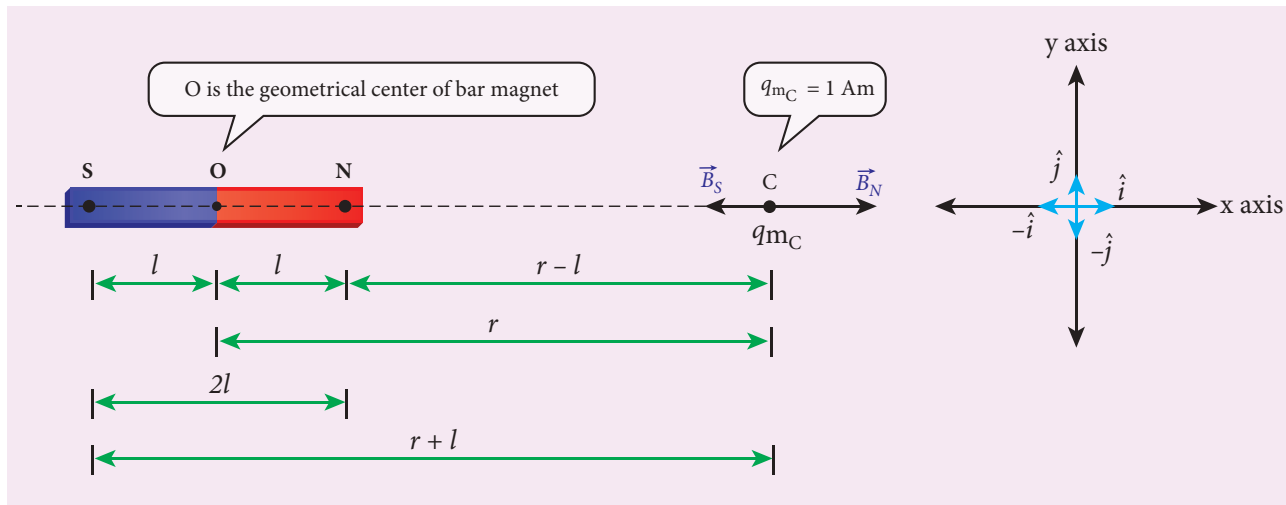


Figure 3.13 Magnetic field at a point along the axial line due to magnetic dipole

at a distance r from the geometrical centre O of the bar magnet can be computed by keeping unit north pole ($q_{m_C} = 1 \text{ Am}$) at C .

The magnetic field at C due to the north pole is

$$\vec{B}_N = \frac{\mu_0}{4\pi} \frac{q_m}{(r-l)^2} \hat{i} \quad (3.9)$$

where $(r-l)$ is the distance between north pole of the bar magnet and unit north pole at C .

The magnetic field at C due to the south pole is

$$\vec{B}_S = -\frac{\mu_0}{4\pi} \frac{q_m}{(r+l)^2} \hat{i} \quad (3.10)$$

where $(r+l)$ is the distance between south pole of the bar magnet and unit north pole at C .

The net magnetic field due to the magnetic dipole at point C

$$\vec{B} = \vec{B}_N + \vec{B}_S$$

$$\begin{aligned} \vec{B} &= \frac{\mu_0}{4\pi} \frac{q_m}{(r-l)^2} \hat{i} + \left(-\frac{\mu_0}{4\pi} \frac{q_m}{(r+l)^2} \hat{i} \right) \\ \vec{B} &= \frac{\mu_0 q_m}{4\pi} \left(\frac{1}{(r-l)^2} - \frac{1}{(r+l)^2} \right) \hat{i} \\ \vec{B} &= \frac{\mu_0 2r}{4\pi} \left(\frac{q_m \cdot (2l)}{(r^2 - l^2)^2} \right) \hat{i} \end{aligned} \quad (3.11)$$

Since the magnitude of magnetic dipole moment is $|\vec{p}_m| = p_m = q_m \cdot 2l$, the magnetic field at a point C can be written as

$$\vec{B}_{axial} = \frac{\mu_0}{4\pi} \left(\frac{2rp_m}{(r^2 - l^2)^2} \right) \hat{i} \quad (3.12)$$

If the distance between two poles in a bar magnet is small (looks like short magnet) when compared to the distance between geometrical centre O of bar magnet and the location of point C ($r \gg l$),

$$(r^2 - l^2)^2 \approx r^4 \quad (3.13)$$

Therefore, using equation (3.13) in equation (3.12), we get

$$\vec{B}_{axial} = \frac{\mu_0}{4\pi} \left(\frac{2p_m}{r^3} \right) \hat{i} = \frac{\mu_0}{4\pi r^3} \vec{p}_m \quad (3.14)$$

where $\vec{p}_m = p_m \hat{i}$.

3.2.2. Magnetic field at a point along the equatorial line due to a magnetic dipole (bar magnet)

Consider a bar magnet NS as shown in Figure 3.14. Let N be the north pole and S be the south pole of the bar magnet, each with pole strength q_m and separated by a distance of $2l$. The magnetic field at a point C (lies along the equatorial line) at a distance r from the geometrical centre O of the bar magnet can be computed by keeping unit north pole ($q_{mC} = 1 \text{ A m}$) at C.

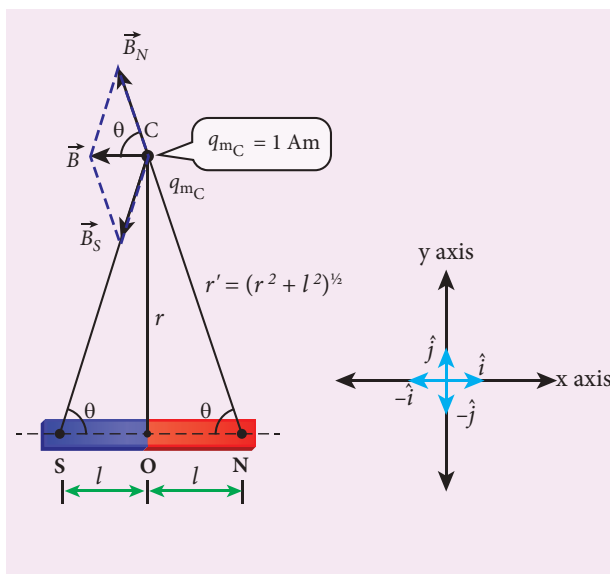


Figure 3.14 Magnetic field at a point along the equatorial line due to a magnetic dipole

The magnetic field at C due to the north pole is

$$\vec{B}_N = -B_N \cos \theta \hat{i} + B_N \sin \theta \hat{j} \quad (3.15)$$

where $B_N = \frac{\mu_0 q_m}{4\pi r'^2}$

Here $r' = (r^2 + l^2)^{\frac{1}{2}}$

The magnetic field at C due to the south pole is

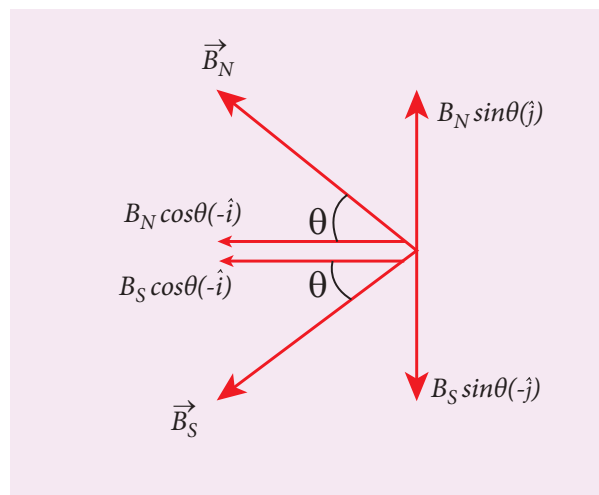


Figure 3.15 Components of magnetic field

$$\vec{B}_S = -B_S \cos \theta \hat{i} - B_S \sin \theta \hat{j} \quad (3.16)$$

where, $B_S = \frac{\mu_0 q_m}{4\pi r'^2}$

From equations (3.15) and (3.16), the net magnetic field at point C due to the dipole is $\vec{B} = \vec{B}_N + \vec{B}_S$.

$$\begin{aligned} \vec{B} &= -(B_N + B_S) \cos \theta \hat{i} \quad \text{Since, } B_N = B_S \\ \vec{B} &= -\frac{2\mu_0 q_m}{4\pi r'^2} \cos \theta \hat{i} = -\frac{2\mu_0 q_m}{4\pi (r^2 + l^2)^{\frac{1}{2}}} \cos \theta \hat{i} \end{aligned} \quad (3.17)$$

In a right angle triangle NOC as shown in Figure 3.14

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{l}{r'} = \frac{l}{(r^2 + l^2)^{\frac{1}{2}}} \quad (3.18)$$

Substituting equation (3.18) in equation (3.17), we get

$$\vec{B} = -\frac{\mu_0 q_m \times (2l)}{4\pi (r^2 + l^2)^{\frac{3}{2}}} \hat{i} \quad (3.19)$$



Since, magnitude of magnetic dipole moment is $|\vec{p}_m| = p_m = q_m \cdot 2l$ and substituting in equation (3.19), the magnetic field at a point C is

$$\vec{B}_{\text{equatorial}} = -\frac{\mu_0}{4\pi} \frac{p_m}{(r^2 + l^2)^{\frac{3}{2}}} \hat{i} \quad (3.20)$$

If the distance between two poles in a bar magnet is small (looks like short magnet) when compared to the distance between geometrical centre O of bar magnet and the location of point C ($r \gg l$),

$$(r^2 + l^2)^{\frac{3}{2}} \approx r^3 \quad (3.21)$$

Therefore, using equation (3.21) in equation (3.20), we get

$$\vec{B}_{\text{equatorial}} = -\frac{\mu_0}{4\pi} \frac{p_m}{r^3} \hat{i}$$

Since $p_m \hat{i} = \vec{p}_m$, the magnetic field at equatorial point is given by

$$\vec{B}_{\text{equatorial}} = -\frac{\mu_0}{4\pi} \frac{\vec{p}_m}{r^3} \quad (3.22)$$

Note that magnitude of B_{axial} is twice that of magnitude of $B_{\text{equatorial}}$ and the direction of B_{axial} and $B_{\text{equatorial}}$ are opposite.

EXAMPLE 3.6

A short bar magnet has a magnetic moment of 0.5 J T^{-1} . Calculate magnitude and direction of the magnetic field produced by the bar magnet which is kept at a distance of 0.1 m from the centre of the bar magnet along (a) axial line of the bar magnet and (b) normal bisector of the bar magnet.

Solution

Given magnetic moment = 0.5 J T^{-1} and distance $r = 0.1 \text{ m}$

(a) When the point lies on the axial line of the bar magnet, the magnetic field for short magnet is given by

$$\vec{B}_{\text{axial}} = \frac{\mu_0}{4\pi} \left(\frac{2p_m}{r^3} \right) \hat{i}$$

$$\vec{B}_{\text{axial}} = 10^{-7} \times \left(\frac{2 \times 0.5}{(0.1)^3} \right) \hat{i} = 1 \times 10^{-4} \hat{i} \text{ T}$$

Hence, the magnitude of the magnetic field along axial is $B_{\text{axial}} = 1 \times 10^{-4} \text{ T}$ and direction is towards South to North.

(b) When the point lies on the normal bisector (equatorial) line of the bar magnet, the magnetic field for short magnet is given by

$$\vec{B}_{\text{equatorial}} = -\frac{\mu_0}{4\pi} \frac{p_m}{r^3} \hat{i}$$

$$\vec{B}_{\text{equatorial}} = -10^{-7} \left(\frac{0.5}{(0.1)^3} \right) \hat{i} = -0.5 \times 10^{-4} \hat{i} \text{ T}$$

Hence, the magnitude of the magnetic field along axial is $B_{\text{equatorial}} = 0.5 \times 10^{-4} \text{ T}$ and direction is towards North to South.

Note that magnitude of B_{axial} is twice that of magnitude of $B_{\text{equatorial}}$ and the direction of B_{axial} and $B_{\text{equatorial}}$ are opposite.

3.3

TORQUE ACTING ON A BAR MAGNET IN UNIFORM MAGNETIC FIELD

Consider a magnet of length $2l$ and pole strength q_m kept in a uniform magnetic field $\vec{B}$ as shown in Figure 3.16. Each pole experiences a force of magnitude $q_m B$ but acting in opposite directions. Therefore, the net force exerted on the magnet is zero and hence, there is no translatory motion. These two equal and opposite forces constitute a couple (about midpoint of bar magnet) tend to align the magnet in the direction of the magnetic field $\vec{B}$.

The force experienced by north pole,

$$\vec{F}_N = q_m \vec{B} \quad (3.23)$$

The force experienced by south pole,

$$\vec{F}_S = -q_m \vec{B} \quad (3.24)$$

Adding equations (3.23) and (3.24), we get the net force acting on the dipole as

$$\vec{F} = \vec{F}_N + \vec{F}_S = \vec{0}$$

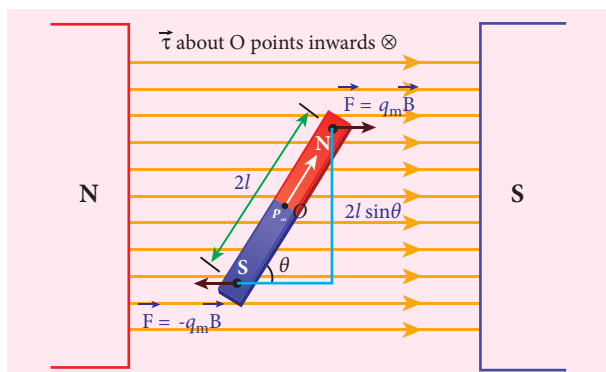


Figure 3.16 Magnetic dipole kept in a uniform magnetic field

The moment of force or torque experienced by north and south pole about point O is

$$\begin{aligned} \vec{\tau} &= \vec{ON} \times \vec{F}_N + \vec{OS} \times \vec{F}_S \\ \vec{\tau} &= \vec{ON} \times q_m \vec{B} + \vec{OS} \times (-q_m \vec{B}) \end{aligned}$$

By using right hand cork screw rule, we conclude that the total torque is pointing into the paper. Since the magnitudes $|\vec{ON}| = |\vec{OS}| = l$ and $|q_m \vec{B}| = |-q_m \vec{B}|$, the magnitude of total torque about point O

$$\begin{aligned} \tau &= l \times q_m B \sin \theta + l \times q_m B \sin \theta \\ &= 2l \times q_m B \sin \theta \\ \tau &= p_m B \sin \theta \quad (\because q_m \times 2l = p_m) \end{aligned}$$

$$\text{In vector notation, } \vec{\tau} = \vec{p}_m \times \vec{B} \quad (3.25)$$



(a) Why a freely suspended bar magnet in your laboratory experiences only torque (rotational motion) but not any translatory motion even though Earth has non-uniform magnetic field?

It is because Earth's magnetic field is locally (physics laboratory) uniform.

(b) Suppose we keep a freely suspended bar magnet in a non-uniform magnetic field. What will happen?

It will undergo translatory motion (net force) and rotational motion (torque).

3.3.1. Potential energy of a bar magnet in a uniform magnetic field

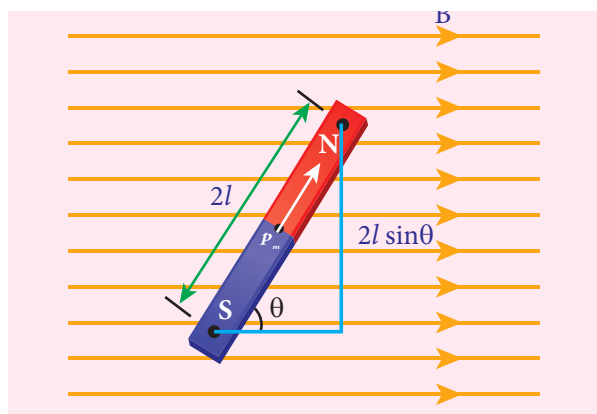


Figure 3.17: A bar magnet (magnetic dipole) in a uniform magnetic field

When a bar magnet (magnetic dipole) of dipole moment $\vec{p}_m$ is held at an angle θ with the direction of a uniform magnetic field $\vec{B}$, as shown in Figure 3.17 the magnitude of the torque acting on the dipole is

$$|\vec{\tau}_B| = |\vec{p}_m| |\vec{B}| \sin \theta$$

If the dipole is rotated through a very small angular displacement $d\theta$ against the

torque τ_B at constant angular velocity, then the work done by external torque ($\vec{\tau}_{ext}$) for this small angular displacement is given by

$$dW = |\vec{\tau}_{ext}| d\theta$$

The bar magnet has to be moved at constant angular velocity, which implies that $|\vec{\tau}_B| = |\vec{\tau}_{ext}|$

$$dW = p_m B \sin \theta d\theta$$

Total work done in rotating the dipole from θ' to θ is

$$W = \int_{\theta'}^{\theta} \tau d\theta = \int_{\theta'}^{\theta} p_m B \sin \theta d\theta = p_m B [-\cos \theta]_{\theta'}^{\theta}$$

$$W = -p_m B (\cos \theta - \cos \theta')$$

This work done is stored as potential energy in bar magnet at an angle θ (when it is rotated from θ' to θ) and it can be written as

$$U = -p_m B (\cos \theta - \cos \theta') \quad (3.26)$$

In fact, the equation (3.26) gives the difference in potential energy between the angular positions θ' and θ . If we choose the reference point as $\theta' = 90^\circ$, so that second term in the equation becomes zero, the equation (3.26) can be written as

$$U = -p_m B (\cos \theta) \quad (3.27)$$

The potential energy stored in a bar magnet in a uniform magnetic field is given by

$$U = -\vec{p}_m \cdot \vec{B} \quad (3.28)$$

Case 1

(i) If $\theta = 0^\circ$, then

$$U = -p_m B (\cos 0^\circ) = -p_m B$$

(ii) If $\theta = 180^\circ$, then

$$U = -p_m B (\cos 180^\circ) = p_m B$$

From the above two results, We infer that the potential energy of the bar magnet is minimum when it is aligned along the external magnetic field and maximum when the bar magnet is aligned anti-parallel to external magnetic field.

EXAMPLE 3.7

Consider a magnetic dipole which on switching ON external magnetic field orient only in two possible ways i.e., one along the direction of the magnetic field (parallel to the field) and another anti-parallel to magnetic field. Compute the energy for the possible orientation.

Solution

Let $\vec{p}_m$ be the dipole and before switching ON the external magnetic field, there is no orientation. Therefore, the energy $U = 0$.

As soon as external magnetic field is switched ON, the magnetic dipole orient parallel ($\theta = 0^\circ$) to the magnetic field with energy,

$$U_{\text{parallel}} = U_{\text{minimum}} = -p_m B \cos 0$$

$$U_{\text{parallel}} = -p_m B$$

$$\text{since } \cos 0^\circ = 1$$

Otherwise, the magnetic dipole orients anti-parallel ($\theta = 180^\circ$) to the magnetic field with energy,

$$U_{\text{anti-parallel}} = U_{\text{maximum}} = -p_m B \cos 180$$

$$\Rightarrow U_{\text{anti-parallel}} = p_m B$$

$$\text{since } \cos 180^\circ = -1$$

3.4

MAGNETIC PROPERTIES

All materials are not magnetic in nature. Further, all the magnetic materials will not behave identically. So, in order to differentiate one magnetic material from another, some basic parameters are used. They are:

(a) Magnetising field

The magnetic field which is used to magnetize a sample or specimen is called the magnetising field. Magnetising field is a vector quantity and is denoted by $\vec{H}$ and its unit is $A\ m^{-1}$.

(b) Magnetic permeability

The magnetic permeability is the measure of ability of the material to allow the passage of magnetic field lines through it or measure of the capacity of the substance to take magnetisation or the degree of penetration of magnetic field through the substance.

In free space, the permeability (or absolute permeability) is denoted by μ_0 and for any other medium it is denoted by μ . The relative permeability μ_r is defined as the ratio between absolute permeability of the medium to the permeability of free space.

$$\mu_r = \frac{\mu}{\mu_0} \quad (3.29)$$

Relative permeability is a dimensionless number and has no units. For free space (air or vacuum), the relative permeability is unity i.e., $\mu_r = 1$.

(c) Intensity of magnetisation

Any bulk material (any object of finite size) contains a large number of atoms. Each atom consists of electrons which undergo orbital motion. Due to orbital motion, electron has magnetic moment which is a vector quantity. In general, these magnetic moments orient randomly, therefore, the net magnetic moment is zero per unit volume of the material.

When such a material is kept in an external magnetic field, atomic dipoles are induced and hence, they will try to align partially or fully along the direction of external field. The net magnetic moment per unit volume of the material is known as intensity of magnetisation. It is a vector quantity. Mathematically,

$$\vec{M} = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{\vec{p}_m}{V} \quad (3.30)$$

The SI unit of intensity of magnetisation is ampere metre⁻¹. For a bar magnet of pole strength q_m , length $2l$ and area of cross-section A , the magnetic moment of the bar magnet is $\vec{p}_m = q_m \vec{2l}$ and volume of the bar magnet is $V = A|\vec{2l}| = 2lA$. The intensity of magnetisation for a bar magnet is

$$\vec{M} = \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{q_m \vec{2l}}{2lA} \quad (3.31)$$

In magnitude, equation (3.31) is

$$|\vec{M}| = M = \frac{q_m \times 2l}{2l \times A} \Rightarrow M = \frac{q_m}{A}$$

This means, for a bar magnet the intensity of magnetisation can be defined as the pole strength per unit area (face area).

(d) Magnetic induction or total magnetic field

When a substance like soft iron bar is placed in a uniform magnetising field $\vec{H}$, the substance gets magnetised. The magnetic induction (total magnetic field) inside the specimen $\vec{B}$ is equal to the sum of the magnetic field $\vec{B}_0$ produced in vacuum due to the magnetising field and the magnetic field $\vec{B}_m$ due to the induced magnetism of the substance.

$$\begin{aligned}\vec{B} &= \vec{B}_0 + \vec{B}_m = \mu_0 \vec{H} + \mu_0 \vec{M} \\ \Rightarrow \vec{B} &= \vec{B}_0 + \vec{B}_m = \mu_0 (\vec{H} + \vec{M})\end{aligned}\quad (3.32)$$

(e) Magnetic susceptibility

When a substance is kept in a magnetising field $\vec{H}$, magnetic susceptibility gives information about how a material responds to the external (applied) magnetic field. In other words, the magnetic susceptibility measures how easily and how strongly a material can be magnetised. It is defined as the ratio of the intensity of magnetisation ($\vec{M}$) induced in the material to the magnetising field ($\vec{H}$)

$$\chi_m = \frac{|\vec{M}|}{|\vec{H}|}\quad (3.33)$$

It is a dimensionless quantity. Magnetic susceptibility for some of the isotropic substances is given in Table 3.1.

Table 3.1 Magnetic susceptibility for various materials

Material	Magnetic susceptibility (χ_m)
Aluminium	2.3×10^{-5}
Copper	-0.98×10^{-5}
Diamond	-2.2×10^{-5}
Gold	-3.6×10^{-5}
Mercury	-3.2×10^{-5}
Silver	-2.6×10^{-5}
Titanium	7.06×10^{-5}
Tungsten	6.8×10^{-5}
Carbon dioxide (1 atm)	-2.3×10^{-9}
Oxygen (1 atm)	2090×10^{-9}

EXAMPLE 3.8

Compute the intensity of magnetisation of the bar magnet whose mass, magnetic moment and density are 200 g, 2 A m² and 8 g cm⁻³, respectively.

Solution

Density of the magnet is

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} \Rightarrow \text{Volume} = \frac{\text{Mass}}{\text{Density}}$$

$$\begin{aligned}\text{Volume} &= \frac{200 \times 10^{-3} \text{ kg}}{(8 \times 10^{-3} \text{ kg}) \times 10^6 \text{ m}^{-3}} \\ &= 25 \times 10^{-6} \text{ m}^3\end{aligned}$$

Magnitude of magnetic moment $p_m = 2 \text{ A m}^2$

Intensity of magnetization,

$$\begin{aligned}M &= \frac{\text{Magnetic moment}}{\text{Volume}} = \frac{2}{25 \times 10^{-6}} \\ M &= 0.8 \times 10^5 \text{ A m}^{-1}\end{aligned}$$

EXAMPLE 3.9

Using the relation $\vec{B} = \mu_0(\vec{H} + \vec{M})$, show that $\chi_m = \mu_r - 1$.

Solution

$$\vec{B} = \mu_0(\vec{H} + \vec{M})$$

But from equation (3.33), in vector form,

$$\vec{M} = \chi_m \vec{H}$$

$$\text{Hence, } \vec{B} = \mu_0(\chi_m + 1)\vec{H} \Rightarrow \vec{B} = \mu\vec{H}$$

$$\text{where, } \mu = \mu_0(\chi_m + 1) \Rightarrow \chi_m + 1 = \frac{\mu}{\mu_0} = \mu_r$$

$$\Rightarrow \chi_m = \mu_r - 1$$

EXAMPLE 3.10

Two materials X and Y are magnetised whose values of intensity of magnetisation are 500 A m^{-1} and 2000 A m^{-1} respectively. If the magnetising field is 1000 A m^{-1} , then which one among these materials can be easily magnetized?

Solution

The susceptibility of material X is

$$\chi_{m,X} = \frac{|\vec{M}|}{|\vec{H}|} = \frac{500}{1000} = 0.5$$

The susceptibility of material Y is

$$\chi_{m,Y} = \frac{|\vec{M}|}{|\vec{H}|} = \frac{2000}{1000} = 2$$

Since, susceptibility of material Y is greater than that of material X, which implies that material Y can be easily magnetized.

3.5

CLASSIFICATION OF MAGNETIC MATERIALS

The magnetic materials are generally classified into three types based on their behaviour in a magnetising field. They are diamagnetic, paramagnetic and ferromagnetic materials.

(a) Diamagnetic materials

The orbital motion of electrons around the nucleus produces a magnetic field perpendicular to the plane of the orbit. Thus each electron orbit has finite orbital magnetic dipole moment. Since the orbital planes of the other electrons are oriented in random manner, the vector sum of magnetic moments is zero and there is no resultant magnetic moment for each atom.

In the presence of a uniform external magnetic field, some electrons are speeded up and some are slowed down. The electrons whose moments were anti-parallel are speeded up according to Lenz's law and this produces an induced magnetic moment in a direction opposite to the field. The induced moment disappears as soon as the external field is removed.

When placed in a non-uniform magnetic field, the interaction between induced magnetic moment and the external field creates a force which tends to move the material from stronger part to weaker part of the external field. It means that diamagnetic material is repelled by the field.

This action is called diamagnetic action and such materials are known as diamagnetic materials. Examples: Bismuth, Copper and Water etc.

The properties of diamagnetic materials are

- Magnetic susceptibility is negative.
- Relative permeability is slightly less than unity.
- The magnetic field lines are repelled or expelled by diamagnetic materials when placed in a magnetic field.
- Susceptibility is nearly temperature independent.



Superconductors are perfect diamagnetic materials. The expulsion of magnetic flux from a superconductor during its transition to the superconducting state is known as Meissner effect. (see figure 3.18)

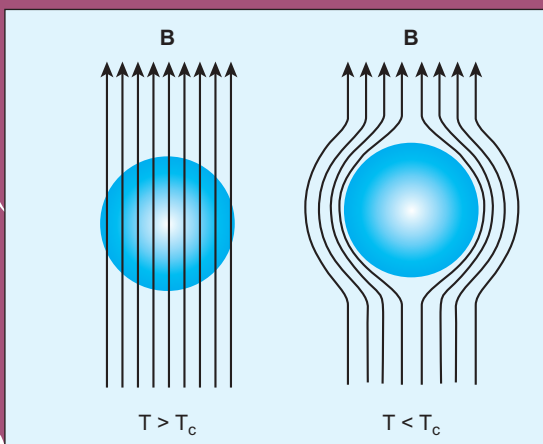


Figure 3.18 Meissner effect – superconductors behave like perfect diamagnetic materials below transition temperature T_c .

(b) Paramagnetic materials

In some magnetic materials, each atom or molecule has net magnetic dipole moment which is the vector sum of orbital and spin magnetic moments of electrons. Due to the random orientation of these magnetic moments, the net magnetic moment of the materials is zero.



Magnetic levitated train

Magnetic levitated train is also called Maglev train. This train floats few centimetres above the guideway because of electromagnet used. Maglev train does not need wheels and also achieve greater speed. The basic mechanism of working of Maglev train involves two sets of magnets. One set is used to repel which makes train to float above the track and another set is used to move the floating train ahead at very great speed. These trains are quieter, smoother and environmental friendly compared conventional trains and have potential for moving with much higher speeds with technology in future.



In the presence of an external magnetic field, the torque acting on the atomic dipoles will align them in the field direction. As a result, there is net magnetic dipole moment induced in the direction of the applied field. The induced dipole moment is present as long as the external field exists.

When placed in a non-uniform magnetic field, the paramagnetic materials will have a tendency to move from weaker to stronger part of the field. Materials which exhibit weak magnetism in the direction of the applied field are known as paramagnetic materials.

Examples: Aluminium, Platinum, Chromium and Oxygen etc.

The properties of paramagnetic materials are:

- Magnetic susceptibility is positive and small.
- Relative permeability is greater than unity.
- The magnetic field lines are attracted into the paramagnetic materials when placed in a magnetic field.
- Susceptibility is inversely proportional to temperature.

Curie's law

When temperature is increased, thermal vibration will upset the alignment of magnetic dipole moments. Therefore, the magnetic susceptibility decreases with increase in temperature. In many cases, the susceptibility of the materials is

$$\chi_m \propto \frac{1}{T} \text{ or } \chi_m = \frac{C}{T}$$

This relation is called Curie's law. Here C is called Curie constant and temperature T is in kelvin. The graph drawn between magnetic susceptibility and temperature is shown in Figure 3.19, which is a rectangular hyperbola.

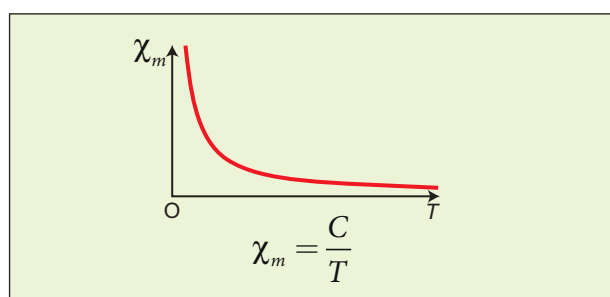


Figure 3.19 Curie's law – susceptibility vs temperature

(c) Ferromagnetic materials

A ferromagnetic material possesses net magnetic dipole moment as in a paramagnetic material. A ferromagnetic material is made up of smaller regions, called ferromagnetic domains (Figure 3.20). Within each domain, the magnetic moments are spontaneously aligned in a direction. This alignment is caused by strong interaction arising from electron spin which depends on the inter-atomic distance. Each domain has net magnetisation in a direction. However the direction of magnetisation varies from domain to domain and thus net magnetisation of the specimen is zero.

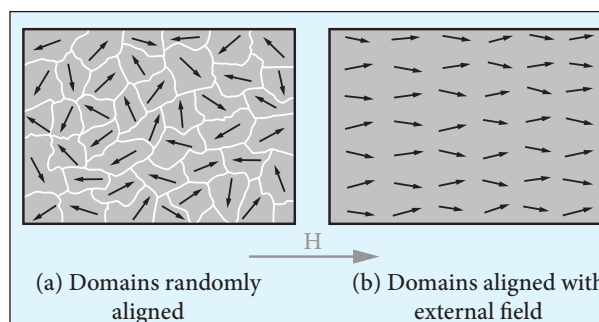


Figure 3.20 Magnetic domains – ferromagnetic materials

In the presence of external magnetic field, two processes take place

- The domains having magnetic moments parallel to the field grow bigger in size
- The other domains (not parallel to field) are rotated so that they are aligned with the field.

As a result of these mechanisms, there is a strong net magnetisation of the material in the direction of the applied field (Figure 3.21).

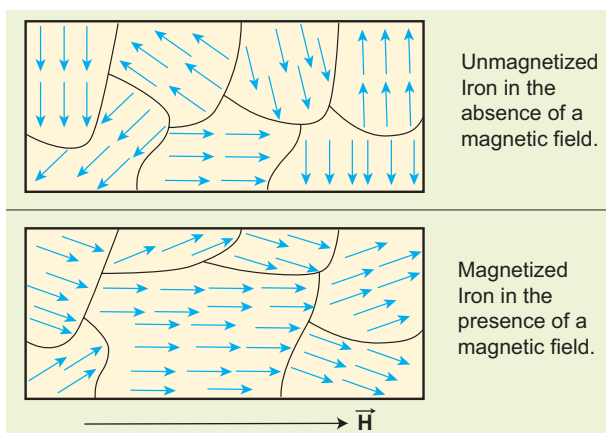


Figure 3.21 Processes of domain magnetization

When placed in a non-uniform magnetic field, the ferromagnetic materials will have a strong tendency to move from weaker to stronger part of the field. Materials which exhibit strong magnetism in the direction of applied field are called ferromagnetic materials. Examples: Iron, Nickel and Cobalt.

The properties of ferromagnetic materials are:



Magnetism plays interesting role in various aspects of life. It has connection with archeological place Keezhadi too. To find whether any archeological structure exists under the surface of a given place, well established technique called 'magnetometer surveying' is used.

In this technique, the variation of the magnetic field in comparison with the neighbouring place is studied. The magnetic field variation is due to the presence of magnetic mineral magnetite and its related minerals present in the archeological structures like buried wall, pottery, bricks, buried tombs, monuments and inhabited sites. Those minerals are either diamagnetic or paramagnetic or ferromagnetic in nature and each type has different range of magnetic susceptibilities.

Indian Institute of Geomagnetism (IIG), Mumbai conducted magnetometer survey on Keezhadi site and found out that there were archeological structures like wall, pottery etc. From the picture (Figure 1), there was magnetic field variation in the range of 10 to 100nT over the particular area (coloured portion). In fact, the existence of massive brick structures at Keezhadi has been revealed through magnetism (Figure 2).



Figure 2

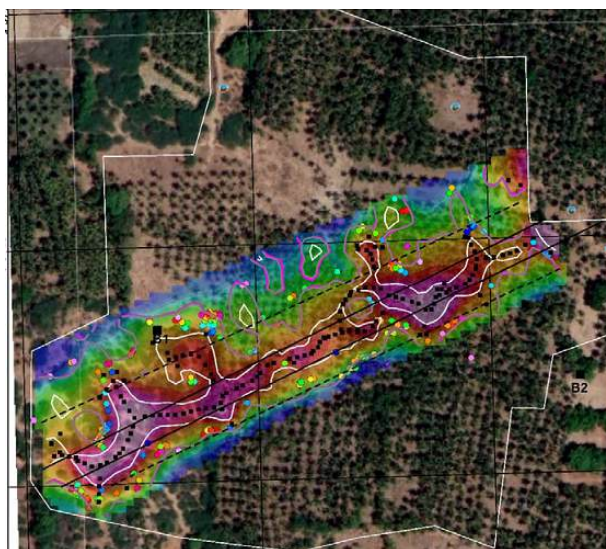


Figure 1



- i) Magnetic susceptibility is positive and large.
- ii) Relative permeability is large.
- iii) The magnetic field lines are strongly attracted into the ferromagnetic materials when placed in a magnetic field.
- iv) Susceptibility is inversely proportional to temperature.

Curie-Weiss law

As temperature increases, the ferromagnetism decreases due to the increased thermal agitation of the atomic dipoles. At a particular temperature, ferromagnetic material becomes paramagnetic. This temperature is known as Curie temperature T_c . The susceptibility of the material above the Curie temperature is given by

$$\chi_m = \frac{C}{T - T_c}$$

This relation is called Curie-Weiss law. The constant C is called Curie constant and temperature T is in kelvin scale. A plot of magnetic susceptibility with temperature is as shown in Figure 3.22.

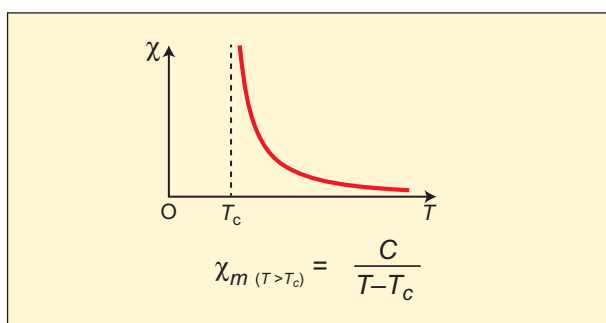
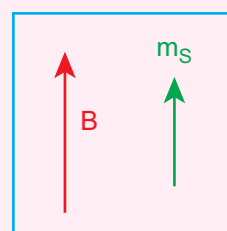


Figure 3.22 Curie-Weiss law – susceptibility vs temperature

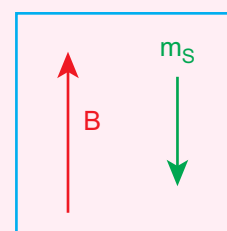
Spin



Like mass and charge for particles, spin is also another important attribute for an elementary particle. Spin is a quantum mechanical phenomenon which is responsible for magnetic properties of the material. Spin in quantum mechanics is entirely different from spin we encounter in classical mechanics. Spin in quantum mechanics does not mean rotation; it is intrinsic angular momentum which does not have classical analogue. For historical reason, the name spin is retained. Spin of a particle takes only positive values but the orientation of the spin vector takes plus or minus values in an external magnetic field. For an example, electron has spin $s = \frac{1}{2}$. In the presence of magnetic field, the spin will orient either parallel or anti-parallel to the direction of magnetic field.

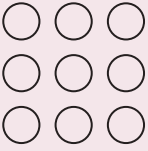
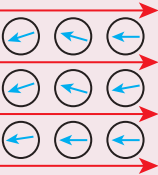
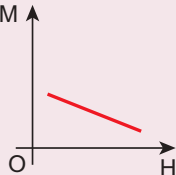
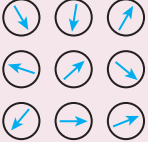
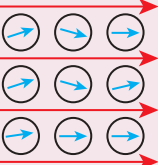
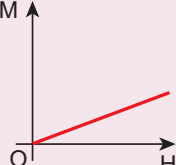
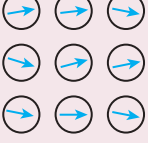
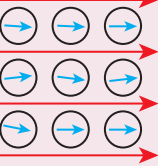
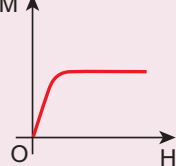


Spin is parallel to the magnetic field direction (Spin up)



Spin is anti-parallel to the magnetic field direction (Spin down)

This implies that the magnetic spin m_s takes two values for an electron, such as $m_s = \frac{1}{2}$ (spin up) and $m_s = -\frac{1}{2}$ (spin down). Spin for proton and neutron is $s = \frac{1}{2}$. For photon, spin $s = 1$.

Comparison of Types of Magnetism (NOT FOR EXAMINATION)					
Type of magnetism	Magnetising field is absent ($H = 0$)	Magnetising field is present ($H \neq 0$)	Magnetisation of the material	Susceptibility	Relative permeability
Diamagnetism	 (Zero magnetic moment)	 (Aligned opposite to the field)		Negative	Less than unity
Paramagnetism	 (Net magnetic moment but random alignment)	 (Aligned with the field)		Positive and small	Greater than unity
Ferromagnetism	 (Net magnetic moment in a domain but they are randomly aligned)	 (Aligned with the field)		Positive and large	Very large

3.6

HYSTERESIS

When a ferromagnetic material is kept in a magnetising field, the material gets magnetised by induction. An important characteristic of ferromagnetic material is that the variation of magnetic induction $\vec{B}$ with magnetising field $\vec{H}$ is not linear. It means that the ratio $\frac{B}{H} = \mu$ is not a constant. Let us study this behaviour in detail.

A ferromagnetic material (example, Iron) is magnetised slowly by a magnetising field $\vec{H}$. The magnetic induction $\vec{B}$ of the material increases from point A with the magnitude of the magnetising field and then attains a saturation level. This

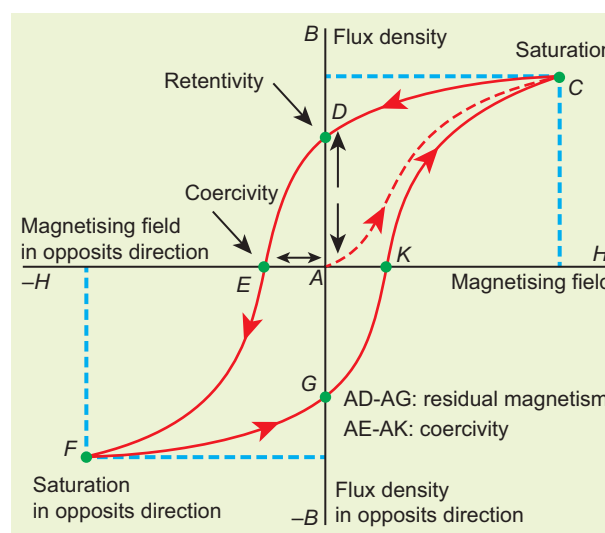


Figure 3.23 Hysteresis – plot for B vs H

response of the material is depicted by the path AC as shown in Figure 3.23. Saturation magnetization is defined as the maximum point up to which the material can be magnetised by applying the magnetising field.



If the magnetising field is now reduced, the magnetic induction also decreases but does not retrace the original path CA. It takes different path CD. When the magnetising field is zero, the magnetic induction is not zero and it has positive value. This implies that some magnetism is left in the specimen even when $H = 0$. The residual magnetism AD present in the specimen is called remanence or retentivity. **Remanence is defined as the ability of the materials to retain the magnetism in them even after the magnetising field disappears.**

In order to demagnetise the material, the magnetising field is gradually increased in the reverse direction. Now the magnetic induction decreases along DE and becomes zero at E. The magnetising field AE in the reverse direction is required to bring residual magnetism to zero. **The magnitude of the reverse magnetising field for which the residual magnetism of the material vanishes is called its coercivity.**

Further increase of $\vec{H}$ in the reverse direction causes the magnetic induction to increase along EF until it reaches saturation at F in the reverse direction. If magnetising field is decreased and then increased with direction reversed, the magnetic induction traces the path FGKC. This closed curve ACDEFGKC is called hysteresis loop and it corresponds to one cycle of magnetisation.

In the entire cycle, the magnetic induction B lags behind the magnetising field H . This **phenomenon of lagging of magnetic induction behind the magnetising field is called hysteresis.** Hysteresis means 'lagging behind'.

Hysteresis loss

During the magnetisation of the specimen through a cycle, there is loss of energy in the form of heat. This loss is attributed to the rotation and orientation of molecular magnets in various directions. It is found that the energy lost (or dissipated) per unit volume of the material when it is carried through one cycle of magnetisation is equal to the area of the hysteresis loop.

Hard and soft magnetic materials

Based on the shape and size of the hysteresis loop, ferromagnetic materials are classified as soft magnetic materials with smaller area and hard magnetic materials with larger area. The comparison of the hysteresis loops for two magnetic materials is shown in Figure 3.24. Properties of soft and hard magnetic materials are compared in Table 3.2.

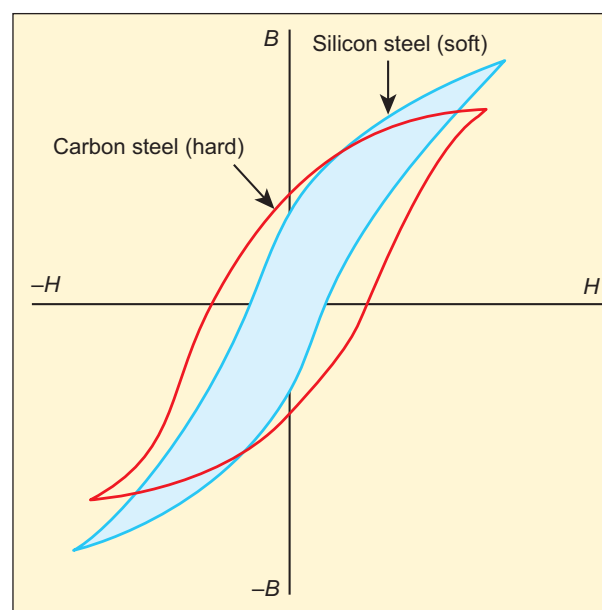


Figure 3.24 Comparison of two ferromagnetic materials based on hysteresis loop

Table 3.2 Differences between soft and hard ferromagnetic materials

S.No.	Properties	Soft ferromagnetic materials	Hard ferromagnetic materials
1	When external field is removed	Magnetisation disappears	Magnetisation persists
2	Area of the loop	Small	Large
3	Retentivity	Low	High
4	Coercivity	Low	High
5	Susceptibility and magnetic permeability	High	Low
6	Hysteresis loss	Less	More
7	Uses	Solenoid core, transformer core and electromagnets	Permanent magnets
8	Examples	Soft iron, Mumetal, Stalloy etc.	Carbon steel, Alnico, Lodestone etc.

Applications of hysteresis loop

The significance of hysteresis loop is that it provides information such as retentivity, coercivity, permeability, susceptibility and energy loss during one cycle of magnetisation for each ferromagnetic material. Therefore, the study of hysteresis loop will help us in selecting proper and suitable material for a given purpose. Some examples:

i) Permanent magnets:

The materials with high retentivity, high coercivity and low permeability are suitable for making permanent magnets.

Examples: Carbon steel and Alnico

ii) Electromagnets:

The materials with high initial permeability, low retentivity, low coercivity and thin hysteresis loop with smaller area are preferred to make electromagnets.

Examples: Soft iron and Mumetal (Nickel Iron alloy).

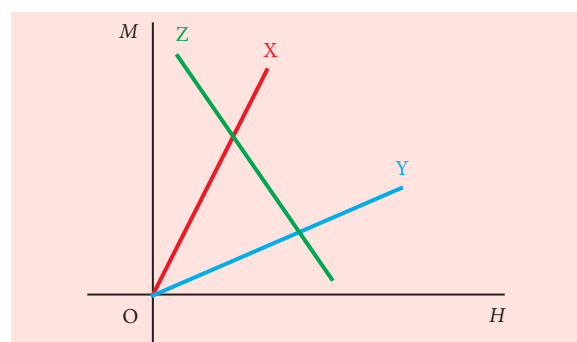
iii) Core of the transformer:

The materials with high initial permeability, large magnetic induction and thin hysteresis loop with smaller area are needed to design transformer cores.

Examples: Soft iron

EXAMPLE 3.11

The following figure shows the variation of intensity of magnetisation with the applied magnetic field intensity for three magnetic materials X, Y and Z. Identify the materials X, Y and Z.



Solution

The slope of M - H graph is a measure of the magnetic susceptibility, which is given by

$$\chi_m = \frac{M}{H}$$

Material X: Slope is positive and larger value. So, it is a ferromagnetic material.

Material Y: Slope is positive and lesser value than X. So, it could be a paramagnetic material.

Material Z: Slope is negative and hence, it is a diamagnetic material.

3.7

MAGNETIC EFFECTS OF CURRENT

3.7.1 Oersted experiment

In 1820 Hans Christian Oersted, while preparing for his lecture in physics, noticed that electric current passing through a wire deflects the nearby magnetic needle in the compass. By proper investigation, he observed that the deflection of magnetic needle is due to the change in magnetic field produced around current carrying conductor (Figure 3.25). When the direction of current is reversed, the magnetic needle is deflected in the opposite direction. This led to the development of the theory 'electromagnetism' which unifies two branches in physics namely, electricity and magnetism.

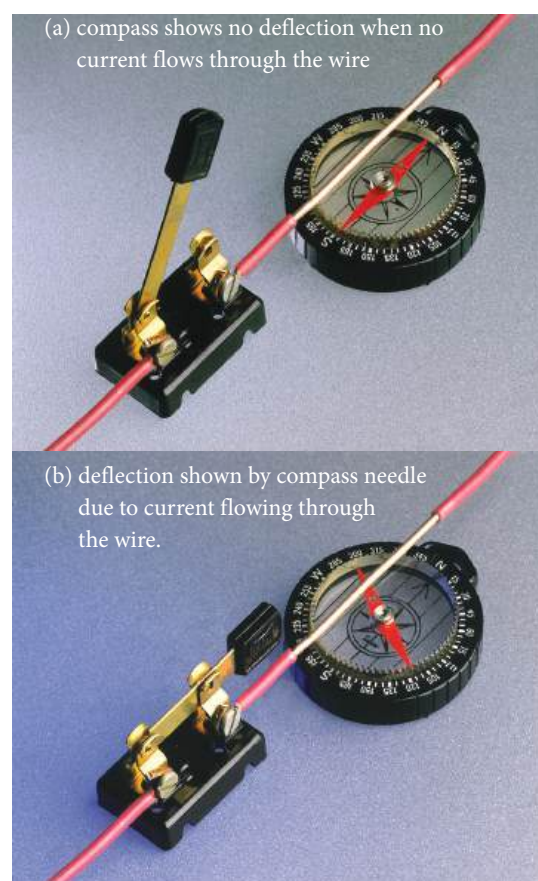
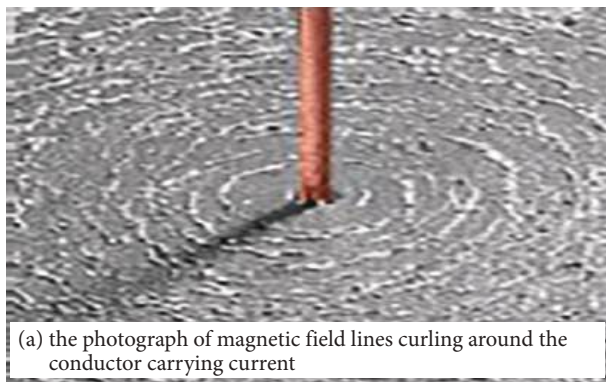


Figure 3.25 Oersted's experiment - current carrying wire and deflection of magnetic needle

3.7.2 Magnetic field around a straight current-carrying conductor and circular loop

(a) Current carrying straight conductor:

Suppose we keep a magnetic compass near a current-carrying straight conductor, then the needle of the magnetic compass experiences a torque and deflects to align in the direction of the magnetic field at that point. Tracing out the direction shown by magnetic needle, we can draw the magnetic field lines at a distance. For a straight current-carrying conductor, the nature of magnetic field is like concentric circles having their common centre on the axis of the conductor as shown in Figure 3.26 (a).



(a) the photograph of magnetic field lines curling around the conductor carrying current

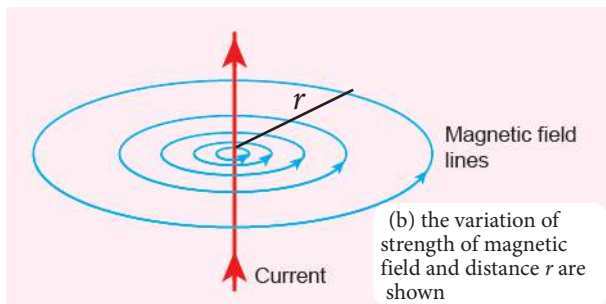


Figure 3.26 Magnetic field lines around straight, long wire carrying current

The direction of circular magnetic field lines will be clockwise or anticlockwise depending on the direction of current in the conductor. If the strength (or magnitude) of the current is increased then the density of the magnetic field will also increase. The strength of the magnetic field (B) decreases as the distance (r) from the conductor increases (Figure 3.26 (b)).

(b) Circular coil carrying current

Suppose we keep a magnetic compass near a current carrying circular conductor, then the needle of the magnetic compass experiences a torque and deflects to align in the direction of the magnetic field at that point. We can notice that at the points A and B in the vicinity of the coil, the magnetic field lines are circular. The magnetic field lines are nearly parallel to each other near the centre of the loop, indicating that the field present near the centre of the coil is almost uniform (Figure 3.27).

The strength of the magnetic field is increased if either the current in the coil or the number of turns or both are increased. The polarity (north pole or south pole) depends on the direction of current in the loop.

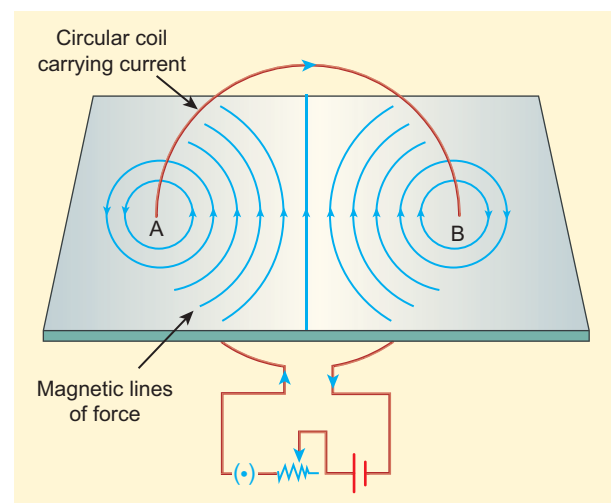
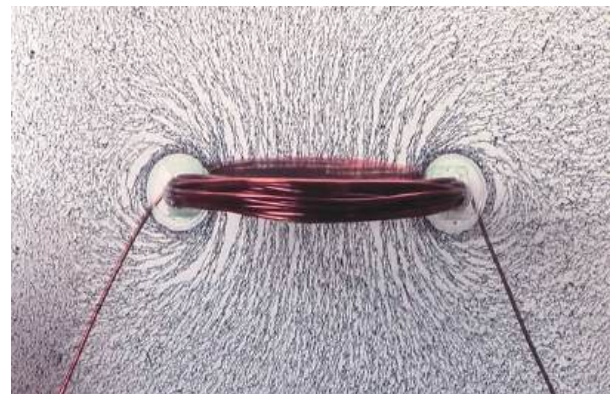


Figure 3.27 The magnetic field lines curling around the circular coil carrying current.

3.7.3 Right hand thumb rule

The right hand rule is used to find the direction of magnetic field when the direction of current in a conductor is known.

Assume that we hold the current carrying conductor in our right hand such that the thumb points in the direction of current flow, then the fingers encircling the conductor point in the direction of the magnetic field lines produced.

The Figure 3.28 shows the right hand rule for current carrying straight conductor and circular coil.

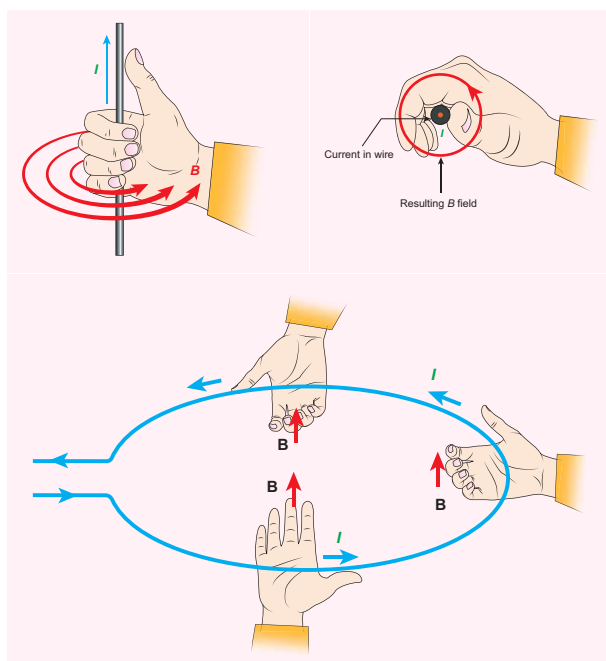


Figure 3.28 Right hand rule – straight conductor and circular loop

3.7.4 Maxwell's right hand cork screw rule

This rule can also be used to find the direction of the magnetic field around the current-carrying conductor. If we rotate a right-handed screw using a screw driver, then the direction of current is same as the direction in which screw advances and the direction of rotation of the screw gives the direction of the magnetic field. (Figure 3.29)

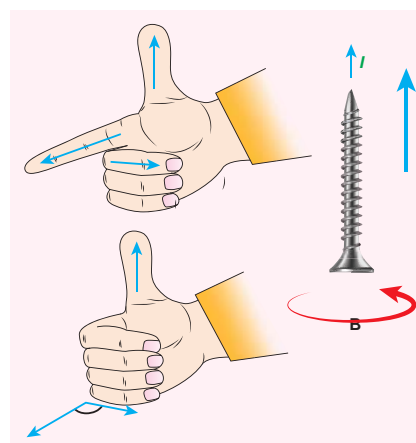
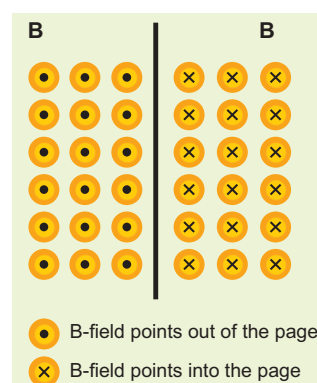


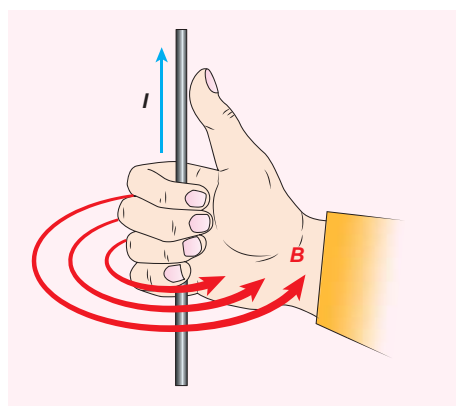
Figure 3.29 Maxwell's right hand cork screw rule

EXAMPLE 3.12

The magnetic field shown in the figure is due to the current carrying wire. In which direction does the current flow in the wire?.



Solution



Using right hand rule, current flows upwards.

3.8

BIOT - SAVART LAW

Soon after Oersted's discovery, both Jean-Baptiste Biot and Felix Savart in 1819 did quantitative experiments on the force experienced by a magnet kept near current carrying wire and arrived at a mathematical expression that gives the magnetic field at some point in space in terms of the current that produces the magnetic field. This is true for any shape of the conductor.

3.8.1 Definition and explanation of Biot- Savart law

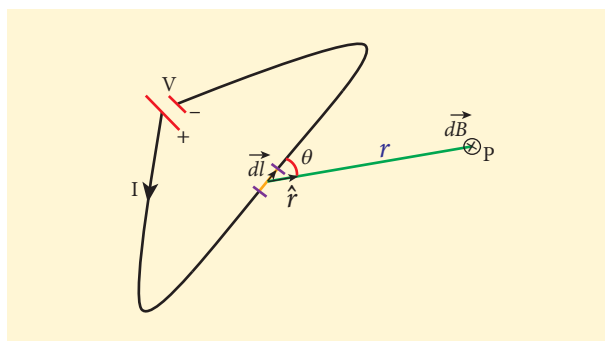


Figure 3.30 Magnetic field at a point P due to current carrying conductor

Biot and Savart experimentally observed that the magnitude of magnetic field $d\vec{B}$ at a point P (Figure 3.30) at a distance r from the small elemental length taken on a conductor carrying current varies

- (i) directly as the strength of the current I
- (ii) directly as the magnitude of the length element $d\vec{l}$
- (iii) directly as the sine of the angle θ between $d\vec{l}$ and $\hat{r}$.

- (iv) inversely as the square of the distance r between the point P and length element $d\vec{l}$.

This is expressed as

$$dB \propto \frac{Idl}{r^2} \sin \theta$$

$$dB = k \frac{I dl}{r^2} \sin \theta$$

where $k = \frac{\mu_0}{4\pi}$ in SI units.

In vector notation,

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2} \quad (3.34)$$

Here vector $d\vec{B}$ is perpendicular to both $I d\vec{l}$ (pointing the direction of current flow) and the unit vector $\hat{r}$ directed from $d\vec{l}$ toward point P (Figure 3.31).

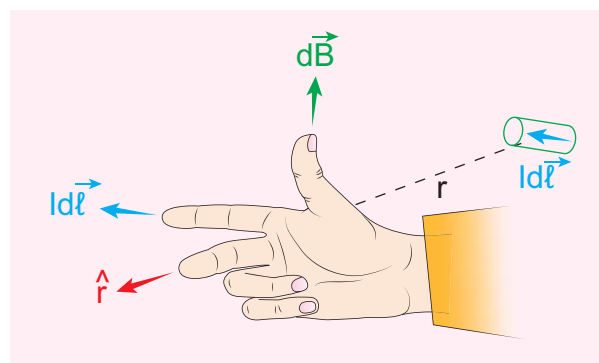


Figure 3.31 The direction of magnetic field using right hand rule

The equation (3.34) is used to compute the magnetic field only due to a small elemental length dl of the conductor. The net magnetic field at P due to the conductor is obtained from principle of superposition by considering the contribution from all current elements $I d\vec{l}$. Hence integrating equation (3.34), we get

$$\vec{B} = \int d\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{l} \times \hat{r}}{r^2} \quad (3.35)$$

where the integral is taken over the entire current distribution.

Cases

1. If the point P lies on the conductor, then $\theta = 0^\circ$. Therefore, $|d\vec{B}|$ is zero.

2. If the point lies perpendicular to the conductor, then $\theta = 90^\circ$. Therefore, $d\vec{B}$ is

$$\text{maximum and is given by } d\vec{B} = \frac{\mu_0}{4\pi} \frac{I dl}{r^2} \hat{n}$$

where $\hat{n}$ is the unit vector perpendicular to both $I d\vec{l}$ and $\hat{r}$



Note Electric current is not a vector quantity. It is a scalar quantity. But electric current in a conductor has direction of flow. Therefore, the electric current flowing in a small elemental conductor can be taken as vector quantity i.e. $I d\vec{l}$

Similarities between electric field (from Coulomb's law) and magnetic field (from Biot-Savart's law)

Electric and magnetic fields

- obey inverse square law, so they are long range fields.
- obey the principle of superposition and are linear with respect to source. In magnitude,

$$E \propto q$$

$$B \propto Idl$$

Differences between electric field (from Coulomb's law) and magnetic field (from Biot-Savart's law)

S. No.	Electric field	Magnetic field
1	Produced by a scalar source i.e., an electric charge q	Produced by a vector source i.e., current element $I d\vec{l}$
2	It is directed along the position vector joining the source and the point at which the field is calculated	It is directed perpendicular to the position vector $\vec{r}$ and the current element $I d\vec{l}$
3	Does not depend on angle	Depends on the angle between the position vector $\vec{r}$ and the current element $I d\vec{l}$



Note The exponent of charge q (source) and exponent of electric field E is unity. Similarly, the exponent of current element $I d\vec{l}$ (source) and exponent of magnetic field B is unity. In other words, electric field $\vec{E}$ is proportional only to charge (source) and not on higher powers of charge (q^2 , q^3 , etc). Similarly, magnetic field $\vec{B}$ is proportional to current element $I d\vec{l}$ (source) and not on square or cube or higher powers of current element. The cause and effect have linear relationship.

3.8.2 Magnetic field due to long straight conductor carrying current

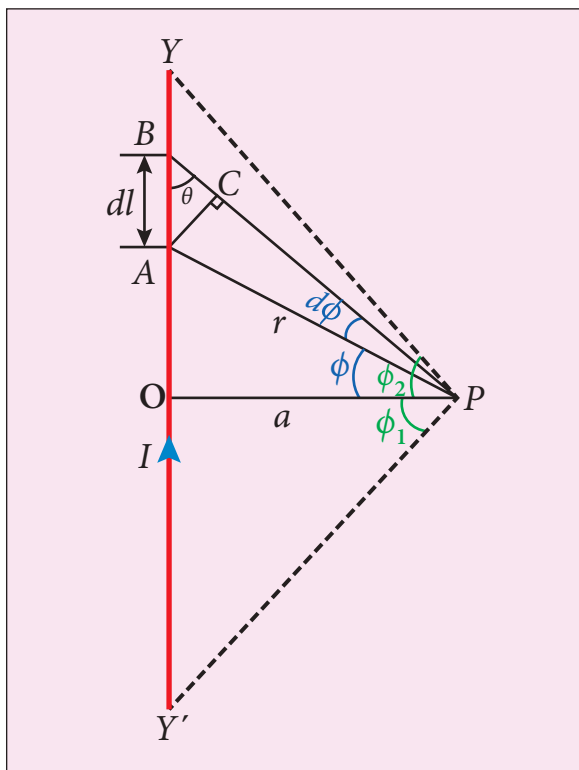


Figure 3.32 Magnetic field due to a long straight current carrying conductor

Let YY' be an infinitely long straight conductor and I be the steady current through the conductor as shown in Figure 3.32. In order to calculate magnetic field at a point P which is at a distance a from the wire, let us consider a small line element dl (segment AB).

The magnetic field at a point P due to current element Idl can be calculated from Biot-Savart's law, which is

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2} \hat{n}$$

where $\hat{n}$ is the unit vector which points into the page at P , θ is the angle between current element Idl and line joining dl and the point P . Let r be the distance between line element at A to the point P .

To apply trigonometry, draw a perpendicular line from A to BP as shown in Figure 3.32.

$$\text{In triangle } \triangle ABC, \sin \theta = \frac{AC}{AB}$$

$$\Rightarrow AC = AB \sin \theta$$

$$\text{But } AB = dl \Rightarrow AC = dl \sin \theta$$

Let $d\phi$ be the angle subtended between AP and BP

$$\text{i.e., } \angle APB = \angle APC = d\phi$$

$$\text{In a triangle } \triangle APC, \sin(d\phi) = \frac{AC}{AP}$$

$$\text{Since } d\phi \text{ is very small, } \sin(d\phi) \simeq d\phi$$

$$\text{But } AP = r \Rightarrow AC = rd\phi$$

$$\therefore AC = dl \sin \theta = rd\phi$$

$$\therefore d\vec{B} = \frac{\mu_0}{4\pi} \frac{I}{r^2} (rd\phi) \cdot \hat{n} = \frac{\mu_0}{4\pi} \frac{Id\phi}{r} \hat{n}$$

Let ϕ be the angle between AP and OP

$$\text{In a } \triangle OPA, \cos \phi = \frac{OP}{AP} = \frac{a}{r}$$

$$\Rightarrow r = \frac{a}{\cos \phi}$$

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I}{a / \cos \phi} d\phi \cdot \hat{n}$$

$$\Rightarrow d\vec{B} = \frac{\mu_0 I}{4\pi a} \cos \phi d\phi \cdot \hat{n}$$

The total magnetic field at P due to the conductor YY' is

$$\begin{aligned} \vec{B} &= \int_{-\phi_1}^{\phi_2} d\vec{B} = \int_{-\phi_1}^{\phi_2} \frac{\mu_0 I}{4\pi a} \cos \phi d\phi \cdot \hat{n} \\ &= \frac{\mu_0 I}{4\pi a} [\sin \phi]_{-\phi_1}^{\phi_2} \hat{n} \end{aligned}$$

$$= \vec{B} = \frac{\mu_0 I}{4\pi a} (\sin \phi_1 + \sin \phi_2) \hat{n}$$

For infinitely long conductor,

$$\phi_1 = \phi_2 = 90^\circ$$

$$\therefore \vec{B} = \frac{\mu_0 I}{4\pi a} \times 2 \hat{n} \Rightarrow \vec{B} = \frac{\mu_0 I}{2\pi a} \hat{n} \quad (3.36)$$

3.8.3 Magnetic field produced along the axis of the current-carrying circular coil

Consider a current carrying circular loop of radius R and let I be the current flowing through the wire in the direction as shown in Figure 3.33.

The magnetic field at a point P on the axis of the circular coil at a distance z from the centre of the coil O is computed by taking two diametrically opposite line elements of the coil each of length $d\vec{l}$ at C and D. Let $\vec{r}$ be the vector joining the current element ($d\vec{l}$) at C and the point P.

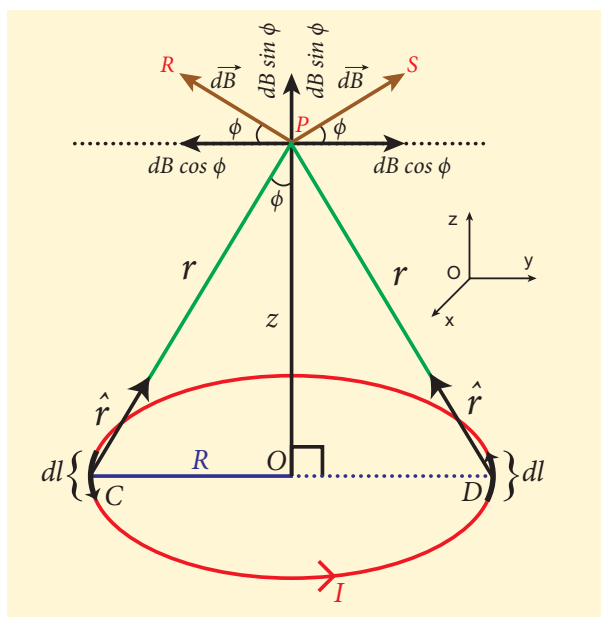


Figure 3.33 Magnetic field due to current-carrying circular loop

According to Biot-Savart's law, the magnetic field at P due to the current element at C is

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

The magnitude of $d\vec{B}$ is

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin \theta}{r^2} = \frac{\mu_0}{4\pi} \frac{I dl}{r^2}$$

where θ is the angle between $I d\vec{l}$ and $\vec{r}$. Here $\theta = 90^\circ$.

The direction of $d\vec{B}$ is perpendicular to the current element $I d\vec{l}$ and CP. It is therefore along PR perpendicular to CP.

The magnitude of magnetic field at P due to current element at D is same as that for the element at C because of equal distances from the coil. But its direction is along PS.

The magnetic field $d\vec{B}$ due to each current element is resolved into two components; $dB \cos \phi$ along y-direction and $dB \sin \phi$ along z-direction. The horizontal components cancel out while the vertical components ($dB \sin \phi \hat{k}$) alone contribute to the net magnetic field $\vec{B}$ at the point P.

$$\begin{aligned} \vec{B} &= \int d\vec{B} = \int dB \sin \phi \hat{k} \\ &= \frac{\mu_0 I}{4\pi} \int \frac{dl}{r^2} \sin \phi \hat{k} \end{aligned}$$

From $\triangle OCP$,

$$\sin \phi = \frac{R}{(R^2 + z^2)^{1/2}} \text{ and } r^2 = R^2 + z^2.$$

Substituting these in the above equation, we get

$$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{R}{(R^2 + z^2)^{3/2}} \hat{k} \left(\int dl \right)$$

If we integrate the line element from 0 to $2\pi R$, we get the net magnetic field $\vec{B}$ at point P due to the current-carrying circular loop.

$$\vec{B} = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}} \hat{k}$$

If the circular coil contains N turns, then the magnetic field is

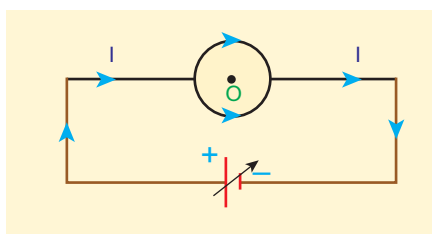
$$\vec{B} = \frac{\mu_0 NI}{2} \frac{R^2}{(R^2 + z^2)^{3/2}} \hat{k} \quad (3.37)$$

The magnetic field at the centre of the coil is

$$\vec{B} = \frac{\mu_0 NI}{2R} \hat{k} \quad \text{since } z = 0 \quad (3.38)$$

EXAMPLE 3.13

What is the magnetic field at the centre of the loop shown in figure?



Solution

The magnetic field due to current in the upper semicircle and lower semicircle of the circular coil are equal in magnitude but opposite in direction. Hence, the net magnetic field at the centre of the loop (at point O) is zero $\vec{B} = \vec{0}$.

3.8.4 Tangent law and Tangent Galvanometer

Tangent galvanometer is a device used to detect very small currents. It is a moving

magnet type galvanometer. Its working is based on tangent law (Figure 3.34).



Figure 3.34 Tangent Galvanometer

Tangent law

When a magnetic needle or magnet is freely suspended in two mutually perpendicular uniform magnetic fields, it will come to rest in the direction of the resultant of the two fields.

Let B be the magnetic field produced by passing current through the coil of the tangent galvanometer and B_H be the horizontal component of Earth's magnetic field. Under the action of two magnetic fields, the needle comes to rest making angle θ with B_H , such that

$$B = B_H \tan \theta \quad (3.39)$$

Construction

Tangent Galvanometer (TG) consists of copper coil of several turns wound on a non-magnetic circular frame. The frame is made up of brass or wood which is mounted vertically on a horizontal base table (turn



table) with three levelling screws. The TG is provided with two or more coils of different number of turns. Most of the equipments we use in laboratory, contains coils of 2 turns, 5 turns and 50 turns which are of different thickness and are used for measuring currents of different strengths.

At the centre of turn table, there is a small upright projection on which a compass box is placed. Compass box consists of a small magnetic needle which is pivoted at its centre, such that the centres of both magnetic needle and circular coil exactly coincide. A thin aluminium pointer attached perpendicular to the magnetic needle moves over a graduated circular scale. The circular scale is divided into four quadrants and they are graduated in degrees, each quadrant being numbered from 0° to 90° . In order to avoid parallax error in measurement, a mirror is placed below the aluminium pointer.

Precautions

1. All the nearby magnets and magnetic materials are kept away from the instrument.
2. Using spirit level, the levelling screws at the base are adjusted so that the small magnetic needle is exactly horizontal and also coil (mounted on the frame) is exactly vertical.
3. The plane of the coil is kept parallel to the small magnetic needle by rotating the coil about its vertical axis. So that, the coil remains in magnetic meridian.
4. The compass box alone is rotated such that the aluminium pointer reads $0^\circ - 0^\circ$.

Theory

In the tangent galvanometer experiment, when no current is passed through the coil, the small magnetic needle lies along horizontal component of Earth's magnetic

field. When the circuit is closed, the electric current will pass through the circular coil and produce magnetic field at the centre of the coil. Now there are two fields which are acting mutually perpendicular to each other. They are:

- (1) the magnetic field (B) due to the electric current in the coil acting normal to the plane of the coil.
- (2) the horizontal component of Earth's magnetic field (B_H)

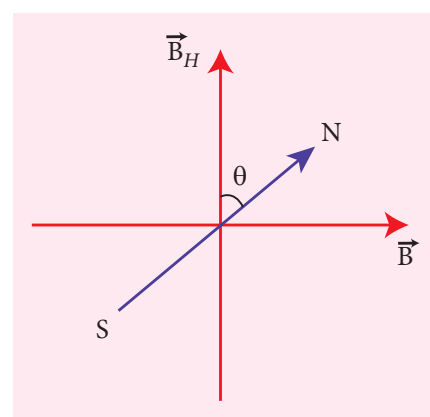


Figure 3.35 Resultant position of pivoted needle

Because of these crossed fields, the pivoted magnetic needle deflects through an angle θ . From tangent law (equation 3.39),

$$B = B_H \tan \theta$$

When an electric current is passed through a circular coil of radius R having N turns, the magnitude of magnetic field at the centre is (from equation (3.38))

$$B = \mu_0 \frac{NI}{2R} \quad (3.40)$$

From equation (3.39) and equation (3.40), we get

$$\mu_0 \frac{NI}{2R} = B_H \tan \theta$$

The horizontal component of Earth's magnetic field is given by

$$B_H = \frac{\mu_0 N}{2R} \frac{I}{\tan \theta} \quad (3.41)$$

EXAMPLE 3.14

A coil of a tangent galvanometer of diameter 0.24 m has 100 turns. If the horizontal component of Earth's magnetic field is 25×10^{-6} T then, calculate the current which gives a deflection of 60° .

Solution

The diameter of the coil is 0.24 m. Therefore, radius of the coil is 0.12 m.

Number of turns is 100 turns.

Earth's magnetic field is 25×10^{-6} T

Deflection is

$$\theta = 60^\circ \Rightarrow \tan 60^\circ = \sqrt{3} = 1.732$$

$$I = \frac{2RB_H}{\mu_0 N} \tan \theta$$

$$= \frac{2 \times 0.12 \times 25 \times 10^{-6}}{4 \times 10^{-7} \times 3.14 \times 100} \times 1.732 = 0.82 \times 10^{-1} \text{ A.}$$

$$I = 0.082 \text{ A}$$

3.8.5 Current loop as a magnetic dipole

The magnetic field at a point on the axis of the current-carrying circular loop of radius R at a distance z from its centre is given by

$$\vec{B} = \frac{\mu_0 I}{2} \frac{R^2}{(R^2 + z^2)^{3/2}} \hat{k} \quad (\text{From eqn. 3.37})$$

At larger distance $z \gg R$, therefore $R^2 + z^2 \approx z^2$, we have

$$\vec{B} = \frac{\mu_0 I R^2}{2 z^3} \hat{k} \text{ or } \vec{B} = \frac{\mu_0 I \pi R^2}{2\pi z^3} \hat{k} \quad (3.42)$$

Let A be the area of the circular loop $A = \pi R^2$. So rewriting the equation (3.42) in terms of area of the loop, we have

$$\begin{aligned} \vec{B} &= \frac{\mu_0 I}{2\pi} \frac{A}{z^3} \hat{k} \\ (\text{or}) \\ \vec{B} &= \frac{\mu_0}{4\pi} \frac{2IA}{z^3} \hat{k} \end{aligned} \quad (3.43)$$

Comparing equation (3.43) with equation (3.14) dimensionally, we get

$$p_m = IA$$

where p_m is called magnetic dipole moment. In vector notation,

$$\vec{p}_m = I \vec{A} \quad (3.44)$$

This implies that a current carrying circular loop behaves as a magnetic dipole of magnetic moment $\vec{p}_m$. So, **the magnetic dipole moment of any current loop is equal to the product of the current and area of the loop.**

Right hand thumb rule

In order to determine the direction of magnetic moment, we use right hand thumb rule which states that

If we curl the fingers of right hand in the direction of current in the loop, then the stretched thumb gives the direction of the magnetic moment associated with the loop.

Table 3.3 End rule – polarity with direction of current in circular loop

Current in circular loop	Polarity	Picture
Anti-clockwise current	North Pole	 Anti-clockwise current Polarity: North Pole
Clockwise current	South Pole	 Clockwise current Polarity: South Pole

3.8.6 Magnetic dipole moment of revolving electron

Suppose an electron undergoes circular motion around the nucleus as shown in Figure 3.36. The circulating electron in a loop is like current in a circular loop (since flow of charge constitutes current). The magnetic dipole moment due to current carrying circular loop is

$$\vec{\mu}_L = I \vec{A} \quad (3.45)$$

In magnitude,

$$\mu_L = I A$$

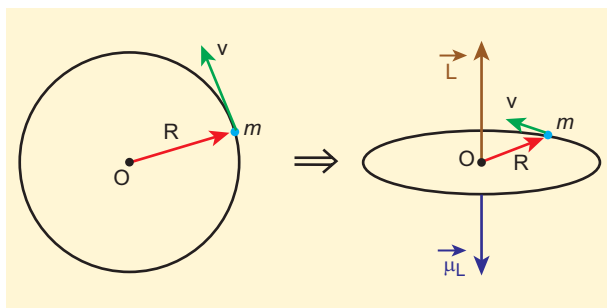


Figure 3.36 (a) Electron revolving in a circular orbit (b) Direction of magnetic dipole moment vector and orbital angular momentum vector are opposite

If T is the time period of revolution of an electron, the current due to circular motion of the electron is

$$I = \frac{-e}{T} \quad (3.46)$$

where $-e$ is the charge of an electron. If R is the radius of the circular orbit and v is the velocity of the electron in the circular orbit, then

$$T = \frac{2\pi R}{v} \quad (3.47)$$

Using equation (3.46) and equation (3.47) in equation (3.45), we get

$$\mu_L = -\frac{e}{\frac{2\pi R}{v}} \pi R^2 = -\frac{evR}{2} \quad (3.48)$$

where $A = \pi R^2$ is the area of the circular loop. By definition, angular momentum of the electron about O is

$$\vec{L} = \vec{R} \times \vec{p}$$

In magnitude,

$$L = R p = m v R \quad (3.49)$$

Using equation (3.48) and equation (3.49), we get

$$\frac{\mu_L}{L} = -\frac{evR/2}{m v R} = -\frac{e}{2m} \Rightarrow \vec{\mu}_L = -\frac{e}{2m} \vec{L} \quad (3.50)$$

The negative sign indicates that the magnetic moment and angular momentum are in opposite direction.

In magnitude,

$$\frac{\mu_L}{L} = \frac{e}{2m} = \frac{1.60 \times 10^{-19}}{2 \times 9.11 \times 10^{-31}} = 0.0878 \times 10^{12} \text{ C kg}^{-1}$$

$$\frac{\mu_L}{L} = 8.78 \times 10^{10} \text{ C kg}^{-1} = \text{constant}$$

The ratio $\frac{\mu_L}{L}$ is a constant known as gyro-magnetic ratio $\left(\frac{e}{2m}\right)$. It must be noted that the gyro-magnetic ratio is a constant of proportionality which connects angular momentum of the electron and the magnetic moment of the electron.

According to Neil's Bohr quantization rule, the angular momentum of an electron moving in a stationary orbit is quantized which means

$$L = n\hbar = n \frac{h}{2\pi}$$

where h is the Planck's constant ($h = 6.63 \times 10^{-34} \text{ J s}$) and number n is the orbit number, i.e., $n = 1, 2, 3, \dots$. Hence,

$$\begin{aligned}\mu_L &= \frac{e}{2m} L = n \frac{eh}{4\pi m} \\ \mu_L &= n \times \frac{(1.60 \times 10^{-19})h}{4\pi m} \text{ A m}^2 \\ &= n \times \frac{(1.60 \times 10^{-19})(6.63 \times 10^{-34})}{4 \times 3.14 \times (9.11 \times 10^{-31})} \\ \mu_L &= n \times 9.27 \times 10^{-24} \text{ A m}^2\end{aligned}$$

The minimum value of magnetic moment can be obtained by substituting $n = 1$,

$$\begin{aligned}\mu_L &= 9.27 \times 10^{-24} \text{ A m}^2 = 9.27 \times 10^{-24} \text{ J T}^{-1} \\ &= (\mu_L)_{\min} = \mu_B\end{aligned}$$

where $\mu_B = \frac{eh}{4\pi m} = 9.27 \times 10^{-24} \text{ A m}^2$ is called

Bohr magneton which is used to measure atomic magnetic moments.

3.9

AMPÈRE'S CIRCUITAL LAW

Ampère's circuital law is used to calculate magnetic field at a point whenever there is a

symmetry in the problem. This is similar to Gauss's law in electrostatics.

3.9.1 Ampère's circuital law

Ampère's law: The line integral of magnetic field over a closed loop is μ_0 times net current enclosed by the loop.

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}} \quad (3.51)$$

where I_{enclosed} is the net current linked by the closed loop C . Note that the line integral does not depend on the shape of the path or the position of the conductor with the magnetic field.



Line integral means integral over a line or curve, symbol used is $\int_C$

Closed line integral means integral over a closed curve (or line), symbol is $\oint$ or $\oint_C$

3.9.2 Magnetic field due to the current carrying wire of infinite length using Ampère's law

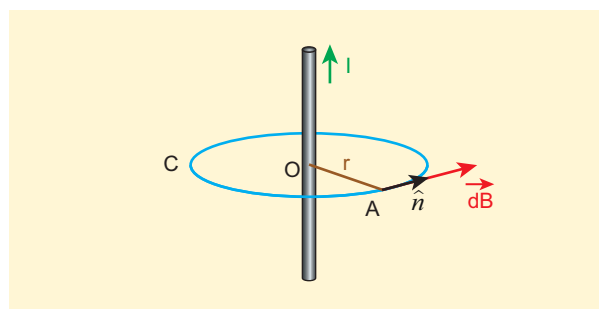


Figure 3.37 Ampèrian loop for current carrying straight wire

Consider a straight conductor of infinite length carrying current I and

the direction of magnetic field lines is shown in Figure 3.37. Since the wire is geometrically cylindrical in shape and symmetrical about its axis, we construct an Ampèrian loop in the form of a circular shape at a distance r from the centre of the conductor as shown in Figure 3.37. From the Ampère's law, we get

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I$$

where $d\vec{l}$ is the line element along the Ampèrian loop (tangent to the circular loop). Hence, the angle between magnetic field vector and line element is zero. Therefore,

$$\oint_C B dl = \mu_0 I$$

where I is the current enclosed by the Ampèrian loop. Due to the symmetry, the magnitude of the magnetic field is uniform over the Ampèrian loop. Hence

$$B \oint_C dl = \mu_0 I$$

For a circular loop, the circumference is $2\pi r$, which implies,

$$B \int_0^{2\pi r} dl = \mu_0 I$$

$$B \cdot 2\pi r = \mu_0 I$$

$$B = \frac{\mu_0 I}{2\pi r}$$

In vector form, the magnetic field is

$$\vec{B} = \frac{\mu_0 I}{2\pi r} \hat{n}$$

where $\hat{n}$ is the unit vector along the tangent to the Ampèrian loop as shown in the Figure 3.37.

EXAMPLE 3.15

Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1 A at distance of 1 m from it. Compare it with Earth's magnetic field.

Solution

Given that $I = 1$ A and radius $r = 1$ m

$$B_{\text{straightwire}} = \frac{\mu_0 I}{2\pi r} = \frac{4\pi \times 10^{-7} \times 1}{2\pi \times 1} = 2 \times 10^{-7} \text{ T}$$

But the Earth's magnetic field is $B_{\text{Earth}} \sim 10^{-5} \text{ T}$

So, $B_{\text{straightwire}}$ is one hundred times smaller than B_{Earth} .

Solenoid

A solenoid is a long coil of wire closely wound in the form of helix as shown in Figure 3.38. When electric current is passed through the solenoid, the magnetic field is produced. The magnetic field of the solenoid is due to the superposition of magnetic fields of each turn of the solenoid. The direction of magnetic field due to solenoid is given by right hand palm-rule.

Inside the solenoid, the magnetic field is nearly uniform and parallel to its axis whereas, outside the solenoid the field is negligibly small. Based on the direction of the current, one end of the solenoid behaves like North Pole and the other end behaves like South Pole.

The current carrying solenoid is held in right hand. If the fingers curl in the direction of current, then extended thumb gives the direction of magnetic field of current carrying solenoid. It is shown in



Figure 3.38 Solenoid

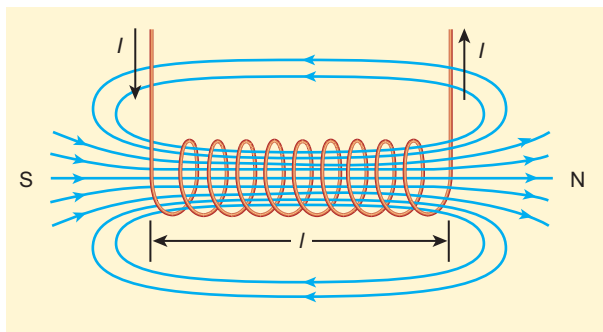


Figure 3.39 Solenoid as a bar magnet

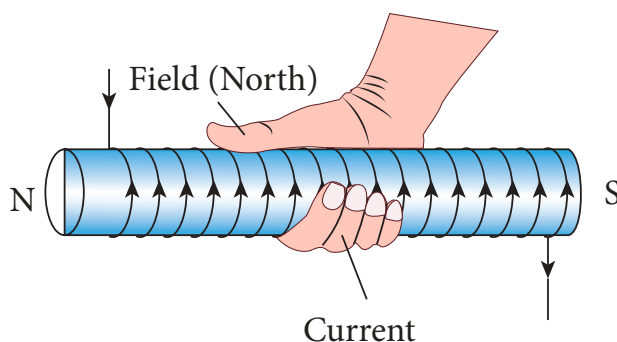


Figure 3.40 The direction of magnetic field of solenoid

Figure 3.39. Hence, the magnetic field of a solenoid looks like the magnetic field of a bar magnet.

The solenoid is assumed to be long which means that the length of the solenoid is large when compared to its diameter. The winding need not to be always circular, it can also be in other shapes. We consider here only circularly wound solenoid as shown in Figure 3.40.

3.9.3 Magnetic field due to a long current carrying solenoid

Consider a solenoid of length L having N turns. The diameter of the solenoid is assumed to be much smaller when compared to its length and the coil is wound very closely.

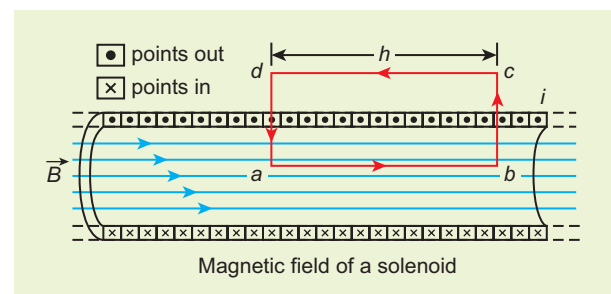


Figure 3.41 Amperian loop for solenoid

In order to calculate the magnetic field at any point inside the solenoid, we use Ampère's circuital law. Consider a rectangular loop abcd as shown in Figure 3.41. Then from Ampère's circuital law,

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}} \\ = \mu_0 \times (\text{total current enclosed by Amperian loop})$$

The left hand side of the equation is

$$\oint_C \vec{B} \cdot d\vec{l} = \int_a^b \vec{B} \cdot d\vec{l} + \int_b^c \vec{B} \cdot d\vec{l} + \int_c^d \vec{B} \cdot d\vec{l} + \int_d^a \vec{B} \cdot d\vec{l}$$

Since the elemental lengths along bc and da are perpendicular to the magnetic field

which is along the axis of the solenoid, the integrals

$$\int_b^c \vec{B} \cdot d\vec{l} = \int_b^c |\vec{B}| |d\vec{l}| \cos 90^\circ = 0$$

Similarly

$$\int_d^a \vec{B} \cdot d\vec{l} = 0$$

Since the magnetic field outside the solenoid is zero, the integral $\int_c^d \vec{B} \cdot d\vec{l} = 0$

For the path along ab , the integral is

$$\int_a^b \vec{B} \cdot d\vec{l} = B \int_a^b dl \cos 0^\circ = B \int_a^b dl$$

where the length of the loop ab as shown in the Figure 3.41 is h . But the choice of length of the loop ab is arbitrary. We can take very large loop such that it is equal to the length of the solenoid L . Therefore the integral is

$$\int_a^b \vec{B} \cdot d\vec{l} = BL$$

Let I be the current passing through the solenoid of N turns, then

$$\int_a^b \vec{B} \cdot d\vec{l} = BL = \mu_0 NI \Rightarrow B = \mu_0 \frac{NI}{L}$$

The number of turns per unit length is given by $\frac{N}{L} = n$, Then

$$B = \mu_0 \frac{nLI}{L} = \mu_0 nI \quad (3.52)$$

Since n is a constant for a given solenoid and μ_0 is also constant. For a fixed current I , the magnetic field inside the solenoid is also a constant.



Note

Solenoid can be used as electromagnet. It produces strong magnetic field that can be turned ON or OFF. This is not possible in case of permanent magnet. Further the strength of the magnetic field can be increased by keeping iron bar inside the solenoid. This is because the magnetic field of the solenoid magnetizes the iron bar and hence the net magnetic field is the sum of magnetic field of the solenoid and magnetic field of magnetised iron. Because of these properties, solenoids are useful in designing variety of electrical appliances.

EXAMPLE 3.16

Calculate the magnetic field inside a solenoid, when

- the length of the solenoid becomes twice with fixed number of turns
- both the length of the solenoid and number of turns are doubled
- the number of turns becomes twice for the fixed length of the solenoid

Compare the results.

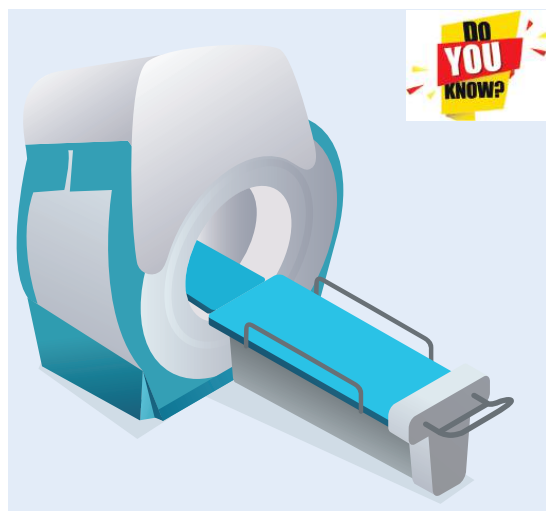
Solution

The magnetic field of a solenoid (inside) is

$$B_{L,N} = \mu_0 \frac{NI}{L}$$

- length of the solenoid becomes twice with fixed number of turns
 $L \rightarrow 2L$ (length becomes twice)
 $N \rightarrow N$ (number of turns remains constant)

The magnetic field is



MRI is Magnetic Resonance Imaging which helps the physicians to diagnose or monitor treatment for a variety of abnormal conditions happening within the head, chest, abdomen and pelvis. It is a non invasive medical test. The patient is placed in a circular opening (actually interior of a solenoid which is made up of superconducting wire) and large current is sent through the superconducting wire to produce a strong magnetic field. So, it uses more powerful magnet, radio frequency pulses and a computer to produce pictures of organs which helps the physicians to examine various parts of the body.

$$B_{2L,N} = \mu_0 \frac{NI}{2L} = \frac{1}{2} B_{L,N}$$

- (b) both the length of the solenoid and number of turns are doubled
 $L \rightarrow 2L$ (length becomes twice)
 $N \rightarrow 2N$ (number of turns becomes twice)

The magnetic field is

$$B_{2L,2N} = \mu_0 \frac{2NI}{2L} = B_{L,N}$$

- (c) the number of turns becomes twice but the length of the solenoid remains same

$L \rightarrow L$ (length is fixed)

$N \rightarrow 2N$ (number of turns becomes twice)

The magnetic field is

$$B_{L,2N} = \mu_0 \frac{2NI}{L} = 2B_{L,N}$$

From the above results,

$$B_{L,2N} > B_{2L,2N} > B_{2L,N}$$

Thus, strength of the magnetic field is increased when we pack more loops into the same length for a given current.

3.9.4 Toroid

A solenoid is bent in such a way its ends are joined together to form a closed ring shape, is called a toroid which is shown in Figure 3.42. The magnetic field has constant magnitude inside the toroid whereas in the interior region (say, at point P) and exterior region (say, at point Q), the magnetic field is zero.



Figure 3.42 Toroid

(a) Open space interior to the toroid

Let us calculate the magnetic field B_p at point P. We construct an Amperian loop 1 of radius r_1 around the point P as shown in

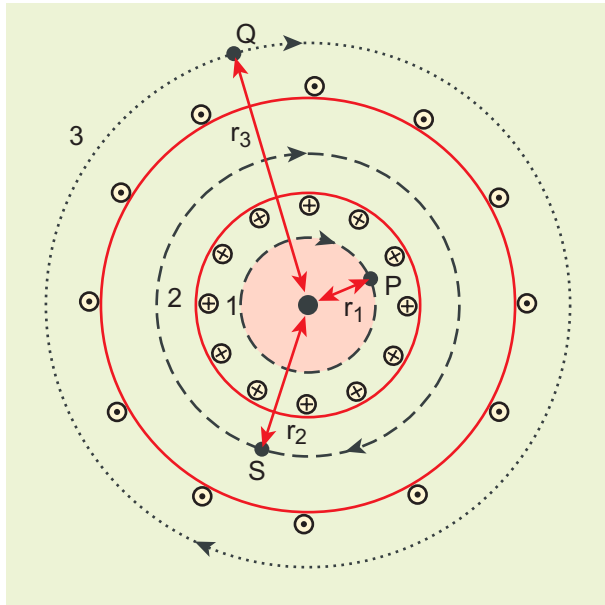


Figure 3.43 Toroid – Amperian loop

Figure 3.43. For simplicity, we take circular loop so that the length of the loop is its circumference.

$$L_1 = 2\pi r_1$$

Ampère's circuital law for the loop 1 is

$$\oint_{\text{loop1}} \vec{B}_p \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

Since the loop 1 encloses no current, $I_{\text{enclosed}} = 0$

$$\oint_{\text{loop1}} \vec{B}_p \cdot d\vec{l} = 0$$

This is possible only if the magnetic field at point P vanishes i.e.

$$\vec{B}_p = 0$$

(b) Open space exterior to the toroid

Let us calculate the magnetic field B_Q at point Q. We construct an Amperian loop 3 of radius r_3 around the point Q as shown in Figure 3.43. The length of the loop is

$$L_3 = 2\pi r_3$$

Ampère's circuital law for the loop 3 is

$$\oint_{\text{loop3}} \vec{B}_Q \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

Since in each turn of the toroid loop, current coming out of the plane of paper is cancelled by the current going into the plane of paper. Thus, $I_{\text{enclosed}} = 0$

$$\oint_{\text{loop3}} \vec{B}_Q \cdot d\vec{l} = 0$$

This is possible only if the magnetic field at point Q vanishes i.e.

$$\vec{B}_Q = 0$$

(c) Inside the toroid

Let us calculate the magnetic field B_S at point S by constructing an Amperian loop 2 of radius r_2 around the point S as shown in Figure 3.43. The length of the loop is

$$L_2 = 2\pi r_2$$

Ampère's circuital law for the loop 2 is

$$\oint_{\text{loop2}} \vec{B}_S \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

Let I be the current passing through the toroid and N be the number of turns of the toroid, then

$$I_{\text{enclosed}} = NI$$

Therefore,

$$\oint_{\text{loop2}} \vec{B}_S \cdot d\vec{l} = \oint_{\text{loop2}} B_S dl \cos\theta = B_S 2\pi r_2$$

$$\oint_{\text{loop2}} \vec{B}_S \cdot d\vec{l} = \mu_0 NI$$

$$B_S = \mu_0 \frac{NI}{2\pi r_2}$$

If the number of turns per unit length $n = \frac{N}{2\pi r_2}$, then the magnetic field at point S is

$$B_S = \mu_0 nI \quad (3.53)$$

3.10

LORENTZ FORCE

When an electric charge q is kept at rest in a magnetic field, no force acts on it. At the same time, if the charge moves in the magnetic field, it experiences a force. This force is different from Coulomb force, studied in unit 1. This force is known as magnetic force. It is given by the equation

$$\vec{F} = q(\vec{v} \times \vec{B}) \quad (3.54)$$

In general, if the charge is moving in both the electric and magnetic fields, the total force experienced by the charge is given by $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. It is known as Lorentz force.

3.10.1 Force on a moving charge in a magnetic field

When an electric charge q is moving with velocity $\vec{v}$ in the magnetic field $\vec{B}$, it experiences a force, called magnetic force $\vec{F}_m$. After careful experiments, Lorentz deduced the force experienced by a moving charge in the magnetic field $\vec{F}_m$

$$\vec{F}_m = q(\vec{v} \times \vec{B}) \quad (3.55)$$

$$\text{In magnitude, } F_m = qvB \sin \theta \quad (3.56)$$

The equations (3.55) and (3.56) imply

1. $\vec{F}_m$ is directly proportional to the magnetic field $\vec{B}$
2. $\vec{F}_m$ is directly proportional to the velocity $\vec{v}$ of the moving charge

3. $\vec{F}_m$ is directly proportional to sine of the angle between the velocity and magnetic field
4. $\vec{F}_m$ is directly proportional to the magnitude of the charge q
5. The direction of $\vec{F}_m$ is always perpendicular to $\vec{v}$ and $\vec{B}$ as $\vec{F}_m$ is the cross product of $\vec{v}$ and $\vec{B}$

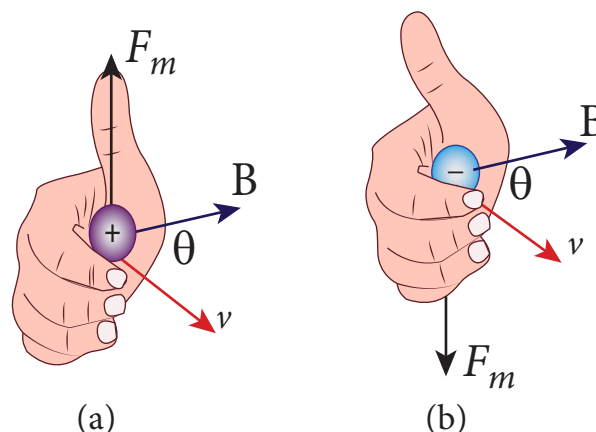


Figure 3.44 Direction of the Lorentz force on (a) positive charge (b) negative charge

6. The direction of $\vec{F}_m$ on negative charge is opposite to the direction of $\vec{F}_m$ on positive charge provided other factors are identical as shown Figure 3.44 (b)
7. If velocity $\vec{v}$ of the charge q is along magnetic field $\vec{B}$ then, $\vec{F}_m$ is zero

Definition of tesla

The strength of the magnetic field is one tesla if a unit charge moving normal to the magnetic field with unit velocity experiences unit force.

$$1 \text{ T} = \frac{1 \text{ N s}}{\text{C m}} = 1 \frac{\text{N}}{\text{A m}} = 1 \text{ N A}^{-1} \text{ m}^{-1}$$

EXAMPLE 3.17

A particle of charge q moves with velocity $\vec{v}$ along positive y - direction in a magnetic field $\vec{B}$. Compute the Lorentz force experienced by the particle (a) when

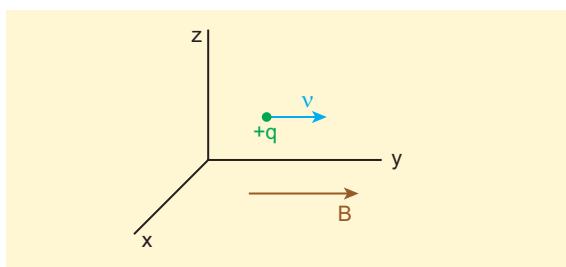


magnetic field is along positive y-direction (b) when magnetic field points in positive z - direction (c) when magnetic field is in zy - plane and making an angle θ with velocity of the particle. Mark the direction of magnetic force in each case.

Solution

Velocity of the particle is $\vec{v} = v \hat{j}$

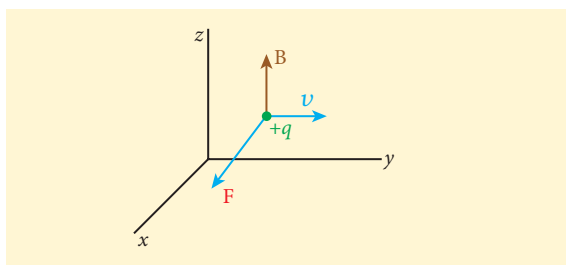
(a) Magnetic field is along positive y - direction, this implies, $\vec{B} = B \hat{j}$



From Lorentz force, $\vec{F}_m = q(\vec{v} \times \vec{B}) = 0$

So, no force acts on the particle when it moves along the direction of magnetic field.

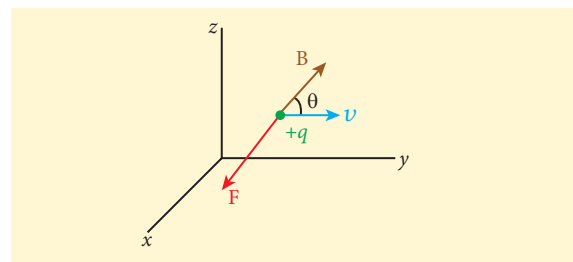
(b) Since the magnetic field points in positive z - direction, this implies, $\vec{B} = B \hat{k}$



From Lorentz force, $\vec{F}_m = q(\vec{v} \times B \hat{k})$
 $= qvB \hat{i}$

Therefore, the magnitude of the Lorentz force is qvB and direction is along positive x - direction.

(c) Magnetic field is in zy - plane and making an angle θ with the velocity of the particle, which implies $\vec{B} = B \cos \theta \hat{j} + B \sin \theta \hat{k}$



From Lorentz force,

$$\vec{F}_m = q(\vec{v} \times (B \cos \theta \hat{j} + B \sin \theta \hat{k}))$$

$$= qvB \sin \theta \hat{i}$$

EXAMPLE 3.18

Compute the work done and power delivered by the Lorentz force on the particle of charge q moving with velocity $\vec{v}$. Calculate the angle between Lorentz force and velocity of the charged particle and also interpret the result.

Solution

For a charged particle moving on a magnetic field, $\vec{F} = q(\vec{v} \times \vec{B})$

The work done by the magnetic field is

$$W = \int \vec{F} \cdot d\vec{r} = \int \vec{F} \cdot \vec{v} dt$$

$$W = q \int (\vec{v} \times \vec{B}) \cdot \vec{v} dt = 0$$

Since $\vec{v} \times \vec{B}$ is perpendicular to $\vec{v}$ and hence $(\vec{v} \times \vec{B}) \cdot \vec{v} = 0$ This means that Lorentz force does no work on the particle. From work-kinetic energy theorem, (Refer section 4.2.6, XI th standard Volume I)

$$\frac{dW}{dt} = P = 0$$

Since, $\vec{F} \cdot \vec{v} = 0 \Rightarrow \vec{F}$ and $\vec{v}$ are perpendicular to each other. The angle between Lorentz force and velocity of the charged particle is 90° . Thus Lorentz force changes the direction of the velocity but not the magnitude of the velocity. Hence Lorentz force does no work and also does not alter kinetic energy of the particle.

3.10.2 Motion of a charged particle in a uniform magnetic field

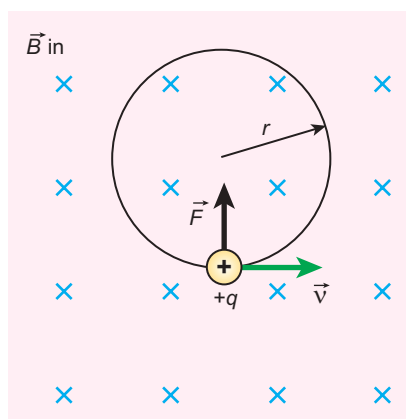


Figure 3.45 Circular motion of a charged particle in a perpendicular uniform magnetic field

Consider a charged particle of charge q having mass m entering into a region of uniform magnetic field $\vec{B}$ with velocity $\vec{v}$ such that velocity is perpendicular to the magnetic field. As soon as the particle enters into the field, Lorentz force acts on it in a direction perpendicular to both magnetic field $\vec{B}$ and velocity $\vec{v}$.

As a result, the charged particle moves in a circular orbit as shown in Figure 3.45. The Lorentz force on the charged particle is given by

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Since Lorentz force alone acts on the particle, the magnitude of the net force on the particle is

$$\sum_i F_i = F_m = qvB$$

This Lorentz force acts as centripetal force for the particle causing it to execute circular motion. Therefore,

$$qvB = m \frac{v^2}{r}$$

The radius of the circular path is

$$r = \frac{mv}{qB} = \frac{p}{qB} \quad (3.57)$$

where $p = mv$ is the magnitude of the linear momentum of the particle. Let T be the time taken by the particle to finish one complete circular motion, then

$$T = \frac{2\pi r}{v} \quad (3.58)$$

Hence substituting (3.57) in (3.58), we get

$$T = \frac{2\pi m}{qB} \quad (3.59)$$

Equation (3.59) is called the **cyclotron period**. The reciprocal of time period is the frequency f , which is

$$f = \frac{1}{T}$$

$$f = \frac{qB}{2\pi m} \quad (3.60)$$

In terms of angular frequency ω ,

$$\omega = 2\pi f = \frac{q}{m} B \quad (3.61)$$

Equations (3.60) and (3.61) are called as **cyclotron frequency or gyro-frequency**.

From equations (3.59), (3.60) and (3.61), we infer that time period and frequency depend only on charge-to-mass ratio (specific charge) but not on velocity or the radius of the circular path.

If a charged particle moves in a region of uniform magnetic field such that its velocity is not perpendicular to the magnetic field, then the velocity of the particle is split up into two components; one component is parallel to the field while the other component perpendicular to the field. The component of velocity parallel to field remains unchanged and the component perpendicular to the field keeps changing due to Lorentz force. Hence the path of the particle is not a circle; it is a helical around the field lines as shown in Figure 3.46.

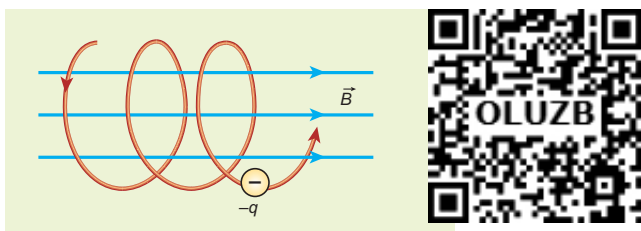


Figure 3.46 Helical path of the electron in a uniform magnetic field

For an example, the helical path of an electron when it moves in a magnetic field is shown in Figure 3.47. Inside the particle detector called cloud chamber, the path is made visible by the condensation of water droplets.

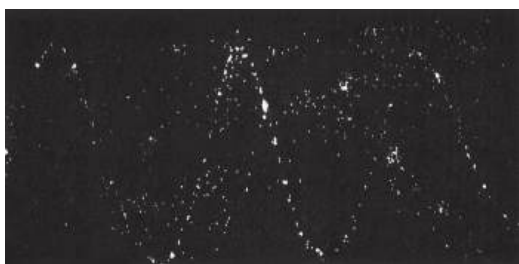


Figure 3.47 Helical path of the electron inside the cloud chamber

EXAMPLE 3.19

An electron moving perpendicular to a uniform magnetic field 0.500 T undergoes circular motion of radius 2.50 mm. What is the speed of electron?

Solution

Charge of an electron $q = -1.60 \times 10^{-19} \text{ C}$
 $\Rightarrow |q| = 1.60 \times 10^{-19} \text{ C}$

Magnitude of magnetic field $B = 0.500 \text{ T}$

Mass of the electron, $m = 9.11 \times 10^{-31} \text{ kg}$

Radius of the orbit, $r = 2.50 \text{ mm}$
 $= 2.50 \times 10^{-3} \text{ m}$

$$\text{Speed of the electron, } v = |q| \frac{rB}{m}$$

$$v = 1.60 \times 10^{-19} \times \frac{2.50 \times 10^{-3} \times 0.500}{9.11 \times 10^{-31}}$$

$$v = 2.195 \times 10^8 \text{ ms}^{-1}$$

EXAMPLE 3.20

A proton moves in a uniform magnetic field of strength 0.500 T magnetic field is directed along the x-axis. At initial time, $t = 0 \text{ s}$, the proton has velocity $\vec{v} = (1.95 \times 10^5 \hat{i} + 2.00 \times 10^5 \hat{k}) \text{ ms}^{-1}$. Find

- At initial time, what is the acceleration of the proton.
- Is the path circular or helical? If helical, calculate the radius of helical trajectory and also calculate the pitch of the helix (Note: Pitch of the helix is the distance travelled along the helix axis per revolution).

Solution

Magnetic field $\vec{B} = 0.500 \hat{i} \text{ T}$



Velocity of the particle

$$\vec{v} = (1.95 \times 10^5 \hat{i} + 2.00 \times 10^5 \hat{k}) \text{ ms}^{-1}$$

Charge of the proton $q = 1.60 \times 10^{-19} \text{ C}$

Mass of the proton $m = 1.67 \times 10^{-27} \text{ kg}$

(a) The force experienced by the proton is

$$\begin{aligned}\vec{F} &= q(\vec{v} \times \vec{B}) \\ &= 1.60 \times 10^{-19} \times ((1.95 \times 10^5 \hat{i} + 2.00 \times 10^5 \hat{k}) \times (0.500 \hat{i})) \\ \vec{F} &= 1.60 \times 10^{-14} \hat{j} \text{ N}\end{aligned}$$

Therefore, from Newton's second law,

$$\begin{aligned}\vec{a} &= \frac{1}{m} \vec{F} = \frac{1}{1.67 \times 10^{-27}} (1.60 \times 10^{-14}) \hat{j} \\ &= 9.58 \times 10^{12} \hat{j} \text{ ms}^{-2}\end{aligned}$$

(b) Trajectory is helical

Radius of helical path is

$$\begin{aligned}R &= \frac{mv_z}{|q|B} = \frac{1.67 \times 10^{-27} \times 2.00 \times 10^5}{1.60 \times 10^{-19} \times 0.500} \\ &= 4.175 \times 10^{-3} \text{ m} = 4.18 \text{ mm}\end{aligned}$$

Pitch of the helix is the distance travelled along x -axis in a time T , which is $P = v_x T$

But time,

$$\begin{aligned}T &= \frac{2\pi}{\omega} = \frac{2\pi m}{|q|B} = \frac{2 \times 3.14 \times 1.67 \times 10^{-27}}{1.60 \times 10^{-19} \times 0.500} \\ &= 13.1 \times 10^{-8} \text{ s}\end{aligned}$$

Hence, pitch of the helix is

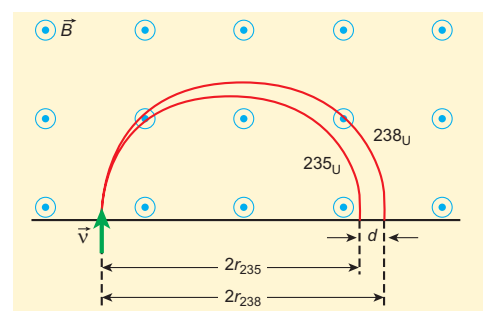
$$\begin{aligned}P &= v_x T = (1.95 \times 10^5) (13.1 \times 10^{-8}) \\ &= 25.5 \times 10^{-3} \text{ m} = 25.5 \text{ mm}\end{aligned}$$

The proton experiences appreciable acceleration in the magnetic field, hence the pitch of the helix is almost six times greater than the radius of the helix.

EXAMPLE 3.21

Two singly ionized isotopes of uranium ${}^{235}_{92}\text{U}$ and ${}^{238}_{92}\text{U}$ (isotopes have same atomic

number but different mass number) are sent with velocity $1.00 \times 10^5 \text{ m s}^{-1}$ into a magnetic field of strength 0.500 T normally. Compute the distance between the two isotopes after they complete a semi-circle. Also compute the time taken by each isotope to complete one semi-circular path. (Given: masses of the isotopes: $m_{235} = 3.90 \times 10^{-25} \text{ kg}$ and $m_{238} = 3.95 \times 10^{-25} \text{ kg}$)



Solution

Since isotopes are singly ionized, they have equal charge which is equal to the charge of an electron, $q = -1.6 \times 10^{-19} \text{ C}$. Mass of uranium ${}^{235}_{92}\text{U}$ and ${}^{238}_{92}\text{U}$ are $3.90 \times 10^{-25} \text{ kg}$ and $3.95 \times 10^{-25} \text{ kg}$ respectively. Magnetic field applied, $B = 0.500 \text{ T}$. Velocity of the electron is $1.00 \times 10^5 \text{ ms}^{-1}$, then

(a) the radius of the path of ${}^{235}_{92}\text{U}$ is r_{235}

$$\begin{aligned}r_{235} &= \frac{m_{235}v}{|q|B} = \frac{3.90 \times 10^{-25} \times 1.00 \times 10^5}{1.6 \times 10^{-19} \times 0.500} \\ &= 48.8 \times 10^{-2} \text{ m} \\ r_{235} &= 48.8 \text{ cm}\end{aligned}$$

The diameter of the semi-circle due to ${}^{235}_{92}\text{U}$ is $d_{235} = 2r_{235} = 97.6 \text{ cm}$

The radius of the path of ${}^{238}_{92}\text{U}$ is r_{238} then

$$\begin{aligned}r_{238} &= \frac{m_{238}v}{|q|B} = \frac{3.95 \times 10^{-25} \times 1.00 \times 10^5}{1.6 \times 10^{-19} \times 0.500} \\ &= 49.4 \times 10^{-2} \text{ m} \\ r_{238} &= 49.4 \text{ cm}\end{aligned}$$

The diameter of the semi-circle due to ${}^{238}_{92}\text{U}$ is $d_{238} = 2r_{238} = 98.8 \text{ cm}$

Therefore the separation distance between the isotopes is $\Delta d = d_{238} - d_{235} = 1.2 \text{ cm}$

(b) The time taken by each isotope to complete one semi-circular path are

$$t_{235} = \frac{\text{magnitude of the displacement}}{\text{velocity}}$$

$$= \frac{97.6 \times 10^{-2}}{1.00 \times 10^5} = 9.76 \times 10^{-6} \text{ s} = 9.76 \mu\text{s}$$

$$t_{238} = \frac{\text{magnitude of the displacement}}{\text{velocity}}$$

$$= \frac{98.8 \times 10^{-2}}{1.00 \times 10^5} = 9.88 \times 10^{-6} \text{ s} = 9.88 \mu\text{s}$$

Note that even though the difference between mass of two isotopes are very small, this arrangement helps us to convert this small difference into an easily measurable distance of separation. This arrangement is known as mass spectrometer. A mass spectrometer is used in many areas in sciences, especially in medicine, in space science, in geology etc. For example, in medicine, anaesthesiologists use it to measure the respiratory gases and biologist use it to determine the reaction mechanisms in photosynthesis.

3.10.3 Motion of a charged particle under crossed electric and magnetic field (velocity selector)

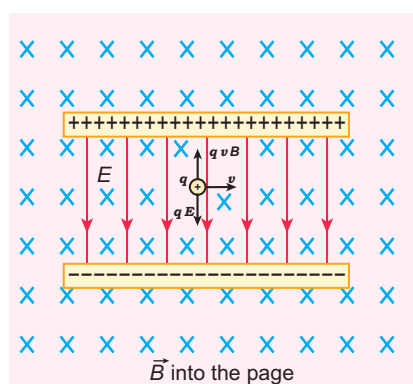


Figure 3.48 Velocity selector

Let us consider an experimental arrangement to illustrate velocity selector as shown in Figure 3.48. In the region of space between the parallel plates of a capacitor which produce uniform electric field $\vec{E}$, a uniform magnetic field $\vec{B}$ is maintained perpendicular to the direction of electric field.

Suppose a charged particle with charge q enters the space from left side with a velocity $\vec{v}$, the net force on the particle is

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

For a positive charge, the electric force on the charge acts in downward direction whereas the Lorentz force acts upwards. When these two forces balance each other, then

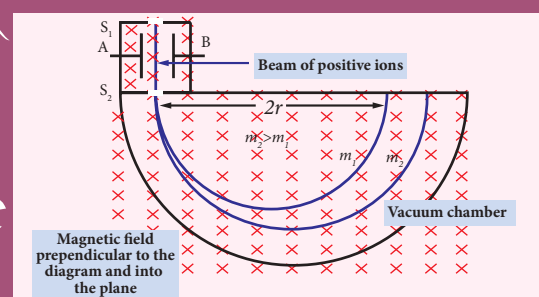
$$qE = qvB$$

$$\Rightarrow v = \frac{E}{B} \quad (3.62)$$



This principle is used in Bainbridge mass spectrograph to separate the isotopes. This concept is explained in Example (3.21).

Schematic diagram of Bainbridge mass spectrometer





This means, for a given magnitude of $\vec{E}$ -field and $\vec{B}$ -field, the forces act only on the particle moving with particular speed $v = \frac{E}{B}$. This speed is independent of mass and charge.

By proper choice of electric and magnetic fields, the particle with particular speed can be selected. Such an arrangement of fields is called a **velocity selector**.



EXAMPLE 3.22

Let E be the electric field of magnitude $6.0 \times 10^6 \text{ N C}^{-1}$ and B be the magnetic field magnitude 0.83 T . Suppose an electron is accelerated with a potential of 200 V , will it show zero deflection?. If not, at what potential will it show zero deflection.

Solution:

Electric field, $E = 6.0 \times 10^6 \text{ N C}^{-1}$ and magnetic field, $B = 0.83 \text{ T}$.

Then

$$v = \frac{E}{B} = \frac{6.0 \times 10^6}{0.83} = 7.23 \times 10^6 \text{ ms}^{-1}$$

When an electron goes with this velocity, it shows null deflection. Since the accelerating potential is 200 V , the electron acquires kinetic energy because of this accelerating potential. Hence,

$$\frac{1}{2}mv^2 = eV$$
$$v = \sqrt{\frac{2eV}{m}}$$

Since the mass of the electron, $m = 9.1 \times 10^{-31} \text{ kg}$ and charge of an electron, $|q| = e = 1.6 \times 10^{-19} \text{ C}$. The velocity acquired by the electron due to accelerating potential of 200 V is

$$v_{200} = \sqrt{\frac{2(1.6 \times 10^{-19})(200)}{(9.1 \times 10^{-31})}} = 8.39 \times 10^6 \text{ ms}^{-1}$$

Since the speed $v_{200} > v$, the electron is deflected towards direction of Lorentz force. So, in order to have null deflection, the potential, we have to supply is

$$V = \frac{1}{2} \frac{mv^2}{e} = \frac{(9.1 \times 10^{-31}) \times (7.23 \times 10^6)^2}{2 \times (1.6 \times 10^{-19})}$$
$$V = 148.65 \text{ V}$$

3.10.4 Cyclotron

Cyclotron (Figure 3.49) is a device used to accelerate the charged particles to gain large kinetic energy. It is also called as high energy accelerator. It was invented by Lawrence and Livingston in 1934.

Principle

When a charged particle moves perpendicular to the magnetic field, it experiences magnetic Lorentz force.

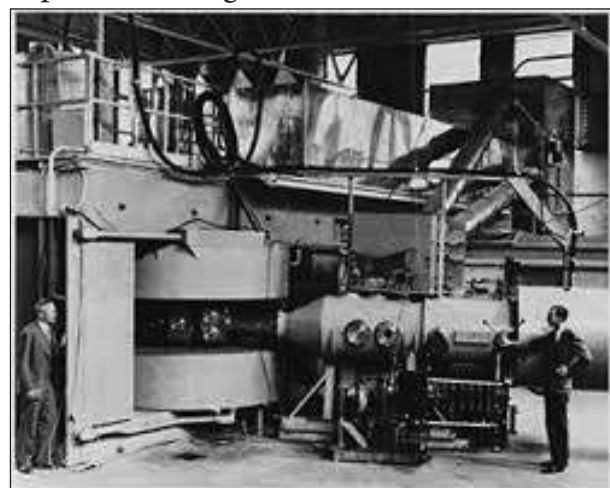


Figure 3.49 Cyclotron invented by Lawrence and Livingston

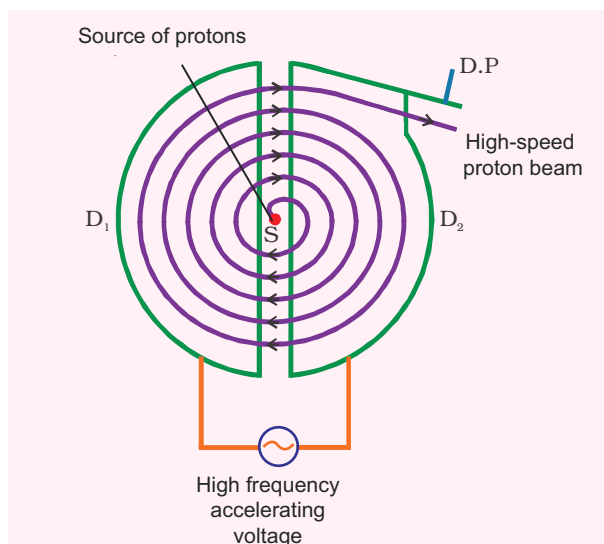


Figure 3.50 Working of cyclotron

Construction

The schematic diagram of a cyclotron is shown in Figure 3.50. The particles are allowed to move in between two semi-circular metal containers called Dees (hollow D - shaped objects). Dees are enclosed in an evacuated chamber and it is kept in a region with uniform magnetic field controlled by an electromagnet. The direction of magnetic field is normal to the plane of the Dees. The two Dees are kept separated with a gap and the source S (which ejects the particle to be accelerated) is placed at the centre in the gap between the Dees. Dees are connected to high frequency alternating potential difference.

Working

Let us assume that the ion ejected from source S is positively charged. As soon as ion is ejected, it is accelerated towards a Dee (say, D_1) which has negative potential at that time. Since the magnetic field is normal to the plane of the Dees, the ion moves in a circular path. After one semi-circular path inside D_1 , the ion reaches the gap between Dees. At this time, the polarities of the Dees

are reversed so that the ion is now accelerated towards D_2 with a greater velocity. For this circular motion, the centripetal force on the charged particle q is provided by Lorentz force.

$$\begin{aligned}\frac{mv^2}{r} &= qvB \\ \Rightarrow r &= \frac{m}{qB}v \\ \Rightarrow r &\propto v\end{aligned}\quad (3.63)$$

From the equation (3.63), the increase in velocity increases the radius of circular path. This process continues and hence the particle moves in spiral path of increasing radius. Once it reaches near the edge, it is taken out with the help of deflector plate and allowed to hit the target T.

The important condition in cyclotron operation is that when the frequency f at which the positive ion circulates in the magnetic field must be equal to the constant frequency of the electrical oscillator f_{osc} . This is called resonance condition.

From equation (3.60), we have

$$f_{osc} = \frac{qB}{2\pi m}$$

The time period of oscillation is

$$T = \frac{2\pi m}{qB}$$

The kinetic energy of the charged particle is

$$KE = \frac{1}{2}mv^2 = \frac{q^2 B^2 r^2}{2m} \quad (3.64)$$

Limitations of cyclotron

- (a) the speed of the ion is limited
- (b) electron cannot be accelerated
- (c) uncharged particles cannot be accelerated



Note

Deutrons (bundles of one proton and one neutron) can be accelerated because it has same charge as that of proton. But neutron (electrically neutral particle) cannot be accelerated by the cyclotron. When a deuteron is bombarded with a beryllium target, a beam of high energy neutrons are produced. These high-energy neutrons are sent into the patient's cancerous region to break the bonds in the DNA of the cancer cells (killing the cells). This is used in treatment of fast-neutron cancer therapy.

EXAMPLE 3.23

Suppose a cyclotron is operated to accelerate protons with a magnetic field of strength 1 T. Calculate the frequency in which the electric field between two Dees could be reversed.

Solution

Magnetic field $B = 1 \text{ T}$

Mass of the proton, $m_p = 1.67 \times 10^{-27} \text{ kg}$

Charge of the proton, $q = 1.60 \times 10^{-19} \text{ C}$

$$f = \frac{qB}{2\pi m_p} = \frac{(1.60 \times 10^{-19})(1)}{2(3.14)(1.67 \times 10^{-27})}$$

$$= 15.3 \times 10^6 \text{ Hz} = 15.3 \text{ MHz}$$

3.10.5 Force on a current carrying conductor placed in a magnetic field

When a current carrying conductor is placed in a magnetic field, the force experienced by the conductor is equal to the sum of Lorentz forces on the individual charge carriers in the conductor. Consider a small segment of conductor of length dl , with cross-sectional area A

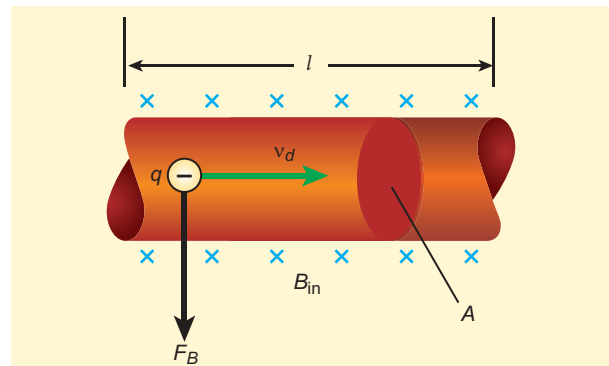


Figure 3.51 Current carrying conductor in a magnetic field

and current I as shown in Figure 3.51. The free electrons drift opposite to the direction of current. So the relation between current I and magnitude of drift velocity v_d (Refer Unit 2) is

$$I = neAv_d \quad (3.65)$$

If the conductor is kept in a magnetic field $\vec{B}$, then average force experienced by the charge (electron) in the conductor is

$$\vec{f} = -e(\vec{v}_d \times \vec{B})$$

If n is the number of free electrons present in unit volume, then

$$n = \frac{N}{V}$$

where N is the number of free electrons in the small element of volume $V = Adl$.

Hence Lorentz force on the elementary section of length dl is the product of the number of the electrons ($N = nAdl$) and the force acting on each electron.

$$d\vec{F} = -enAdl(\vec{v}_d \times \vec{B})$$

The current element in the conductor is $I d\vec{l} = -enA\vec{v}_d dl$. Therefore the force on the small elemental section of the current-carrying conductor is

$$d\vec{F} = (I d\vec{l} \times \vec{B}) \quad (3.66)$$

Thus the force on a straight current carrying conductor of length l placed in a uniform magnetic field is

$$\vec{F}_{\text{total}} = (I\vec{l} \times \vec{B}) \quad (3.67)$$

In magnitude,

$$F_{\text{total}} = BIl \sin \theta$$

- If the conductor is placed along the direction of the magnetic field, the angle then $\theta = 0^\circ$. Hence, the force experienced by the conductor is zero.
- If the conductor is placed perpendicular to the magnetic field, then the angle $\theta = 90^\circ$. Hence, the force experienced by the conductor is maximum, which is $F_{\text{total}} = BIl$.

Fleming's left hand rule

When a current carrying conductor is placed in a magnetic field, the direction of the force experienced by it is given by Fleming's Left Hand Rule (FLHR) as shown in Figure 3.52.

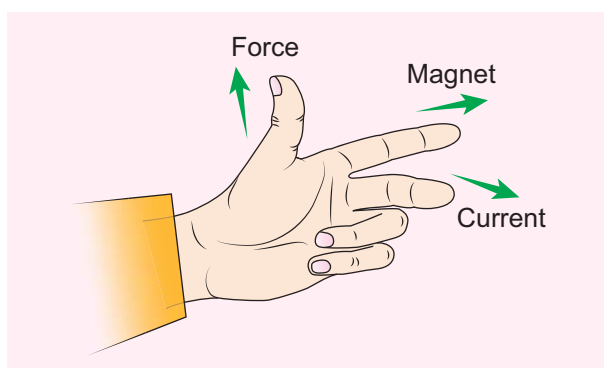


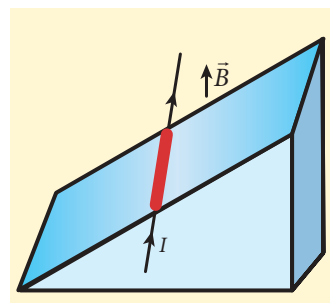
Figure 3.52 Fleming's Left Hand Rule (FLHR)

Stretch out forefinger, the middle finger and the thumb of the left hand such that they are in three mutually perpendicular directions. If the forefinger points in the direction of magnetic field,

the middle finger in the direction of the electric current, then thumb will point in the direction of the force experienced by the conductor.

EXAMPLE 3.24

A metallic rod of linear density is 0.25 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 45° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of strength 0.25 T is acting on it in the vertical direction. Calculate the electric current flowing in the rod to keep it stationary.

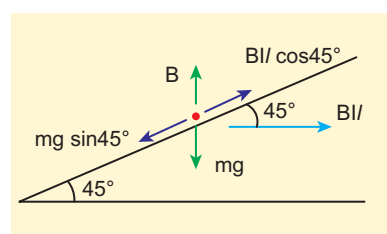


Solution

The linear density of the rod i.e., mass per unit length of the rod is 0.25 kg m^{-1}

$$\Rightarrow \frac{m}{l} = 0.25 \text{ kg m}^{-1}$$

Let I be the current flowing in the metallic rod. The direction of electric current is into the plane of the paper. The direction of magnetic force IBl is given by Fleming's left hand rule.





For equilibrium of the rod,

$$\begin{aligned}
 mg \sin 45^\circ &= IBl \cos 45^\circ \\
 \Rightarrow I &= \frac{1}{B} \frac{m}{l} g \tan 45^\circ \\
 &= \frac{0.25 \text{ kg m}^{-1}}{0.25 \text{ T}} \times 1 \times 9.8 \text{ m s}^{-2} \\
 \Rightarrow I &= 9.8 \text{ A}
 \end{aligned}$$

So, we need to supply current of 9.8 A to keep the metallic rod stationary.

3.10.6 Force between two long parallel current carrying conductors

Let two long straight parallel current carrying conductors separated by a distance r be kept in air medium as shown in Figure 3.53. Let I_1 and I_2 be the electric currents passing through the conductors A and B in same direction (i.e. along z - direction) respectively. The net magnetic field at a distance r due to current I_1 in conductor A is

$$\vec{B}_1 = \frac{\mu_0 I_1}{2\pi r} (-\hat{i}) = -\frac{\mu_0 I_1}{2\pi r} \hat{i}$$

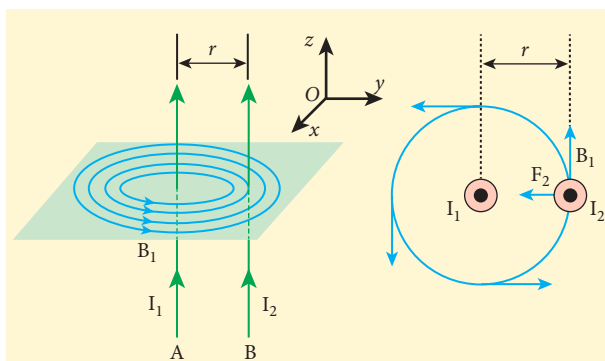


Figure 3.53 Two long straight parallel wires

From thumb rule, the direction of magnetic field is perpendicular to the plane of the paper and inwards (arrow into the page $\otimes$) i.e. along negative $\hat{i}$ direction.

Let us consider a small elemental length dl in conductor B at which the magnetic field $\vec{B}_1$ is present. From equation 3.66, Lorentz force on the element dl of conductor B is

$$\begin{aligned}
 d\vec{F} &= (I_2 d\vec{l} \times \vec{B}_1) = -I_2 dl \frac{\mu_0 I_1}{2\pi r} (\hat{k} \times \hat{i}) \\
 &= -\frac{\mu_0 I_1 I_2 dl}{2\pi r} \hat{j}
 \end{aligned}$$

Therefore the force on dl of the wire B is directed towards the conductor A. So the element of length dl in B is attracted towards the conductor A. Hence the force per unit length of the conductor B due to current in the conductor A is

$$\frac{\vec{F}}{l} = -\frac{\mu_0 I_1 I_2}{2\pi r} \hat{j}$$

Similarly, the net magnetic induction due to current I_2 (in conductor B) at a distance r in the elemental length dl of conductor A is

$$\vec{B}_2 = \frac{\mu_0 I_2}{2\pi r} \hat{i}$$

From the thumb rule, direction of magnetic field is perpendicular to the plane of the paper and outwards (arrow out of the page $\odot$) i.e., along positive $\hat{i}$ direction. Hence, the magnetic force acting on element dl of the conductor A is

$$\begin{aligned}
 d\vec{F} &= (I_1 d\vec{l} \times \vec{B}_2) = I_1 dl \frac{\mu_0 I_2}{2\pi r} (\hat{k} \times \hat{i}) \\
 &= \frac{\mu_0 I_1 I_2 dl}{2\pi r} \hat{j}
 \end{aligned} \quad (3.68)$$

Therefore the force on dl of conductor A is directed towards the conductor B. So the length dl is attracted towards the conductor B as shown in Figure (3.54).

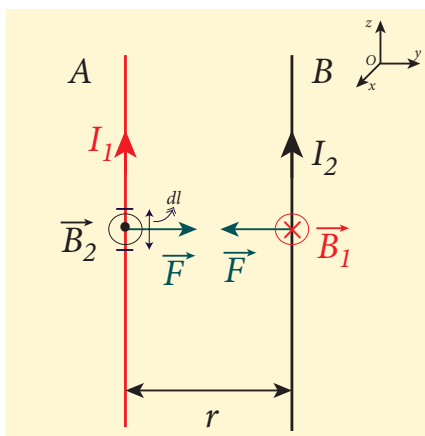


Figure 3.54 Current in both the conductors are in the same direction - attracts each other

The force acting per unit length of the conductor A due to the current in conductor B is

$$\frac{\vec{F}}{l} = \frac{\mu_0 I_1 I_2}{2\pi r} \hat{j}$$

Thus the force between two parallel current carrying conductors is attractive if they carry current in the same direction. (Figure 3.55)

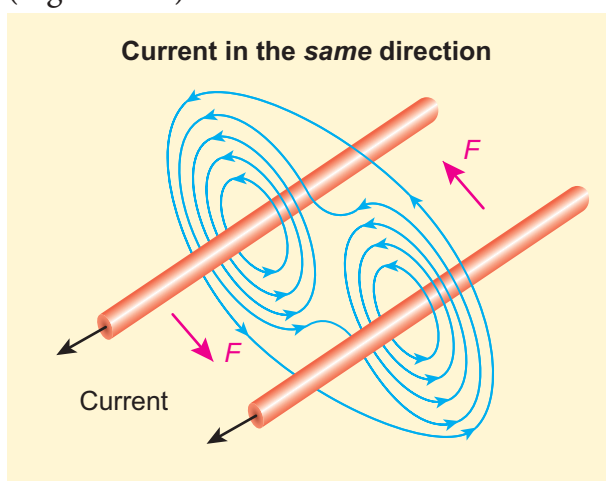


Figure 3.55 Two parallel conductors carrying current in same direction experience an attractive force

The force between two parallel current carrying conductors is repulsive if they carry current in opposite directions (Figure 3.56).

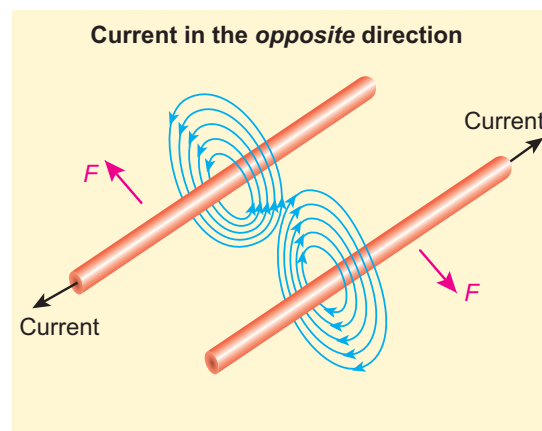


Figure 3.56 Two parallel conductors carrying current in opposite direction experience a repulsive force

Definition of ampere

One ampere is defined as that constant current which when passed through each of the two infinitely long parallel straight conductors kept side by side parallelly at a distance of one metre apart in air or vacuum causes each conductor to experience a force of 2×10^{-7} newton per metre length of conductor.



In November 2018, however, the redefinition of the ampere — along with three other SI base units: the kilogram (mass), kelvin (temperature) and mole (amount of substance) — was approved. Starting on May 20, 2019, the ampere is based on a fundamental physical constant: the elementary charge (e). Thus, **one ampere is defined as the electric current corresponding to the flow of $1/(1.602\,176\,634 \times 10^{-19})$ elementary charges per second.**

3.11

TORQUE ON A CURRENT LOOP

The force on a current carrying wire in a magnetic field is responsible for the motor operation.

3.11.1 Torque on a current loop placed in a magnetic field

Consider a rectangular loop PQRS carrying current I is placed in a uniform magnetic field $\vec{B}$. Let a and b be the length and breadth of rectangular loop respectively. The unit vector $\hat{n}$ normal to the plane of the loop makes an angle θ with the magnetic field as shown in Figure 3.57.

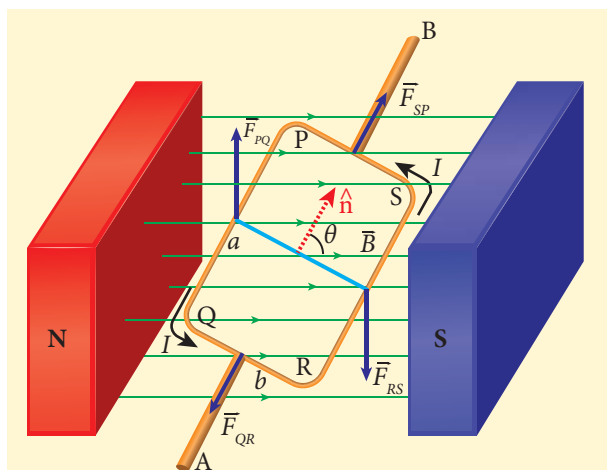


Figure 3.57 Rectangular coil placed in a magnetic field

The magnitude of the magnetic force acting on the current-carrying arm PQ is $F_{PQ} = IaB \sin\left(\frac{\pi}{2}\right) = IaB$. The direction of the force is found using right hand cork screw rule and its direction is upwards.

The magnitude of the force on the arm QR is $F_{QR} = IbB \sin\left(\frac{\pi}{2} - \theta\right) = IbB \cos \theta$ and its direction is as shown in Figure 3.57.

The magnitude of the force on the arm RS is $F_{RS} = IaB \sin\left(\frac{\pi}{2}\right) = IaB$ and its direction is downwards.

The magnitude of the force acting on the arm SP is $F_{SP} = IbB \sin\left(\frac{\pi}{2} + \theta\right) = IbB \cos \theta$ and its direction is also as shown in the Figure 3.57.

Since the forces F_{QR} and F_{SP} are equal, opposite and collinear, they cancel each other. But the forces F_{PQ} and F_{RS} , which are equal in magnitude and opposite in direction, are not acting along same straight line. Therefore, F_{PQ} and F_{RS} constitute a couple which exerts a torque on the loop.

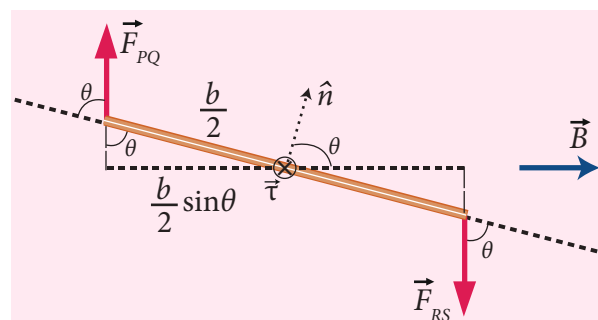


Figure 3.58 Side view of current loop

The magnitude of torque acting on the arm PQ about AB is $\tau_{PQ} = \left(\frac{b}{2} \sin \theta\right) IaB$ and it points in the direction of AB. The magnitude of the torque acting on the arm RS about AB is $\tau_{RS} = \left(\frac{b}{2} \sin \theta\right) IaB$ and points also in the same direction AB as shown in Figure 3.58.

The total torque acting on the entire loop about an axis AB is given by

$$\begin{aligned} \tau &= \left(\frac{b}{2} \sin \theta\right) F_{PQ} + \left(\frac{b}{2} \sin \theta\right) F_{RS} \\ &= Ia(b \sin \theta)B \end{aligned}$$

$$\tau = IAB \sin \theta \text{ along the direction AB}$$



In vector form, $\vec{\tau} = (I\vec{A}) \times \vec{B}$

The above equation can also be written in terms of magnetic dipole moment

$\vec{\tau} = \vec{p}_m \times \vec{B}$ where $\vec{p}_m = I\vec{A}$

The tendency of the torque is to rotate the loop so as to align its normal vector with the direction of the magnetic field.

If there are N turns in the rectangular loop, then the torque is given by $\tau = NIAB \sin \theta$

Special cases:

(a) When $\theta = 90^\circ$ or the plane of the loop is parallel to the magnetic field, the torque on the current loop is maximum.

$$\tau_{\max} = IAB$$

(b) When $\theta = 0^\circ/180^\circ$ or the plane of the loop is perpendicular to the magnetic field, the torque on the current loop is zero.

3.11.2 Moving coil galvanometer

Moving coil galvanometer is a device which is used to detect the flow of current in an electrical circuit.

Principle When a current carrying loop is placed in a uniform magnetic field, it experiences a torque.

Construction

A moving coil galvanometer consists of a rectangular coil PQRS of insulated thin copper wire. The coil contains a large number of turns wound over a light metallic frame. A cylindrical soft-iron core is placed symmetrically inside the coil as shown in Figure 3.59. The rectangular coil is suspended freely between two pole pieces of a horse-shoe magnet.

The upper end of the rectangular coil is attached to one end of fine strip of phosphor bronze and the lower end of the coil is connected to a hair spring which is also made up of phosphor bronze. In a fine suspension strip, a small plane mirror is attached in order

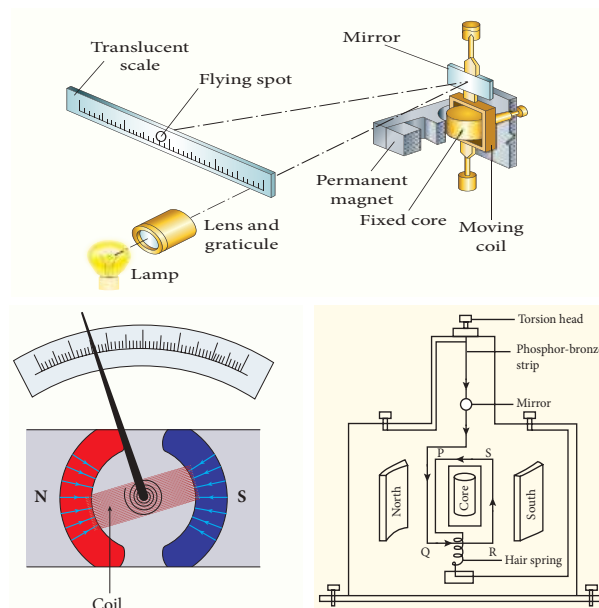


Figure 3.59 Moving coil galvanometer – its parts

to measure the deflection of the coil with the help of lamp and scale arrangement. The other end of the mirror is connected to a torsion head. In order to pass electric current through the galvanometer, the suspension strip and the spring are connected to terminals.

Working

Consider a single turn of the rectangular coil PQRS whose length is l and breadth b . $PQ = RS = l$ and $QR = SP = b$. Let I be the electric current flowing through the rectangular coil PQRS as shown in Figure 3.60. The horse-shoe magnet has hemi-spherical magnetic poles which produces a radial magnetic field. Due to this radial field, the sides QR and SP are always parallel to the magnetic field B and experience no force. The sides PQ and RS are always perpendicular to the magnetic field and experience equal forces in opposite directions. Due to this, torque is produced.

For single turn, the deflecting torque is

$$\tau = bF = bBIl = (lb) BI = ABI$$

since area of the coil, $A = lb$

For coil with N turns, we get

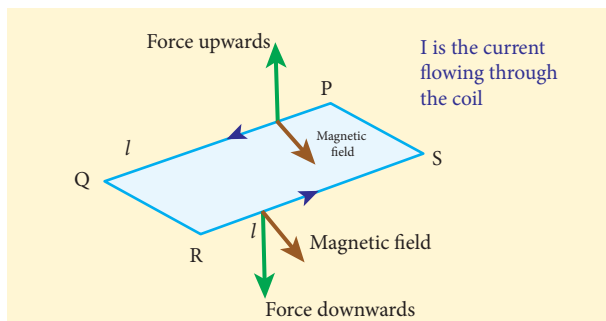


Figure 3.60 Force acting on current carrying coil

$$\tau = NAB I \quad (3.69)$$

Due to this deflecting torque, the coil gets twisted and restoring torque (also known as restoring couple) is developed. Hence the moment of the restoring torque is proportional to the amount of twist θ . Thus

$$\tau = K \theta \quad (3.70)$$

where K is the restoring couple per unit twist.

At equilibrium, the deflecting couple must be equal to the restoring couple. Therefore we get,

$$\begin{aligned} NAB I &= K \theta \\ \Rightarrow I &= \frac{K}{NAB} \theta \\ \text{(or) } I &= G \theta \end{aligned} \quad (3.71)$$

where $G = \frac{K}{NAB}$ is called galvanometer constant or current reduction factor of the galvanometer.

Since the suspended moving coil galvanometer is very sensitive, we have to handle with high care while doing experiments. Most of the galvanometer we use are pointer type moving coil galvanometer.

Figure of merit of a galvanometer

It is defined as the current required to produce a deflection of one scale division in the galvanometer.

Sensitivity of a galvanometer

The galvanometer is said to be sensitive if it shows large scale deflection even for a small current passed through it or a small voltage applied across it.

Current sensitivity

It is defined as the deflection produced per unit current flowing through galvanometer.

$$I_s = \frac{\theta}{I} = \frac{NAB}{K} \Rightarrow I_s = \frac{1}{G} \quad (3.72)$$

The current sensitivity of a galvanometer can be increased by

- (i) increasing the number of turns, N
- (ii) increasing the magnetic induction, B
- (iii) increasing the area of the coil, A
- (iv) decreasing the couple per unit twist of the suspension wire, K .

Phosphor - bronze wire is used as the suspension wire because the couple per unit twist is very small.

Voltage sensitivity

It is defined as the deflection produced per unit voltage applied across galvanometer.

$$\begin{aligned} V_s &= \frac{\theta}{V} \\ V_s &= \frac{\theta}{IR_g} = \frac{NAB}{K R_g} \quad \text{(or)} \\ V_s &= \frac{1}{G R_g} = \frac{I_s}{R_g} \end{aligned} \quad (3.73)$$

where R_g is the resistance of galvanometer.

EXAMPLE 3.25

The coil of a moving coil galvanometer has 5 turns and each turn has an effective area of $2 \times 10^{-2} \text{ m}^2$. It is suspended in a magnetic field whose strength is $4 \times 10^{-2} \text{ Wb m}^{-2}$. If the torsional constant K of the suspension fibre is $4 \times 10^{-9} \text{ N m deg}^{-1}$.

- Find its current sensitivity in division per micro - ampere.
- Calculate the voltage sensitivity of the galvanometer for it to have full scale deflection of 50 divisions for 25 mV.
- Compute the resistance of the galvanometer.

Solution

$$N = 5 \text{ turns}$$

$$A = 2 \times 10^{-2} \text{ m}^2$$

$$B = 4 \times 10^{-2} \text{ Wb m}^{-2}$$

$$K = 4 \times 10^{-9} \text{ N m deg}^{-1}$$

(a) Current sensitivity

$$I_s = \frac{NAB}{K} = \frac{5 \times 2 \times 10^{-2} \times 4 \times 10^{-2}}{4 \times 10^{-9}} \\ = 10^6 \text{ divisions per ampere}$$

$$1 \mu\text{A} = 1 \text{ microampere} = 10^{-6} \text{ ampere}$$

Therefore,

$$I_s = 10^6 \frac{\text{div}}{\text{A}} = 1 \frac{\text{div}}{10^{-6} \text{ A}} = 1 \frac{\text{div}}{\mu\text{A}} \\ I_s = 1 \text{ div}(\mu\text{A})^{-1}$$

(b) Voltage sensitivity

$$V_s = \frac{\theta}{V} = \frac{50 \text{ div}}{25 \text{ mV}} = 2 \times 10^3 \text{ div V}^{-1}$$

(c) The resistance of the galvanometer is

$$R_g = \frac{I_s}{V_s} = \frac{10^6 \frac{\text{div}}{\text{A}}}{2 \times 10^3 \frac{\text{div}}{\text{V}}} = 0.5 \times 10^3 \frac{\text{V}}{\text{A}} = 0.5 \text{ k}\Omega$$

EXAMPLE 3.26

The resistance of a moving coil galvanometer is made twice its original value in order to increase current sensitivity by 50%. Find the percentage change in voltage sensitivity.

Solution

$$\text{Voltage sensitivity is } V_s = \frac{I_s}{R_g}$$

When the resistance is doubled, then new resistance is $R_g' = 2R_g$

Increase in current sensitivity is

$$I_s' = \left(1 + \frac{50}{100}\right) I_s = \frac{3}{2} I_s$$

The new voltage sensitivity is

$$V_s' = \frac{\frac{3}{2} I_s}{2R_g} = \frac{3}{4} V_s$$

Hence the voltage sensitivity decreases. The percentage decrease in voltage sensitivity is

$$\frac{V_s - V_s'}{V_s} \times 100\% = 25\%$$

Conversion of galvanometer into ammeter and voltmeter

A galvanometer is very sensitive instrument to detect the current. It can be easily converted into ammeter and voltmeter.

Galvanometer to an Ammeter

Ammeter is an instrument used to measure current flowing in the electrical circuit. The ammeter must offer low resistance such that it will not change the current passing through it. So ammeter is connected in series to measure the circuit current.

A galvanometer is converted into an ammeter by connecting a low resistance in parallel with the galvanometer. This low resistance is called shunt resistance S . The scale is now calibrated in ampere and the range of ammeter depends on the values of the shunt resistance.

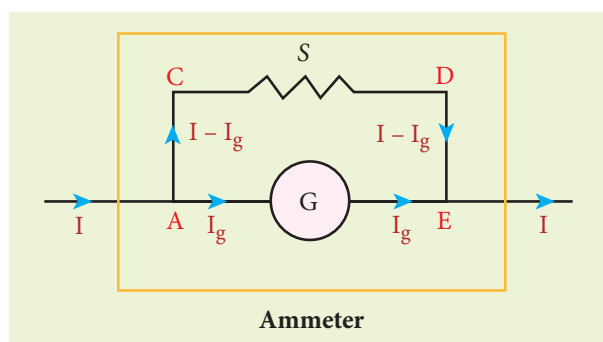


Figure 3.61 Shunt resistance connected in parallel

Let I be the current passing through the circuit as shown in Figure 3.61. When current I reaches the junction A, it divides into two components. Let I_g be the current passing through the galvanometer of resistance R_g through a path AGE and the remaining current $(I - I_g)$ passes along the path ACDE through shunt resistance S . The value of shunt resistance is so adjusted that current I_g produces full scale deflection in the galvanometer. The potential difference across galvanometer is same as the potential difference across shunt resistance.

$$V_{\text{galvanometer}} = V_{\text{shunt}}$$

$$\Rightarrow I_g R_g = (I - I_g) S$$

$$S = \frac{I_g}{(I - I_g)} R_g \text{ or}$$

$$I_g = \frac{S}{S + R_g} I$$

Since, the deflection in the galvanometer is proportional to the current passing through it,

$$\theta = \frac{1}{G} I_g \Rightarrow \theta \propto I_g \Rightarrow \theta \propto I \text{ So,}$$

the deflection produced in the galvanometer is a measure of the current I passing through the circuit.

Shunt resistance is connected in parallel to galvanometer. Therefore, resistance of ammeter (R_a) can be determined by computing the effective resistance, which is

$$\frac{1}{R_{\text{eff}}} = \frac{1}{R_g} + \frac{1}{S} \Rightarrow R_{\text{eff}} = \frac{R_g S}{R_g + S} = R_a$$

Since, the shunt resistance is a very low resistance and the ratio $\frac{S}{R_g}$ is also small. This means, R_a is also small, i.e., the resistance offered by the ammeter is small. So, when we connect ammeter in series, the ammeter will not change appreciably the current in the circuit. For an ideal ammeter, the resistance must be equal to zero. But in reality, the reading in ammeter is always less than the actual current in the circuit. Let I_{ideal} be the current measured by ideal ammeter and I_{actual} be the actual current in the circuit. Then, the percentage error in measuring a current through an ammeter is

$$\frac{\Delta I}{I} \times 100\% = \frac{I_{\text{ideal}} - I_{\text{actual}}}{I_{\text{ideal}}} \times 100\%$$

Key points

1. An ammeter is a low resistance instrument and it is always connected in series to the circuit
2. An ideal ammeter has zero resistance

3. In order to increase the range of an ammeter n times, the value of shunt resistance to be connected in parallel is

$$S = \frac{R_g}{n-1}$$

Galvanometer to a voltmeter

A voltmeter is an instrument used to measure potential difference across any two points in the electrical circuits. It should not draw any current from the circuit otherwise the value of potential difference to be measured will change.

Voltmeter must have high resistance and when it is connected in parallel, it will not draw appreciable current so that it will indicate the true potential difference.

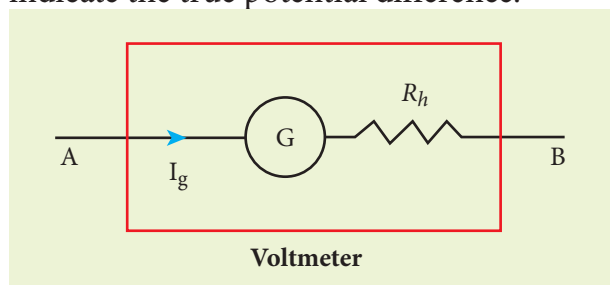


Figure 3.62 High resistance connected in series

A galvanometer is converted into a voltmeter by connecting high resistance R_h in series with galvanometer as shown in Figure 3.62. The scale is now calibrated in volt and the range of voltmeter depends on the values of the resistance R_h connected in series i.e. the value of resistance is so adjusted so that current I_g produces full scale deflection in the galvanometer.

Let R_g be the resistance of galvanometer and I_g be the current with which the galvanometer produces full scale deflection. Since the galvanometer is connected in series with high resistance, the current in the electrical circuit is same as the current passing through the galvanometer.

$$I = I_g$$

$$I = I_g \Rightarrow I_g = \frac{\text{potential difference}}{\text{total resistance}}$$

Since the galvanometer and high resistance are connected in series, the total resistance or effective resistance in the circuit is the sum of their resistances. This gives the resistance of voltmeter. Thus the voltmeter resistance is

$$R_v = R_g + R_h$$

Therefore,

$$I_g = \frac{V}{R_g + R_h}$$

$$\Rightarrow R_h = \frac{V}{I_g} - R_g$$

Note that $I_g \propto V$

The deflection in the galvanometer is proportional to current I_g . But current I_g is proportional to the potential difference. Hence the deflection in the galvanometer is a measure of potential difference. Since the resistance of voltmeter is very large, a voltmeter connected in parallel in an electrical circuit will draw least current in the circuit. An ideal voltmeter is one which has infinite resistance.

Key points

1. Voltmeter is a high resistance instrument and it is always connected in parallel with the circuit element across which the potential difference is to be measured.
2. An ideal voltmeter has infinite resistance
3. In order to increase the range of voltmeter n times the value of resistance to be connected in series with galvanometer is

$$R_h = (n-1) R_g$$

SUMMARY

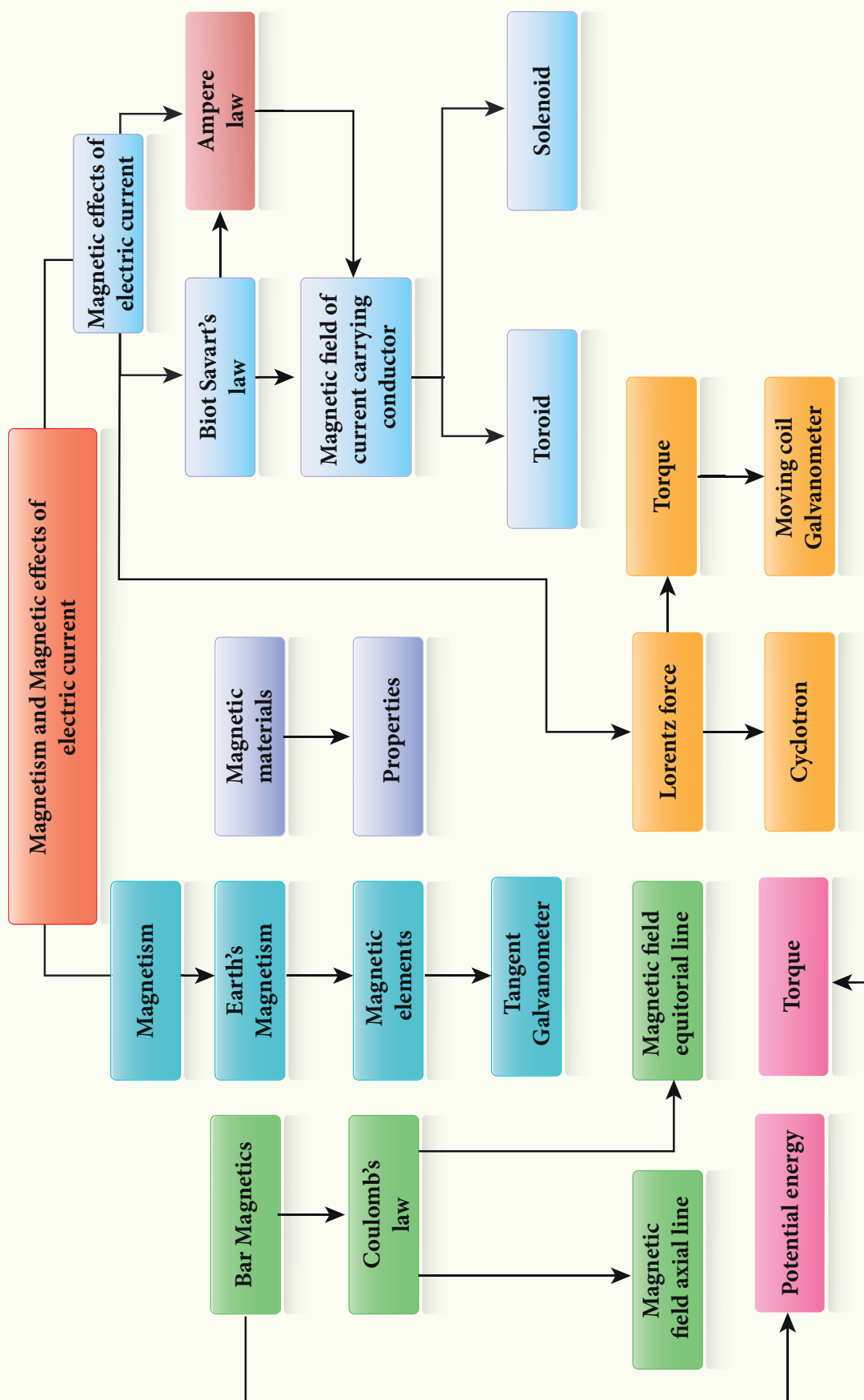
- A vertical plane passing through geographic axis is called geographic meridian.
- A vertical plane passing through magnetic axis is called magnetic meridian.
- The angle between magnetic meridian at a point with the geographic meridian is called the declination or magnetic declination.
- The angle subtended by the Earth's total magnetic field $\vec{B}$ with the horizontal direction in the magnetic meridian is called dip or magnetic inclination at that point.
- The magnetic moment is defined as the product of its pole strength and magnetic length. It is a vector quantity, denoted by $\vec{P}_m$.
- The region surrounding magnet where magnetic pole of strength unity experiences a force is known as magnetic field. It is a vector quantity and denoted by $\vec{B}$. Its unit is $\text{N A}^{-1} \text{m}^{-1}$.
- The number of magnetic field lines crossing normally through a given area is called magnetic flux Φ_B . It is a scalar quantity. In SI unit, magnetic flux Φ_B is weber, symbol Wb.
- Statement of Coulomb's law in magnetism "The force of attraction or repulsion between two magnetic poles is proportional to the product of their pole strengths and inversely proportional to the square of distance between them".
- Magnetic dipole kept in a uniform magnetic field experiences torque.
- Tangent galvanometer is a device used to measure very small currents. It is a moving magnet type galvanometer. Its working is based on tangent law.
- Tangent law is $B = B_H \tan \theta$.
- The magnetic field which is used to magnetize a sample or specimen is called the magnetising field. It is a vector quantity and denoted by $\vec{H}$ and its unit is A m^{-1} .
- The measure of ability of the material to allow the passage of magnetic lines of force through it is known as magnetic permeability.
- The net magnetic moment per unit volume of material is known as intensity of magnetisation.
- Magnetic susceptibility is defined as the ratio of the intensity of magnetisation ($\vec{I}$) induced in the material due to the magnetising field ($\vec{H}$).
- Magnetic materials are classified into three categories: diamagnetic, paramagnetic and ferromagnetic materials.
- The lagging of magnetic induction $\vec{B}$ behind the cyclic variation in magnetising field $\vec{H}$ is defined as "Hysteresis", which means "lagging behind".
- The right hand thumb rule "Assume that we hold the current carrying conductor in our right hand such that the thumb points in the direction of current flow, then the rest of the fingers encircling the wire point in the direction of the magnetic field lines produced".
- Maxwell right hand cork screw rule "If we rotate a screw by a screw driver, then the direction of current is same as the direction in which screw advances, and the direction of rotation of the screw will determine the direction of the magnetic field".



- Ampère's circuital law is $\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$.
- Magnetic field inside the solenoid is $B = \mu_0 nI$, where n is the number of turns per unit length.
- Magnetic field interior to the toroid is $B = \mu_0 nI$, where n is the number of turns per unit length.
- Lorentz force is $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$.
- Charged particle moving in a uniform magnetic field will undergo circular motion.
- Fleming's Left Hand Rule: Stretch forefinger, the middle finger and the thumb of the left hand such that they are in mutually perpendicular directions. If we keep the forefinger in the direction of magnetic field, the middle finger in the direction of the electric current, then the thumb points in the direction the force experienced by the conductors.
- One ampere is defined as that constant current when it is passed through each of the two infinitely long parallel straight conductors kept at a distance of one meter apart in vacuum causes each conductor to experience a force of 2×10^{-7} newton per metre length of the conductor.
- When a current carrying coil is placed in a uniform magnetic field, the net force on it is always zero but net torque is not zero. The magnitude of net torque is $\tau = NAB I \sin \theta$.
- Moving coil galvanometer is an instrument used for the detection and measurement of small currents.
- In moving coil galvanometer, current passing through the galvanometer is directly proportional to the deflection. Mathematically, $I = G\theta$, where $G = \frac{K}{NAB}$ is called galvanometer constant or current reduction factor of the galvanometer.
- Current sensitivity is defined as the deflection produced per unit current flowing through it, $I_s = \frac{\theta}{I} = \frac{NAB}{K} \Rightarrow I_s = \frac{1}{G}$.
- Voltage sensitivity is defined as the deflection produced per unit voltage which is applied across it, $V_s = \frac{\theta}{V} = \frac{1}{GR_g} = \frac{I_s}{R_g}$, where, R_g is the resistance of galvanometer.
- Ammeter is an instrument used to measure current in an electrical circuit.
- A galvanometer can be converted into an ammeter of given range by connecting a suitable low resistance S called shunt in parallel to the given galvanometer.
- An ideal ammeter has zero resistance.
- Voltmeter is an instrument used to measure potential difference across any element in an electrical circuit.
- A galvanometer can be converted into suitable voltmeter of given range by connecting a suitable high resistance R_h in series with the given galvanometer.
- An ideal voltmeter has infinite resistance.



CONCEPT MAP



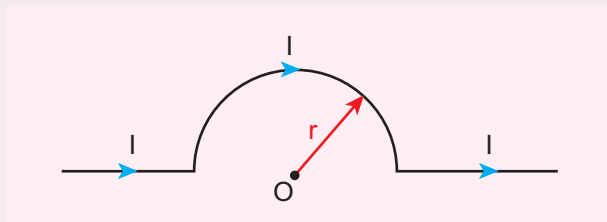


EVALUATION

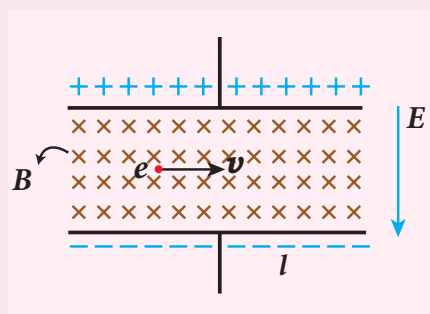


I Multiple choice questions

1. The magnetic field at the centre O of the following current loop is



- (a) $\frac{\mu_0 I}{4r} \otimes$ (b) $\frac{\mu_0 I}{4r} \odot$
(c) $\frac{\mu_0 I}{2r} \otimes$ (d) $\frac{\mu_0 I}{2r} \odot$
2. An electron moves in a straight line inside a charged parallel plate capacitor of uniform charge density σ . The time taken by the electron to cross the parallel plate capacitor undeflected when the plates of the capacitor are kept under constant magnetic field of induction ($\vec{B}$) is



- (a) $\epsilon_0 \frac{elB}{\sigma}$ (b) $\epsilon_0 \frac{lB}{\sigma l}$
(c) $\epsilon_0 \frac{lB}{e\sigma}$ (d) $\epsilon_0 \frac{lB}{\sigma}$
3. A particle having mass m and charge q accelerated through a potential difference V . Find the force experienced when it is kept under perpendicular magnetic field $\vec{B}$.

- (a) $\sqrt{\frac{2q^3 BV}{m}}$ (b) $\sqrt{\frac{q^3 B^2 V}{2m}}$
(c) $\sqrt{\frac{2q^3 B^2 V}{m}}$ (d) $\sqrt{\frac{2q^3 BV}{m^3}}$

4. A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is nearly
(a) 1.0 A m² (b) 1.2 A m²
(c) 0.5 A m² (d) 0.8 A m²
5. A thin insulated wire forms a plane spiral of $N = 100$ tight turns carrying a current $I = 8$ m A (milli ampere). The radii of inside and outside turns are $a = 50$ mm and $b = 100$ mm respectively. The magnetic induction at the centre of the spiral is
(a) 5 μ T (b) 7 μ T
(c) 8 μ T (d) 10 μ T

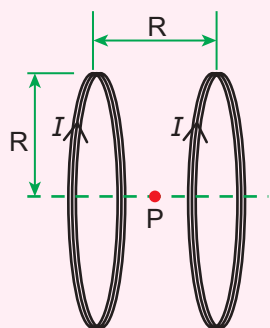
6. Three wires of equal lengths are bent in the form of loops. One of the loops is circle, another is a semi-circle and the third one is a square. They are placed in a uniform magnetic field and same electric current is passed through them. Which of the following loop configuration will experience greater torque?

- (a) Circle (b) Semi-circle
(c) Square (d) All of them

7. Two identical coils, each with N turns and radius R are placed coaxially at a distance R as shown in the figure. If I is the current passing through the loops in the same direction, then the



magnetic field at a point P at a distance of $R/2$ from the centre of each coil is



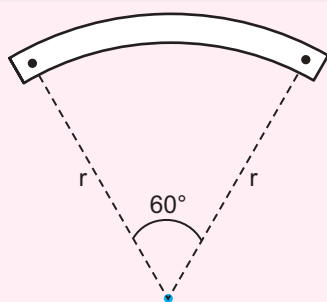
- (a) $\frac{8N\mu_0 I}{\sqrt{5}R}$ (b) $\frac{8N\mu_0 I}{5^{3/2}R}$
 (c) $\frac{8N\mu_0 I}{5R}$ (d) $\frac{4N\mu_0 I}{\sqrt{5}R}$

8. A wire of length l carrying a current I along the Y direction is kept in a magnetic field given by $\vec{B} = \frac{\beta}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})T$. The magnitude of Lorentz force acting on the wire is

- (a) $\sqrt{\frac{2}{3}}\beta Il$ (b) $\sqrt{\frac{1}{3}}\beta Il$
 (c) $\sqrt{2}\beta Il$ (d) $\sqrt{\frac{1}{2}}\beta Il$

9. A bar magnet of length l and magnetic moment p_m is bent in the form of an arc as shown in figure. The new magnetic dipole moment will be

(NEET 2013)

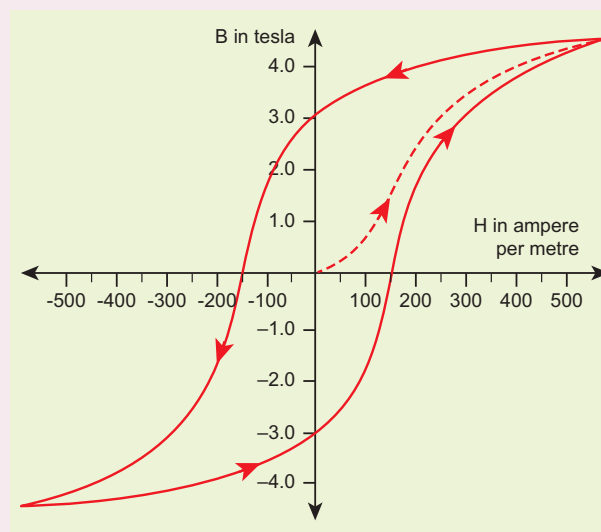


- (a) p_m (b) $\frac{3}{\pi}p_m$
 (c) $\frac{2}{\pi}p_m$ (d) $\frac{1}{2}p_m$

10. A non-conducting charged ring carrying a charge of q , mass m and radius r is rotated about its axis with constant angular speed ω . Find the ratio of its magnetic moment with angular momentum is

- (a) $\frac{q}{m}$ (b) $\frac{2q}{m}$
 (c) $\frac{q}{2m}$ (d) $\frac{q}{4m}$

11. The BH curve for a ferromagnetic material is shown in the figure. The material is placed inside a long solenoid which contains 1000 turns/cm. The current that should be passed in the solenoid to demagnetize the ferromagnet completely is



- (a) 1.00 mA (b) 1.25 mA
 (c) 1.50 mA (d) 1.75 mA



12. Two short bar magnets have magnetic moments 1.20 Am^2 and 1.00 Am^2 respectively. They are kept on a horizontal table parallel to each other with their north poles pointing towards south. They have a common magnetic equator and are separated by a distance of 20.0 cm . The value of the resultant horizontal magnetic induction at the mid-point O of the line joining their centres is (Horizontal components of Earth's magnetic induction is $3.6 \times 10^{-5} \text{ Wb m}^{-2}$)

(NSEP 2000-2001)

- (a) $3.60 \times 10^{-5} \text{ Wb m}^{-2}$
(b) $3.5 \times 10^{-5} \text{ Wb m}^{-2}$
(c) $2.56 \times 10^{-4} \text{ Wb m}^{-2}$
(d) $2.2 \times 10^{-4} \text{ Wb m}^{-2}$
13. The vertical component of Earth's magnetic field at a place is equal to the horizontal component. What is the value of angle of dip at this place?
(a) 30° (b) 45°
(c) 60° (d) 90°
14. A flat dielectric disc of radius R carries an excess charge on its surface. The surface charge density is σ . The disc rotates about an axis perpendicular to its plane passing through the centre with angular velocity ω . Find the magnitude of the torque on the disc if it is placed in a uniform magnetic field whose strength is B which is directed perpendicular to the axis of rotation
(a) $\frac{1}{4}\sigma\omega\pi BR$ (b) $\frac{1}{2}\sigma\omega\pi BR^2$
(c) $\frac{1}{4}\sigma\omega\pi BR^3$ (d) $\frac{1}{4}\sigma\omega\pi BR^4$

15. The potential energy of magnetic dipole whose dipole moment is $\vec{p}_m = (-0.5\hat{i} + 0.4\hat{j}) \text{ Am}^2$ kept in uniform magnetic field $\vec{B} = 0.2\hat{i} \text{ T}$
(a) -0.1 J (b) -0.8 J
(c) 0.1 J (d) 0.8 J

Answers

- 1) a 2) d 3) c 4) b 5) b
6) a 7) b 8) a 9) b 10) c
11) c 12) c 13) b 14) d 15) c

II Short answer questions:

1. What is magnetic field?
2. Define magnetic flux.
3. Define magnetic dipole moment.
4. State Coulomb's inverse law.
5. What is magnetic susceptibility?
6. State Biot-Savart's law.
7. What is magnetic permeability?
8. State Ampere's circuital law.
9. Compare dia, para and ferro-magnetism.
10. What is meant by hysteresis?
11. Define magnetic declination and inclination.
12. What is resonance condition in cyclotron?
13. Define ampere.
14. State Fleming's left hand rule.
15. Is an ammeter connected in series or parallel in a circuit? Why?
16. Explain the concept of velocity selector.
17. Why is the path of a charged particle not a circle when its velocity is not perpendicular to the magnetic field?
18. Give the properties of dia / para / ferromagnetic materials.

19. What happens to the domains in a ferromagnetic material in the presence of external magnetic field?
20. How is a galvanometer converted into (i) an ammeter and (ii) a voltmeter?

III Long answer questions

1. Discuss Earth's magnetic field in detail.
2. Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot-Savart law.
3. Obtain a relation for the magnetic field at a point along the axis of a circular coil carrying current using Biot-Savart law.
4. Compute the torque experienced by a magnetic needle in a uniform magnetic field.
5. Calculate the magnetic field at a point on the axial line of a bar magnet.
6. Obtain the magnetic field at a point on the equatorial line of a bar magnet.
7. Find the magnetic field due to a long straight conductor using Ampere's circuital law.
8. Discuss the working of cyclotron in detail.
9. What is tangent law? Discuss in detail.
10. Derive the expression for the torque on a current-carrying coil in a magnetic field.
11. Discuss the conversion of galvanometer into an ammeter and also a voltmeter.
12. Calculate the magnetic field inside and outside of the long solenoid using Ampere's circuital law.
13. Derive the expression for the force between two parallel, current-carrying conductors.
14. Give an account of magnetic Lorentz force.

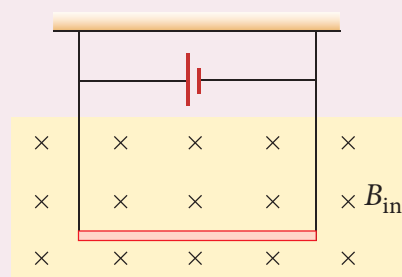
15. Compare the properties of soft and hard ferromagnetic materials.
16. Derive the expression for the force on a current-carrying conductor in a magnetic field.
17. Explain the principle and working of a moving coil galvanometer.

IV. Numerical problems

1. A bar magnet having a magnetic moment $\vec{p}_m$ is cut into four pieces i.e., first cut into two pieces along the axis of the magnet and each piece is further cut along the axis into two pieces. Compute the magnetic moment of each piece.

Answer $\vec{p}_{m_{new}} = \frac{1}{4} \vec{p}_m$

2. A conductor of linear mass density 0.2 g m^{-1} suspended by two flexible wire as shown in figure. Suppose the tension in the supporting wires is zero when it is kept inside the magnetic field of 1 T whose direction is into the page. Compute the current inside the conductor and also the direction of the current. Assume $g = 10 \text{ m s}^{-2}$



Answer 2 mA

3. A circular coil with cross-sectional area 0.1 cm^2 is kept in a uniform magnetic field of strength 0.2 T . If the current passing in the coil is 3 A and plane of the loop is perpendicular to the direction of magnetic field. Calculate



- (a) total torque on the coil
(b) total force on the coil
(c) average force on each electron in the coil due to the magnetic field. (The free electron density for the material of the wire is 10^{28} m^{-3}).

Answer (a) zero (b) zero (c) $0.6 \times 10^{-23} \text{ N}$

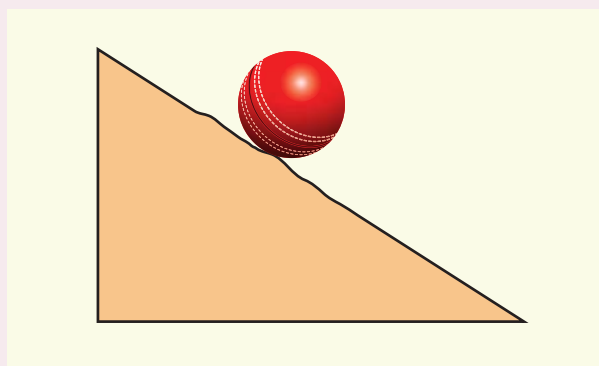
4. A bar magnet is placed in a uniform magnetic field whose strength is 0.8 T. If the bar magnet is oriented at an angle 30° with the external field experiences a torque of 0.2 Nm. Calculate:
(i) the magnetic moment of the magnet
(ii) the work done by the applied force in moving it from most stable configuration to the most unstable configuration and also compute the work done by the applied magnetic field in this case.

Answer (i) 0.5 A m^2 (ii) $W = 0.8 \text{ J}$ and $W_{\text{mag}} = -0.8 \text{ J}$

5. A non - conducting sphere has a mass of 100 g and radius 20 cm. A flat compact coil of wire with turns 5 is

wrapped tightly around it with each turns concentric with the sphere. This sphere is placed on an inclined plane such that plane of coil is parallel to the inclined plane. A uniform magnetic field of 0.5 T exists in the region in vertically upward direction. Compute the current I required to rest the sphere in equilibrium.

Answer $\frac{2}{\pi} \text{ A}$



6. Calculate the magnetic field at the centre of a square loop which carries a current of 1.5 A, length of each side being 50 cm. Answer $3.4 \times 10^{-6} \text{ T}$

BOOKS FOR REFERENCE

1. H. C. Verma, *Concepts of Physics – Volume 2*, Bharati Bhawan Publisher.
2. Halliday, Resnick and Walker, *Fundamentals of Physics*, Wiley Publishers, 10th edition.
3. Serway and Jewett, *Physics for scientist and engineers with modern physics*, Brook/Cooler publishers, Eighth edition.
4. David J. Griffiths, *Introduction to Electrodynamics*, Pearson publishers.
5. Rita John, *Solid State Physics (Magnetism chapter)*, McGraw Hill Education (India) Pvt. Ltd.
6. Paul Tipler and Gene Mosca, *Physics for scientist and engineers with modern physics*, Sixth edition, W.H. Freeman and Company.





ICT CORNER

Magnetism

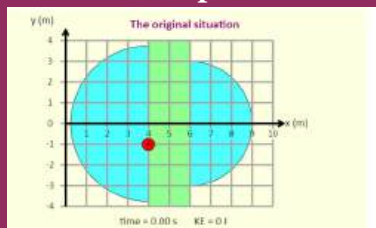
Topic: Cyclotron

In this activity you will be able to visualize and understand the working of cyclotron.

STEPS:

- Open the browser and type 'physics.bu.edu/~duffy/HTML5/cyclotron.html' in the address bar.
- Click 'play' to release the positively charged particle between the D-shaped sections.
- Observe trajectory of positively charged particle under the magnetic field between D-shaped sections.
- Note the kinetic energy of the particle after some time (say $t = 20$ s)

Step1



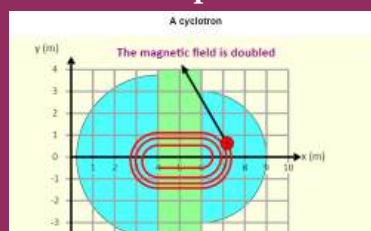
Step2



Step3



Step4



Double the electric and magnetic fields by clicking corresponding buttons and observe the change in kinetic energy for a particular given time t .

URL:

<http://physics.bu.edu/~duffy/HTML5/cyclotron.html>

- * Pictures are indicative only.
- * If browser requires, allow **Flash Player** or **Java Script** to load the page.

